Supplementary appendix

Examining the bidirectional relationship between sleep and physical activity using pooled individual data

Table of contents

Methods . . . 1
Data screening . . . 1
MVPA cut-points . . . 2
Protocol deviations . . . 2
Results . . . 3
Study characteristics . . . 3
Model diagnostics . . . 9
Fixed Effect Model . . . 21
Log Transforming PA Volume . . . 31
All Eligible Participants Regardless of Valid Wear Days . . . 38
Participants With at Least Seven Valid Days . . . 47
Pre-registered Model Without the Within–Between Split . . . 56
Cross-lagged Model . . . 61
Minutes of Moderate-to-Vigorous Physical Activity . . . 70
Negative Control for Sleep Regularity . . . 78
Moderation . . . 79
BMI z-score . . . 80
SES . . . 91
sex . . . 106
weekday . . . 119
season . . . 145
region . . . 162
daylight hours . . . 184
wear location . . . 195
most active hour . . . 208
ethnicity . . . 219

Methods

Data screening

We screened all data for implausibly low or high measurements, as these likely indicate measurement device failures. Screening had four stages. First, we excluded values falling outside fixed physiological limits. For average acceleration this was 200 mg, a value above any sustainable 24-hour average. Second, because a miscalibrated accelerometer corrupts every variable derived from acceleration magnitude rather than isolated days, we excluded all physical activity variables for files whose mean average acceleration exceeded that limit (76 participants). Third, we excluded remaining values more than 4 SD from the mean of their own study, rather than of the pooled sample, so that the threshold reflected implausible measurement rather than between-study differences in device, protocol, and population. Fourth, for participants under 18 years, we excluded heights, weights, and body mass index values that the CDC classifies as biologically implausible for the child's age and sex against its 2000 growth reference (height-for-age z <

−5 or > 4, weight-for-age z < −5 or > 8, and BMI-for-age z < −4 or > 8). Values excluded at any stage were treated as missing data.

MVPA cut-points

Minutes of moderate-to-vigorous physical activity (MVPA) were the waking minutes above the moderate cut-point. *sleepIPD* assigned cut-points by each study's age group, device, and wear location (Table 1), from Hildebrand and colleagues (2014) for moderate and vigorous activity and from Hurter and colleagues (2018) or Hildebrand and colleagues (2016) for light activity. The moderate cut-point differed more than two-fold across groups. Because cut-points were assigned by study, one ActiGraph study and one Axivity study were processed with GENEActiv cut-points, and one study applied child cut-points to participants up to 19 years old.

Table 1: Physical activity cut-points applied by *GGIR*, by age group, device, and wear location

Age group	Device	Wear location	Light (mg)	Moderate (mg)	Vigorous (mg)
Children	ActiGraph	Hip	32.6	142.6	464.6
Children	ActiGraph	Wrist	48.1	201.4	707.0
Children	GENEActiv	Wrist	51.6	191.6	695.8
Adults	ActiGraph	Wrist	44.8	100.6	428.8
Adults	GENEActiv	Wrist	45.8	93.2	418.3

Protocol deviations

We made the following changes to our protocol while working with the holdout data.

Using study ID as a fixed effect

We initially planned to use Study ID as a fixed effect, given that our study selection was not random. But, as we aimed for our findings to generalise to other studies and populations, we instead opted to include random intercepts for Study ID which can better model generalisable patterns. This approach allows us to account for variability between studies without fully partialling out study effects, thereby better modelling common phenomena. While this change lessened the strength of model intercepts, it did not significantly alter model slopes. The results using Study ID as a fixed effect (i.e., those that align with the protocol) are in the Fixed Effect Model section below and are qualitatively similar to the main results.

Sleep indicator variable

Our research examines various metrics of sleep: onset, duration, regularity, and efficiency. We initially planned to merge these variables into a single indicator to simplify analysis. However, no validated indicator existed which could be estimated from our measured variables. We decided against creating one as we could not validate it with our data. Consequently, our models illustrate the relationship between each aspect of sleep and physical activity, instead of describing a straightforward link between overall sleep and physical activity.

Log transformation

We aimed to improve model diagnostics using log transformations of skewed variables. Pilot analysis using holdout data revealed that while diagnostics improved on the log scale, model predictions followed extreme exponential patterns near the edge of the data range when back-transformed. Accordingly, we do not use log transformations in the main paper. These models are in the Log Transforming PA Volume section below.

Changes after co-author review

The protocol did not specify how to treat day-level predictors. After the full analysis, we split each predictor into the participant's mean and that day's deviation from it, so that day-to-day (within-person)

associations are not confounded by stable differences between people. The pre-registered models, which combine the two, are in the Pre-registered Model Without the Within–Between Split section below. We also added minutes of MVPA as a secondary exposure, and, because *GGIR*'s sleep regularity index for a day spans that day and the next, we used the index for the two preceding days in models of next-day activity.

Results

Study characteristics

Table 2 describes each study in the primary analytic sample, and Table 3 shows how each study's measure of socioeconomic status was harmonised.

Valid observations were not uniform across the days of the week ($\chi^2(6) = 343.32$, $p < .001$). Weekends and Fridays were over-represented (Saturday: $z = 11.24$; Sunday: $z = 7.84$; Friday: $z = 5.57$), while Wednesdays ($z = -9.30$) and Tuesdays ($z = -8.22$) were under-represented.

Table 2: Characteristics of the included studies (primary analytic sample)

Study ID	Study	Reference	Country	Device	Wear location	N	Age, mean (range)	Female, %	Valid days, median
100			12 countries	GT3X	Hip	6,360	10.4 (9-11)	54.5	7
101			Switzerland	GENEActiv	Wrist	2,928	61.9 (45-86)	54.0	13
102			Denmark	GT3X	Hip	691	10.0 (8-11)	48.2	7
103			New Zealand	Actigraph	Wrist	651	19.3 (8-58)	60.3	6
104			Brazil	GT3X+	Wrist	737	16.4 (14-22)	50.9	6
105			Czech Republic	GT3X+, GT9X Link, wGT3X-BT, wGT3X+	Wrist	1,170	14.0 (8-19)	54.5	6
106			Australia	GENEActiv	Wrist	301	42.0 (18-65)	72.4	8
107			Spain	GT3X+	Wrist	2,423	72.6 (64-94)	53.1	6
108			Australia	GENEActiv	Wrist	1,038	8.8 (7-11)	51.6	8
110			Finland	Actisleep	Wrist	801	62.6 (58-64)	84.8	7
111			United Kingdom	Axivity AX3	Wrist	692	44.7 (20-70)	72.8	6
112			Chile	GENEActiv	Wrist	144	9.5 (8-12)	38.9	6
113			Spain	Actisleep	Wrist	353	8.4 (8-9)	47.3	6
114			United States	WGT3X-BT	Wrist	154	3.5 (2-4)	56.5	5
115			Ecuador	wGT3X-BT	Hip	1,019	8.8 (7-13)	48.0	6
117			Spain	GENEActiv	Wrist	175	8.8 (4-11)	48.6	6
118			Australia	Actigraph	Wrist	304	16.6 (15-18)	51.0	8

Study ID	Study	Reference	Country	Device	Wear location	N	Age, mean (range)	Female, %	Valid days, median
212			Spain	WGT3X	Hip	285	4.7 (2-6)	46.5	7
221			United Kingdom	GENEActiv	Wrist	4,071	69.4 (59-82)	26.0	8
222			Brazil	GT3X	Wrist	2,939	3.8 (3-4)	49.4	6

Table 3: Harmonisation of socioeconomic status

Study ID	Original category	Harmonised category
100	1	Low
100	2	Medium
100	4	High
100	3	High
101	High school	Medium
101	Apprenticeship	Low
101	Mandatory education	Low
101	University education	High
102	Vocational basic training/ vocational internship and main course	Medium
102	Short higher education	High
102	Long higher education/ Research degree	High
102	Regular secondary education and vocational secondary education	Medium
102	Medium higher education	High
102	Primary school	Low
102	Unanswered	Missing
104	Incomplete elementary school	Low
104	Complete Bachelor degree	High
104	Incomplete bachelor degree	Medium
104	Complete high school	Medium
104	Complete elementary school	Low
104	Unknown	Missing
104	Has not studied	Low
105	vocational education	Medium
105	high school education	Medium
105	master's degree or higher	High
105	bachelor's degree or equivalent	High
105	10th grade or less	Low
105	short tertiary education	Medium
105	26-100 books	Medium
105	11-25 books	Low

Study ID	Original category	Harmonised category
105	>200 books	High
105	101-200 books	High
105	0-10 books	Low
106	High school or less	Medium
106	Technical or further education institutions (e.g., TAFE)	Medium
106	University degree	High
107	Secondary education 2nd stage (i.e., High shool graduate, complete 12th grade or equivalent	Medium
107	Incomplete primary education (i.e., less than 6th grade or equivalent)	Low
107	Complete primary education (i.e., complete 6th grade or equivalent)	Low
107	Medium university studies (i.e., university diplomature or equivalent)	High
107	Higher university studies (i.e., Bachelor degree, Master degree, Doctoral degree or equivalent)	High
107	Secondary education 1st stage (i.e., complete 10th grade or equivalent)	Medium
107	Illiterate (i.e., not able to read or write)	Low
108	Enough to fill two bookcases (101-200 Books)	High
108	Enough to fill one shelf (11-25 Books)	Low
108	Enough to fill 3 or more bookcases (more than 200 Books)	High
108	Enough to fill one bookcases (26-100 books)	Medium
108	None or very few (0-10 Books)	Low
110	lower-level non-manual occupation	Medium

Study ID	Original category	Harmonised category
110	higher-level non-manual occupation	High
110	manual occupation	Low
110	service occupation	Medium
111	10	Medium
113	primary school	Low
113	university degree	High
113	bachelor's degree	High
113	high school	Medium
113	master's degree or higher	High
113	vocational training	Medium
114	poor	Low
114	O level	Low
114	A/S level	Low
114	ba/bsc	Low
114	higher degree	Low
114	No academic quals	Low
115	Medium	Medium
115	High	High
115	Low	Low
117	secondary education	Medium
117	higher education	High
117	primary education	Low
118	2	Low
118	7	Medium
118	5	Medium
118	3	Low
118	10	High
118	8	High
118	9	High
118	6	Medium
118	4	Medium
118	1	Low
221	O level	Medium

Study ID	Original category	Harmonised category
221	A/S level	Medium
221	ba/bsc	High
221	higher degree	High
221	No academic quals	Low
222	q5-wealthiest	High
222	q4	Medium
222	q3	Medium
222	q2	Medium
222	q1-poorest	Low

Model diagnostics

The model diagnostics for the main analysis, and for all models presented in this document are in Table 4

Table 4: Model diagnostics

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Models moderated by age				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.67	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.36	100.00%	0.00%
Scale sleep regularity by PA volume	−0.70	2.07	100.00%	0.00%
Scale sleep duration by PA intensity	−0.05	2.15	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.40	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.10	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	5.01	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA volume by sleep regularity(lagged)	0.77	5.21	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.18	1.52	100.00%	0.00%
Models moderated by age fixedef				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.67	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.36	100.00%	0.00%
Scale sleep regularity by PA volume	−0.70	2.07	100.00%	0.00%
Scale sleep duration by PA intensity	−0.05	2.15	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.40	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.10	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	5.00	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.77	5.21	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.18	1.52	100.00%	0.00%
Models moderated by age log				
Scale sleep duration by log PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by log PA volume	−0.91	3.67	100.00%	0.00%
Scale sleep onset by log PA volume	0.16	9.40	100.00%	0.00%
Scale sleep regularity by log PA volume	−0.72	2.13	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep duration by PA intensity	−0.05	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.40	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.09	100.00%	0.00%
Log PA volume by sleep duration(lagged)	−0.46	2.16	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Log PA volume by sleep efficiency(lagged)	−0.46	2.18	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Log PA volume by sleep onset(lagged)	−0.46	2.18	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Log PA volume by sleep regularity(lagged)	−0.33	1.71	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.22	1.56	100.00%	0.00%
Models moderated by age allwear				
Scale sleep duration by PA volume	−0.05	2.16	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.92	3.74	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.35	100.00%	0.00%
Scale sleep regularity by PA volume	−0.70	2.07	100.00%	0.00%
Scale sleep duration by PA intensity	−0.05	2.15	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.92	3.73	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.39	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.10	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	5.00	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.65	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.77	5.22	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.18	1.52	100.00%	0.00%
Models moderated by age 7days				
Scale sleep duration by PA volume	0.01	2.79	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.83	3.00	100.00%	0.00%
Scale sleep onset by PA volume	0.09	12.13	100.00%	0.00%
Scale sleep regularity by PA volume	−0.78	2.61	100.00%	0.00%
Scale sleep duration by PA intensity	0.02	2.79	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.83	3.00	100.00%	0.00%
Scale sleep onset by PA intensity	0.10	12.22	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.80	2.65	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.54	5.11	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.23	1.54	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.53	5.16	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.52	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.53	5.11	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.52	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.61	5.92	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.13	1.28	100.00%	0.00%
Models moderated by bmi				
Scale sleep duration by PA volume	−0.12	2.48	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.92	3.84	100.00%	0.00%
Scale sleep onset by PA volume	0.12	11.82	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep regularity by PA volume	−0.70	2.09	100.00%	0.00%
Scale sleep duration by PA intensity	−0.12	2.47	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.92	3.83	100.00%	0.00%
Scale sleep onset by PA intensity	0.15	11.86	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.10	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	5.01	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.17	1.53	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.68	5.02	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.17	1.52	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.69	5.01	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.16	1.51	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.77	5.23	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.18	1.50	100.00%	0.00%
Models moderated by ses				
Scale sleep duration by PA volume	−0.03	2.15	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.90	3.56	100.00%	0.00%
Scale sleep onset by PA volume	0.15	9.61	100.00%	0.00%
Scale sleep regularity by PA volume	−0.70	2.18	100.00%	0.00%
Scale sleep duration by PA intensity	−0.03	2.12	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.90	3.55	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.60	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.18	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	5.01	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.26	1.67	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.02	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA intensity by sleep efficiency(lagged)	0.26	1.66	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	5.01	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.26	1.66	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.74	5.21	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.20	1.54	100.00%	0.00%
Models moderated by weekday				
Scale sleep duration by PA volume	−0.08	2.18	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.90	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.09	9.84	100.00%	0.00%
Scale sleep regularity by PA volume	−0.78	2.63	100.00%	0.00%
Scale sleep duration by PA intensity	−0.08	2.17	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.90	3.68	100.00%	0.00%
Scale sleep onset by PA intensity	0.13	9.75	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.80	2.71	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.73	5.08	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.31	1.73	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.73	5.10	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.30	1.69	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.73	5.08	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.32	1.75	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.82	5.28	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.24	1.61	100.00%	0.00%
Models moderated by season				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale sleep efficiency by PA volume	−0.91	3.70	100.00%	0.00%
Scale sleep onset by PA volume	0.17	9.35	100.00%	0.00%
Scale sleep regularity by PA volume	−0.70	2.07	100.00%	0.00%
Scale sleep duration by PA intensity	−0.06	2.15	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.38	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.71	2.08	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.97	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.23	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.77	5.20	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.18	1.53	100.00%	0.00%
Models moderated by region				
Scale sleep duration by PA volume	−0.06	2.18	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.70	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.40	100.00%	0.00%
Scale sleep regularity by PA volume	−0.70	2.08	100.00%	0.00%
Scale sleep duration by PA intensity	−0.06	2.15	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.38	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.71	2.07	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.69	4.98	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.68	4.96	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.78	5.21	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.18	1.52	100.00%	0.00%
Models moderated by daylight				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.70	100.00%	0.00%
Scale sleep onset by PA volume	0.17	9.35	100.00%	0.00%
Scale sleep regularity by PA volume	−0.70	2.07	100.00%	0.00%
Scale sleep duration by PA intensity	−0.06	2.15	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.38	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.08	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.77	5.20	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.18	1.52	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Models moderated by wear location				
Scale sleep duration by PA volume	−0.05	2.16	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.37	100.00%	0.00%
Scale sleep regularity by PA volume	−0.71	2.10	100.00%	0.00%
Scale sleep duration by PA intensity	−0.05	2.15	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.68	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.41	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.10	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.99	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.78	5.21	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.18	1.52	100.00%	0.00%
Models moderated by PA mostactivehr				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.18	9.35	100.00%	0.00%
Scale sleep regularity by PA volume	−0.70	2.07	100.00%	0.00%
Scale sleep duration by PA intensity	−0.06	2.15	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.20	9.38	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.08	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.67	4.99	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.66	5.01	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.23	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.77	5.20	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.18	1.52	100.00%	0.00%
Models moderated by sex				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.68	100.00%	0.00%
Scale sleep onset by PA volume	0.17	9.36	100.00%	0.00%
Scale sleep regularity by PA volume	−0.70	2.07	100.00%	0.00%
Scale sleep duration by PA intensity	−0.06	2.15	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.68	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.39	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.08	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.77	5.20	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA intensity by sleep regularity(lagged)	0.18	1.53	100.00%	0.00%
Models moderated by ethnicity				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA volume	0.17	9.38	100.00%	0.00%
Scale sleep regularity by PA volume	−0.70	2.07	100.00%	0.00%
Scale sleep duration by PA intensity	−0.05	2.15	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.69	100.00%	0.00%
Scale sleep onset by PA intensity	0.19	9.37	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.71	2.07	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.66	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	4.98	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.23	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.77	5.20	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.18	1.52	100.00%	0.00%
Models moderated by age blended				
Scale sleep duration by PA volume	−0.05	2.17	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.91	3.66	100.00%	0.00%
Scale sleep onset by PA volume	0.16	9.38	100.00%	0.00%
Scale sleep regularity by PA volume	−0.71	2.11	100.00%	0.00%
Scale sleep duration by PA intensity	−0.05	2.14	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.91	3.65	100.00%	0.00%
Scale sleep onset by PA intensity	0.18	9.40	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.72	2.09	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA volume by sleep duration(lagged)	0.68	4.99	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.24	1.65	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.67	5.00	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.24	1.64	100.00%	0.00%
Scale PA volume by sleep regularity(lagged)	0.76	5.14	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.22	1.56	100.00%	0.00%
Models moderated by age clpm				
Scale sleep duration by PA volume	−0.07	1.97	100.00%	0.00%
Scale sleep efficiency by PA volume	−0.79	2.87	100.00%	0.00%
Scale sleep onset by PA volume	0.26	7.74	100.00%	0.00%
Scale sleep regularity by PA volume	−0.69	2.94	100.00%	0.00%
Scale sleep duration by PA intensity	−0.07	1.94	100.00%	0.00%
Scale sleep efficiency by PA intensity	−0.80	2.87	100.00%	0.00%
Scale sleep onset by PA intensity	0.28	7.74	100.00%	0.00%
Scale sleep regularity by PA intensity	−0.70	3.02	100.00%	0.00%
Scale PA volume by sleep duration(lagged)	0.66	5.01	100.00%	0.00%
Scale PA intensity by sleep duration(lagged)	0.22	1.60	100.00%	0.00%
Scale PA volume by sleep efficiency(lagged)	0.64	5.02	100.00%	0.00%
Scale PA intensity by sleep efficiency(lagged)	0.22	1.59	100.00%	0.00%
Scale PA volume by sleep onset(lagged)	0.65	5.01	100.00%	0.00%
Scale PA intensity by sleep onset(lagged)	0.22	1.58	100.00%	0.00%

Model name	Skewness	Kurtosis	Converged (%)	Singular (%)
Scale PA volume by sleep regularity(lagged)	0.73	5.26	100.00%	0.00%
Scale PA intensity by sleep regularity(lagged)	0.16	1.46	100.00%	0.00%
Models moderated by age mvPA				
Scale sleep duration by PA mvPA	−0.07	2.14	100.00%	0.00%
Scale sleep efficiency by PA mvPA	−0.91	3.67	100.00%	0.00%
Scale sleep onset by PA mvPA	0.20	9.43	100.00%	0.00%
Scale sleep regularity by PA mvPA	−0.70	2.07	100.00%	0.00%
Scale PA mvPA by sleep duration(lagged)	0.65	4.73	100.00%	0.00%
Scale PA mvPA by sleep efficiency(lagged)	0.63	4.82	100.00%	0.00%
Scale PA mvPA by sleep onset(lagged)	0.65	4.69	100.00%	0.00%
Scale PA mvPA by sleep regularity(lagged)	0.77	2.79	100.00%	0.00%

Fixed Effect Model

In this document we show the results using another analytic technique. Namely treating study ID as a fixed effect. This is what we wrote in the protocol originally.

The association between physical activity volume and sleep

We estimated the associations between physical activity and sleep (RQ1) using fixed-effects models. We estimated the association between physical activity volume and sleep by age, and the results are presented in Table 5 and Figure 1.

Table 5: Physical activity predicting sleep controlling for SES, sex, BMI z-score, and the fixed effects of study IDs.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.68 [0.64, 0.73]	< .001	0.57 [0.52, 0.61]	< .001
Physical activity (within)	0.10 [0.09, 0.11]	< .001	0.06 [0.05, 0.07]	< .001
Age	−0.04 [−0.05, −0.03]	< .001	−0.02 [−0.04, −0.01]	.002
Physical activity (within)²	0.01 [0.00, 0.03]	.113	0.00 [−0.01, 0.01]	.910
Physical activity (between)	0.07 [0.05, 0.09]	< .001	0.11 [0.09, 0.13]	< .001
Physical activity (between)²	0.00 [−0.01, 0.02]	.925	−0.02 [−0.03, 0.00]	.031
Physical activity (within) × Age	0.00 [−0.01, 0.00]	.260	−0.01 [−0.01, 0.00]	< .001
Age × Physical activity (within)²	0.01 [0.00, 0.01]	.002	0.00 [−0.01, 0.00]	.012
Age × Physical activity (between)	−0.04 [−0.05, −0.03]	< .001	−0.03 [−0.04, −0.02]	< .001
Age × Physical activity (between)²	0.01 [0.00, 0.01]	.003	0.00 [0.00, 0.00]	.706
Sleep efficiency (z)				
(Intercept)	−0.44 [−0.49, −0.39]	< .001	−0.56 [−0.61, −0.51]	< .001
Physical activity (within)	−0.01 [−0.02, 0.00]	.165	0.00 [−0.01, 0.01]	.987
Age	0.01 [−0.01, 0.02]	.473	−0.01 [−0.02, 0.01]	.391
Physical activity (within)²	−0.03 [−0.04, −0.02]	< .001	−0.03 [−0.04, −0.02]	< .001
Physical activity (between)	0.10 [0.08, 0.12]	< .001	0.07 [0.05, 0.09]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (between)²	−0.04 [−0.06, −0.03]	< .001	−0.05 [−0.07, −0.03]	< .001
Physical activity (within) × Age	−0.02 [−0.02, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Physical activity (within)²	−0.01 [−0.01, 0.00]	.019	0.01 [0.00, 0.01]	< .001
Age × Physical activity (between)	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.02, −0.01]	< .001
Age × Physical activity (between)²	0.00 [0.00, 0.01]	.649	0.01 [0.01, 0.01]	< .001
Sleep onset (z)				
(Intercept)	−1.11 [−1.15, −1.08]	< .001	−0.99 [−1.03, −0.96]	< .001
Physical activity (within)	−0.13 [−0.13, −0.12]	< .001	−0.09 [−0.09, −0.08]	< .001
Age	0.00 [−0.01, 0.02]	.510	0.01 [−0.01, 0.02]	.267
Physical activity (within)²	0.00 [−0.01, 0.01]	.963	0.00 [−0.01, 0.01]	.861
Physical activity (between)	−0.10 [−0.12, −0.09]	< .001	−0.07 [−0.09, −0.05]	< .001
Physical activity (between)²	−0.02 [−0.03, 0.00]	.025	−0.02 [−0.03, 0.00]	.021
Physical activity (within) × Age	−0.01 [−0.01, −0.01]	< .001	0.00 [0.00, 0.00]	.179
Age × Physical activity (within)²	−0.01 [−0.01, 0.00]	.005	0.01 [0.01, 0.01]	< .001
Age × Physical activity (between)	0.00 [0.00, 0.01]	.359	0.02 [0.01, 0.03]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Physical activity (between)²	0.00 [−0.01, 0.00]	.044	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.12 [0.06, 0.18]	< .001	−0.44 [−0.50, −0.38]	< .001
Physical activity (within)	0.24 [0.23, 0.26]	< .001	0.09 [0.08, 0.09]	< .001
Age	0.04 [0.02, 0.06]	< .001	−0.07 [−0.10, −0.05]	< .001
Physical activity (within)²	−0.31 [−0.33, −0.28]	< .001	−0.13 [−0.14, −0.12]	< .001
Physical activity (between)	0.44 [0.41, 0.47]	< .001	0.15 [0.13, 0.18]	< .001
Physical activity (between)²	−0.19 [−0.21, −0.16]	< .001	−0.13 [−0.15, −0.11]	< .001
Physical activity (within) × Age	0.03 [0.02, 0.03]	< .001	−0.02 [−0.02, −0.02]	< .001
Age × Physical activity (within)²	−0.07 [−0.08, −0.06]	< .001	0.03 [0.03, 0.03]	< .001
Age × Physical activity (between)	0.01 [0.00, 0.02]	.024	−0.07 [−0.08, −0.06]	< .001
Age × Physical activity (between)²	−0.04 [−0.05, −0.03]	< .001	0.00 [0.00, 0.01]	.075

Adjusted for SES, sex, BMI z-score, and the fixed effects of study IDs. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days). Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

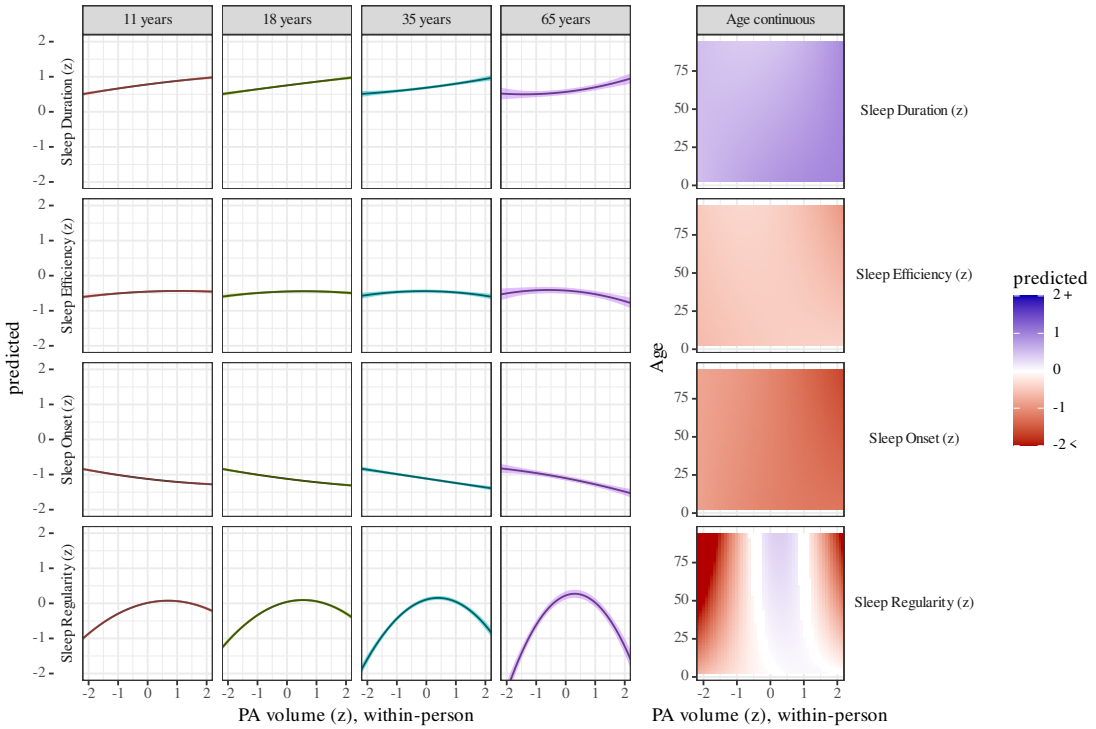


Figure 1: Sleep metrics on Physical activity volume

The association between physical activity intensity and sleep

We estimated how physical activity intensity is associated with sleep across different age groups. We present the results controlling for SES, sex, BMI z-score, and the fixed effects of study ID, in Table 5 and Figure 2.

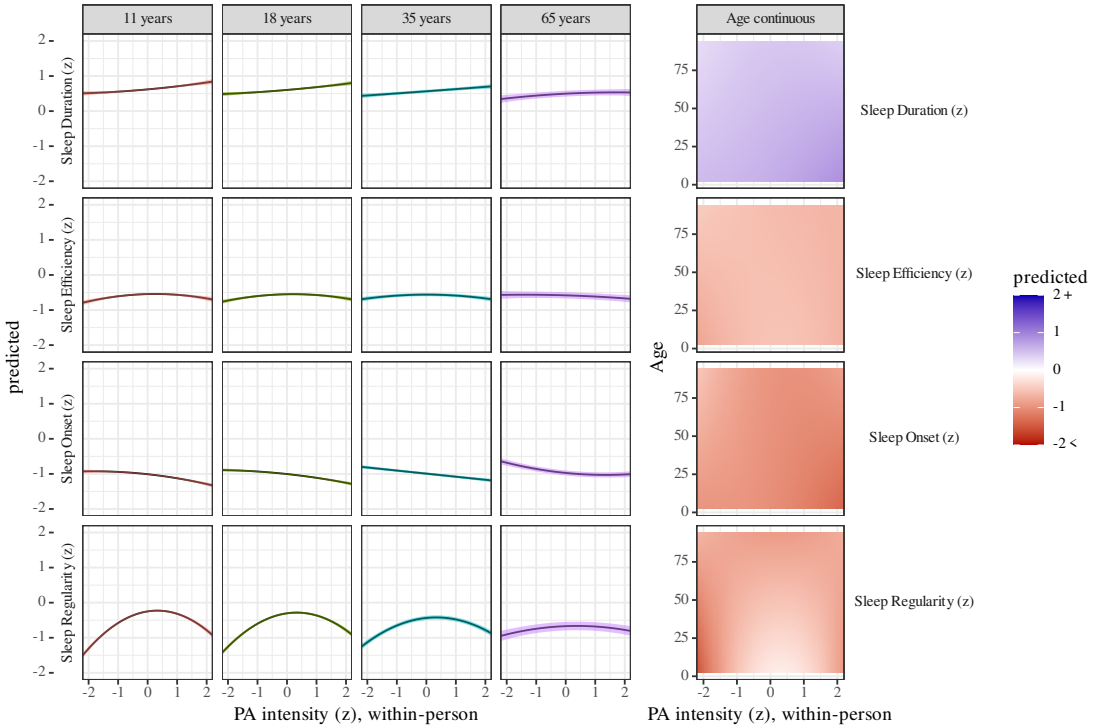


Figure 2: Sleep metrics on Physical activity intensity

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 6 and Figure 3.

Table 6: Sleep predicting physical activity controlling for SES, sex, BMI z-score, and the fixed effects of study IDs

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−1.14 [−1.18, −1.10]	< .001	−0.45 [−0.48, −0.41]	< .001
Sleep duration (within, prior)	0.01 [0.01, 0.02]	< .001	0.02 [0.01, 0.02]	< .001
Age	−0.23 [−0.24, −0.22]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep duration (within, prior)²	−0.04 [−0.05, −0.04]	< .001	−0.02 [−0.03, −0.02]	< .001
Sleep duration (between)	0.07 [0.06, 0.08]	< .001	0.06 [0.05, 0.07]	< .001
Sleep duration (between)²	−0.05 [−0.06, −0.04]	< .001	−0.01 [−0.02, 0.00]	.013
Sleep duration (within, prior) × Age	−0.01 [−0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.010
Age × Sleep duration (within, prior)²	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	< .001
Age × Sleep duration (between)	−0.03 [−0.04, −0.03]	< .001	−0.01 [−0.02, −0.01]	< .001
Age × Sleep duration (between)²	0.00 [0.00, 0.00]	.880	−0.01 [−0.01, −0.01]	< .001
Sleep efficiency (z)				
(Intercept)	−1.08 [−1.12, −1.05]	< .001	−0.38 [−0.41, −0.35]	< .001
Sleep efficiency (within, prior)	0.01 [0.00, 0.01]	.003	0.01 [0.00, 0.01]	.003
Age	−0.23 [−0.24, −0.22]	< .001	−0.28 [−0.29, −0.26]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency (within, prior)²	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (between)	−0.02 [−0.03, 0.00]	.007	0.01 [−0.01, 0.02]	.274
Sleep efficiency (between)²	−0.03 [−0.04, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (within, prior) × Age	0.00 [0.00, 0.00]	.903	0.00 [−0.01, 0.00]	< .001
Age × Sleep efficiency (within, prior)²	0.00 [0.00, 0.00]	.070	0.00 [0.00, 0.00]	.326
Age × Sleep efficiency (between)	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.015
Age × Sleep efficiency (between)²	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.01]	.111
Sleep onset (z)				
(Intercept)	−1.27 [−1.32, −1.23]	< .001	−0.54 [−0.58, −0.50]	< .001
Sleep onset (within, prior)	0.00 [−0.01, 0.00]	.122	0.00 [0.00, 0.01]	.202
Age	−0.23 [−0.24, −0.21]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep onset (within, prior)²	−0.03 [−0.03, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep onset (between)	−0.11 [−0.12, −0.09]	< .001	−0.07 [−0.08, −0.06]	< .001
Sleep onset (between)²	−0.01 [−0.01, 0.00]	.164	0.01 [0.00, 0.01]	.092
Sleep onset (within, prior) × Age	0.00 [−0.01, 0.00]	< .001	0.01 [0.00, 0.01]	< .001
Age × Sleep onset (within, prior)²	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	.001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Sleep onset (between)	0.03 [0.02, 0.03]	< .001	0.02 [0.01, 0.02]	< .001
Age × Sleep onset (between)²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep regularity (z)				
(Intercept)	−1.08 [−1.13, −1.04]	< .001	−0.53 [−0.58, −0.49]	< .001
Sleep regularity (within, prior)	−0.01 [−0.01, 0.00]	.067	0.01 [0.00, 0.01]	.053
Age	−0.23 [−0.25, −0.21]	< .001	−0.33 [−0.34, −0.31]	< .001
Sleep regularity (within, prior)²	0.00 [−0.01, 0.01]	.641	0.00 [−0.01, 0.00]	.250
Sleep regularity (between)	0.17 [0.16, 0.18]	< .001	0.07 [0.06, 0.08]	< .001
Sleep regularity (between)²	0.00 [−0.01, 0.01]	.622	−0.02 [−0.03, −0.01]	< .001
Sleep regularity (within, prior) × Age	0.00 [0.00, 0.00]	.878	0.00 [−0.01, 0.00]	.002
Age × Sleep regularity (within, prior)²	0.00 [0.00, 0.00]	.209	0.00 [−0.01, 0.00]	< .001
Age × Sleep regularity (between)	0.00 [0.00, 0.01]	.102	0.00 [−0.01, 0.00]	.279
Age × Sleep regularity (between)²	0.00 [0.00, 0.00]	.387	0.00 [−0.01, 0.00]	.004

Adjusted for SES, sex, BMI z-score, and the fixed effects of study IDs. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days). Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

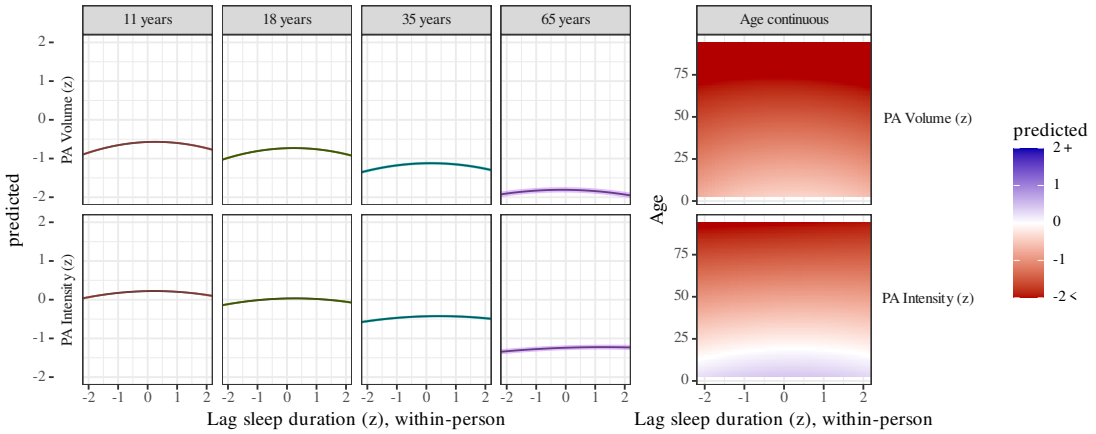


Figure 3: Physical activity by sleep duration

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 6 and Figure 4.

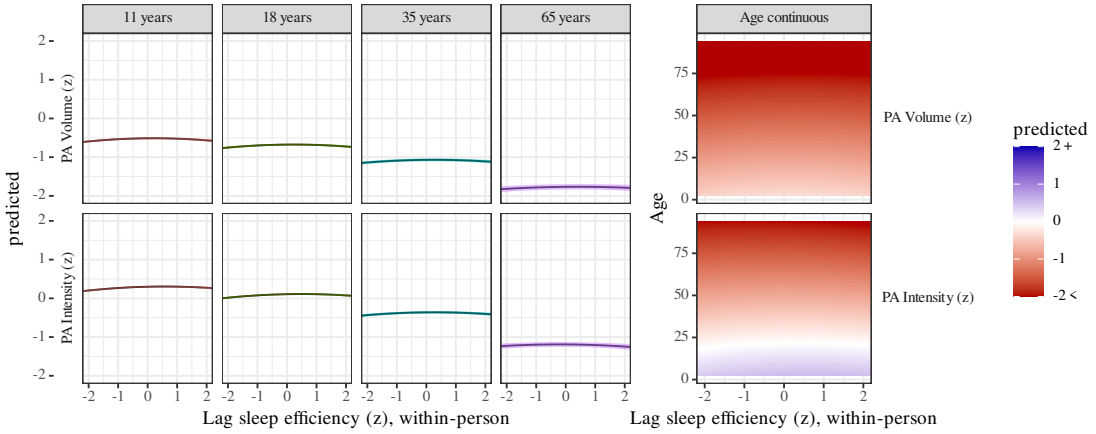


Figure 4: Physical activity by sleep efficiency

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 6 and Figure 5.

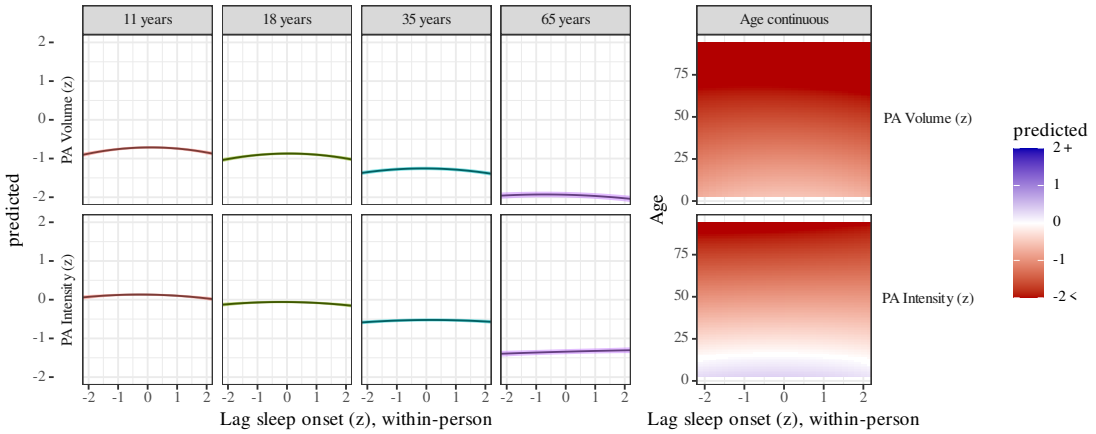


Figure 5: Physical activity by sleep onset

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by age. Results, controlling for SES, sex, BMI z-score, and the fixed effects of study ID are presented in Table 6 and Figure 6.

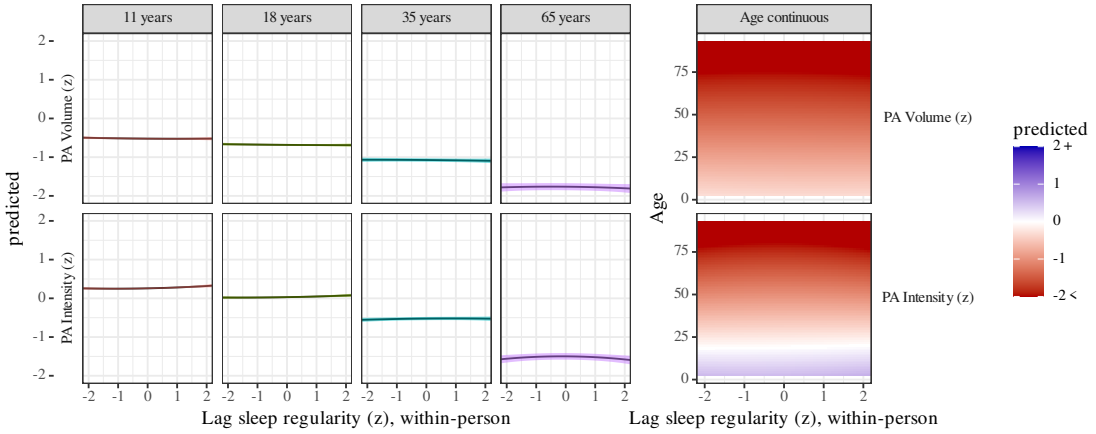


Figure 6: Physical activity by sleep regularity

Log Transforming PA Volume

Based on preliminary modelling on a subset of the data, we chose to leave physical activity volume in its natural form to enhance interpretability. The log scale meant that the variable became uninterpretable towards the edge of its range. Below, we present the results had we not made this decision. These results are all moderated by age, as this was our primary analysis, and use the pre-registered specification without the within–between split. The associated figures are back-transformed to the scale range.

The association between physical activity volume and sleep

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by age, and the results are presented in Table 7 and Figure 7.

Table 7: Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.17 [−0.08, 0.42]	.185	0.00 [−0.15, 0.14]	.995
Log pa volume	−0.27 [−0.38, −0.16]	< .001	0.05 [0.05, 0.06]	< .001
Age	−0.06 [−0.13, 0.01]	.081	−0.02 [−0.03, −0.01]	.003
Log pa volume²	0.06 [0.04, 0.08]	< .001	0.00 [0.00, 0.01]	.612
Log pa volume × Age	0.05 [0.01, 0.09]	.016	−0.01 [−0.01, −0.01]	< .001
Age × Log pa volume²	−0.01 [−0.02, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	−1.12 [−1.37, −0.87]	< .001	0.15 [−0.02, 0.32]	.086
Log pa volume	0.67 [0.56, 0.78]	< .001	0.02 [0.02, 0.03]	< .001
Age	0.22 [0.15, 0.29]	< .001	−0.01 [−0.03, 0.00]	.067
Log pa volume²	−0.09 [−0.10, −0.07]	< .001	−0.02 [−0.02, −0.01]	< .001
Log pa volume × Age	−0.09 [−0.13, −0.06]	< .001	−0.01 [−0.02, −0.01]	< .001
Age × Log pa volume²	0.01 [0.00, 0.01]	.003	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	−0.37 [−0.68, −0.06]	.018	0.12 [−0.14, 0.37]	.379
Log pa volume	0.47 [0.39, 0.56]	< .001	−0.06 [−0.06, −0.05]	< .001
Age	0.31 [0.26, 0.37]	< .001	0.00 [−0.02, 0.01]	.524
Log pa volume²	−0.09 [−0.10, −0.08]	< .001	−0.01 [−0.02, −0.01]	< .001
Log pa volume × Age	−0.17 [−0.20, −0.14]	< .001	0.01 [0.00, 0.01]	< .001

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Log pa volume²	0.02 [0.02, 0.03]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	−4.20 [−4.48, −3.93]	< .001	0.06 [−0.14, 0.25]	.569
Log pa volume	1.98 [1.84, 2.12]	< .001	0.14 [0.13, 0.15]	< .001
Age	−0.39 [−0.47, −0.30]	< .001	−0.08 [−0.10, −0.06]	< .001
Log pa volume²	−0.22 [−0.24, −0.20]	< .001	−0.09 [−0.09, −0.08]	< .001
Log pa volume × Age	0.22 [0.17, 0.27]	< .001	−0.05 [−0.05, −0.05]	< .001
Age × Log pa volume²	−0.03 [−0.04, −0.02]	< .001	0.01 [0.01, 0.01]	< .001

Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised and (ln) natural-log transformed; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

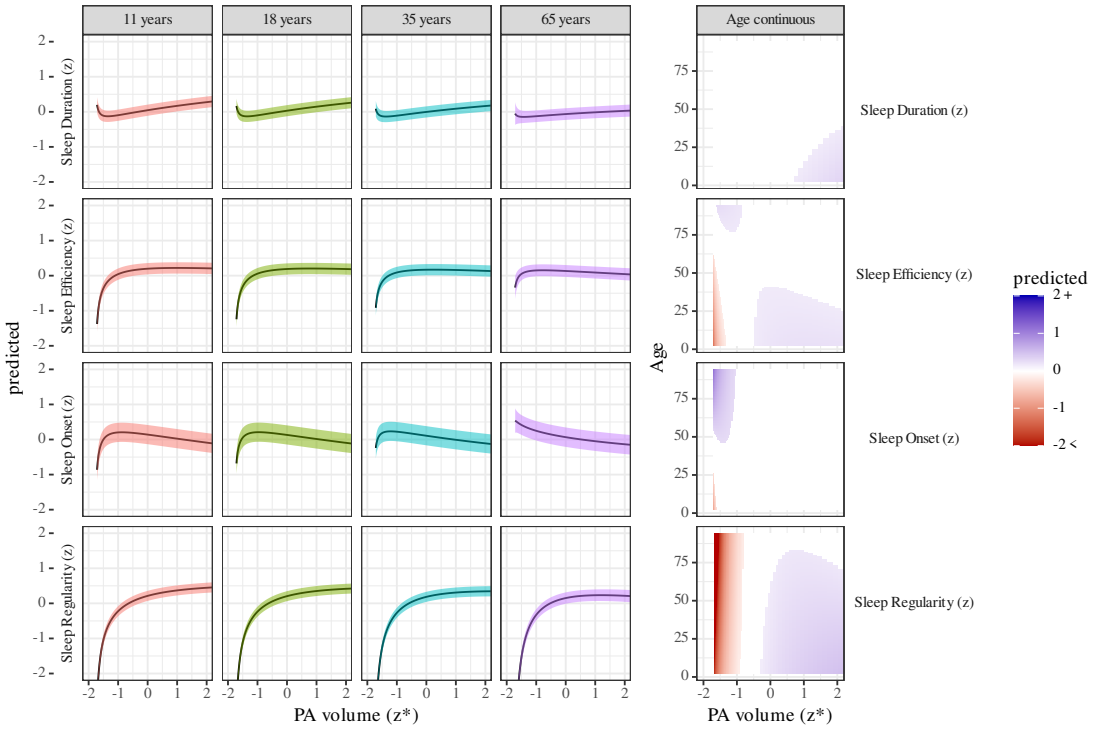


Figure 7: Sleep metrics on Physical activity volume

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 8 and Figure 8.

Table 8: Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	3.66 [3.51, 3.80]	< .001	−0.07 [−0.19, 0.05]	.254
Sleep duration	0.01 [0.00, 0.01]	< .001	0.03 [0.02, 0.03]	< .001
Age	−0.13 [−0.14, −0.12]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep duration²	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep duration × Age	−0.01 [−0.01, −0.01]	< .001	0.00 [0.00, 0.00]	.051
Age × Sleep duration²	0.00 [0.00, 0.00]	.992	0.00 [0.00, 0.00]	.326
Sleep efficiency (z)				
(Intercept)	3.65 [3.51, 3.79]	< .001	−0.07 [−0.20, 0.05]	.242
Sleep efficiency	0.01 [0.00, 0.01]	< .001	0.01 [0.00, 0.01]	.009
Age	−0.13 [−0.14, −0.12]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep efficiency²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.00]	.010	0.00 [0.00, 0.00]	.200
Age × Sleep efficiency²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.227
Sleep onset (z)				
(Intercept)	3.66 [3.51, 3.80]	< .001	−0.08 [−0.21, 0.04]	.202
Sleep onset	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.02, −0.01]	< .001
Age	−0.13 [−0.14, −0.12]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep onset²	−0.01 [−0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.811
Sleep onset × Age	0.00 [0.00, 0.00]	.001	0.01 [0.01, 0.01]	< .001

Term	PA volume (ln) β [95% CI]	PA volume (ln) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Sleep onset²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.020
Sleep regularity (z)				
(Intercept)	3.61 [3.47, 3.75]	< .001	−0.18 [−0.30, −0.05]	.006
Sleep regularity	0.08 [0.08, 0.09]	< .001	0.06 [0.06, 0.07]	< .001
Age	−0.13 [−0.14, −0.13]	< .001	−0.32 [−0.34, −0.31]	< .001
Sleep regularity²	−0.01 [−0.01, −0.01]	< .001	0.00 [0.00, 0.00]	.414
Sleep regularity × Age	0.00 [0.00, 0.00]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Sleep regularity²	0.00 [0.00, 0.00]	.481	0.00 [0.00, 0.00]	.023

Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised and (ln) natural-log transformed; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

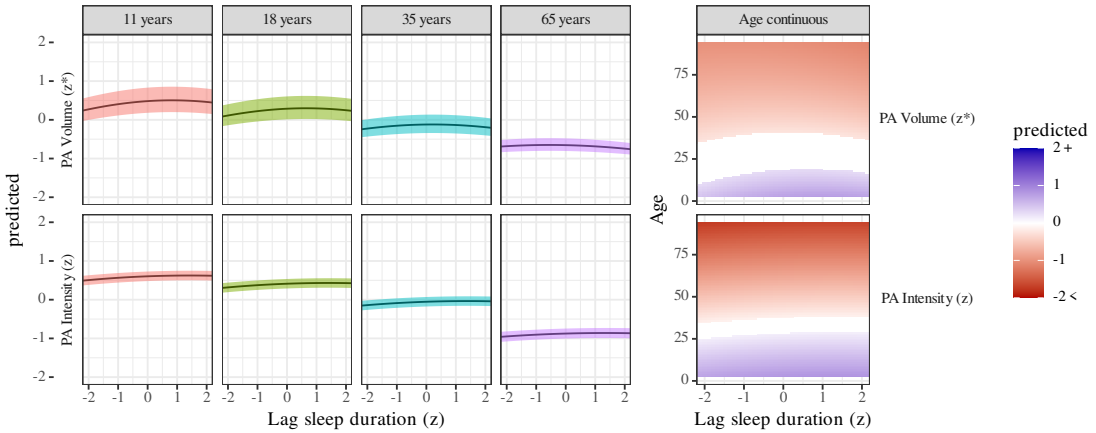


Figure 8: Physical activity by sleep duration

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 8 and Figure 9.

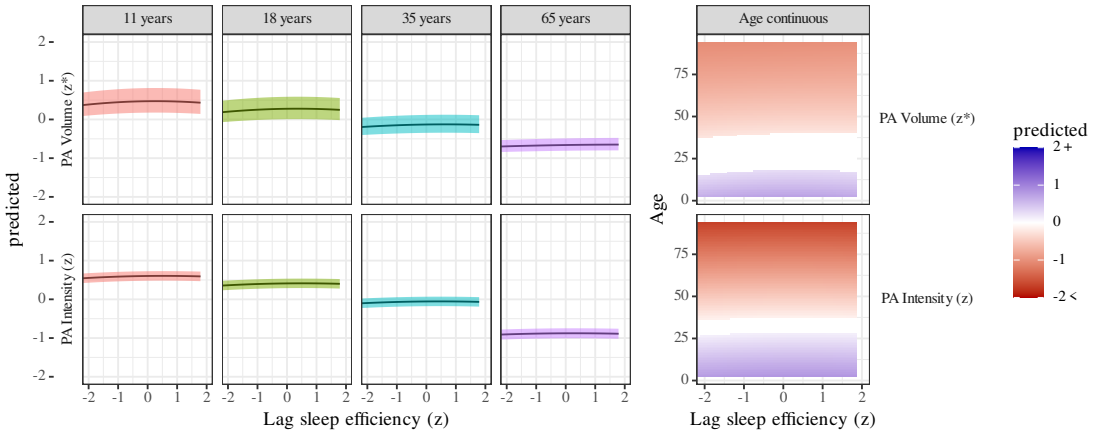


Figure 9: Physical activity by sleep efficiency

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 8 and Figure 10.

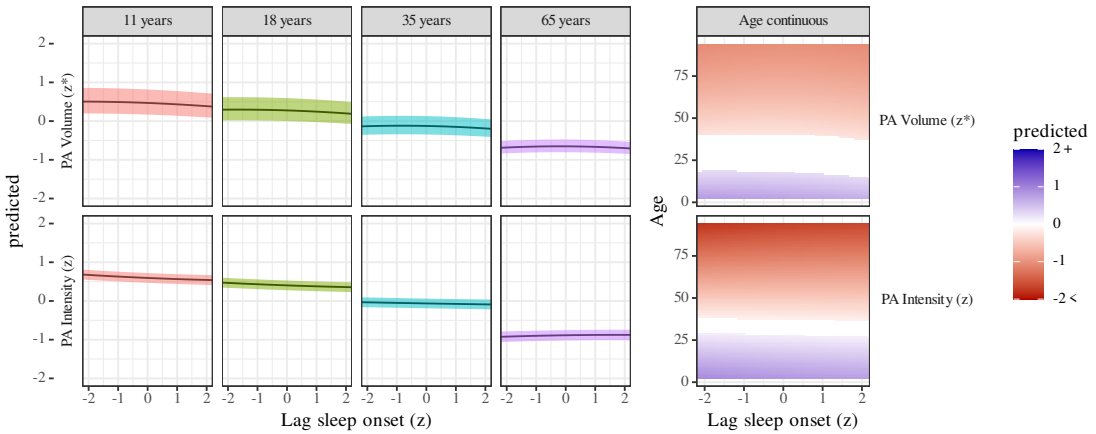


Figure 10: Physical activity by sleep onset

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by age. Results, controlling for SES, sex, and BMI z-score are presented in Table 8 and Figure 11.

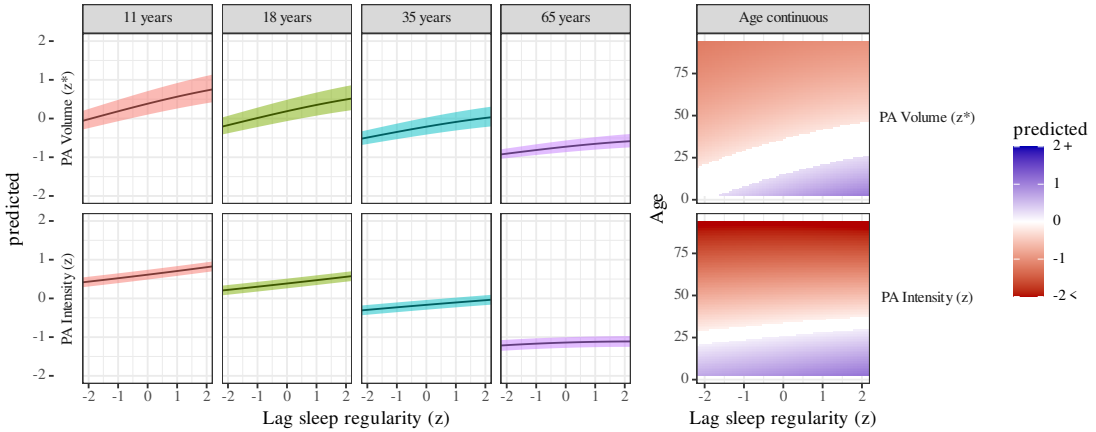


Figure 11: Physical activity by sleep regularity

All Eligible Participants Regardless of Valid Wear Days

The primary analysis required at least 4 days of valid wear time. Here we refit the primary models to every eligible participant, regardless of how many valid days they contributed. Both analyses share the same imputations and standardisation constants. Results, controlling for SES, sex, and BMI z-score, are presented in Tables 9 and 10 and Figures 12 to 17.

Table 9: Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.05 [−0.21, 0.11]	.549	−0.04 [−0.18, 0.10]	.558
Physical activity (within)	0.10 [0.09, 0.11]	< .001	0.06 [0.05, 0.07]	< .001
Age	−0.04 [−0.06, −0.03]	< .001	−0.03 [−0.04, −0.01]	< .001
Physical activity (within)²	0.01 [0.00, 0.03]	.099	0.00 [−0.01, 0.01]	.875
Physical activity (between)	0.07 [0.05, 0.09]	< .001	0.11 [0.09, 0.13]	< .001
Physical activity (between)²	0.00 [−0.01, 0.02]	.888	−0.02 [−0.04, −0.01]	.007
Physical activity (within) × Age	0.00 [−0.01, 0.00]	.283	−0.01 [−0.01, 0.00]	< .001
Age × Physical activity (within)²	0.01 [0.00, 0.01]	.002	0.00 [−0.01, 0.00]	.011
Age × Physical activity (between)	−0.04 [−0.05, −0.04]	< .001	−0.03 [−0.04, −0.02]	< .001
Age × Physical activity (between)²	0.01 [0.00, 0.01]	.001	0.00 [0.00, 0.00]	.828
Sleep efficiency (z)				
(Intercept)	0.18 [0.03, 0.33]	.017	0.17 [0.00, 0.34]	.048
Physical activity (within)	−0.01 [−0.02, 0.00]	.144	0.00 [−0.01, 0.01]	.898
Age	0.01 [0.00, 0.03]	.112	0.00 [−0.02, 0.01]	.638
Physical activity (within)²	−0.03 [−0.04, −0.02]	< .001	−0.03 [−0.04, −0.02]	< .001
Physical activity (between)	0.10 [0.08, 0.12]	< .001	0.07 [0.05, 0.10]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (between)²	−0.05 [−0.06, −0.03]	< .001	−0.06 [−0.08, −0.04]	< .001
Physical activity (within) × Age	−0.02 [−0.02, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Physical activity (within)²	−0.01 [−0.01, 0.00]	.020	0.01 [0.00, 0.01]	< .001
Age × Physical activity (between)	−0.01 [−0.02, −0.01]	< .001	−0.02 [−0.02, −0.01]	< .001
Age × Physical activity (between)²	0.00 [−0.01, 0.01]	.885	0.01 [0.01, 0.01]	< .001
Sleep onset (z)				
(Intercept)	0.13 [−0.15, 0.40]	.366	0.16 [−0.10, 0.41]	.227
Physical activity (within)	−0.13 [−0.13, −0.12]	< .001	−0.09 [−0.09, −0.08]	< .001
Age	0.01 [−0.01, 0.02]	.261	0.01 [0.00, 0.02]	.213
Physical activity (within)²	0.00 [−0.01, 0.01]	.948	0.00 [−0.01, 0.01]	.781
Physical activity (between)	−0.10 [−0.12, −0.09]	< .001	−0.06 [−0.08, −0.05]	< .001
Physical activity (between)²	−0.02 [−0.03, 0.00]	.010	−0.02 [−0.04, −0.01]	< .001
Physical activity (within) × Age	−0.01 [−0.01, −0.01]	< .001	0.00 [0.00, 0.00]	.182
Age × Physical activity (within)²	−0.01 [−0.01, 0.00]	.005	0.01 [0.01, 0.01]	< .001
Age × Physical activity (between)	0.00 [0.00, 0.01]	.437	0.02 [0.01, 0.02]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Physical activity (between)²	0.00 [−0.01, 0.00]	.042	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.26 [0.12, 0.40]	< .001	0.08 [−0.10, 0.27]	.378
Physical activity (within)	0.24 [0.23, 0.25]	< .001	0.09 [0.08, 0.09]	< .001
Age	0.04 [0.03, 0.06]	< .001	−0.06 [−0.09, −0.04]	< .001
Physical activity (within)²	−0.31 [−0.33, −0.28]	< .001	−0.13 [−0.14, −0.12]	< .001
Physical activity (between)	0.44 [0.41, 0.47]	< .001	0.15 [0.13, 0.18]	< .001
Physical activity (between)²	−0.19 [−0.21, −0.16]	< .001	−0.13 [−0.15, −0.11]	< .001
Physical activity (within) × Age	0.03 [0.02, 0.03]	< .001	−0.02 [−0.02, −0.02]	< .001
Age × Physical activity (within)²	−0.07 [−0.08, −0.06]	< .001	0.03 [0.03, 0.03]	< .001
Age × Physical activity (between)	0.01 [0.00, 0.02]	.016	−0.07 [−0.08, −0.06]	< .001
Age × Physical activity (between)²	−0.04 [−0.05, −0.03]	< .001	0.00 [0.00, 0.01]	.131

Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days). Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 10: Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [−0.17, 0.44]	.382	−0.07 [−0.20, 0.05]	.247
Sleep duration (within, prior)	0.01 [0.01, 0.02]	< .001	0.02 [0.01, 0.02]	< .001
Age	−0.23 [−0.24, −0.22]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep duration (within, prior)²	−0.04 [−0.05, −0.04]	< .001	−0.02 [−0.03, −0.02]	< .001
Sleep duration (between)	0.07 [0.06, 0.08]	< .001	0.06 [0.05, 0.07]	< .001
Sleep duration (between)²	−0.05 [−0.06, −0.04]	< .001	−0.01 [−0.02, 0.00]	.006
Sleep duration (within, prior) × Age	−0.01 [−0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.011
Age × Sleep duration (within, prior)²	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	< .001
Age × Sleep duration (between)	−0.03 [−0.04, −0.03]	< .001	−0.01 [−0.02, −0.01]	< .001
Age × Sleep duration (between)²	0.00 [0.00, 0.00]	.893	−0.01 [−0.01, −0.01]	< .001
Sleep efficiency (z)				
(Intercept)	0.13 [−0.17, 0.43]	.396	−0.07 [−0.19, 0.05]	.246
Sleep efficiency (within, prior)	0.01 [0.00, 0.01]	.003	0.01 [0.00, 0.01]	.002
Age	−0.23 [−0.24, −0.22]	< .001	−0.28 [−0.29, −0.27]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency (within, prior)²	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (between)	−0.02 [−0.03, 0.00]	.014	0.01 [0.00, 0.02]	.247
Sleep efficiency (between)²	−0.03 [−0.03, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (within, prior) × Age	0.00 [0.00, 0.00]	.899	0.00 [−0.01, 0.00]	< .001
Age × Sleep efficiency (within, prior)²	0.00 [0.00, 0.00]	.070	0.00 [0.00, 0.00]	.328
Age × Sleep efficiency (between)	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.017
Age × Sleep efficiency (between)²	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	.138
Sleep onset (z)				
(Intercept)	0.13 [−0.20, 0.45]	.446	−0.08 [−0.22, 0.05]	.216
Sleep onset (within, prior)	0.00 [−0.01, 0.00]	.117	0.00 [0.00, 0.01]	.208
Age	−0.23 [−0.24, −0.21]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep onset (within, prior)²	−0.03 [−0.03, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep onset (between)	−0.11 [−0.12, −0.09]	< .001	−0.07 [−0.08, −0.05]	< .001
Sleep onset (between)²	−0.01 [−0.01, 0.00]	.142	0.01 [0.00, 0.01]	.152
Sleep onset (within, prior) × Age	0.00 [−0.01, 0.00]	< .001	0.01 [0.00, 0.01]	< .001
Age × Sleep onset (within, prior)²	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	.002

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Sleep onset (between)	0.03 [0.02, 0.03]	< .001	0.02 [0.01, 0.02]	< .001
Age × Sleep onset (between)²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.02 [−0.27, 0.32]	.873	−0.17 [−0.30, −0.04]	.009
Sleep regularity (within, prior)	−0.01 [−0.01, 0.00]	.066	0.01 [0.00, 0.01]	.054
Age	−0.23 [−0.25, −0.21]	< .001	−0.32 [−0.34, −0.31]	< .001
Sleep regularity (within, prior)²	0.00 [−0.01, 0.01]	.643	0.00 [−0.01, 0.00]	.248
Sleep regularity (between)	0.17 [0.16, 0.18]	< .001	0.07 [0.06, 0.08]	< .001
Sleep regularity (between)²	0.00 [−0.01, 0.01]	.553	−0.02 [−0.03, −0.01]	< .001
Sleep regularity (within, prior) × Age	0.00 [0.00, 0.00]	.893	0.00 [−0.01, 0.00]	.002
Age × Sleep regularity (within, prior)²	0.00 [0.00, 0.00]	.213	0.00 [−0.01, 0.00]	.001
Age × Sleep regularity (between)	0.00 [0.00, 0.01]	.101	0.00 [−0.01, 0.00]	.272
Age × Sleep regularity (between)²	0.00 [0.00, 0.00]	.408	0.00 [−0.01, 0.00]	.002

Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days). Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

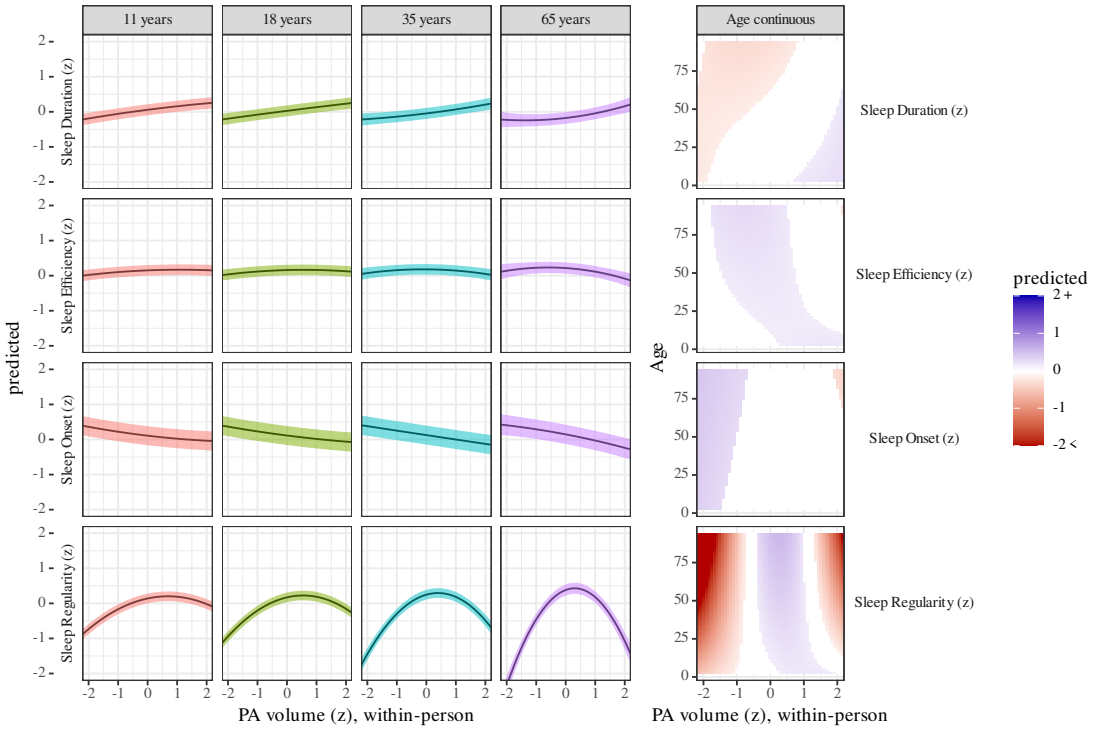


Figure 12: Sleep metrics on physical activity volume (all wear days)

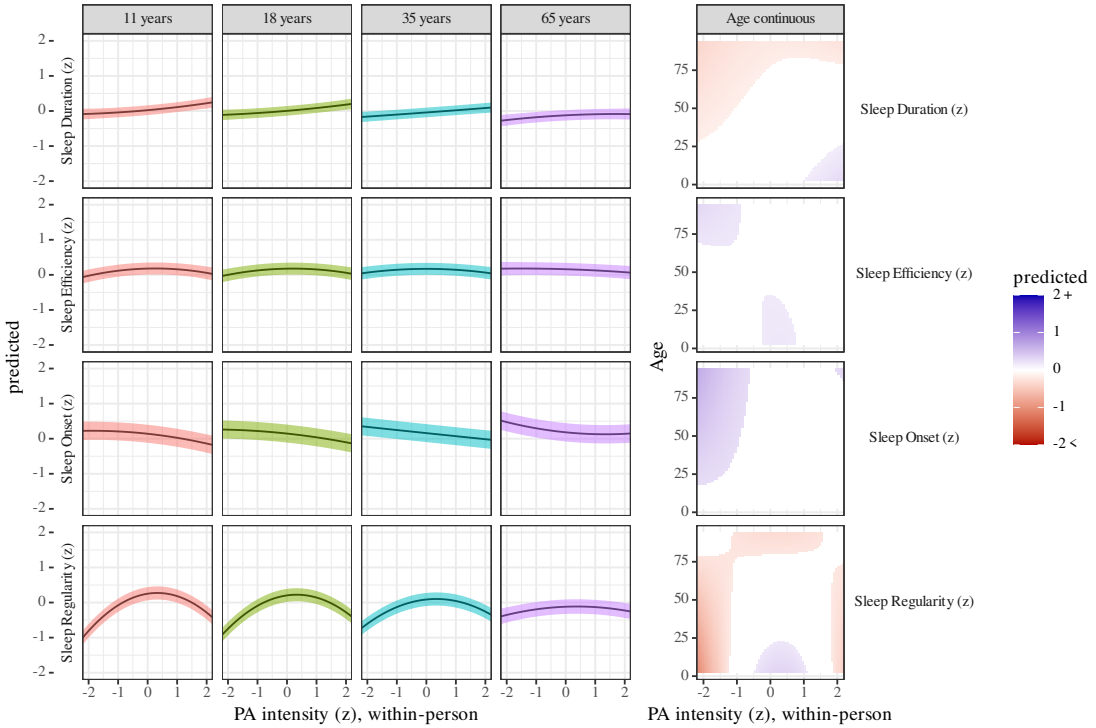


Figure 13: Sleep metrics on physical activity intensity (all wear days)

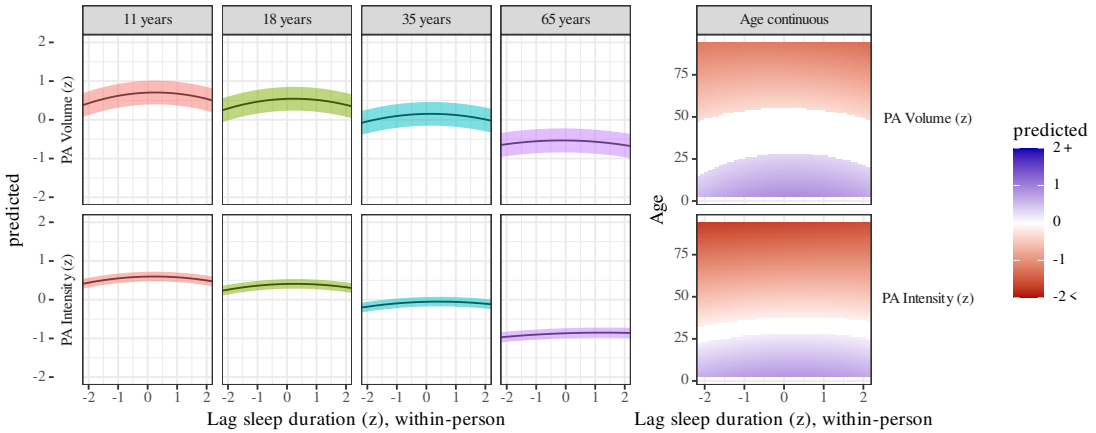


Figure 14: Physical activity by previous night sleep duration (all wear days)

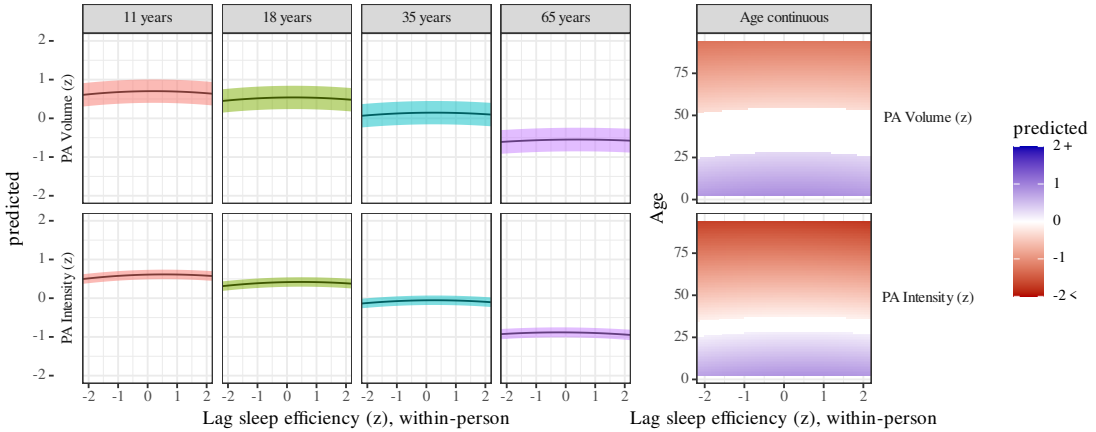


Figure 15: Physical activity by previous night sleep efficiency (all wear days)

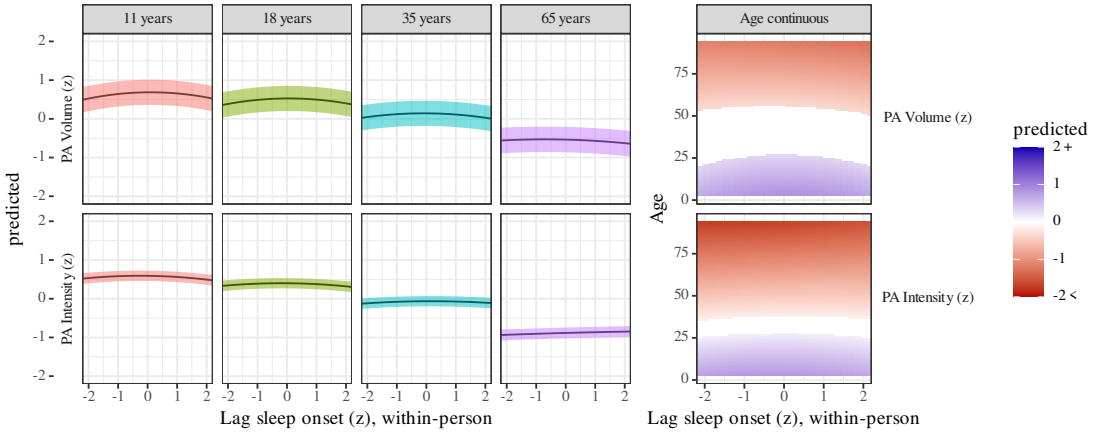


Figure 16: Physical activity by previous night sleep onset (all wear days)

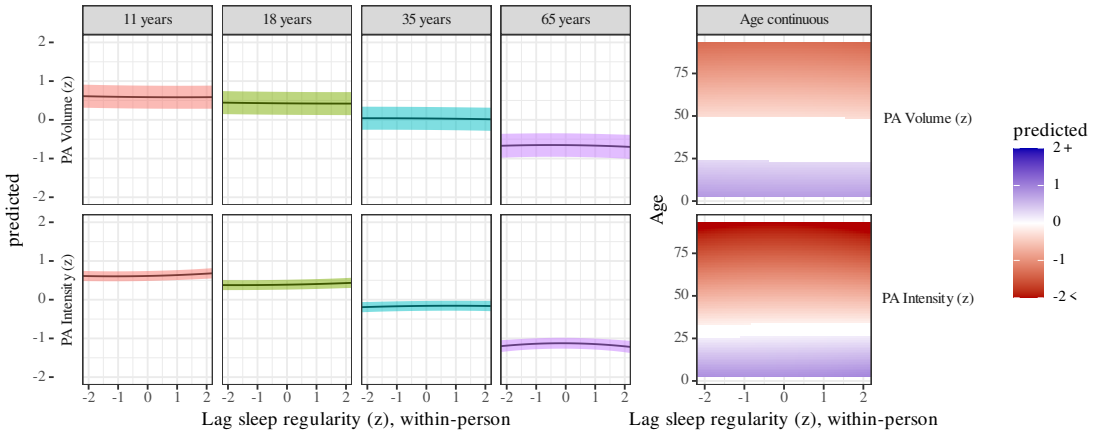


Figure 17: Physical activity by previous night sleep regularity (all wear days)

Participants With at Least Seven Valid Days

Some recommend at least seven nights to estimate sleep regularity. Here we refit the primary models to the 14,981 participants with at least seven days of valid wear time. Results, controlling for SES, sex, and BMI z-score, are presented in Tables 11 and 12 and Figures 18 to 23.

Table 11: Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.05 [−0.30, 0.19]	.678	−0.05 [−0.29, 0.18]	.660
Physical activity (within)	0.08 [0.07, 0.09]	< .001	0.06 [0.05, 0.07]	< .001
Age	0.01 [−0.01, 0.02]	.521	0.04 [0.02, 0.05]	< .001
Physical activity (within)²	0.01 [0.00, 0.03]	.061	0.00 [−0.01, 0.01]	.527
Physical activity (between)	0.08 [0.06, 0.10]	< .001	0.08 [0.05, 0.11]	< .001
Physical activity (between)²	0.00 [−0.02, 0.02]	.762	0.01 [−0.01, 0.03]	.405
Physical activity (within) × Age	0.00 [0.00, 0.01]	.048	0.00 [−0.01, 0.00]	.028
Age × Physical activity (within)²	0.01 [0.00, 0.01]	.024	0.00 [−0.01, 0.00]	.030
Age × Physical activity (between)	−0.03 [−0.04, −0.02]	< .001	−0.01 [−0.03, 0.00]	.008
Age × Physical activity (between)²	0.01 [0.00, 0.01]	.032	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (z)				
(Intercept)	0.15 [−0.14, 0.44]	.307	0.12 [−0.21, 0.45]	.476
Physical activity (within)	0.01 [0.00, 0.03]	.014	0.01 [0.00, 0.02]	.027
Age	−0.03 [−0.05, −0.01]	.007	−0.04 [−0.06, −0.02]	< .001
Physical activity (within)²	−0.03 [−0.05, −0.02]	< .001	−0.04 [−0.05, −0.03]	< .001
Physical activity (between)	0.14 [0.12, 0.17]	< .001	0.11 [0.08, 0.14]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (between)²	−0.06 [−0.08, −0.04]	< .001	−0.05 [−0.08, −0.03]	< .001
Physical activity (within) × Age	−0.02 [−0.03, −0.02]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Physical activity (within)²	−0.01 [−0.01, 0.00]	.006	0.01 [0.00, 0.01]	< .001
Age × Physical activity (between)	−0.03 [−0.04, −0.02]	< .001	−0.03 [−0.04, −0.02]	< .001
Age × Physical activity (between)²	0.00 [−0.01, 0.01]	.989	0.01 [0.00, 0.01]	.063
Sleep onset (z)				
(Intercept)	0.02 [−0.37, 0.41]	.909	0.03 [−0.35, 0.41]	.865
Physical activity (within)	−0.10 [−0.10, −0.09]	< .001	−0.07 [−0.08, −0.07]	< .001
Age	−0.05 [−0.07, −0.04]	< .001	−0.06 [−0.07, −0.04]	< .001
Physical activity (within)²	−0.01 [−0.02, 0.00]	.043	0.00 [−0.01, 0.01]	.947
Physical activity (between)	−0.07 [−0.09, −0.05]	< .001	−0.01 [−0.03, 0.02]	.534
Physical activity (between)²	−0.02 [−0.04, −0.01]	.002	−0.05 [−0.07, −0.03]	< .001
Physical activity (within) × Age	−0.02 [−0.03, −0.02]	< .001	0.00 [−0.01, 0.00]	< .001
Age × Physical activity (within)²	0.00 [−0.01, 0.00]	.146	0.01 [0.01, 0.01]	< .001
Age × Physical activity (between)	−0.02 [−0.02, −0.01]	< .001	0.00 [−0.01, 0.00]	.295

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Physical activity (between)²	0.00 [−0.01, 0.00]	.529	0.02 [0.01, 0.02]	< .001
Sleep regularity (z)				
(Intercept)	0.32 [0.19, 0.45]	< .001	0.09 [−0.19, 0.36]	.543
Physical activity (within)	0.26 [0.25, 0.28]	< .001	0.09 [0.08, 0.10]	< .001
Age	0.07 [0.04, 0.10]	< .001	−0.10 [−0.14, −0.06]	< .001
Physical activity (within)²	−0.32 [−0.35, −0.29]	< .001	−0.15 [−0.16, −0.13]	< .001
Physical activity (between)	0.42 [0.38, 0.46]	< .001	0.15 [0.12, 0.19]	< .001
Physical activity (between)²	−0.24 [−0.28, −0.20]	< .001	−0.17 [−0.20, −0.14]	< .001
Physical activity (within) × Age	0.02 [0.01, 0.02]	< .001	−0.02 [−0.02, −0.01]	< .001
Age × Physical activity (within)²	−0.07 [−0.08, −0.05]	< .001	0.03 [0.03, 0.04]	< .001
Age × Physical activity (between)	0.00 [−0.01, 0.02]	.619	−0.08 [−0.09, −0.06]	< .001
Age × Physical activity (between)²	−0.06 [−0.08, −0.05]	< .001	0.01 [0.00, 0.02]	.042

Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days). Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 12: Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.01 [−0.37, 0.35]	.971	−0.11 [−0.26, 0.05]	.170
Sleep duration (within, prior)	0.01 [0.01, 0.02]	< .001	0.03 [0.02, 0.03]	< .001
Age	−0.19 [−0.21, −0.18]	< .001	−0.23 [−0.25, −0.22]	< .001
Sleep duration (within, prior)²	−0.05 [−0.05, −0.04]	< .001	−0.03 [−0.03, −0.02]	< .001
Sleep duration (between)	0.06 [0.04, 0.08]	< .001	0.08 [0.06, 0.10]	< .001
Sleep duration (between)²	−0.04 [−0.06, −0.03]	< .001	−0.02 [−0.03, 0.00]	.071
Sleep duration (within, prior) × Age	−0.01 [−0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.839
Age × Sleep duration (within, prior)²	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Age × Sleep duration (between)	−0.03 [−0.04, −0.02]	< .001	−0.02 [−0.02, −0.01]	< .001
Age × Sleep duration (between)²	0.00 [−0.01, 0.00]	.260	−0.01 [−0.01, 0.00]	.003
Sleep efficiency (z)				
(Intercept)	−0.01 [−0.36, 0.34]	.952	−0.11 [−0.25, 0.04]	.154
Sleep efficiency (within, prior)	0.01 [0.01, 0.02]	< .001	0.01 [0.00, 0.02]	< .001
Age	−0.19 [−0.20, −0.17]	< .001	−0.23 [−0.24, −0.22]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency (within, prior)²	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (between)	0.01 [0.00, 0.03]	.150	0.03 [0.01, 0.05]	< .001
Sleep efficiency (between)²	−0.02 [−0.03, −0.01]	< .001	−0.01 [−0.02, 0.00]	.077
Sleep efficiency (within, prior) × Age	0.00 [0.00, 0.00]	.002	0.00 [−0.01, 0.00]	< .001
Age × Sleep efficiency (within, prior)²	0.00 [0.00, 0.00]	.888	0.00 [0.00, 0.00]	.047
Age × Sleep efficiency (between)	0.01 [0.00, 0.01]	.006	0.00 [−0.01, 0.01]	.950
Age × Sleep efficiency (between)²	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.835
Sleep onset (z)				
(Intercept)	−0.05 [−0.45, 0.35]	.804	−0.10 [−0.26, 0.06]	.218
Sleep onset (within, prior)	−0.03 [−0.04, −0.02]	< .001	0.00 [−0.01, 0.00]	.376
Age	−0.20 [−0.21, −0.19]	< .001	−0.23 [−0.25, −0.22]	< .001
Sleep onset (within, prior)²	−0.03 [−0.03, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep onset (between)	−0.12 [−0.14, −0.09]	< .001	−0.06 [−0.08, −0.03]	< .001
Sleep onset (between)²	−0.04 [−0.05, −0.02]	< .001	0.00 [−0.02, 0.01]	.777
Sleep onset (within, prior) × Age	0.00 [0.00, 0.01]	.001	0.01 [0.01, 0.01]	< .001
Age × Sleep onset (within, prior)²	0.00 [0.00, 0.01]	.001	0.01 [0.00, 0.01]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Sleep onset (between)	0.03 [0.02, 0.04]	< .001	0.02 [0.01, 0.02]	< .001
Age × Sleep onset (between)²	0.00 [0.00, 0.01]	.396	−0.01 [−0.01, 0.00]	.006
Sleep regularity (z)				
(Intercept)	−0.15 [−0.49, 0.19]	.384	−0.22 [−0.39, −0.05]	.012
Sleep regularity (within, prior)	0.00 [0.00, 0.01]	.217	0.01 [0.00, 0.02]	.005
Age	−0.18 [−0.20, −0.16]	< .001	−0.31 [−0.33, −0.28]	< .001
Sleep regularity (within, prior)²	0.00 [−0.01, 0.01]	.723	0.00 [−0.01, 0.01]	.690
Sleep regularity (between)	0.14 [0.13, 0.15]	< .001	0.06 [0.04, 0.07]	< .001
Sleep regularity (between)²	−0.01 [−0.02, 0.01]	.353	−0.02 [−0.04, −0.01]	< .001
Sleep regularity (within, prior) × Age	0.00 [−0.01, 0.00]	.002	−0.01 [−0.01, 0.00]	< .001
Age × Sleep regularity (within, prior)²	0.00 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Age × Sleep regularity (between)	0.01 [0.01, 0.02]	< .001	0.00 [−0.01, 0.00]	.837
Age × Sleep regularity (between)²	0.00 [0.00, 0.00]	.880	0.00 [0.00, 0.00]	.846

Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days). Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

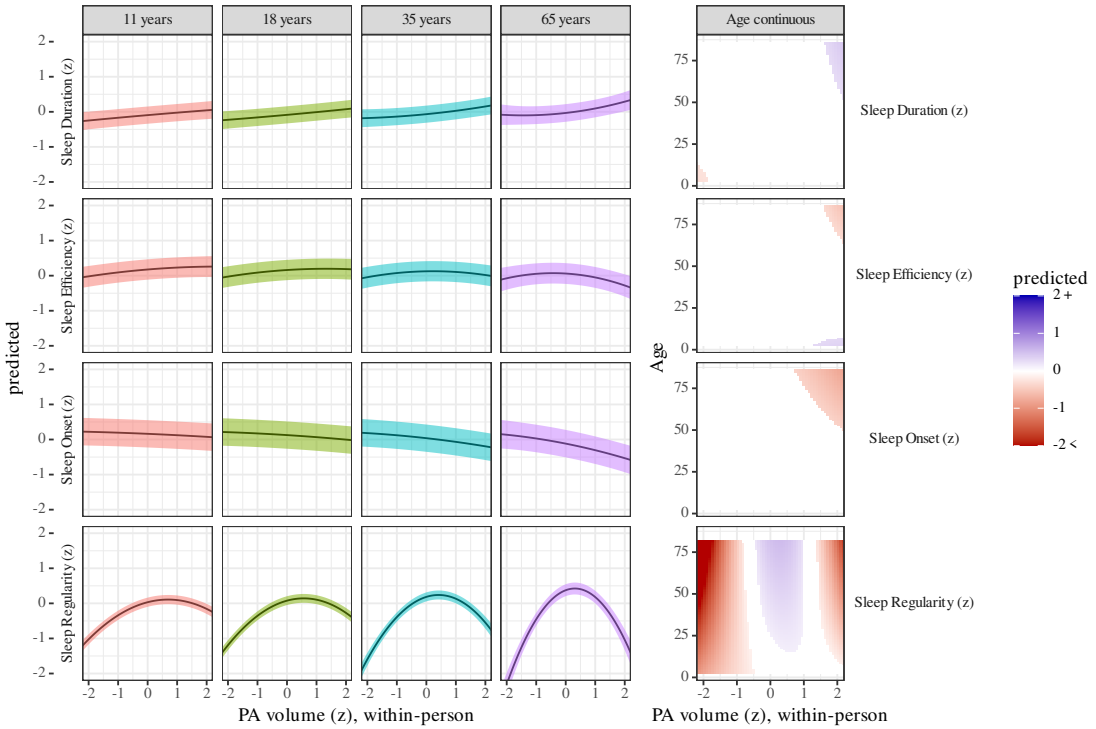


Figure 18: Sleep metrics on physical activity volume (at least seven valid days)

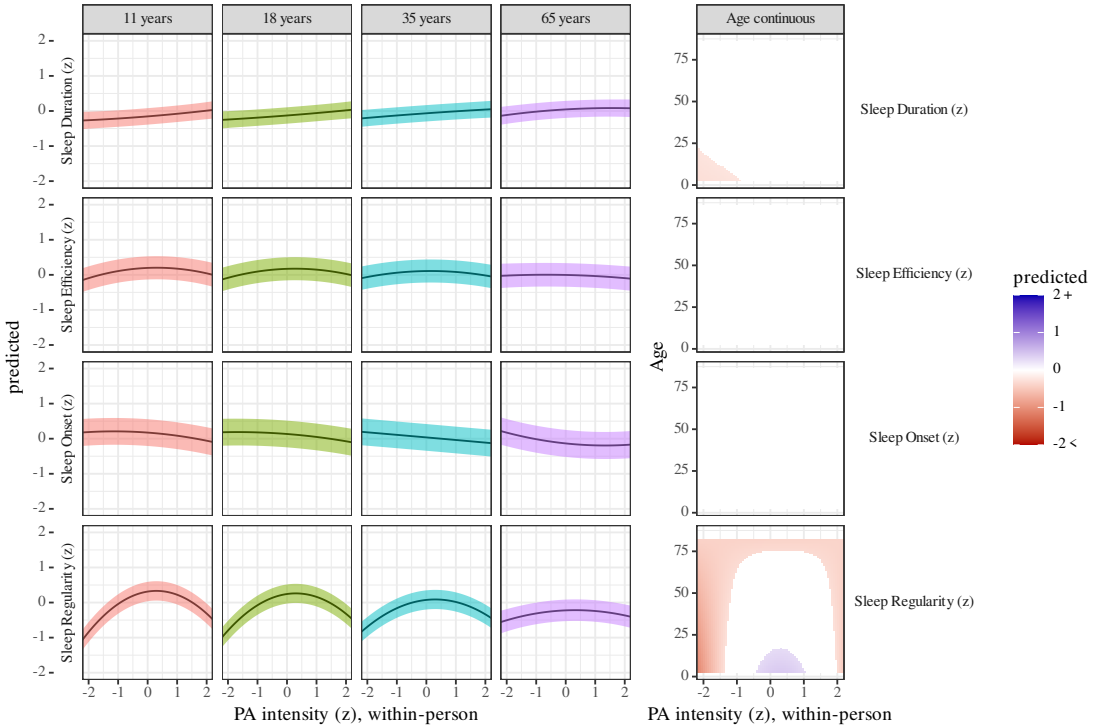


Figure 19: Sleep metrics on physical activity intensity (at least seven valid days)

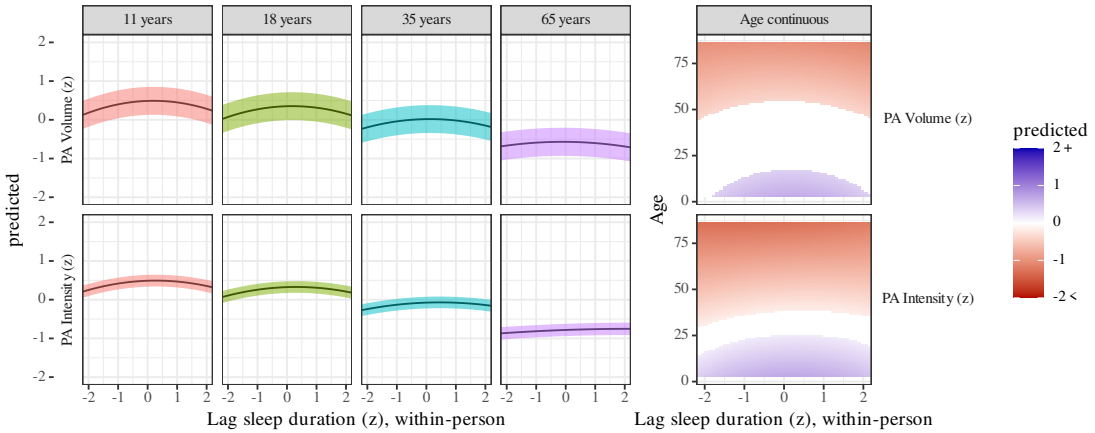


Figure 20: Physical activity by previous night sleep duration (at least seven valid days)

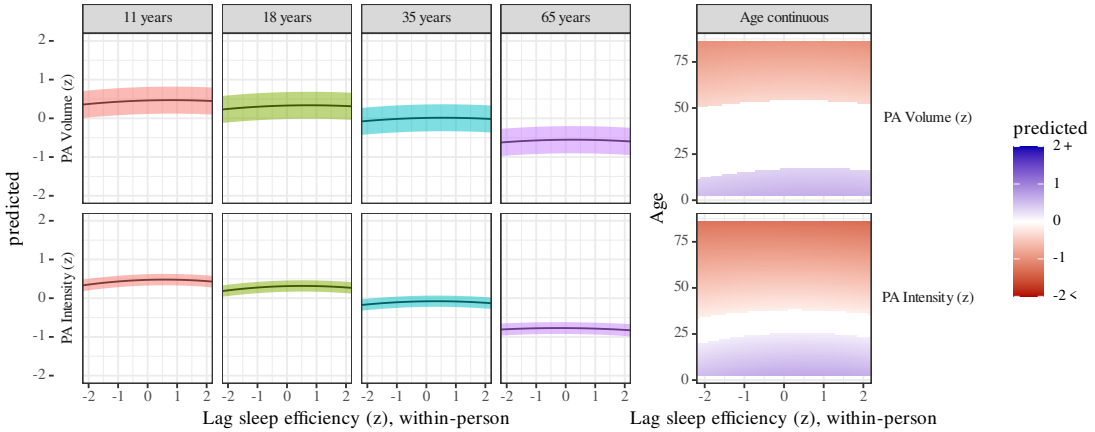


Figure 21: Physical activity by previous night sleep efficiency (at least seven valid days)

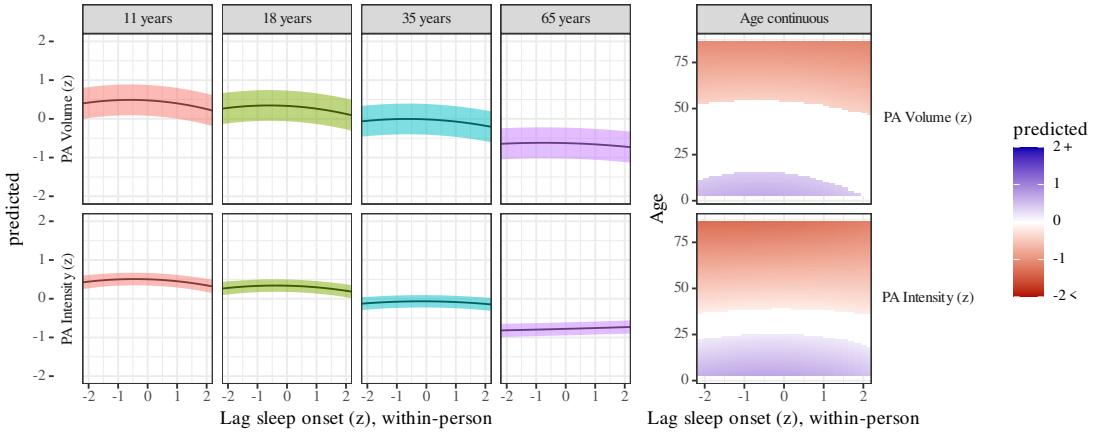


Figure 22: Physical activity by previous night sleep onset (at least seven valid days)

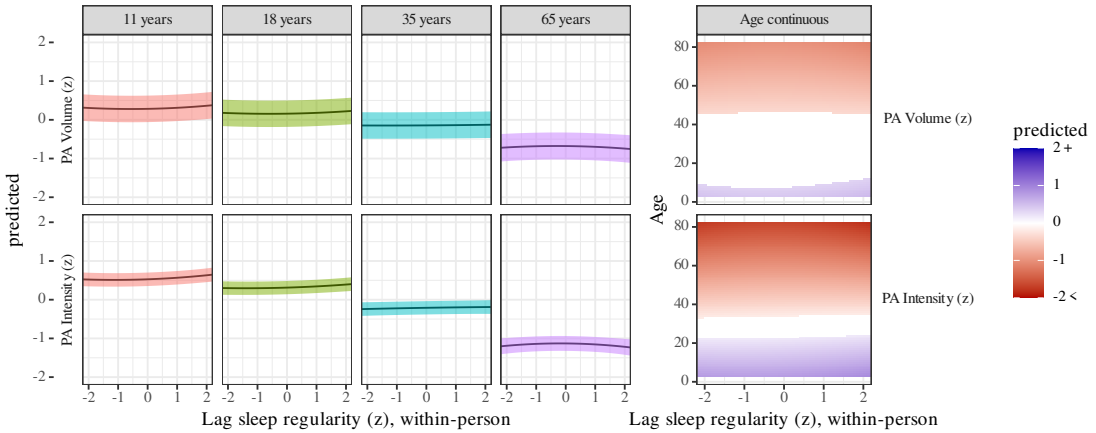


Figure 23: Physical activity by previous night sleep regularity (at least seven valid days)

Pre-registered Model Without the Within–Between Split

The pre-registered models entered each predictor as a single standardised term, so their estimates combine within- and between-person associations. Results are presented in Tables 13 and 14.

Table 13: Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.00 [−0.16, 0.16]	.996	0.00 [−0.15, 0.14]	.995
Physical activity	0.10 [0.09, 0.11]	< .001	0.05 [0.05, 0.06]	< .001
Age	−0.02 [−0.03, 0.00]	.007	−0.02 [−0.03, −0.01]	.003
Physical activity²	−0.02 [−0.02, −0.01]	< .001	0.00 [0.00, 0.01]	.612
Physical activity × Age	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Physical activity²	0.00 [−0.01, 0.00]	.001	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency (z)				
(Intercept)	0.15 [−0.01, 0.31]	.062	0.15 [−0.02, 0.32]	.086
Physical activity	0.03 [0.03, 0.04]	< .001	0.02 [0.02, 0.03]	< .001
Age	−0.01 [−0.02, 0.01]	.323	−0.01 [−0.03, 0.00]	.067
Physical activity²	−0.02 [−0.02, −0.01]	< .001	−0.02 [−0.02, −0.01]	< .001
Physical activity × Age	−0.02 [−0.02, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Age × Physical activity²	0.00 [0.00, 0.00]	.095	0.01 [0.00, 0.01]	< .001
Sleep onset (z)				
(Intercept)	0.09 [−0.18, 0.37]	.503	0.12 [−0.14, 0.37]	.379
Physical activity	−0.12 [−0.12, −0.11]	< .001	−0.06 [−0.06, −0.05]	< .001
Age	−0.01 [−0.02, 0.00]	.036	0.00 [−0.02, 0.01]	.524
Physical activity²	0.02 [0.02, 0.03]	< .001	−0.01 [−0.02, −0.01]	< .001
Physical activity × Age	−0.01 [−0.01, −0.01]	< .001	0.01 [0.00, 0.01]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Physical activity²	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.18 [0.04, 0.33]	.014	0.06 [−0.14, 0.25]	.569
Physical activity	0.29 [0.27, 0.30]	< .001	0.14 [0.13, 0.15]	< .001
Age	0.00 [−0.02, 0.02]	.730	−0.08 [−0.10, −0.06]	< .001
Physical activity²	−0.15 [−0.16, −0.14]	< .001	−0.09 [−0.09, −0.08]	< .001
Physical activity × Age	0.00 [−0.01, 0.00]	.049	−0.05 [−0.05, −0.05]	< .001
Age × Physical activity²	−0.03 [−0.04, −0.03]	< .001	0.01 [0.01, 0.01]	< .001

Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 14: Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [−0.17, 0.44]	.378	−0.07 [−0.19, 0.05]	.254
Sleep duration	0.02 [0.01, 0.02]	< .001	0.03 [0.02, 0.03]	< .001
Age	−0.23 [−0.24, −0.22]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep duration²	−0.02 [−0.03, −0.02]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep duration × Age	−0.01 [−0.01, −0.01]	< .001	0.00 [0.00, 0.00]	.051
Age × Sleep duration²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.326
Sleep efficiency (z)				
(Intercept)	0.13 [−0.17, 0.42]	.412	−0.07 [−0.20, 0.05]	.242
Sleep efficiency	0.00 [0.00, 0.01]	.279	0.01 [0.00, 0.01]	.009
Age	−0.23 [−0.24, −0.21]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep efficiency²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Sleep efficiency × Age	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.200
Age × Sleep efficiency²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.227
Sleep onset (z)				
(Intercept)	0.13 [−0.18, 0.43]	.410	−0.08 [−0.21, 0.04]	.202
Sleep onset	−0.02 [−0.03, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Age	−0.22 [−0.23, −0.21]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep onset²	0.00 [−0.01, 0.00]	< .001	0.00 [0.00, 0.00]	.811
Sleep onset × Age	0.00 [0.00, 0.00]	.012	0.01 [0.01, 0.01]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Sleep onset²	0.00 [0.00, 0.00]	.002	0.00 [0.00, 0.00]	.020
Sleep regularity (z)				
(Intercept)	0.04 [−0.27, 0.34]	.817	−0.18 [−0.30, −0.05]	.006
Sleep regularity	0.12 [0.11, 0.12]	< .001	0.06 [0.06, 0.07]	< .001
Age	−0.24 [−0.26, −0.22]	< .001	−0.32 [−0.34, −0.31]	< .001
Sleep regularity²	0.00 [0.00, 0.00]	.448	0.00 [0.00, 0.00]	.414
Sleep regularity × Age	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Sleep regularity²	0.00 [0.00, 0.00]	.145	0.00 [0.00, 0.00]	.023

Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Cross-lagged Model

These models add the outcome's own within-person value on the previous day (for sleep regularity, the index for the two preceding days) to the primary models. Results are presented in Tables 15 and 16.

Table 15: Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.02 [−0.19, 0.14]	.779	−0.02 [−0.16, 0.13]	.825
Physical activity (within)	0.10 [0.09, 0.11]	< .001	0.07 [0.06, 0.08]	< .001
Age	−0.04 [−0.05, −0.02]	< .001	−0.02 [−0.03, 0.00]	.017
Physical activity (within)²	0.02 [0.00, 0.03]	.048	−0.01 [−0.02, 0.00]	.190
Physical activity (between)	0.06 [0.04, 0.08]	< .001	0.11 [0.09, 0.14]	< .001
Physical activity (between)²	0.01 [−0.01, 0.02]	.317	−0.01 [−0.03, 0.01]	.198
Sleep duration (within, prior)	−0.19 [−0.20, −0.18]	< .001	−0.19 [−0.19, −0.18]	< .001
Physical activity (within) × Age	0.00 [−0.01, 0.00]	.237	0.00 [−0.01, 0.00]	.033
Age × Physical activity (within)²	0.01 [0.00, 0.01]	.001	0.00 [−0.01, 0.00]	.110
Age × Physical activity (between)	−0.04 [−0.04, −0.03]	< .001	−0.03 [−0.04, −0.02]	< .001
Age × Physical activity (between)²	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.906
Sleep efficiency (z)				
(Intercept)	0.21 [0.06, 0.35]	.006	0.19 [0.03, 0.35]	.021
Physical activity (within)	0.00 [−0.01, 0.01]	.831	0.00 [−0.01, 0.01]	.585
Age	0.01 [0.00, 0.03]	.152	0.00 [−0.01, 0.02]	.798
Physical activity (within)²	−0.03 [−0.05, −0.02]	< .001	−0.03 [−0.04, −0.02]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (between)	0.08 [0.06, 0.10]	< .001	0.07 [0.04, 0.09]	< .001
Physical activity (between)²	−0.03 [−0.05, −0.02]	< .001	−0.03 [−0.05, −0.01]	< .001
Sleep efficiency (within, prior)	−0.15 [−0.16, −0.15]	< .001	−0.15 [−0.16, −0.15]	< .001
Physical activity (within) × Age	−0.02 [−0.02, −0.01]	< .001	−0.01 [−0.01, −0.01]	< .001
Age × Physical activity (within)²	−0.01 [−0.01, 0.00]	.006	0.01 [0.00, 0.01]	< .001
Age × Physical activity (between)	−0.01 [−0.01, 0.00]	.027	−0.01 [−0.02, 0.00]	.017
Age × Physical activity (between)²	0.00 [0.00, 0.01]	.450	0.01 [0.00, 0.01]	.002
Sleep onset (z)				
(Intercept)	0.14 [−0.13, 0.42]	.302	0.18 [−0.08, 0.43]	.179
Physical activity (within)	−0.13 [−0.14, −0.12]	< .001	−0.09 [−0.09, −0.08]	< .001
Age	0.00 [−0.01, 0.01]	.690	0.01 [−0.01, 0.02]	.231
Physical activity (within)²	0.01 [0.00, 0.02]	.129	0.00 [−0.01, 0.01]	.567
Physical activity (between)	−0.12 [−0.13, −0.10]	< .001	−0.07 [−0.09, −0.05]	< .001
Physical activity (between)²	−0.01 [−0.02, 0.01]	.239	−0.01 [−0.02, 0.01]	.306
Sleep onset (within, prior)	−0.15 [−0.16, −0.14]	< .001	−0.15 [−0.15, −0.14]	< .001
Physical activity (within) × Age	−0.02 [−0.02, −0.01]	< .001	0.00 [0.00, 0.00]	.549

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Physical activity (within)²	0.00 [−0.01, 0.00]	.370	0.01 [0.01, 0.01]	< .001
Age × Physical activity (between)	0.00 [0.00, 0.01]	.110	0.02 [0.01, 0.03]	< .001
Age × Physical activity (between)²	0.00 [−0.01, 0.00]	.073	0.01 [0.00, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.25 [0.10, 0.40]	.001	0.07 [−0.13, 0.27]	.478
Physical activity (within)	0.22 [0.20, 0.23]	< .001	0.09 [0.08, 0.10]	< .001
Age	0.05 [0.02, 0.07]	< .001	−0.06 [−0.08, −0.04]	< .001
Physical activity (within)²	−0.29 [−0.32, −0.26]	< .001	−0.11 [−0.12, −0.10]	< .001
Physical activity (between)	0.43 [0.40, 0.46]	< .001	0.15 [0.13, 0.18]	< .001
Physical activity (between)²	−0.18 [−0.21, −0.15]	< .001	−0.12 [−0.14, −0.10]	< .001
Sleep regularity (within, prior)	−0.30 [−0.31, −0.29]	< .001	−0.30 [−0.31, −0.29]	< .001
Physical activity (within) × Age	0.02 [0.01, 0.03]	< .001	−0.02 [−0.03, −0.02]	< .001
Age × Physical activity (within)²	−0.07 [−0.08, −0.06]	< .001	0.02 [0.02, 0.03]	< .001
Age × Physical activity (between)	0.01 [0.00, 0.02]	.023	−0.07 [−0.08, −0.06]	< .001
Age × Physical activity (between)²	−0.04 [−0.05, −0.03]	< .001	0.00 [0.00, 0.01]	.320

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p

Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days). Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 16: Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [−0.17, 0.44]	.379	−0.07 [−0.20, 0.05]	.256
Sleep duration (within, prior)	0.02 [0.01, 0.02]	< .001	0.02 [0.02, 0.03]	< .001
Age	−0.23 [−0.24, −0.22]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep duration (within, prior)²	−0.04 [−0.05, −0.04]	< .001	−0.02 [−0.03, −0.02]	< .001
Sleep duration (between)	0.07 [0.06, 0.08]	< .001	0.06 [0.05, 0.07]	< .001
Sleep duration (between)²	−0.05 [−0.06, −0.04]	< .001	−0.01 [−0.02, 0.00]	.011
Physical activity (within, prior)	−0.12 [−0.12, −0.11]	< .001	−0.13 [−0.13, −0.12]	< .001
Sleep duration (within, prior) × Age	−0.01 [−0.01, −0.01]	< .001	0.00 [0.00, 0.00]	.032
Age × Sleep duration (within, prior)²	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	< .001
Age × Sleep duration (between)	−0.03 [−0.04, −0.03]	< .001	−0.01 [−0.02, −0.01]	< .001
Age × Sleep duration (between)²	0.00 [0.00, 0.00]	.900	−0.01 [−0.01, −0.01]	< .001
Sleep efficiency (z)				
(Intercept)	0.13 [−0.17, 0.43]	.391	−0.07 [−0.19, 0.05]	.255
Sleep efficiency (within, prior)	0.01 [0.00, 0.01]	.001	0.01 [0.00, 0.01]	.004

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age	−0.23 [−0.24, −0.22]	< .001	−0.28 [−0.29, −0.27]	< .001
Sleep efficiency (within, prior)²	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (between)	−0.02 [−0.03, 0.00]	.009	0.01 [0.00, 0.02]	.263
Sleep efficiency (between)²	−0.03 [−0.04, −0.02]	< .001	−0.01 [−0.02, −0.01]	.001
Physical activity (within, prior)	−0.12 [−0.12, −0.11]	< .001	−0.13 [−0.13, −0.12]	< .001
Sleep efficiency (within, prior) × Age	0.00 [0.00, 0.00]	.326	0.00 [−0.01, 0.00]	< .001
Age × Sleep efficiency (within, prior)²	0.00 [0.00, 0.00]	.109	0.00 [0.00, 0.00]	.309
Age × Sleep efficiency (between)	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.018
Age × Sleep efficiency (between)²	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.01]	.111
Sleep onset (z)				
(Intercept)	0.13 [−0.20, 0.45]	.443	−0.08 [−0.22, 0.05]	.224
Sleep onset (within, prior)	−0.02 [−0.02, −0.01]	< .001	−0.01 [−0.01, 0.00]	.027
Age	−0.23 [−0.24, −0.21]	< .001	−0.27 [−0.28, −0.26]	< .001
Sleep onset (within, prior)²	−0.03 [−0.03, −0.02]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep onset (between)	−0.11 [−0.12, −0.09]	< .001	−0.07 [−0.08, −0.06]	< .001
Sleep onset (between)²	−0.01 [−0.01, 0.00]	.163	0.01 [0.00, 0.01]	.096

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (within, prior)	−0.12 [−0.12, −0.11]	< .001	−0.13 [−0.13, −0.12]	< .001
Sleep onset (within, prior) × Age	0.00 [−0.01, 0.00]	.003	0.01 [0.00, 0.01]	< .001
Age × Sleep onset (within, prior)²	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	.002
Age × Sleep onset (between)	0.03 [0.02, 0.03]	< .001	0.02 [0.01, 0.02]	< .001
Age × Sleep onset (between)²	−0.01 [−0.01, 0.00]	< .001	−0.01 [−0.01, −0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.03 [−0.27, 0.32]	.859	−0.17 [−0.30, −0.04]	.008
Sleep regularity (within, prior)	0.01 [0.00, 0.02]	.001	0.02 [0.01, 0.02]	< .001
Age	−0.23 [−0.24, −0.21]	< .001	−0.32 [−0.34, −0.31]	< .001
Sleep regularity (within, prior)²	0.00 [−0.01, 0.01]	.675	0.00 [−0.01, 0.00]	.392
Sleep regularity (between)	0.17 [0.16, 0.18]	< .001	0.07 [0.05, 0.08]	< .001
Sleep regularity (between)²	−0.01 [−0.02, 0.00]	.166	−0.02 [−0.03, −0.01]	< .001
Physical activity (within, prior)	−0.17 [−0.18, −0.16]	< .001	−0.17 [−0.18, −0.16]	< .001
Sleep regularity (within, prior) × Age	0.00 [−0.01, 0.00]	.019	−0.01 [−0.01, 0.00]	< .001
Age × Sleep regularity (within, prior)²	0.00 [−0.01, 0.00]	.041	−0.01 [−0.01, 0.00]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Age × Sleep regularity (between)	0.00 [0.00, 0.01]	.137	0.00 [−0.01, 0.00]	.060
Age × Sleep regularity (between)²	0.00 [0.00, 0.00]	.726	0.00 [−0.01, 0.00]	.066

Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days). Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Minutes of Moderate-to-Vigorous Physical Activity

These models replace physical activity volume and intensity with daily minutes of moderate-to-vigorous physical activity (MVPA), from *GGIR*'s cut-points for each age group, device, and wear location. Results are presented in Tables 17 and 18 and Figures 24 to 28.

Table 17: Physical activity predicting sleep controlling for SES, sex, and BMI z-score.

Term	MVPA (z) β [95% CI]	MVPA (z) p
Sleep duration (z)		
(Intercept)	0.02 [−0.12, 0.15]	.807
Physical activity (within)	−0.01 [−0.02, 0.00]	.055
Age	−0.08 [−0.09, −0.06]	< .001
Physical activity (within)²	0.00 [−0.01, 0.01]	.607
Physical activity (between)	−0.05 [−0.07, −0.04]	< .001
Physical activity (between)²	−0.03 [−0.04, −0.02]	< .001
Physical activity (within) × Age	−0.02 [−0.02, −0.01]	< .001
Age × Physical activity (within)²	0.01 [0.00, 0.01]	< .001
Age × Physical activity (between)	−0.02 [−0.03, −0.02]	< .001
Age × Physical activity (between)²	0.02 [0.02, 0.02]	< .001
Sleep efficiency (z)		
(Intercept)	0.21 [0.05, 0.36]	.009
Physical activity (within)	0.07 [0.06, 0.07]	< .001
Age	0.00 [−0.02, 0.01]	.973
Physical activity (within)²	−0.01 [−0.02, 0.00]	.012
Physical activity (between)	0.08 [0.06, 0.09]	< .001
Physical activity (between)²	−0.04 [−0.05, −0.03]	< .001
Physical activity (within) × Age	−0.02 [−0.02, −0.01]	< .001
Age × Physical activity (within)²	0.00 [0.00, 0.00]	.902

Term	MVPA (z) β [95% CI]	MVPA (z) p
Age × Physical activity (between)	−0.01 [−0.02, 0.00]	< .001
Age × Physical activity (between)²	0.01 [0.01, 0.02]	< .001
Sleep onset (z)		
(Intercept)	0.10 [−0.15, 0.36]	.420
Physical activity (within)	0.07 [0.06, 0.08]	< .001
Age	0.03 [0.02, 0.05]	< .001
Physical activity (within)²	0.02 [0.02, 0.03]	< .001
Physical activity (between)	0.01 [−0.01, 0.02]	.258
Physical activity (between)²	0.00 [−0.01, 0.01]	.732
Physical activity (within) × Age	0.00 [0.00, 0.00]	.131
Age × Physical activity (within)²	−0.01 [−0.02, −0.01]	< .001
Age × Physical activity (between)	0.00 [0.00, 0.01]	.302
Age × Physical activity (between)²	−0.01 [−0.01, 0.00]	.006
Sleep regularity (z)		
(Intercept)	0.28 [0.12, 0.43]	< .001
Physical activity (within)	0.22 [0.21, 0.23]	< .001
Age	−0.05 [−0.07, −0.03]	< .001
Physical activity (within)²	−0.24 [−0.25, −0.22]	< .001
Physical activity (between)	0.31 [0.29, 0.33]	< .001
Physical activity (between)²	−0.11 [−0.12, −0.09]	< .001
Physical activity (within) × Age	−0.03 [−0.03, −0.03]	< .001

Term	MVPA (z) β [95% CI]	MVPA (z) p
Age × Physical activity (within)²	0.03 [0.03, 0.04]	< .001
Age × Physical activity (between)	0.00 [−0.01, 0.01]	.845
Age × Physical activity (between)²	0.01 [0.01, 0.02]	< .001

Adjusted for SES, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days). Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

Table 18: Sleep predicting physical activity controlling for SES, sex, and BMI z-score

Term	MVPA (z) β [95% CI]	MVPA (z) p
Sleep duration (z)		
(Intercept)	−0.32 [−0.76, 0.12]	.155
Sleep duration (within, prior)	−0.04 [−0.05, −0.04]	< .001
Age	−0.28 [−0.29, −0.27]	< .001
Sleep duration (within, prior)²	−0.05 [−0.05, −0.04]	< .001
Sleep duration (between)	−0.09 [−0.11, −0.08]	< .001
Sleep duration (between)²	−0.03 [−0.04, −0.02]	< .001
Sleep duration (within, prior) × Age	−0.01 [−0.01, −0.01]	< .001
Age × Sleep duration (within, prior)²	0.00 [0.00, 0.00]	.001
Age × Sleep duration (between)	−0.02 [−0.03, −0.02]	< .001
Age × Sleep duration (between)²	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (z)		
(Intercept)	−0.35 [−0.80, 0.09]	.121
Sleep efficiency (within, prior)	0.02 [0.02, 0.03]	< .001
Age	−0.29 [−0.30, −0.27]	< .001
Sleep efficiency (within, prior)²	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (between)	0.03 [0.02, 0.05]	< .001
Sleep efficiency (between)²	−0.01 [−0.02, 0.00]	.224
Sleep efficiency (within, prior) × Age	0.00 [0.00, 0.00]	.016
Age × Sleep efficiency (within, prior)²	0.00 [0.00, 0.00]	.899

Term	MVPA (z) β [95% CI]	MVPA (z) p
Age × Sleep efficiency (between)	0.01 [0.00, 0.01]	.001
Age × Sleep efficiency (between)²	0.00 [0.00, 0.01]	.175
Sleep onset (z)		
(Intercept)	−0.32 [−0.77, 0.12]	.157
Sleep onset (within, prior)	−0.08 [−0.09, −0.08]	< .001
Age	−0.28 [−0.30, −0.27]	< .001
Sleep onset (within, prior)²	−0.04 [−0.04, −0.03]	< .001
Sleep onset (between)	0.03 [0.01, 0.05]	.002
Sleep onset (between)²	−0.03 [−0.04, −0.02]	< .001
Sleep onset (within, prior) × Age	−0.01 [−0.01, −0.01]	< .001
Age × Sleep onset (within, prior)²	0.00 [−0.01, 0.00]	.005
Age × Sleep onset (between)	0.00 [−0.01, 0.01]	.893
Age × Sleep onset (between)²	−0.01 [−0.01, 0.00]	< .001
Sleep regularity (z)		
(Intercept)	−0.40 [−0.73, −0.07]	.017
Sleep regularity (within, prior)	0.00 [0.00, 0.01]	.441
Age	−0.18 [−0.20, −0.16]	< .001
Sleep regularity (within, prior)²	0.00 [−0.01, 0.00]	.168
Sleep regularity (between)	0.16 [0.15, 0.17]	< .001
Sleep regularity (between)²	−0.01 [−0.02, 0.00]	.063
Sleep regularity (within, prior) × Age	0.00 [0.00, 0.00]	.073

Term	MVPA (z) β [95% CI]	MVPA (z) p
Age × Sleep regularity (within, prior)²	0.00 [−0.01, 0.00]	.003
Age × Sleep regularity (between)	0.03 [0.03, 0.03]	< .001
Age × Sleep regularity (between)²	0.00 [−0.01, 0.00]	.168

Adjusted for SES, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days). Age was centred at 35.6 years (the mean across person-days) and scaled per 10 years, so main effects are associations at age 35.6 and age interactions are the change per 10 years.

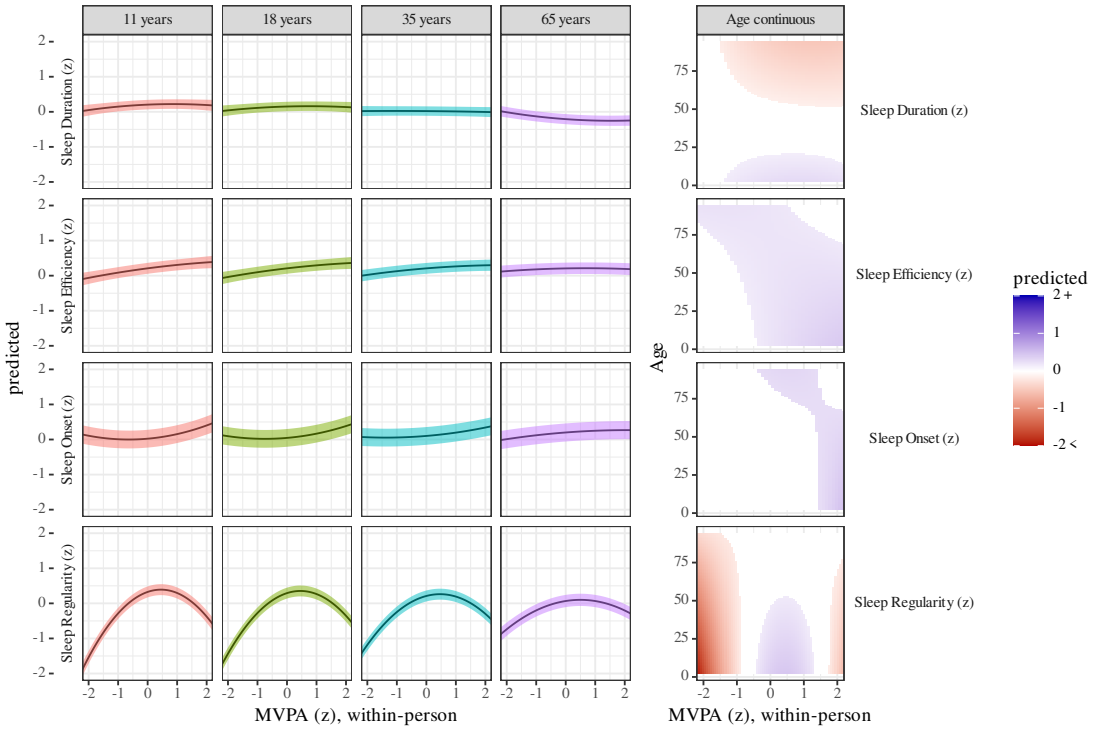


Figure 24: Sleep metrics on MVPA

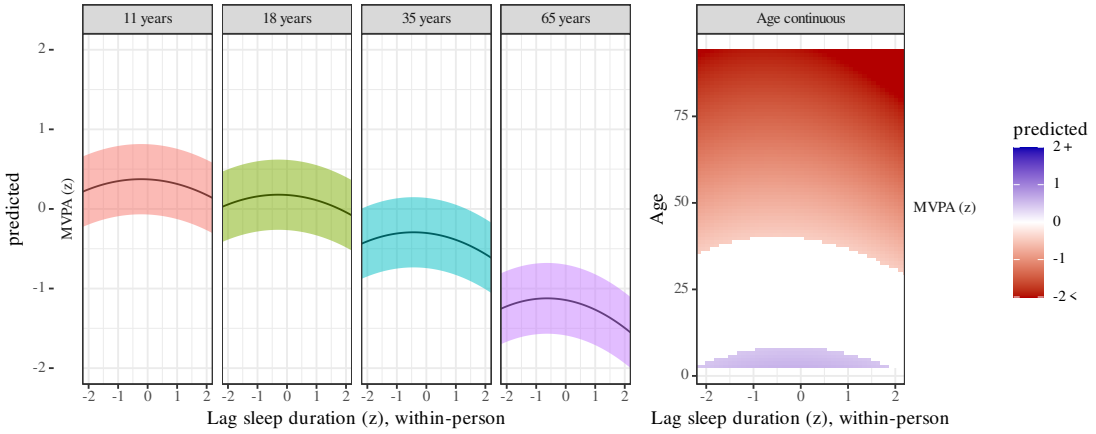


Figure 25: MVPA by previous night sleep duration

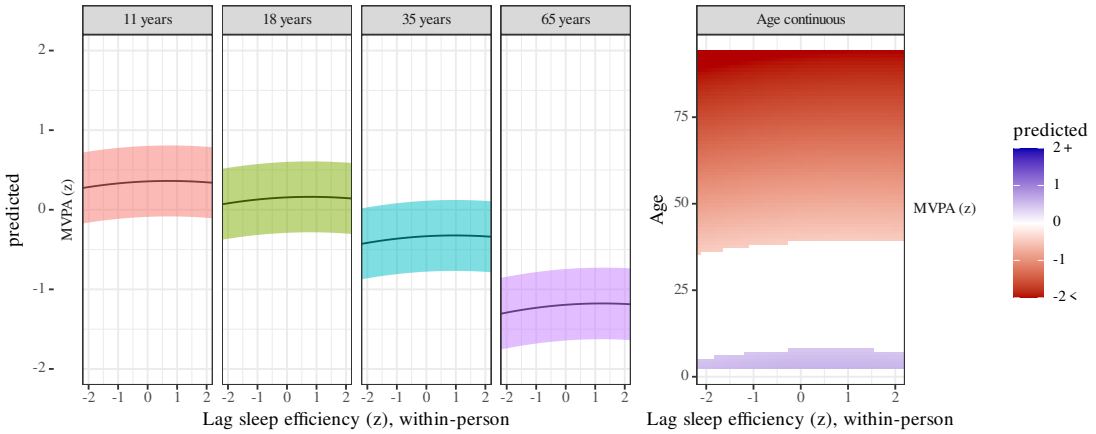


Figure 26: MVPA by previous night sleep efficiency

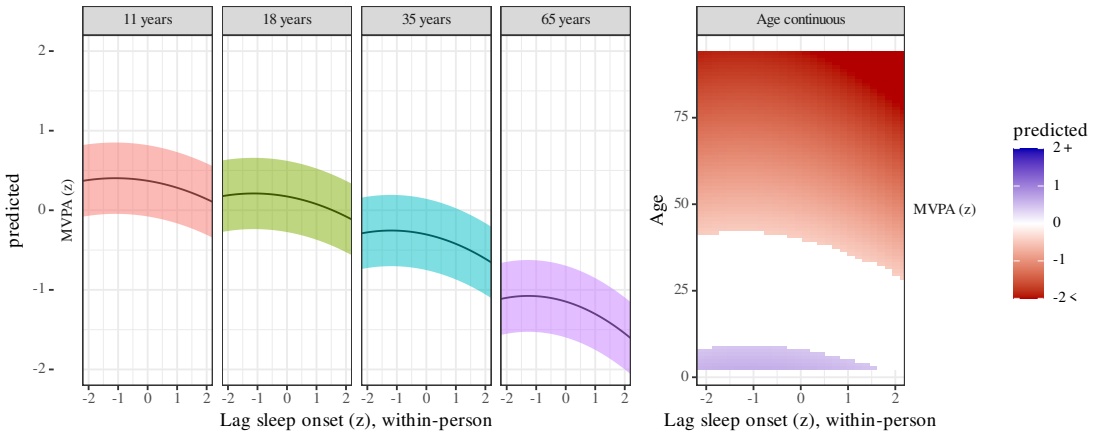


Figure 27: MVPA by previous night sleep onset

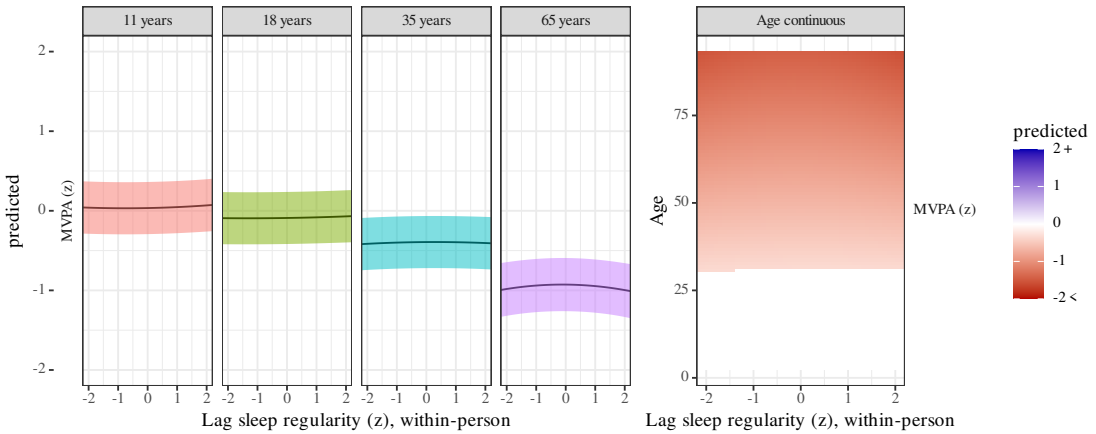


Figure 28: MVPA by sleep regularity over the two preceding days

Negative Control for Sleep Regularity

GGIR's sleep regularity index for a day compares that calendar day with the next, and counts daytime as well as night-time rest. To check whether this couples it to physical activity, we predicted each sleep outcome from within-person physical activity volume on the previous day, the same day, and the next day, adjusting for the person mean, SES, age, sex, and BMI z-score (Table 19). Next-day activity cannot affect the night's sleep, so an association with it indicates overlap in measurement.

Table 19: Sleep outcomes predicted by within-person physical activity volume on the previous, same, and next day

Outcome	Activity	β [95% CI]	p
Regularity	Previous day	−0.01 [−0.02, 0.00]	.006
Regularity	Same day	0.19 [0.18, 0.20]	< .001
Regularity	Next day	0.23 [0.22, 0.25]	< .001
Duration	Previous day	0.00 [−0.01, 0.01]	.363
Duration	Same day	0.10 [0.09, 0.11]	< .001
Duration	Next day	0.03 [0.03, 0.04]	< .001
Onset	Previous day	0.01 [0.00, 0.02]	.002
Onset	Same day	−0.10 [−0.11, −0.09]	< .001
Onset	Next day	−0.02 [−0.03, −0.02]	< .001

Moderation

In the following section we present models for a range of moderators which were pre-defined in our protocol.

BMI z-score

The association between physical activity volume and sleep by BMI z-score

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by BMI z-score, and the results are presented in Table 20 and Figure 29

Table 20: Physical activity predicting sleep controlling for SES, age, and sex.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.05 [−0.20, 0.10]	.504	−0.05 [−0.19, 0.09]	.456
Physical activity (within)	0.11 [0.10, 0.12]	< .001	0.05 [0.05, 0.06]	< .001
BMI z-score	−0.01 [−0.02, 0.00]	.161	−0.02 [−0.03, −0.01]	< .001
Physical activity (within)²	−0.01 [−0.01, 0.00]	.163	0.00 [−0.01, 0.01]	.795
Physical activity (between)	0.07 [0.05, 0.09]	< .001	0.09 [0.08, 0.11]	< .001
Physical activity (between)²	0.01 [0.00, 0.02]	.008	0.03 [0.02, 0.04]	< .001
Physical activity (within) × BMI z-score	0.00 [0.00, 0.01]	.218	0.00 [−0.01, 0.01]	.648
BMI z-score × Physical activity (within)²	−0.01 [−0.01, 0.00]	.159	0.00 [−0.01, 0.01]	.542
BMI z-score × Physical activity (between)	0.01 [0.00, 0.02]	.017	0.03 [0.02, 0.03]	< .001
BMI z-score × Physical activity (between)²	0.00 [−0.01, 0.00]	.698	0.01 [0.00, 0.01]	.021
Sleep efficiency (z)				
(Intercept)	0.23 [0.08, 0.38]	.002	0.22 [0.07, 0.38]	.005
Physical activity (within)	0.03 [0.02, 0.04]	< .001	0.00 [0.00, 0.01]	.366
BMI z-score	−0.01 [−0.02, 0.00]	.042	−0.02 [−0.03, −0.01]	< .001
Physical activity (within)²	−0.01 [−0.02, −0.01]	< .001	−0.03 [−0.04, −0.02]	< .001
Physical activity (between)	0.08 [0.06, 0.10]	< .001	0.03 [0.01, 0.05]	.001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (between)²	−0.03 [−0.04, −0.03]	< .001	−0.01 [−0.02, 0.00]	.071
Physical activity (within) × BMI z-score	0.00 [−0.01, 0.01]	.664	0.00 [−0.01, 0.01]	.866
BMI z-score × Physical activity (within)²	0.00 [−0.01, 0.01]	.630	0.00 [−0.01, 0.00]	.336
BMI z-score × Physical activity (between)	0.03 [0.02, 0.04]	< .001	0.03 [0.02, 0.04]	< .001
BMI z-score × Physical activity (between)²	−0.01 [−0.01, 0.00]	.088	0.01 [0.00, 0.01]	.180
Sleep onset (z)				
(Intercept)	0.15 [−0.12, 0.43]	.276	0.17 [−0.09, 0.43]	.207
Physical activity (within)	−0.10 [−0.11, −0.09]	< .001	−0.08 [−0.08, −0.07]	< .001
BMI z-score	0.00 [−0.01, 0.00]	.259	0.00 [−0.01, 0.01]	.763
Physical activity (within)²	0.01 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.421
Physical activity (between)	−0.09 [−0.11, −0.08]	< .001	−0.09 [−0.10, −0.07]	< .001
Physical activity (between)²	−0.01 [−0.01, 0.00]	.048	−0.03 [−0.04, −0.03]	< .001
Physical activity (within) × BMI z-score	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.02, 0.00]	< .001
BMI z-score × Physical activity (within)²	0.00 [0.00, 0.01]	.199	−0.01 [−0.01, 0.00]	.064
BMI z-score × Physical activity (between)	−0.01 [−0.01, 0.00]	.243	−0.02 [−0.02, −0.01]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
BMI z-score × Physical activity (between)²	0.00 [−0.01, 0.00]	.916	0.00 [−0.01, 0.00]	.569
Sleep regularity (z)				
(Intercept)	0.19 [0.05, 0.33]	.008	0.17 [0.00, 0.34]	.051
Physical activity (within)	0.18 [0.17, 0.19]	< .001	0.10 [0.09, 0.11]	< .001
BMI z-score	−0.04 [−0.06, −0.03]	< .001	−0.07 [−0.08, −0.06]	< .001
Physical activity (within)²	−0.12 [−0.13, −0.11]	< .001	−0.14 [−0.15, −0.13]	< .001
Physical activity (between)	0.47 [0.45, 0.49]	< .001	0.13 [0.11, 0.15]	< .001
Physical activity (between)²	−0.09 [−0.10, −0.09]	< .001	−0.02 [−0.03, −0.01]	< .001
Physical activity (within) × BMI z-score	0.01 [0.00, 0.02]	.009	0.01 [0.00, 0.02]	.032
BMI z-score × Physical activity (within)²	0.01 [0.00, 0.01]	.141	−0.03 [−0.04, −0.02]	< .001
BMI z-score × Physical activity (between)	0.04 [0.03, 0.06]	< .001	0.05 [0.04, 0.06]	< .001
BMI z-score × Physical activity (between)²	−0.01 [−0.02, 0.00]	.005	0.00 [0.00, 0.01]	.287

Adjusted for SES, age, and sex. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

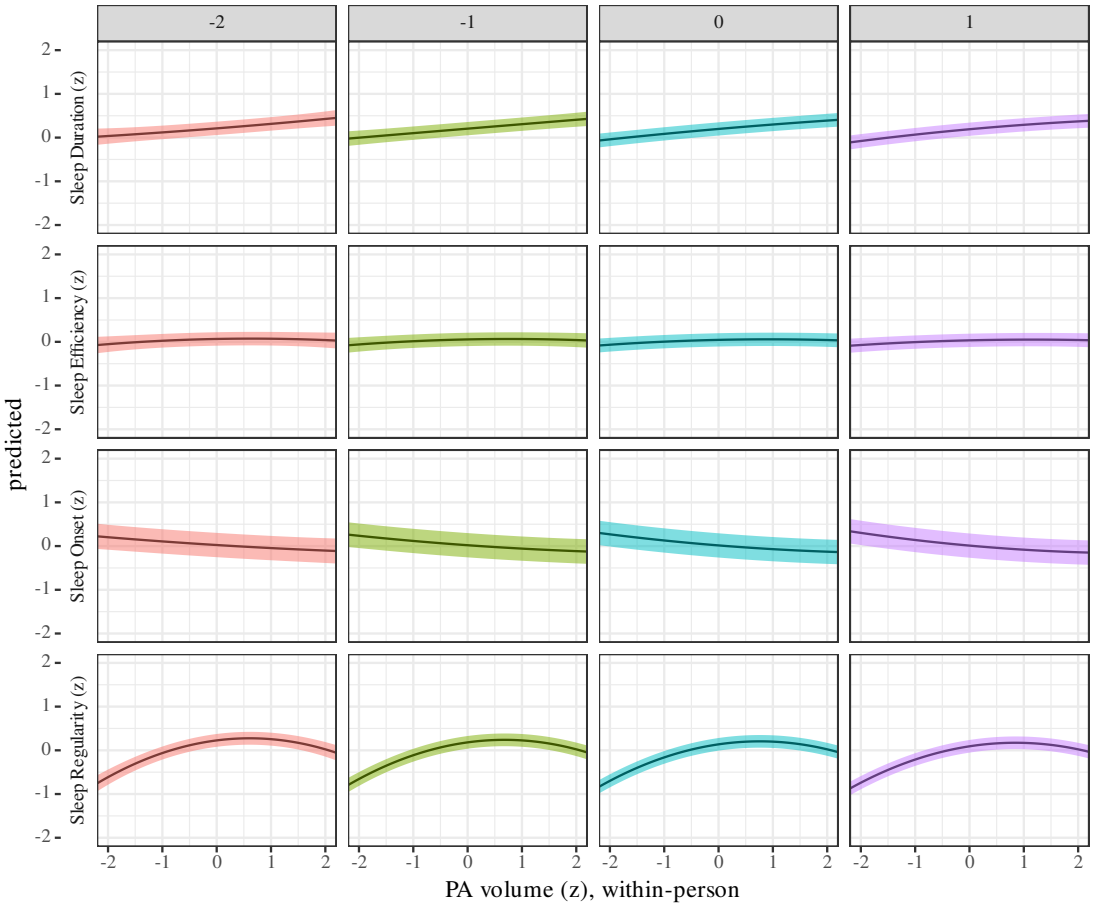


Figure 29: Sleep metrics on Physical activity volume by BMI z-score

The association between physical activity intensity and sleep by BMI z-score

We estimated how physical activity intensity is associated with sleep across BMI z-score. We present the results in Table 20 and Figure 30.

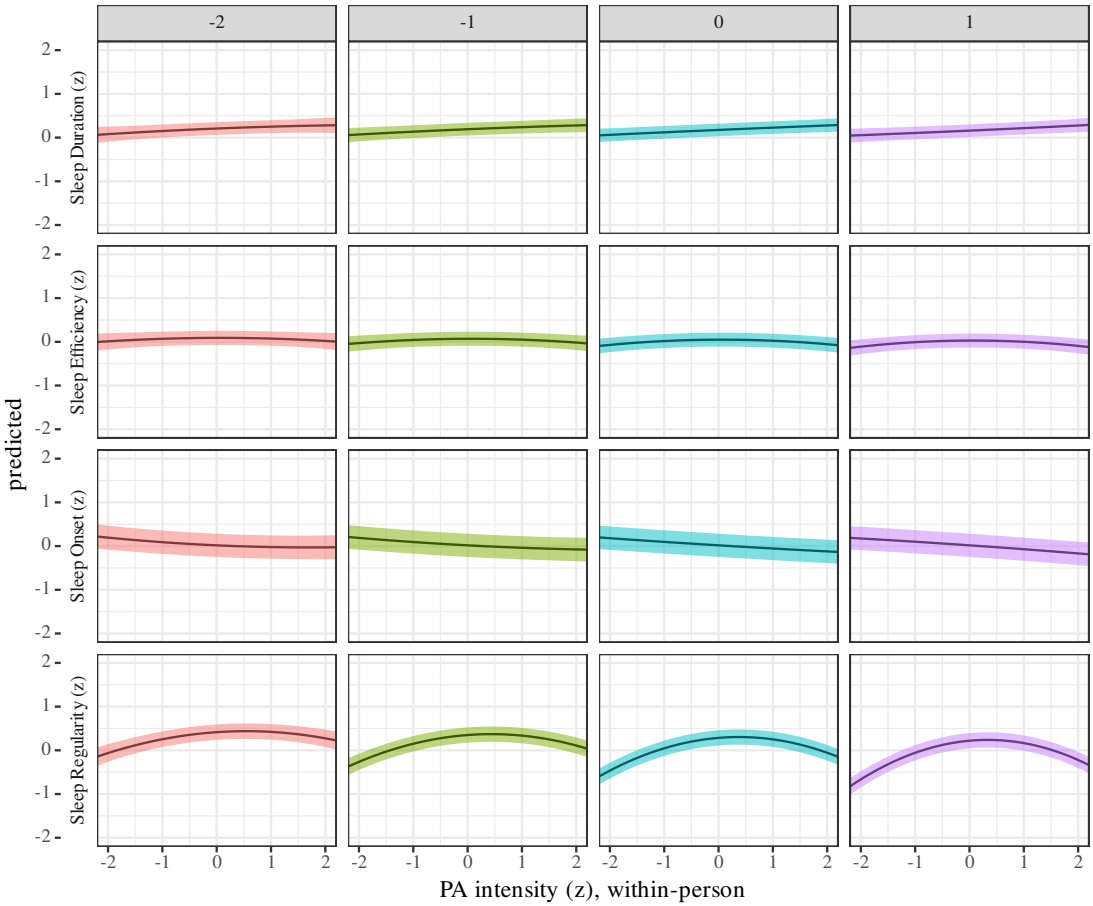


Figure 30: Sleep metrics on Physical activity intensity moderated by BMI z-score

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 21 and Figure 31.

Table 21: Sleep predicting physical activity controlling for SES, age, and sex

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.08 [−0.24, 0.40]	.620	−0.15 [−0.29, −0.01]	.032
Sleep duration (within, prior)	0.02 [0.01, 0.02]	< .001	0.01 [0.01, 0.02]	< .001
BMI z-score	−0.05 [−0.06, −0.04]	< .001	−0.06 [−0.07, −0.05]	< .001
Sleep duration (within, prior)²	−0.05 [−0.05, −0.04]	< .001	−0.03 [−0.03, −0.02]	< .001
Sleep duration (between)	0.06 [0.05, 0.07]	< .001	0.06 [0.05, 0.07]	< .001
Sleep duration (between)²	−0.01 [−0.02, 0.00]	.021	0.00 [−0.01, 0.01]	.511
Sleep duration (within, prior) × BMI z-score	0.00 [0.00, 0.01]	.248	0.01 [0.00, 0.01]	.012
BMI z-score × Sleep duration (within, prior)²	0.00 [0.00, 0.01]	.291	0.01 [0.00, 0.01]	.002
BMI z-score × Sleep duration (between)	0.01 [0.00, 0.02]	.013	0.00 [−0.01, 0.01]	.723
BMI z-score × Sleep duration (between)²	−0.01 [−0.02, 0.00]	.114	−0.01 [−0.01, 0.00]	.196
Sleep efficiency (z)				
(Intercept)	0.07 [−0.25, 0.39]	.666	−0.16 [−0.29, −0.02]	.024
Sleep efficiency (within, prior)	0.01 [0.00, 0.01]	.018	0.01 [0.01, 0.02]	< .001
BMI z-score	−0.05 [−0.06, −0.04]	< .001	−0.06 [−0.06, −0.05]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency (within, prior)²	−0.02 [−0.02, −0.01]	< .001	−0.02 [−0.02, −0.01]	< .001
Sleep efficiency (between)	−0.01 [−0.02, 0.01]	.279	0.01 [0.00, 0.02]	.147
Sleep efficiency (between)²	−0.02 [−0.03, −0.01]	< .001	−0.01 [−0.02, 0.00]	.006
Sleep efficiency (within, prior) × BMI z-score	0.00 [−0.01, 0.00]	.618	0.00 [0.00, 0.01]	.467
BMI z-score × Sleep efficiency (within, prior)²	0.00 [0.00, 0.01]	.297	0.00 [0.00, 0.01]	.051
BMI z-score × Sleep efficiency (between)	−0.02 [−0.03, −0.01]	< .001	−0.01 [−0.02, 0.00]	.011
BMI z-score × Sleep efficiency (between)²	−0.01 [−0.02, 0.00]	.042	−0.01 [−0.01, 0.00]	.086
Sleep onset (z)				
(Intercept)	0.07 [−0.26, 0.41]	.673	−0.16 [−0.30, −0.02]	.030
Sleep onset (within, prior)	0.00 [0.00, 0.01]	.504	0.00 [−0.01, 0.01]	.998
BMI z-score	−0.05 [−0.06, −0.05]	< .001	−0.06 [−0.06, −0.05]	< .001
Sleep onset (within, prior)²	−0.03 [−0.03, −0.02]	< .001	−0.02 [−0.02, −0.01]	< .001
Sleep onset (between)	−0.11 [−0.13, −0.10]	< .001	−0.08 [−0.10, −0.07]	< .001
Sleep onset (between)²	0.01 [0.00, 0.01]	.233	0.01 [0.01, 0.02]	< .001
Sleep onset (within, prior) × BMI z-score	0.00 [−0.01, 0.01]	.948	0.00 [−0.01, 0.00]	.217
BMI z-score × Sleep onset (within, prior)²	0.01 [0.00, 0.01]	.032	0.01 [0.00, 0.01]	.053

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
BMI z-score × Sleep onset (between)	−0.01 [−0.01, 0.00]	.083	0.01 [0.00, 0.01]	.089
BMI z-score × Sleep onset (between)²	0.00 [0.00, 0.01]	.580	0.00 [−0.01, 0.00]	.098
Sleep regularity (z)				
(Intercept)	0.03 [−0.27, 0.33]	.852	−0.17 [−0.30, −0.04]	.011
Sleep regularity (within, prior)	0.00 [−0.01, 0.00]	.241	0.01 [0.01, 0.02]	< .001
BMI z-score	−0.03 [−0.04, −0.02]	< .001	−0.05 [−0.06, −0.05]	< .001
Sleep regularity (within, prior)²	0.00 [0.00, 0.01]	.436	0.00 [−0.01, 0.01]	.778
Sleep regularity (between)	0.17 [0.16, 0.18]	< .001	0.06 [0.05, 0.07]	< .001
Sleep regularity (between)²	0.00 [−0.01, 0.01]	.668	−0.02 [−0.03, −0.01]	< .001
Sleep regularity (within, prior) × BMI z-score	−0.01 [−0.01, 0.00]	.014	0.00 [−0.01, 0.01]	.907
BMI z-score × Sleep regularity (within, prior)²	−0.01 [−0.01, 0.00]	.040	0.00 [0.00, 0.01]	.407
BMI z-score × Sleep regularity (between)	−0.02 [−0.03, −0.01]	< .001	0.00 [−0.01, 0.01]	.629
BMI z-score × Sleep regularity (between)²	−0.01 [−0.02, −0.01]	.001	0.00 [−0.01, 0.00]	.287

Adjusted for SES, age, and sex. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

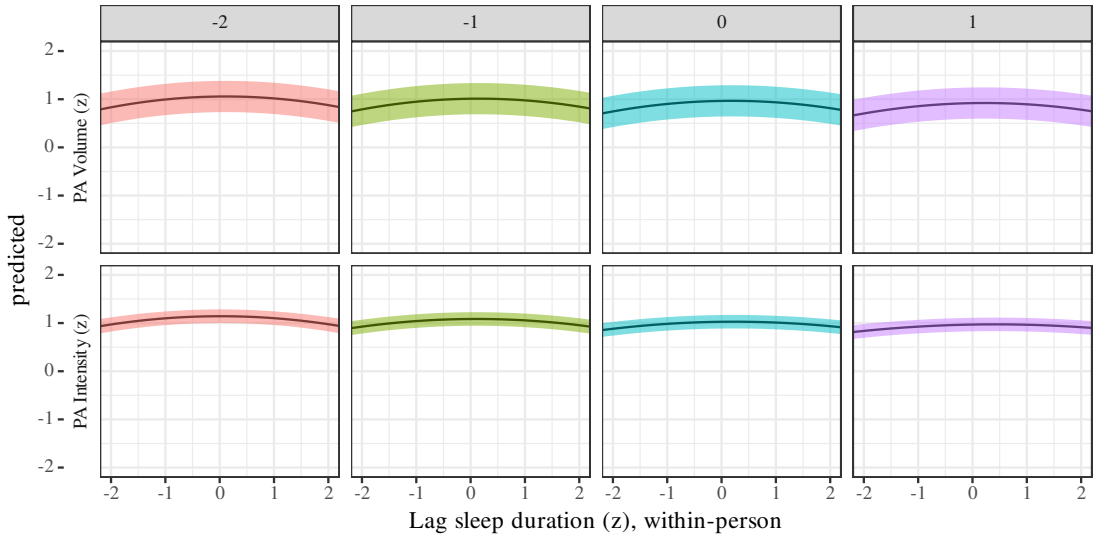


Figure 31: Physical activity by sleep duration moderated by BMI z-score

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 21 and Figure 32.

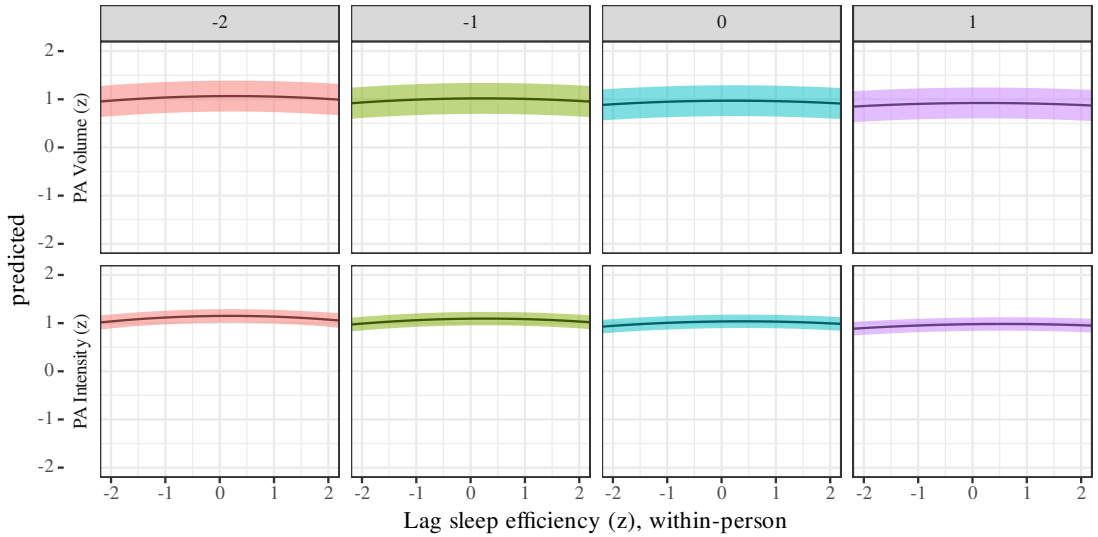


Figure 32: Physical activity by sleep efficiency moderated by BMI z-score

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by BMI z-score. Results, controlling for SES, age and sex are presented in Table 21 and Figure 33.

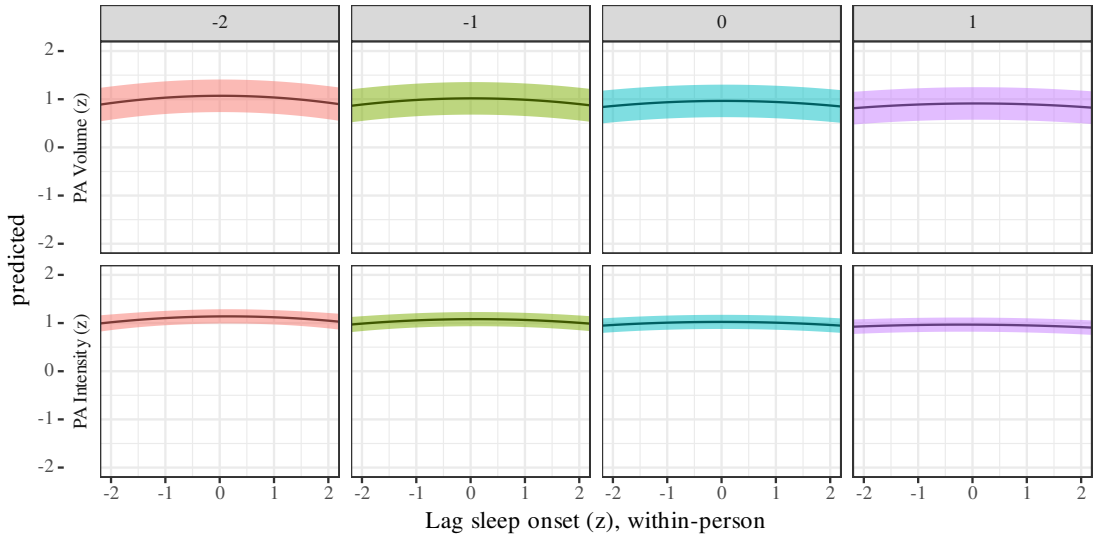


Figure 33: Physical activity by sleep onset moderated by BMI z-score

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by BMI z-score. Results are presented in Table 21 and Figure 34.

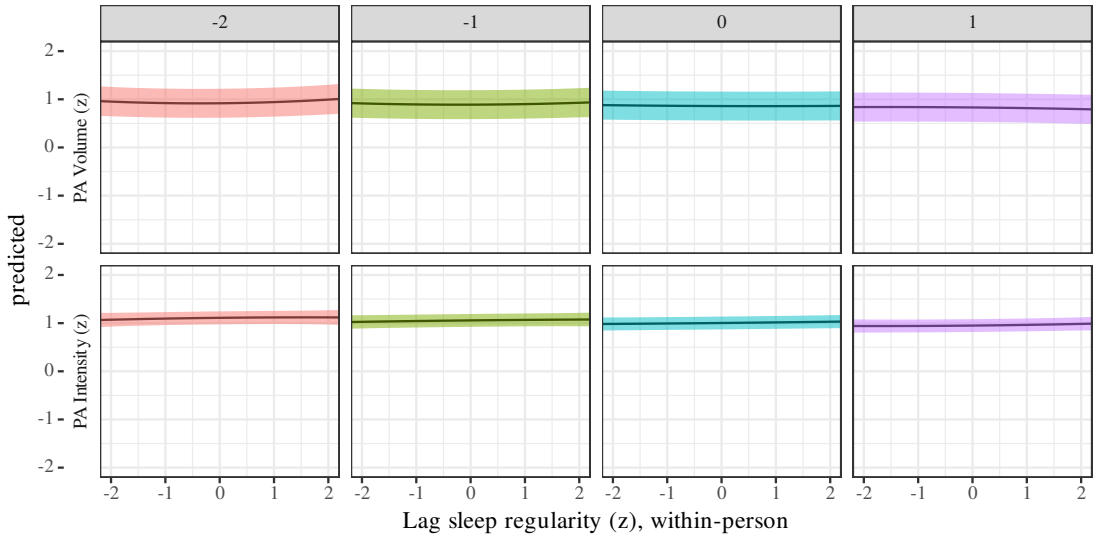


Figure 34: Physical activity by sleep regularity moderated by BMI z-score

SES

The association between physical activity volume and sleep by SES

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by SES, and the results are presented in Table 22 and Figure 35

Table 22: Physical activity predicting sleep controlling for age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.01 [−0.20, 0.18]	.896	0.00 [−0.18, 0.18]	.980
Physical activity (within)	0.12 [0.10, 0.13]	< .001	0.08 [0.06, 0.09]	< .001
Medium	0.02 [−0.01, 0.04]	.143	0.02 [0.00, 0.05]	.064
High	0.04 [0.02, 0.07]	< .001	−0.01 [−0.03, 0.02]	.624
Physical activity (within)²	−0.01 [−0.02, 0.01]	.430	−0.02 [−0.04, 0.00]	.102
Physical activity (between)	0.01 [−0.01, 0.04]	.369	0.07 [0.05, 0.10]	< .001
Physical activity (between)²	0.03 [0.01, 0.04]	< .001	0.02 [0.01, 0.04]	.003
Physical activity (within) × Medium	0.01 [−0.02, 0.03]	.607	−0.02 [−0.04, 0.00]	.072
Physical activity (within) × High	−0.03 [−0.05, −0.01]	.012	−0.04 [−0.06, −0.02]	< .001
Medium × Physical activity (within)²	0.00 [−0.02, 0.02]	.785	0.01 [−0.02, 0.03]	.612
High × Physical activity (within)²	0.00 [−0.02, 0.03]	.660	0.02 [−0.01, 0.04]	.234
Medium × Physical activity (between)	0.04 [0.01, 0.07]	.013	0.03 [0.00, 0.05]	.026
High × Physical activity (between)	0.09 [0.06, 0.12]	< .001	0.05 [0.02, 0.07]	< .001
Medium × Physical activity (between)²	−0.01 [−0.02, 0.01]	.444	−0.02 [−0.04, 0.00]	.023

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
High × Physical activity (between)²	−0.03 [−0.05, −0.01]	< .001	0.02 [0.00, 0.04]	.047
Sleep efficiency (z)				
(Intercept)	0.16 [−0.02, 0.34]	.083	0.15 [−0.04, 0.34]	.131
Physical activity (within)	0.00 [−0.02, 0.01]	.662	−0.02 [−0.04, −0.01]	.005
Medium	0.03 [0.00, 0.06]	.024	0.04 [0.01, 0.07]	.004
High	0.05 [0.02, 0.08]	< .001	0.03 [0.00, 0.05]	.081
Physical activity (within)²	−0.03 [−0.04, −0.01]	< .001	−0.03 [−0.05, −0.01]	< .001
Physical activity (between)	0.08 [0.05, 0.11]	< .001	0.04 [0.02, 0.06]	.001
Physical activity (between)²	−0.03 [−0.04, −0.01]	< .001	0.00 [−0.02, 0.01]	.850
Physical activity (within) × Medium	0.02 [0.00, 0.04]	.015	0.03 [0.01, 0.05]	.009
Physical activity (within) × High	0.04 [0.02, 0.06]	< .001	0.04 [0.02, 0.06]	< .001
Medium × Physical activity (within)²	0.02 [0.00, 0.04]	.072	0.00 [−0.02, 0.03]	.948
High × Physical activity (within)²	0.01 [−0.01, 0.03]	.240	0.01 [−0.02, 0.03]	.562
Medium × Physical activity (between)	−0.01 [−0.04, 0.02]	.467	0.00 [−0.03, 0.02]	.909
High × Physical activity (between)	0.05 [0.01, 0.08]	.007	0.01 [−0.01, 0.04]	.364

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Medium × Physical activity (between)²	0.00 [−0.02, 0.02]	.973	−0.01 [−0.03, 0.01]	.158
High × Physical activity (between)²	−0.02 [−0.04, −0.01]	.007	0.00 [−0.02, 0.02]	.845
Sleep onset (z)				
(Intercept)	0.07 [−0.25, 0.39]	.683	0.06 [−0.24, 0.37]	.678
Physical activity (within)	−0.15 [−0.16, −0.14]	< .001	−0.12 [−0.13, −0.10]	< .001
Medium	0.03 [0.01, 0.05]	.005	0.04 [0.02, 0.06]	< .001
High	0.02 [0.00, 0.04]	.045	0.05 [0.03, 0.07]	< .001
Physical activity (within)²	0.00 [−0.01, 0.02]	.424	0.01 [0.00, 0.03]	.153
Physical activity (between)	−0.10 [−0.12, −0.08]	< .001	−0.06 [−0.08, −0.04]	< .001
Physical activity (between)²	0.00 [−0.01, 0.01]	.777	−0.01 [−0.02, 0.00]	.040
Physical activity (within) × Medium	0.03 [0.01, 0.04]	< .001	0.03 [0.01, 0.04]	< .001
Physical activity (within) × High	0.09 [0.07, 0.10]	< .001	0.07 [0.05, 0.08]	< .001
Medium × Physical activity (within)²	0.02 [0.01, 0.04]	.004	0.01 [−0.01, 0.02]	.552
High × Physical activity (within)²	0.00 [−0.02, 0.02]	.962	−0.01 [−0.03, 0.01]	.162
Medium × Physical activity (between)	0.00 [−0.03, 0.03]	.956	−0.02 [−0.04, 0.00]	.109

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
High × Physical activity (between)	0.01 [−0.02, 0.04]	.527	−0.04 [−0.06, −0.02]	< .001
Medium × Physical activity (between)²	0.00 [−0.01, 0.01]	.974	0.00 [−0.01, 0.02]	.734
High × Physical activity (between)²	−0.01 [−0.03, 0.00]	.112	−0.03 [−0.05, −0.01]	.001
Sleep regularity (z)				
(Intercept)	0.19 [0.04, 0.35]	.016	0.19 [0.00, 0.37]	.046
Physical activity (within)	0.18 [0.16, 0.20]	< .001	0.09 [0.07, 0.11]	< .001
Medium	0.02 [−0.01, 0.05]	.190	−0.01 [−0.04, 0.03]	.708
High	0.06 [0.03, 0.10]	< .001	0.05 [0.02, 0.09]	.005
Physical activity (within)²	−0.12 [−0.14, −0.10]	< .001	−0.17 [−0.19, −0.14]	< .001
Physical activity (between)	0.50 [0.46, 0.53]	< .001	0.12 [0.09, 0.15]	< .001
Physical activity (between)²	−0.10 [−0.12, −0.09]	< .001	−0.02 [−0.04, 0.00]	.012
Physical activity (within) × Medium	−0.01 [−0.03, 0.01]	.419	−0.01 [−0.04, 0.01]	.225
Physical activity (within) × High	0.01 [−0.01, 0.04]	.344	0.02 [0.00, 0.05]	.087
Medium × Physical activity (within)²	0.01 [−0.02, 0.03]	.522	0.07 [0.04, 0.10]	< .001
High × Physical activity (within)²	−0.02 [−0.04, 0.00]	.117	0.00 [−0.03, 0.03]	.777

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Medium × Physical activity (between)	−0.02 [−0.05, 0.02]	.428	0.01 [−0.02, 0.04]	.620
High × Physical activity (between)	0.00 [−0.04, 0.04]	.908	0.01 [−0.02, 0.04]	.612
Medium × Physical activity (between)²	0.00 [−0.02, 0.02]	.840	0.00 [−0.03, 0.02]	.717
High × Physical activity (between)²	0.00 [−0.02, 0.02]	.823	−0.02 [−0.04, 0.00]	.104

Adjusted for age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

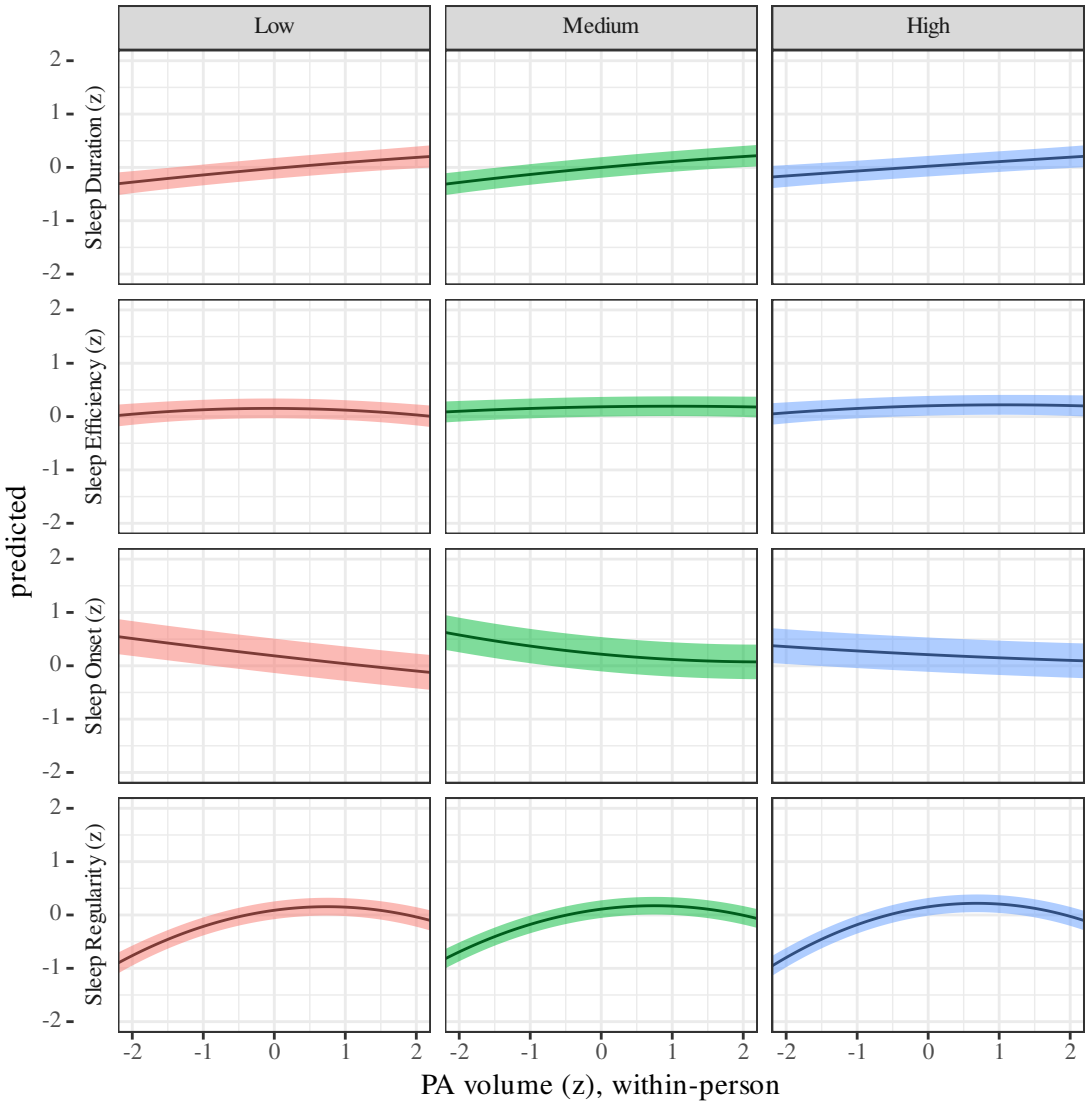


Figure 35: Sleep metrics on Physical activity volume by SES

The association between physical activity intensity and sleep by SES

We estimated how physical activity intensity is associated with sleep across SES. We present the results in Table 22 and Figure 36.

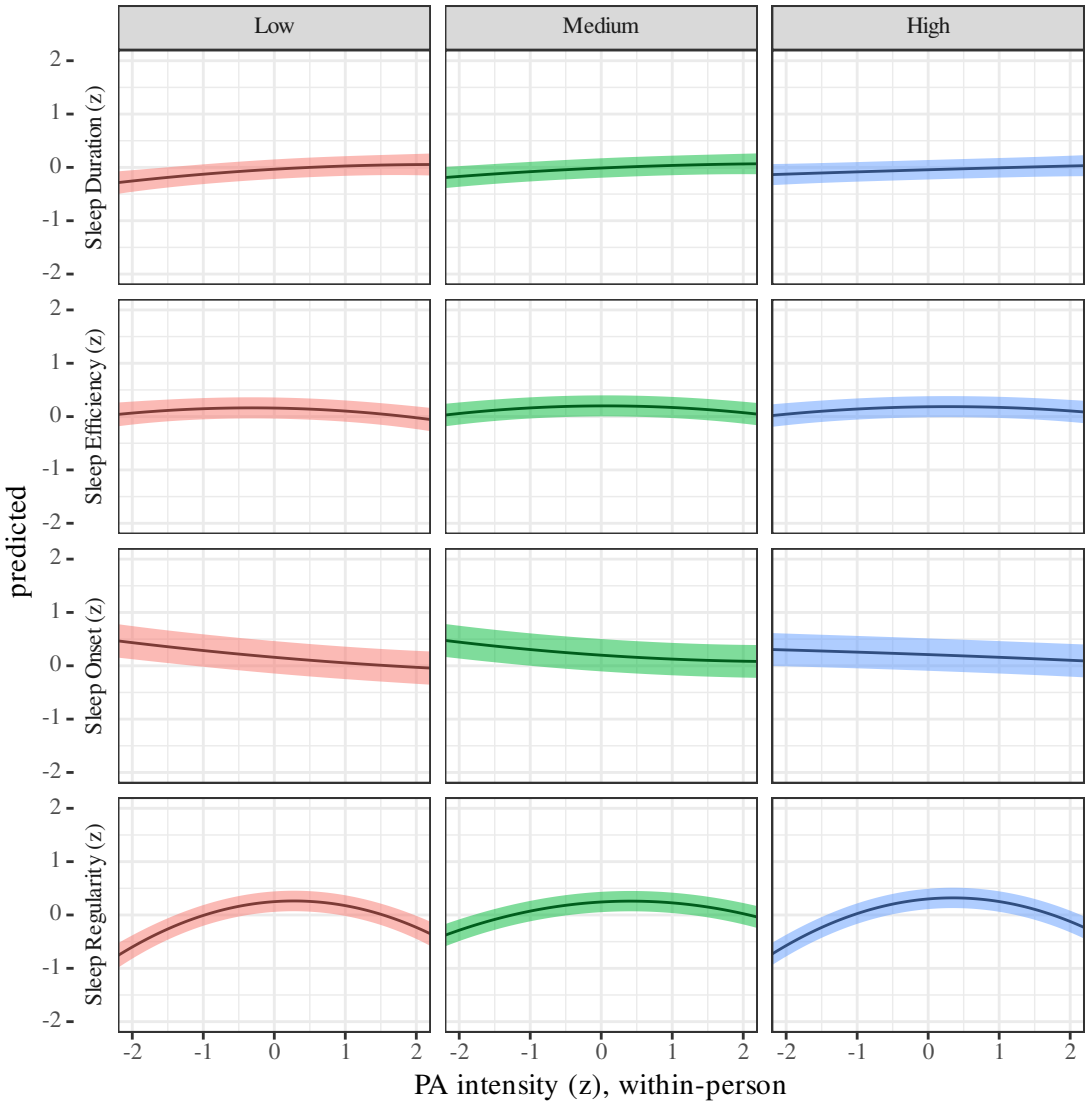


Figure 36: Sleep metrics on Physical activity intensity moderated by SES

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 23 and Figure 37.

Table 23: Sleep predicting physical activity controlling for age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.19 [−0.13, 0.50]	.242	−0.05 [−0.18, 0.07]	.390
Sleep duration (within, prior)	0.00 [−0.01, 0.00]	.474	0.02 [0.02, 0.03]	< .001
Medium	−0.07 [−0.10, −0.05]	< .001	0.00 [−0.02, 0.02]	.832
High	−0.10 [−0.13, −0.08]	< .001	−0.01 [−0.03, 0.02]	.593
Sleep duration (within, prior)²	−0.04 [−0.05, −0.04]	< .001	−0.02 [−0.02, −0.01]	< .001
Sleep duration (between)	0.02 [0.00, 0.04]	.031	0.03 [0.01, 0.05]	< .001
Sleep duration (between)²	−0.04 [−0.06, −0.02]	< .001	−0.03 [−0.04, −0.01]	< .001
Sleep duration (within, prior) × Medium	0.02 [0.01, 0.03]	< .001	−0.01 [−0.02, 0.01]	.312
Sleep duration (within, prior) × High	0.02 [0.01, 0.03]	.001	0.00 [−0.01, 0.01]	.716
Medium × Sleep duration (within, prior)²	0.00 [−0.01, 0.01]	.515	0.00 [−0.01, 0.01]	.444
High × Sleep duration (within, prior)²	0.00 [−0.01, 0.01]	.887	−0.01 [−0.02, 0.00]	.133
Medium × Sleep duration (between)	0.03 [0.00, 0.05]	.023	0.03 [0.01, 0.05]	.007
High × Sleep duration (between)	0.06 [0.04, 0.08]	< .001	0.05 [0.03, 0.07]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Medium × Sleep duration (between)²	0.02 [0.00, 0.04]	.054	0.01 [−0.01, 0.03]	.180
High × Sleep duration (between)²	0.03 [0.01, 0.06]	.004	0.04 [0.02, 0.06]	< .001
Sleep efficiency (z)				
(Intercept)	0.15 [−0.15, 0.46]	.328	−0.07 [−0.20, 0.06]	.277
Sleep efficiency (within, prior)	0.00 [−0.01, 0.01]	.670	0.00 [−0.01, 0.01]	.754
Medium	−0.05 [−0.08, −0.03]	< .001	0.01 [−0.01, 0.03]	.313
High	−0.07 [−0.09, −0.05]	< .001	0.03 [0.00, 0.05]	.015
Sleep efficiency (within, prior)²	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.02, 0.00]	.005
Sleep efficiency (between)	0.03 [0.01, 0.06]	< .001	0.02 [0.00, 0.04]	.015
Sleep efficiency (between)²	−0.01 [−0.02, 0.01]	.542	−0.01 [−0.02, 0.01]	.501
Sleep efficiency (within, prior) × Medium	0.00 [−0.01, 0.01]	.711	0.01 [0.00, 0.02]	.090
Sleep efficiency (within, prior) × High	0.01 [0.00, 0.03]	.015	0.02 [0.01, 0.03]	.004
Medium × Sleep efficiency (within, prior)²	0.00 [−0.01, 0.01]	.856	0.00 [−0.01, 0.01]	.844
High × Sleep efficiency (within, prior)²	0.00 [−0.01, 0.01]	.783	−0.01 [−0.02, 0.00]	.062

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Medium × Sleep efficiency (between)	−0.04 [−0.06, −0.01]	.004	0.00 [−0.03, 0.02]	.776
High × Sleep efficiency (between)	−0.04 [−0.07, −0.02]	.001	−0.02 [−0.04, 0.01]	.153
Medium × Sleep efficiency (between)²	−0.01 [−0.03, 0.01]	.398	0.00 [−0.02, 0.02]	.857
High × Sleep efficiency (between)²	−0.02 [−0.05, 0.00]	.025	−0.01 [−0.03, 0.01]	.400
Sleep onset (z)				
(Intercept)	0.18 [−0.14, 0.50]	.275	−0.06 [−0.19, 0.07]	.350
Sleep onset (within, prior)	−0.02 [−0.03, −0.01]	< .001	0.01 [0.00, 0.02]	.028
Medium	−0.08 [−0.10, −0.05]	< .001	0.00 [−0.02, 0.03]	.648
High	−0.11 [−0.13, −0.09]	< .001	0.00 [−0.03, 0.02]	.776
Sleep onset (within, prior)²	−0.04 [−0.05, −0.03]	< .001	−0.02 [−0.03, −0.01]	< .001
Sleep onset (between)	−0.08 [−0.10, −0.06]	< .001	−0.05 [−0.07, −0.03]	< .001
Sleep onset (between)²	−0.02 [−0.04, −0.01]	.001	0.00 [−0.02, 0.01]	.629
Sleep onset (within, prior) × Medium	0.03 [0.02, 0.05]	< .001	0.00 [−0.02, 0.01]	.773
Sleep onset (within, prior) × High	0.02 [0.00, 0.03]	.034	−0.01 [−0.03, 0.00]	.088
Medium × Sleep onset (within, prior)²	0.02 [0.01, 0.03]	.002	0.01 [0.00, 0.02]	.205

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
High × Sleep onset (within, prior)²	0.03 [0.02, 0.04]	< .001	0.01 [0.00, 0.03]	.049
Medium × Sleep onset (between)	−0.01 [−0.03, 0.01]	.184	−0.01 [−0.03, 0.01]	.325
High × Sleep onset (between)	−0.03 [−0.05, −0.01]	.011	−0.03 [−0.05, −0.01]	.002
Medium × Sleep onset (between)²	0.02 [0.01, 0.04]	.007	0.01 [−0.01, 0.02]	.460
High × Sleep onset (between)²	0.03 [0.01, 0.05]	< .001	0.02 [0.01, 0.04]	.006
Sleep regularity (z)				
(Intercept)	0.08 [−0.23, 0.39]	.610	−0.15 [−0.28, −0.03]	.013
Sleep regularity (within, prior)	−0.01 [−0.02, 0.00]	.140	0.00 [−0.01, 0.01]	.932
Medium	−0.06 [−0.09, −0.04]	< .001	0.02 [−0.01, 0.04]	.136
High	−0.08 [−0.11, −0.06]	< .001	0.03 [0.01, 0.06]	.010
Sleep regularity (within, prior)²	0.00 [−0.01, 0.02]	.680	0.01 [−0.01, 0.02]	.438
Sleep regularity (between)	0.15 [0.13, 0.18]	< .001	0.07 [0.05, 0.09]	< .001
Sleep regularity (between)²	−0.02 [−0.04, 0.00]	.051	−0.01 [−0.03, 0.01]	.374
Sleep regularity (within, prior) × Medium	−0.01 [−0.02, 0.01]	.486	0.00 [−0.01, 0.02]	.564
Sleep regularity (within, prior) × High	0.02 [0.00, 0.04]	.015	0.02 [0.01, 0.04]	.011

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Medium × Sleep regularity (within, prior)²	−0.01 [−0.03, 0.00]	.174	−0.01 [−0.03, 0.01]	.241
High × Sleep regularity (within, prior)²	0.01 [−0.01, 0.02]	.285	0.00 [−0.01, 0.02]	.813
Medium × Sleep regularity (between)	0.03 [0.00, 0.06]	.034	0.00 [−0.03, 0.02]	.761
High × Sleep regularity (between)	−0.01 [−0.04, 0.02]	.608	−0.02 [−0.05, 0.01]	.171
Medium × Sleep regularity (between)²	0.02 [−0.01, 0.04]	.124	0.00 [−0.03, 0.02]	.826
High × Sleep regularity (between)²	0.02 [−0.01, 0.04]	.177	−0.03 [−0.05, 0.00]	.047

Adjusted for age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

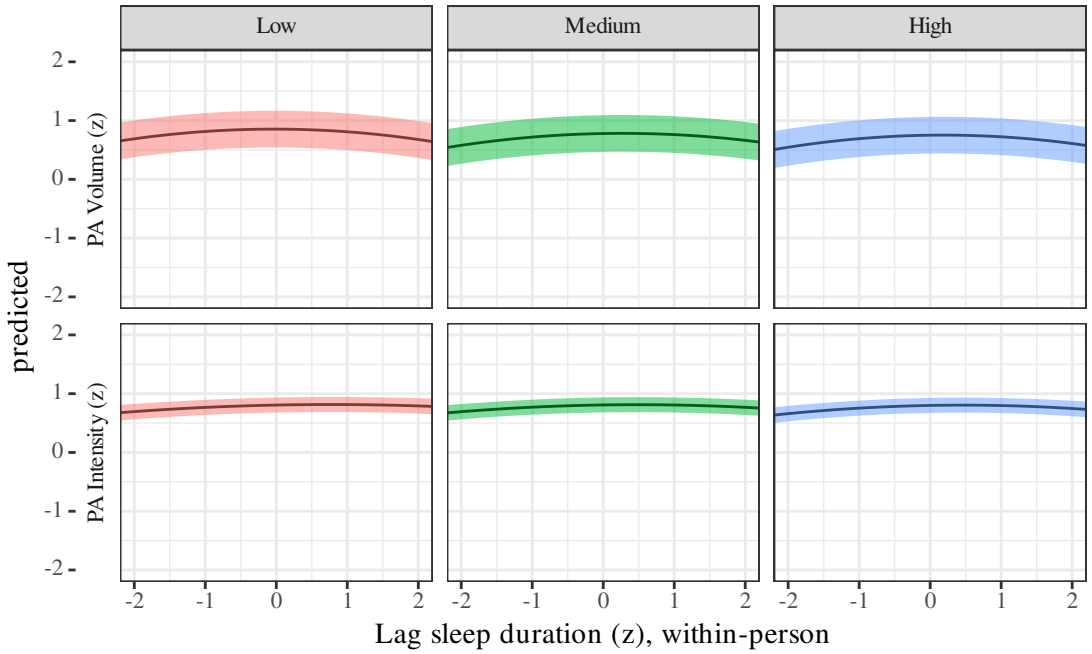


Figure 37: Physical activity by sleep duration moderated by SES

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 23 and Figure 38.

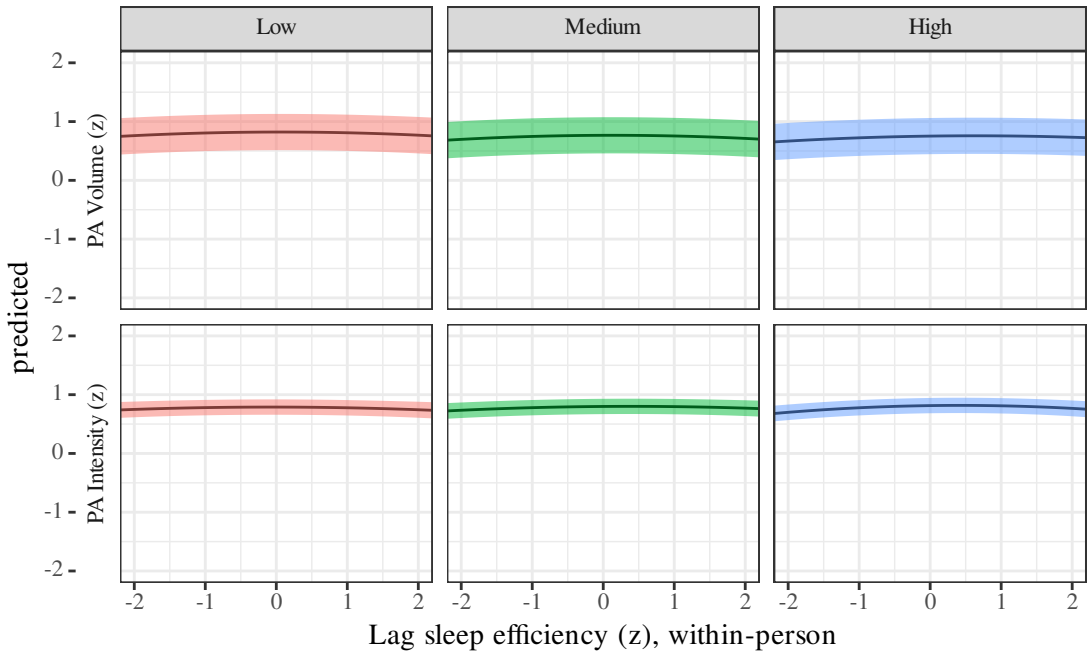


Figure 38: Physical activity by sleep efficiency moderated by SES

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by SES. Results, controlling for age, sex and BMI z-score are presented in Table 23 and Figure 39.

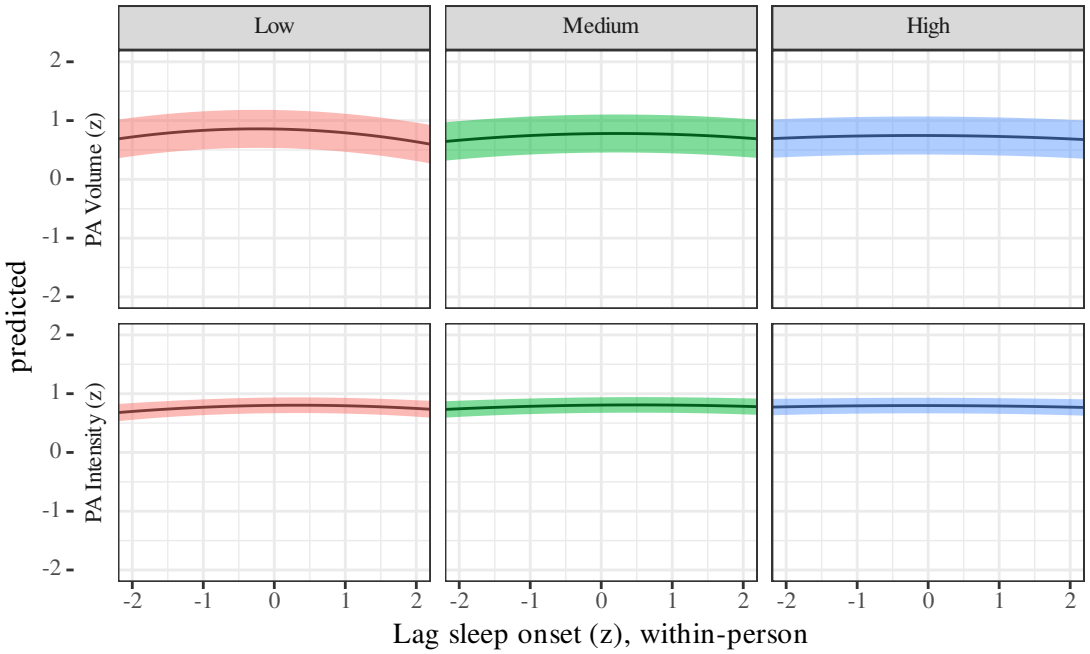


Figure 39: Physical activity by sleep onset moderated by SES

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by SES. Results are presented in Table 23 and Figure 40.

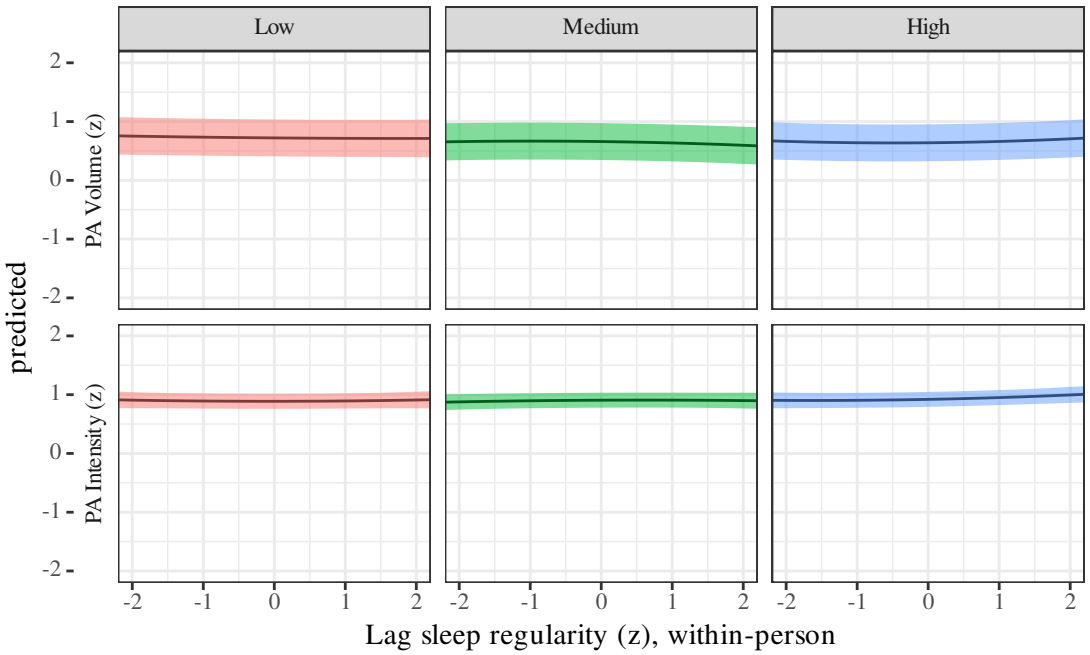


Figure 40: Physical activity by sleep regularity moderated by SES

sex

The association between physical activity volume and sleep by sex

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by sex, and the results are presented in Table 24 and Figure 41

Table 24: Physical activity predicting sleep controlling for SES, age, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.01 [−0.14, 0.17]	.879	0.00 [−0.14, 0.15]	.981
Physical activity (within)	0.12 [0.11, 0.13]	< .001	0.07 [0.06, 0.08]	< .001
Male	−0.15 [−0.16, −0.13]	< .001	−0.14 [−0.16, −0.12]	< .001
Physical activity (within)²	−0.02 [−0.03, 0.00]	.023	−0.01 [−0.03, 0.00]	.066
Physical activity (between)	0.10 [0.08, 0.12]	< .001	0.10 [0.08, 0.12]	< .001
Physical activity (between)²	0.00 [−0.01, 0.01]	.733	0.03 [0.02, 0.04]	< .001
Physical activity (within) × Male	−0.02 [−0.03, 0.00]	.019	−0.02 [−0.03, 0.00]	.029
Male × Physical activity (within)²	0.01 [0.00, 0.03]	.081	0.01 [−0.01, 0.03]	.270
Male × Physical activity (between)	−0.06 [−0.08, −0.03]	< .001	0.01 [0.00, 0.03]	.153
Male × Physical activity (between)²	0.02 [0.01, 0.03]	.007	−0.01 [−0.02, 0.01]	.431
Sleep efficiency (z)				
(Intercept)	0.22 [0.07, 0.37]	.005	0.19 [0.03, 0.35]	.024
Physical activity (within)	0.02 [0.01, 0.03]	.003	0.00 [−0.01, 0.01]	.677
Male	−0.20 [−0.22, −0.18]	< .001	−0.16 [−0.18, −0.14]	< .001
Physical activity (within)²	−0.02 [−0.03, −0.01]	.005	−0.03 [−0.04, −0.02]	< .001
Physical activity (between)	0.11 [0.09, 0.13]	< .001	0.00 [−0.02, 0.02]	.679

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (between)²	−0.06 [−0.08, −0.05]	< .001	−0.01 [−0.02, 0.00]	.109
Physical activity (within) × Male	0.00 [−0.01, 0.02]	.922	0.01 [−0.01, 0.02]	.385
Male × Physical activity (within)²	0.00 [−0.01, 0.02]	.627	0.00 [−0.02, 0.01]	.758
Male × Physical activity (between)	−0.02 [−0.05, 0.00]	.070	0.06 [0.04, 0.07]	< .001
Male × Physical activity (between)²	0.04 [0.02, 0.05]	< .001	−0.02 [−0.03, 0.00]	.056
Sleep onset (z)				
(Intercept)	0.12 [−0.15, 0.39]	.385	0.13 [−0.13, 0.38]	.328
Physical activity (within)	−0.12 [−0.13, −0.12]	< .001	−0.09 [−0.10, −0.08]	< .001
Male	0.01 [0.00, 0.03]	.075	0.03 [0.01, 0.05]	< .001
Physical activity (within)²	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.02]	.008
Physical activity (between)	−0.11 [−0.13, −0.09]	< .001	−0.11 [−0.12, −0.09]	< .001
Physical activity (between)²	−0.01 [−0.02, 0.00]	.160	−0.03 [−0.04, −0.02]	< .001
Physical activity (within) × Male	0.02 [0.01, 0.03]	< .001	0.02 [0.01, 0.03]	.001
Male × Physical activity (within)²	−0.01 [−0.02, 0.00]	.100	−0.01 [−0.02, 0.00]	.108
Male × Physical activity (between)	0.02 [0.00, 0.04]	.061	0.02 [0.01, 0.04]	.002

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Male × Physical activity (between)²	0.00 [−0.01, 0.01]	.848	−0.01 [−0.02, 0.00]	.069
Sleep regularity (z)				
(Intercept)	0.21 [0.07, 0.35]	.003	0.18 [0.01, 0.35]	.041
Physical activity (within)	0.21 [0.20, 0.23]	< .001	0.11 [0.09, 0.12]	< .001
Male	−0.20 [−0.22, −0.18]	< .001	−0.15 [−0.18, −0.13]	< .001
Physical activity (within)²	−0.17 [−0.18, −0.16]	< .001	−0.16 [−0.17, −0.14]	< .001
Physical activity (between)	0.49 [0.46, 0.51]	< .001	0.11 [0.09, 0.14]	< .001
Physical activity (between)²	−0.11 [−0.13, −0.10]	< .001	−0.02 [−0.04, −0.01]	< .001
Physical activity (within) × Male	−0.05 [−0.06, −0.03]	< .001	0.00 [−0.02, 0.01]	.824
Male × Physical activity (within)²	0.08 [0.06, 0.10]	< .001	0.03 [0.01, 0.05]	.003
Male × Physical activity (between)	0.00 [−0.03, 0.03]	.961	0.05 [0.03, 0.07]	< .001
Male × Physical activity (between)²	0.02 [0.00, 0.03]	.024	−0.01 [−0.03, 0.01]	.414

Adjusted for SES, age, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

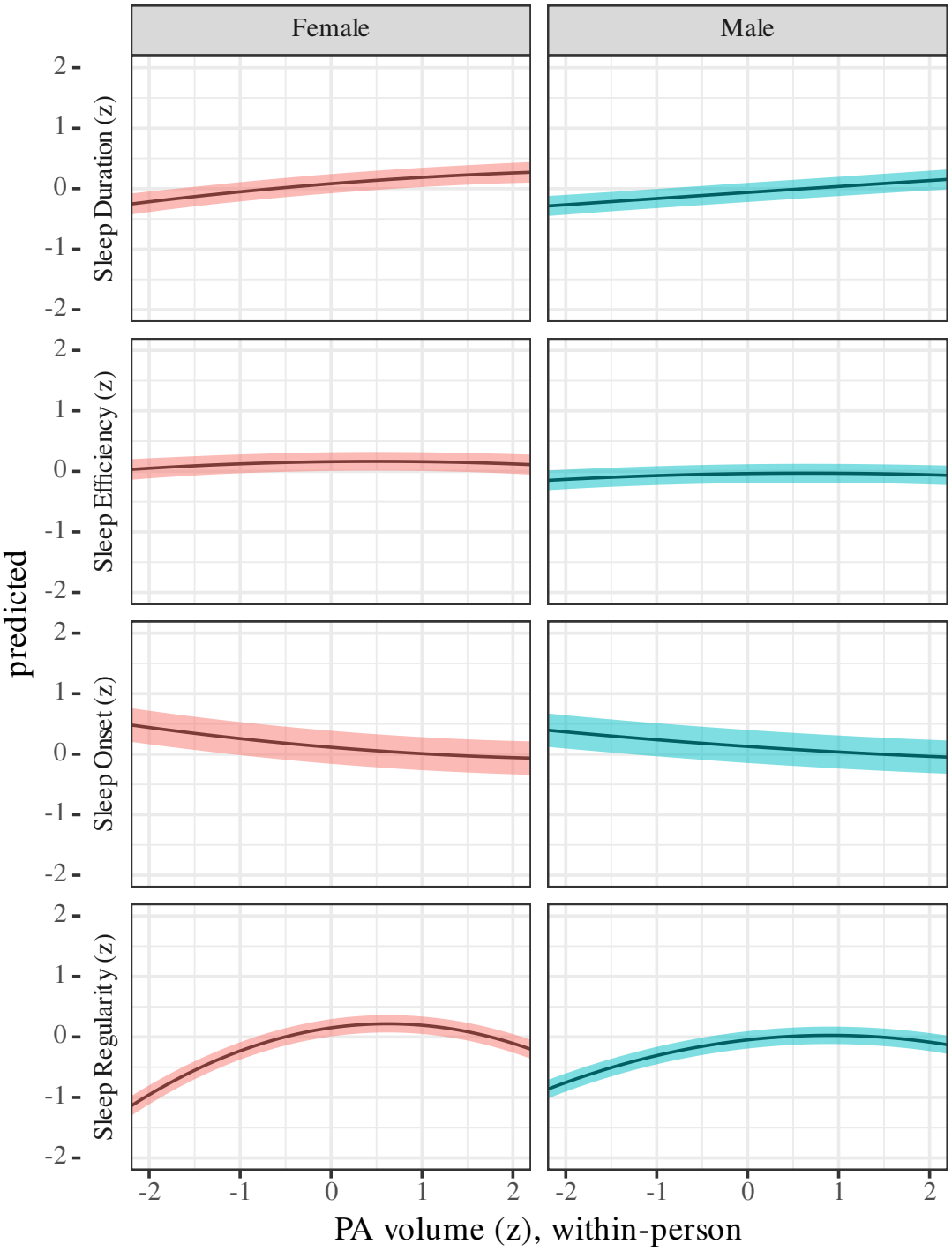


Figure 41: Sleep metrics on Physical activity volume by sex

The association between physical activity intensity and sleep by sex

We estimated how physical activity intensity is associated with sleep across sex. We present the results in Table 24 and Figure 42.

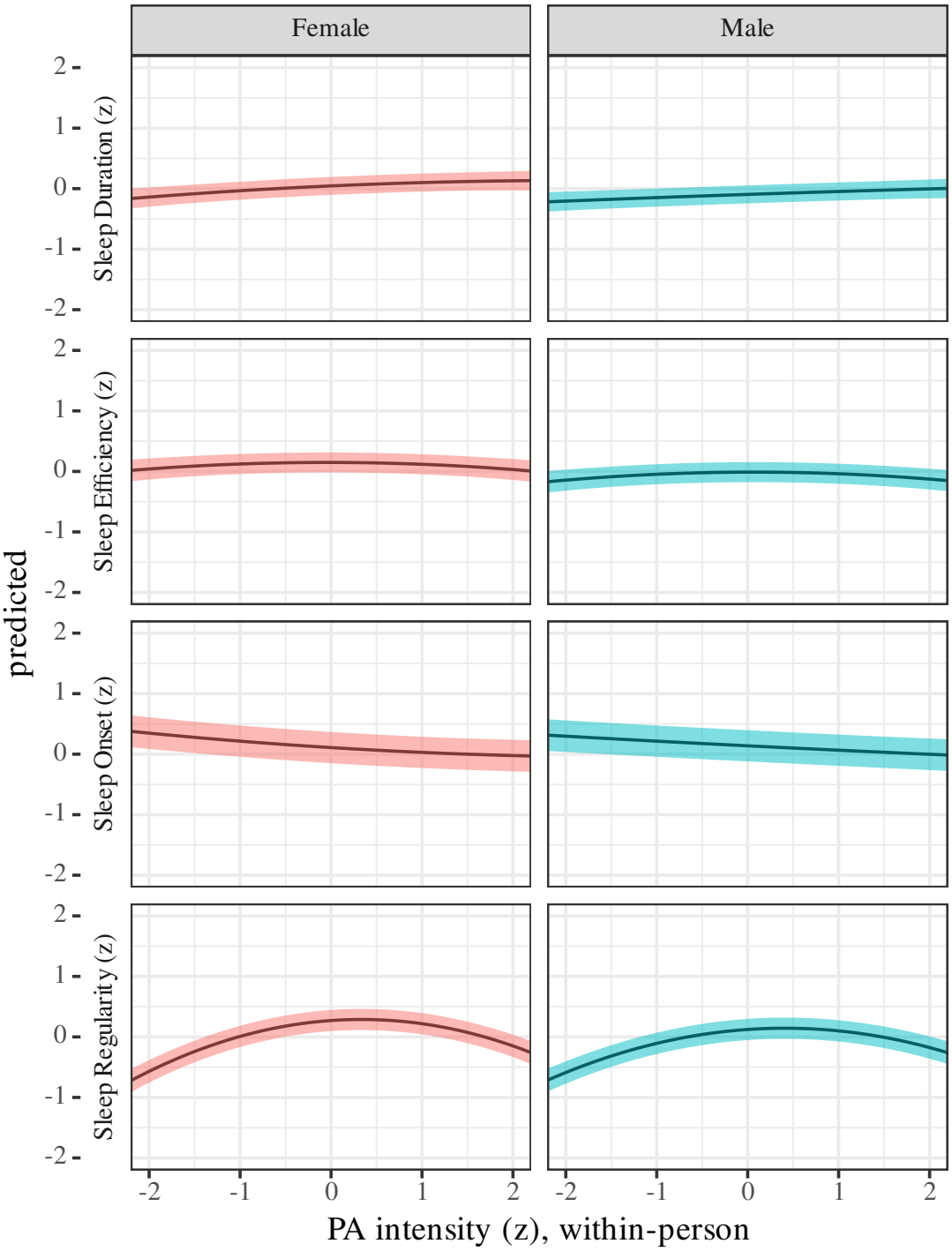


Figure 42: Sleep metrics on Physical activity intensity moderated by sex

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 25 and Figure 43.

Table 25: Sleep predicting physical activity controlling for SES, age, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.15 [−0.16, 0.45]	.342	−0.07 [−0.19, 0.06]	.286
Sleep duration (within, prior)	0.01 [0.00, 0.02]	< .001	0.02 [0.01, 0.03]	< .001
Male	0.16 [0.14, 0.17]	< .001	0.22 [0.21, 0.24]	< .001
Sleep duration (within, prior)²	−0.04 [−0.04, −0.03]	< .001	−0.02 [−0.03, −0.02]	< .001
Sleep duration (between)	0.01 [−0.01, 0.02]	.232	0.06 [0.04, 0.07]	< .001
Sleep duration (between)²	−0.02 [−0.04, −0.01]	.001	−0.01 [−0.02, 0.00]	.143
Sleep duration (within, prior) × Male	0.00 [0.00, 0.01]	.375	0.00 [−0.01, 0.01]	.667
Male × Sleep duration (within, prior)²	−0.01 [−0.02, −0.01]	< .001	0.00 [−0.01, 0.01]	.985
Male × Sleep duration (between)	0.10 [0.08, 0.12]	< .001	0.01 [0.00, 0.03]	.126
Male × Sleep duration (between)²	0.02 [0.00, 0.04]	.041	0.01 [−0.01, 0.02]	.499
Sleep efficiency (z)				
(Intercept)	0.13 [−0.17, 0.42]	.407	−0.07 [−0.19, 0.05]	.277
Sleep efficiency (within, prior)	0.01 [0.00, 0.02]	.007	0.01 [0.00, 0.02]	< .001
Male	0.17 [0.15, 0.18]	< .001	0.21 [0.20, 0.23]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency (within, prior)²	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.02, −0.01]	< .001
Sleep efficiency (between)	0.05 [0.04, 0.07]	< .001	0.03 [0.01, 0.04]	< .001
Sleep efficiency (between)²	−0.02 [−0.03, 0.00]	.006	−0.02 [−0.03, −0.01]	< .001
Sleep efficiency (within, prior) × Male	0.00 [−0.01, 0.01]	.375	0.00 [−0.01, 0.01]	.459
Male × Sleep efficiency (within, prior)²	−0.01 [−0.01, 0.00]	.056	0.00 [−0.01, 0.00]	.277
Male × Sleep efficiency (between)	−0.10 [−0.12, −0.08]	< .001	−0.02 [−0.04, −0.01]	.008
Male × Sleep efficiency (between)²	−0.02 [−0.03, 0.00]	.066	0.02 [0.00, 0.03]	.014
Sleep onset (z)				
(Intercept)	0.14 [−0.18, 0.46]	.392	−0.07 [−0.20, 0.06]	.288
Sleep onset (within, prior)	0.00 [−0.01, 0.01]	.734	0.00 [0.00, 0.01]	.301
Male	0.15 [0.13, 0.17]	< .001	0.21 [0.20, 0.23]	< .001
Sleep onset (within, prior)²	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.01, 0.00]	.024
Sleep onset (between)	−0.07 [−0.08, −0.05]	< .001	−0.07 [−0.09, −0.06]	< .001
Sleep onset (between)²	0.00 [−0.01, 0.01]	.808	0.01 [0.00, 0.01]	.183
Sleep onset (within, prior) × Male	−0.02 [−0.03, 0.00]	.009	0.00 [−0.01, 0.01]	.976
Male × Sleep onset (within, prior)²	−0.02 [−0.03, −0.01]	< .001	−0.01 [−0.02, 0.00]	.062

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Male × Sleep onset (between)	−0.11 [−0.12, −0.09]	< .001	−0.02 [−0.03, 0.00]	.023
Male × Sleep onset (between)²	0.01 [0.00, 0.02]	.082	0.01 [0.00, 0.02]	.047
Sleep regularity (z)				
(Intercept)	0.02 [−0.28, 0.32]	.896	−0.18 [−0.31, −0.05]	.007
Sleep regularity (within, prior)	−0.01 [−0.02, 0.00]	.021	0.01 [0.00, 0.02]	.112
Male	0.21 [0.19, 0.23]	< .001	0.25 [0.23, 0.27]	< .001
Sleep regularity (within, prior)²	0.00 [−0.01, 0.00]	.420	0.00 [−0.01, 0.01]	.616
Sleep regularity (between)	0.16 [0.15, 0.18]	< .001	0.08 [0.07, 0.09]	< .001
Sleep regularity (between)²	0.00 [−0.01, 0.02]	.638	−0.01 [−0.02, 0.00]	.190
Sleep regularity (within, prior) × Male	0.01 [0.00, 0.02]	.195	0.01 [0.00, 0.02]	.193
Male × Sleep regularity (within, prior)²	0.01 [0.00, 0.02]	.129	0.00 [−0.01, 0.01]	.987
Male × Sleep regularity (between)	0.01 [−0.01, 0.03]	.257	−0.03 [−0.05, −0.01]	.003
Male × Sleep regularity (between)²	−0.01 [−0.03, 0.01]	.372	−0.02 [−0.04, 0.00]	.038

Adjusted for SES, age, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

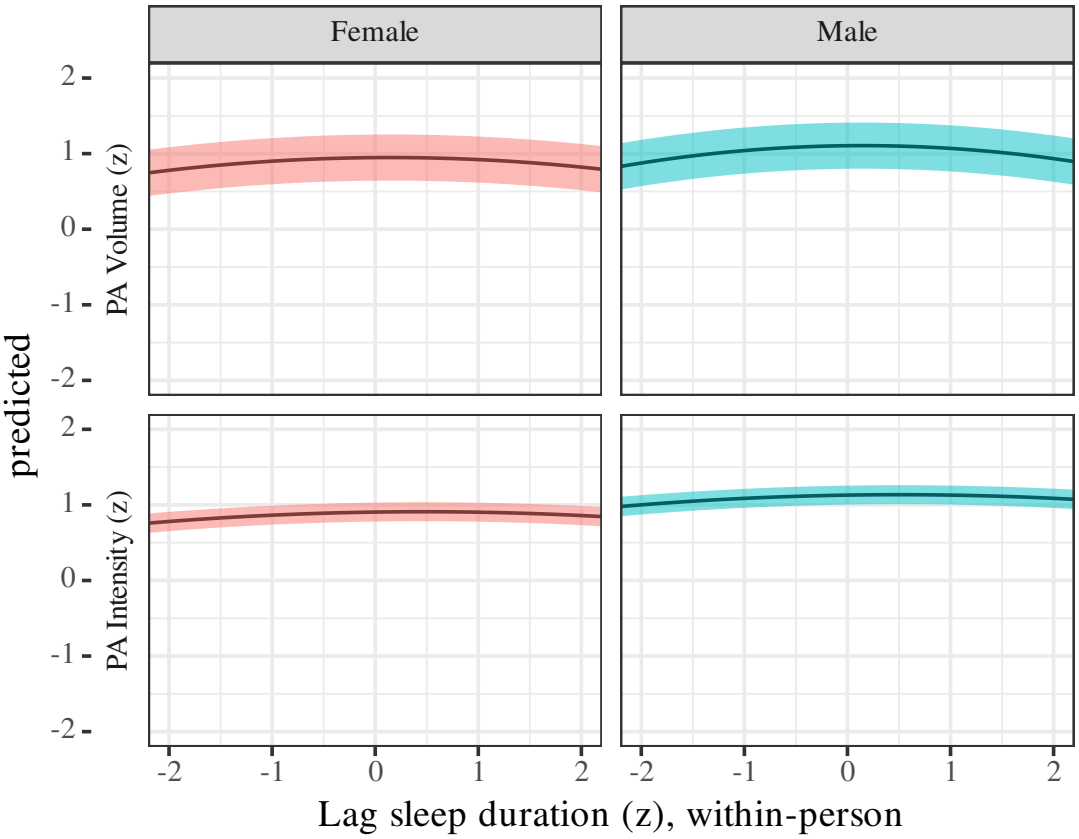


Figure 43: Physical activity by sleep duration moderated by sex

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 25 and Figure 44.

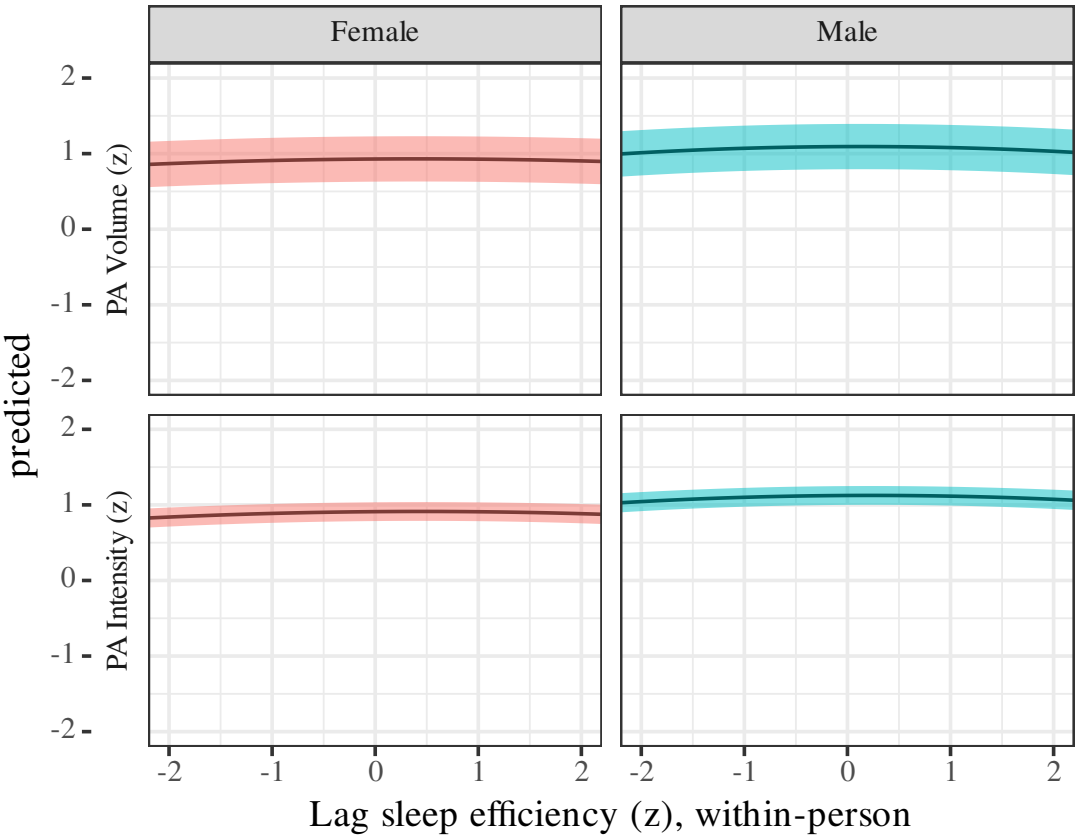


Figure 44: Physical activity by sleep efficiency moderated by sex

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by sex. Results, controlling for SES, age and BMI z-score are presented in Table 25 and Figure 45.

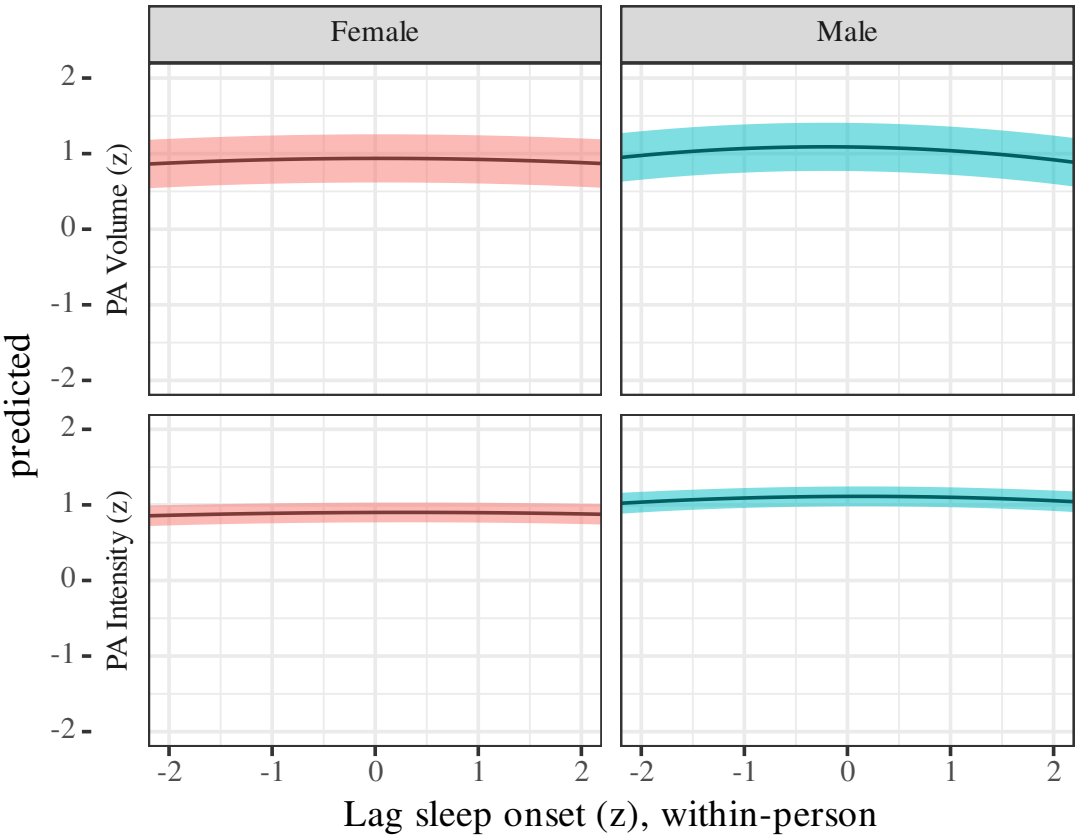


Figure 45: Physical activity by sleep onset moderated by sex

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by sex. Results are presented in Table 25 and Figure 46.

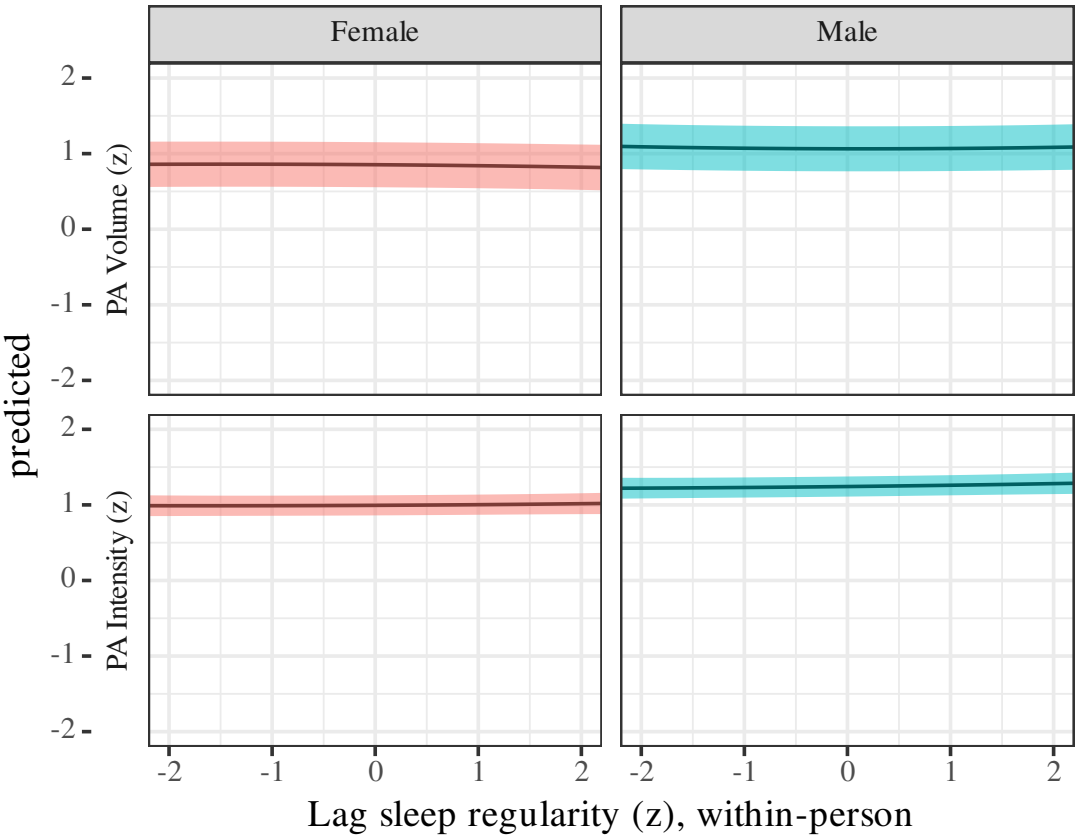


Figure 46: Physical activity by sleep regularity moderated by sex

weekday

The association between physical activity volume and sleep by weekday

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by weekday, and the results are presented in Table 26 and Figure 47

Table 26: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.10 [−0.05, 0.26]	.194	0.10 [−0.04, 0.25]	.154
Physical activity (within)	0.11 [0.09, 0.13]	< .001	0.04 [0.01, 0.06]	.001
Monday	−0.13 [−0.15, −0.11]	< .001	−0.13 [−0.15, −0.11]	< .001
Saturday	−0.02 [−0.03, 0.00]	.047	−0.02 [−0.04, −0.01]	.009
Sunday	−0.15 [−0.17, −0.13]	< .001	−0.16 [−0.18, −0.15]	< .001
Thursday	−0.14 [−0.15, −0.12]	< .001	−0.13 [−0.15, −0.12]	< .001
Tuesday	−0.14 [−0.16, −0.13]	< .001	−0.15 [−0.17, −0.13]	< .001
Wednesday	−0.15 [−0.17, −0.14]	< .001	−0.15 [−0.17, −0.13]	< .001
Physical activity (within)²	−0.04 [−0.06, −0.02]	< .001	0.00 [−0.02, 0.03]	.769
Physical activity (between)	0.08 [0.05, 0.10]	< .001	0.05 [0.04, 0.07]	< .001
Physical activity (between)²	−0.01 [−0.02, 0.00]	.014	−0.01 [−0.02, 0.00]	.217
Physical activity (within) × Monday	−0.02 [−0.05, 0.01]	.218	0.02 [−0.01, 0.05]	.146
Physical activity (within) × Saturday	0.00 [−0.03, 0.03]	.844	0.08 [0.05, 0.11]	< .001
Physical activity (within) × Sunday	−0.01 [−0.04, 0.02]	.359	−0.02 [−0.05, 0.01]	.234
Physical activity (within) × Thursday	0.01 [−0.02, 0.05]	.444	0.01 [−0.03, 0.04]	.727
Physical activity (within) × Tuesday	0.01 [−0.02, 0.05]	.420	0.03 [0.00, 0.06]	.074

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (within) × Wednesday	0.03 [−0.01, 0.06]	.123	0.04 [0.01, 0.07]	.019
Monday × Physical activity (within)²	0.04 [0.01, 0.07]	.019	−0.02 [−0.05, 0.01]	.200
Saturday × Physical activity (within)²	0.03 [0.00, 0.05]	.047	−0.01 [−0.04, 0.02]	.563
Sunday × Physical activity (within)²	0.05 [0.02, 0.08]	< .001	0.00 [−0.03, 0.03]	.869
Thursday × Physical activity (within)²	0.02 [−0.01, 0.05]	.196	0.01 [−0.03, 0.04]	.664
Tuesday × Physical activity (within)²	0.01 [−0.02, 0.04]	.476	−0.02 [−0.06, 0.01]	.223
Wednesday × Physical activity (within)²	0.02 [−0.01, 0.05]	.123	−0.02 [−0.05, 0.02]	.283
Monday × Physical activity (between)	−0.01 [−0.03, 0.02]	.548	0.09 [0.07, 0.10]	< .001
Saturday × Physical activity (between)	0.04 [0.01, 0.06]	.001	−0.01 [−0.03, 0.00]	.078
Sunday × Physical activity (between)	−0.01 [−0.04, 0.01]	.200	0.06 [0.05, 0.08]	< .001
Thursday × Physical activity (between)	−0.03 [−0.05, −0.01]	.008	0.08 [0.06, 0.09]	< .001
Tuesday × Physical activity (between)	−0.01 [−0.03, 0.02]	.516	0.10 [0.08, 0.12]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Wednesday × Physical activity (between)	−0.04 [−0.06, −0.02]	.001	0.07 [0.05, 0.09]	< .001
Monday × Physical activity (between)²	0.03 [0.02, 0.05]	< .001	0.05 [0.03, 0.06]	< .001
Saturday × Physical activity (between)²	−0.01 [−0.02, 0.00]	.059	0.01 [0.00, 0.03]	.091
Sunday × Physical activity (between)²	0.04 [0.03, 0.05]	< .001	0.05 [0.04, 0.07]	< .001
Thursday × Physical activity (between)²	0.04 [0.02, 0.05]	< .001	0.04 [0.03, 0.05]	< .001
Tuesday × Physical activity (between)²	0.03 [0.02, 0.05]	< .001	0.05 [0.04, 0.07]	< .001
Wednesday × Physical activity (between)²	0.04 [0.03, 0.05]	< .001	0.05 [0.03, 0.06]	< .001
Sleep efficiency (z)				
(Intercept)	0.18 [0.02, 0.33]	.023	0.16 [0.00, 0.32]	.053
Physical activity (within)	0.01 [−0.01, 0.03]	.393	−0.04 [−0.06, −0.02]	< .001
Monday	0.05 [0.04, 0.07]	< .001	0.07 [0.06, 0.09]	< .001
Saturday	−0.04 [−0.06, −0.02]	< .001	−0.05 [−0.07, −0.04]	< .001
Sunday	0.00 [−0.02, 0.02]	.950	0.00 [−0.02, 0.02]	.823
Thursday	0.05 [0.03, 0.07]	< .001	0.08 [0.06, 0.09]	< .001
Tuesday	0.05 [0.03, 0.07]	< .001	0.07 [0.05, 0.09]	< .001
Wednesday	0.05 [0.04, 0.07]	< .001	0.08 [0.06, 0.10]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (within)²	−0.01 [−0.03, 0.01]	.346	0.00 [−0.03, 0.02]	.710
Physical activity (between)	0.10 [0.08, 0.12]	< .001	0.02 [0.00, 0.04]	.014
Physical activity (between)²	−0.03 [−0.04, −0.02]	< .001	0.00 [−0.01, 0.01]	.803
Physical activity (within) × Monday	−0.01 [−0.05, 0.02]	.396	0.04 [0.01, 0.07]	.021
Physical activity (within) × Saturday	0.05 [0.02, 0.08]	.002	0.07 [0.04, 0.10]	< .001
Physical activity (within) × Sunday	−0.02 [−0.05, 0.01]	.277	0.00 [−0.03, 0.03]	.920
Physical activity (within) × Thursday	0.00 [−0.04, 0.03]	.883	0.03 [0.00, 0.06]	.071
Physical activity (within) × Tuesday	0.01 [−0.03, 0.04]	.746	0.04 [0.01, 0.07]	.011
Physical activity (within) × Wednesday	−0.01 [−0.05, 0.02]	.422	0.02 [−0.01, 0.05]	.258
Monday × Physical activity (within)²	0.00 [−0.03, 0.03]	.997	−0.02 [−0.05, 0.01]	.258
Saturday × Physical activity (within)²	−0.02 [−0.04, 0.01]	.175	−0.03 [−0.06, 0.00]	.026
Sunday × Physical activity (within)²	0.00 [−0.02, 0.03]	.751	−0.01 [−0.04, 0.02]	.568
Thursday × Physical activity (within)²	0.01 [−0.02, 0.04]	.653	−0.01 [−0.04, 0.03]	.724

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Tuesday × Physical activity (within)²	−0.01 [−0.04, 0.02]	.470	−0.02 [−0.06, 0.01]	.166
Wednesday × Physical activity (within)²	0.01 [−0.01, 0.04]	.318	−0.02 [−0.05, 0.02]	.298
Monday × Physical activity (between)	−0.03 [−0.05, −0.01]	.008	0.02 [0.01, 0.04]	.002
Saturday × Physical activity (between)	0.04 [0.02, 0.07]	< .001	0.00 [−0.01, 0.02]	.939
Sunday × Physical activity (between)	0.01 [−0.01, 0.03]	.460	0.01 [−0.01, 0.02]	.294
Thursday × Physical activity (between)	−0.04 [−0.06, −0.02]	< .001	0.01 [−0.01, 0.02]	.253
Tuesday × Physical activity (between)	−0.02 [−0.05, 0.00]	.041	0.02 [0.00, 0.03]	.032
Wednesday × Physical activity (between)	−0.02 [−0.04, 0.00]	.096	0.02 [0.01, 0.04]	.007
Monday × Physical activity (between)²	0.00 [−0.02, 0.01]	.458	−0.03 [−0.04, −0.01]	< .001
Saturday × Physical activity (between)²	−0.01 [−0.02, 0.00]	.239	0.02 [0.00, 0.03]	.014
Sunday × Physical activity (between)²	−0.01 [−0.02, 0.00]	.091	−0.01 [−0.02, 0.01]	.258
Thursday × Physical activity (between)²	0.00 [−0.01, 0.01]	.984	−0.03 [−0.04, −0.01]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Tuesday × Physical activity (between)²	0.00 [−0.02, 0.01]	.412	−0.02 [−0.04, −0.01]	< .001
Wednesday × Physical activity (between)²	0.00 [−0.02, 0.01]	.551	−0.03 [−0.04, −0.01]	< .001
Sleep onset (z)				
(Intercept)	0.21 [−0.06, 0.48]	.126	0.21 [−0.05, 0.46]	.113
Physical activity (within)	−0.11 [−0.13, −0.09]	< .001	−0.10 [−0.12, −0.08]	< .001
Monday	−0.17 [−0.18, −0.16]	< .001	−0.14 [−0.15, −0.12]	< .001
Saturday	0.08 [0.07, 0.09]	< .001	0.08 [0.07, 0.09]	< .001
Sunday	−0.15 [−0.16, −0.14]	< .001	−0.13 [−0.14, −0.12]	< .001
Thursday	−0.13 [−0.15, −0.12]	< .001	−0.11 [−0.12, −0.10]	< .001
Tuesday	−0.15 [−0.16, −0.14]	< .001	−0.12 [−0.13, −0.11]	< .001
Wednesday	−0.15 [−0.16, −0.14]	< .001	−0.12 [−0.13, −0.11]	< .001
Physical activity (within)²	0.04 [0.02, 0.05]	< .001	0.01 [−0.01, 0.03]	.195
Physical activity (between)	−0.04 [−0.06, −0.02]	< .001	−0.07 [−0.09, −0.06]	< .001
Physical activity (between)²	−0.02 [−0.03, −0.01]	< .001	−0.02 [−0.03, −0.01]	< .001
Physical activity (within) × Monday	−0.01 [−0.04, 0.01]	.260	0.04 [0.02, 0.07]	< .001
Physical activity (within) × Saturday	0.04 [0.02, 0.06]	< .001	0.03 [0.01, 0.05]	.005
Physical activity (within) × Sunday	0.01 [−0.01, 0.04]	.188	0.03 [0.00, 0.05]	.016

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (within) × Thursday	−0.02 [−0.05, 0.00]	.060	0.03 [0.01, 0.06]	.005
Physical activity (within) × Tuesday	−0.01 [−0.04, 0.01]	.391	0.03 [0.01, 0.05]	.013
Physical activity (within) × Wednesday	−0.04 [−0.06, −0.01]	.004	0.02 [0.00, 0.05]	.077
Monday × Physical activity (within)²	−0.03 [−0.05, −0.01]	.004	0.00 [−0.02, 0.03]	.825
Saturday × Physical activity (within)²	−0.04 [−0.06, −0.02]	< .001	−0.04 [−0.06, −0.01]	.001
Sunday × Physical activity (within)²	−0.06 [−0.08, −0.04]	< .001	−0.02 [−0.04, 0.01]	.175
Thursday × Physical activity (within)²	−0.01 [−0.03, 0.01]	.315	−0.01 [−0.03, 0.02]	.555
Tuesday × Physical activity (within)²	−0.03 [−0.05, 0.00]	.031	0.00 [−0.02, 0.03]	.881
Wednesday × Physical activity (within)²	0.00 [−0.03, 0.02]	.658	0.00 [−0.03, 0.02]	.863
Monday × Physical activity (between)	−0.12 [−0.13, −0.10]	< .001	−0.04 [−0.05, −0.03]	< .001
Saturday × Physical activity (between)	0.05 [0.03, 0.06]	< .001	0.01 [−0.01, 0.02]	.352
Sunday × Physical activity (between)	−0.08 [−0.10, −0.06]	< .001	−0.03 [−0.04, −0.02]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Thursday × Physical activity (between)	−0.09 [−0.11, −0.07]	< .001	−0.04 [−0.05, −0.03]	< .001
Tuesday × Physical activity (between)	−0.10 [−0.12, −0.08]	< .001	−0.04 [−0.05, −0.03]	< .001
Wednesday × Physical activity (between)	−0.10 [−0.12, −0.09]	< .001	−0.04 [−0.05, −0.03]	< .001
Monday × Physical activity (between)²	0.03 [0.02, 0.04]	< .001	−0.01 [−0.02, 0.00]	.008
Saturday × Physical activity (between)²	−0.01 [−0.02, −0.01]	.001	−0.01 [−0.02, 0.00]	.216
Sunday × Physical activity (between)²	0.02 [0.01, 0.03]	< .001	−0.01 [−0.02, 0.00]	.144
Thursday × Physical activity (between)²	0.02 [0.01, 0.03]	< .001	−0.02 [−0.03, −0.01]	.001
Tuesday × Physical activity (between)²	0.02 [0.01, 0.03]	< .001	−0.02 [−0.03, −0.01]	< .001
Wednesday × Physical activity (between)²	0.02 [0.01, 0.03]	< .001	−0.02 [−0.03, −0.01]	.003
Sleep regularity (z)				
(Intercept)	0.00 [−0.13, 0.13]	.997	−0.06 [−0.23, 0.11]	.482
Physical activity (within)	0.08 [0.05, 0.10]	< .001	−0.02 [−0.04, 0.01]	.128
Monday	0.41 [0.39, 0.43]	< .001	0.48 [0.46, 0.50]	< .001
Saturday	−0.06 [−0.08, −0.04]	< .001	−0.03 [−0.05, −0.01]	.004

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sunday	−0.06 [−0.08, −0.05]	< .001	−0.07 [−0.09, −0.04]	< .001
Thursday	0.33 [0.31, 0.35]	< .001	0.38 [0.36, 0.40]	< .001
Tuesday	0.41 [0.39, 0.43]	< .001	0.48 [0.45, 0.50]	< .001
Wednesday	0.39 [0.37, 0.41]	< .001	0.47 [0.45, 0.49]	< .001
Physical activity (within)²	−0.10 [−0.12, −0.08]	< .001	−0.08 [−0.11, −0.05]	< .001
Physical activity (between)	0.52 [0.49, 0.54]	< .001	0.08 [0.06, 0.11]	< .001
Physical activity (between)²	−0.10 [−0.11, −0.09]	< .001	0.02 [0.01, 0.04]	< .001
Physical activity (within) × Monday	0.08 [0.05, 0.12]	< .001	0.07 [0.04, 0.11]	< .001
Physical activity (within) × Saturday	0.07 [0.04, 0.10]	< .001	0.09 [0.06, 0.12]	< .001
Physical activity (within) × Sunday	0.08 [0.04, 0.11]	< .001	0.04 [0.01, 0.07]	.022
Physical activity (within) × Thursday	0.12 [0.08, 0.15]	< .001	0.09 [0.06, 0.13]	< .001
Physical activity (within) × Tuesday	0.12 [0.08, 0.16]	< .001	0.09 [0.06, 0.13]	< .001
Physical activity (within) × Wednesday	0.10 [0.06, 0.13]	< .001	0.07 [0.04, 0.11]	< .001
Monday × Physical activity (within)²	0.01 [−0.02, 0.04]	.475	−0.02 [−0.05, 0.02]	.313
Saturday × Physical activity (within)²	0.02 [−0.01, 0.04]	.255	−0.01 [−0.05, 0.02]	.377

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sunday × Physical activity (within)²	0.02 [0.00, 0.05]	.067	−0.03 [−0.06, 0.01]	.119
Thursday × Physical activity (within)²	−0.02 [−0.05, 0.01]	.112	−0.02 [−0.05, 0.02]	.433
Tuesday × Physical activity (within)²	0.01 [−0.02, 0.05]	.388	−0.02 [−0.06, 0.02]	.308
Wednesday × Physical activity (within)²	−0.02 [−0.06, 0.01]	.114	−0.04 [−0.08, 0.00]	.033
Monday × Physical activity (between)	−0.07 [−0.09, −0.04]	< .001	0.11 [0.09, 0.12]	< .001
Saturday × Physical activity (between)	−0.03 [−0.06, −0.01]	.003	0.00 [−0.01, 0.02]	.934
Sunday × Physical activity (between)	−0.02 [−0.04, 0.01]	.182	0.07 [0.05, 0.08]	< .001
Thursday × Physical activity (between)	−0.05 [−0.07, −0.02]	< .001	0.06 [0.04, 0.08]	< .001
Tuesday × Physical activity (between)	−0.06 [−0.09, −0.04]	< .001	0.11 [0.09, 0.12]	< .001
Wednesday × Physical activity (between)	−0.07 [−0.09, −0.05]	< .001	0.07 [0.05, 0.09]	< .001
Monday × Physical activity (between)²	−0.02 [−0.03, −0.01]	.005	−0.10 [−0.12, −0.09]	< .001
Saturday × Physical activity (between)²	0.02 [0.00, 0.03]	.010	−0.01 [−0.03, 0.00]	.043

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sunday × Physical activity (between)²	0.02 [0.01, 0.03]	.002	0.01 [0.00, 0.03]	.077
Thursday × Physical activity (between)²	−0.01 [−0.02, 0.00]	.168	−0.08 [−0.09, −0.06]	< .001
Tuesday × Physical activity (between)²	−0.01 [−0.02, 0.00]	.136	−0.09 [−0.10, −0.07]	< .001
Wednesday × Physical activity (between)²	−0.01 [−0.02, 0.01]	.418	−0.10 [−0.11, −0.08]	< .001

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

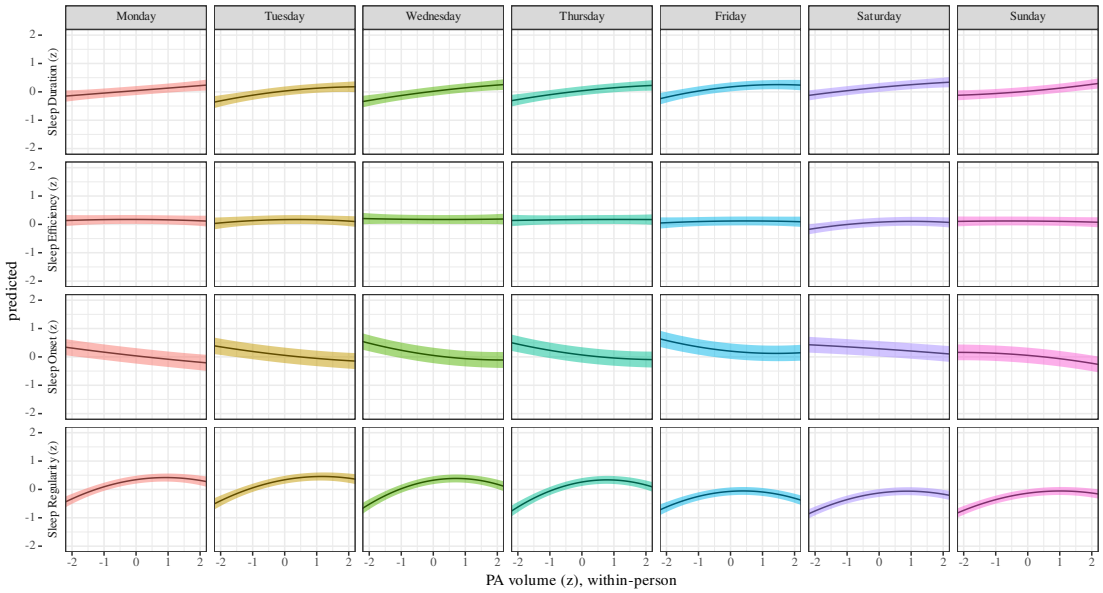


Figure 47: Sleep metrics on Physical activity volume by weekday

The association between physical activity intensity and sleep by weekday

We estimated how physical activity intensity is associated with sleep across weekday. We present the results in Table 26 and Figure 48.

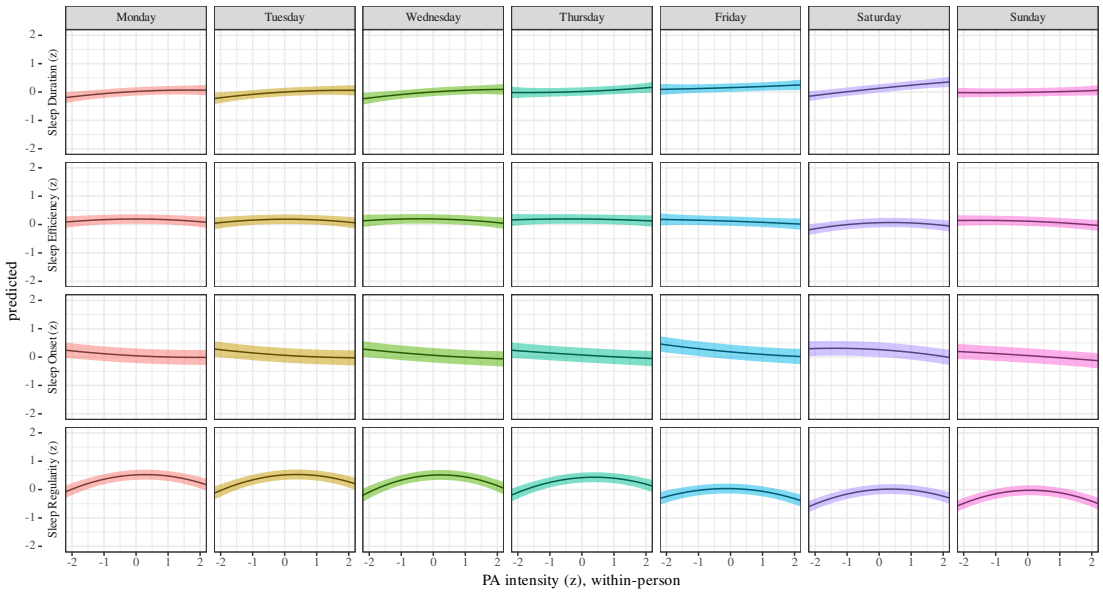


Figure 48: Sleep metrics on Physical activity intensity moderated by weekday

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 49.

Table 27: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.21 [−0.09, 0.52]	.169	−0.04 [−0.16, 0.09]	.556
Sleep duration (within, prior)	0.01 [0.00, 0.03]	.035	0.02 [0.01, 0.03]	< .001
Monday	−0.07 [−0.09, −0.06]	< .001	−0.01 [−0.03, 0.00]	.057
Saturday	−0.12 [−0.14, −0.11]	< .001	−0.12 [−0.13, −0.11]	< .001
Sunday	−0.18 [−0.19, −0.16]	< .001	−0.12 [−0.14, −0.11]	< .001
Thursday	0.00 [−0.02, 0.01]	.709	0.03 [0.01, 0.04]	< .001
Tuesday	−0.02 [−0.04, −0.01]	.003	0.02 [0.00, 0.03]	.010
Wednesday	−0.05 [−0.06, −0.03]	< .001	0.02 [0.00, 0.03]	.020
Sleep duration (within, prior)^2	−0.04 [−0.05, −0.03]	< .001	−0.02 [−0.03, −0.01]	< .001
Sleep duration (between)	0.08 [0.07, 0.10]	< .001	0.11 [0.10, 0.12]	< .001
Sleep duration (between)^2	−0.03 [−0.04, −0.02]	< .001	0.00 [−0.01, 0.01]	.762
Sleep duration (within, prior) × Monday	0.00 [−0.02, 0.01]	.592	−0.01 [−0.03, 0.01]	.365
Sleep duration (within, prior) × Saturday	−0.01 [−0.02, 0.01]	.336	−0.01 [−0.03, 0.01]	.211
Sleep duration (within, prior) × Sunday	0.00 [−0.01, 0.02]	.875	0.01 [−0.01, 0.03]	.215
Sleep duration (within, prior) × Thursday	0.01 [−0.01, 0.03]	.234	0.01 [−0.01, 0.03]	.282

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (within, prior) × Tuesday	0.01 [−0.01, 0.03]	.462	0.00 [−0.02, 0.02]	.917
Sleep duration (within, prior) × Wednesday	0.02 [0.00, 0.04]	.053	0.01 [−0.01, 0.03]	.385
Monday × Sleep duration (within, prior)²	0.01 [−0.01, 0.02]	.301	0.00 [−0.01, 0.01]	.857
Saturday × Sleep duration (within, prior)²	0.00 [−0.01, 0.02]	.556	0.01 [−0.01, 0.02]	.423
Sunday × Sleep duration (within, prior)²	0.00 [−0.01, 0.02]	.418	0.01 [0.00, 0.03]	.025
Thursday × Sleep duration (within, prior)²	0.00 [−0.01, 0.02]	.909	−0.01 [−0.02, 0.01]	.259
Tuesday × Sleep duration (within, prior)²	0.00 [−0.02, 0.01]	.800	0.00 [−0.01, 0.02]	.724
Wednesday × Sleep duration (within, prior)²	0.01 [−0.01, 0.02]	.206	−0.01 [−0.02, 0.01]	.424
Monday × Sleep duration (between)	−0.02 [−0.04, −0.01]	.003	−0.03 [−0.04, −0.02]	< .001
Saturday × Sleep duration (between)	−0.08 [−0.09, −0.07]	< .001	−0.12 [−0.14, −0.11]	< .001
Sunday × Sleep duration (between)	−0.05 [−0.07, −0.04]	< .001	−0.14 [−0.15, −0.12]	< .001
Thursday × Sleep duration (between)	0.01 [0.00, 0.03]	.059	0.01 [−0.01, 0.03]	.203

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Tuesday × Sleep duration (between)	0.00 [−0.01, 0.02]	.570	−0.02 [−0.03, 0.00]	.053
Wednesday × Sleep duration (between)	0.00 [−0.02, 0.01]	.539	0.01 [0.00, 0.03]	.182
Monday × Sleep duration (between)²	0.02 [0.00, 0.03]	.015	0.00 [−0.01, 0.02]	.853
Saturday × Sleep duration (between)²	0.02 [0.00, 0.03]	.020	−0.02 [−0.03, 0.00]	.021
Sunday × Sleep duration (between)²	0.01 [−0.01, 0.02]	.497	−0.02 [−0.04, −0.01]	.004
Thursday × Sleep duration (between)²	0.00 [−0.01, 0.02]	.636	−0.01 [−0.03, 0.00]	.087
Tuesday × Sleep duration (between)²	0.00 [−0.01, 0.02]	.673	−0.02 [−0.04, 0.00]	.023
Wednesday × Sleep duration (between)²	0.02 [0.01, 0.04]	.005	0.01 [−0.01, 0.02]	.396
Sleep efficiency (z)				
(Intercept)	0.20 [−0.10, 0.50]	.196	−0.03 [−0.16, 0.09]	.608
Sleep efficiency (within, prior)	0.00 [−0.01, 0.01]	.978	0.01 [0.00, 0.02]	.120
Monday	−0.07 [−0.08, −0.06]	< .001	−0.01 [−0.03, 0.00]	.101
Saturday	−0.12 [−0.13, −0.11]	< .001	−0.13 [−0.15, −0.12]	< .001
Sunday	−0.17 [−0.18, −0.16]	< .001	−0.13 [−0.15, −0.12]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Thursday	0.01 [−0.01, 0.02]	.445	0.02 [0.01, 0.04]	.002
Tuesday	−0.01 [−0.02, 0.01]	.266	0.01 [0.00, 0.03]	.042
Wednesday	−0.03 [−0.05, −0.02]	< .001	0.02 [0.01, 0.03]	.007
Sleep efficiency (within, prior)²	0.00 [−0.01, 0.01]	.436	0.00 [−0.01, 0.01]	.771
Sleep efficiency (between)	−0.02 [−0.03, 0.00]	.057	−0.03 [−0.05, −0.02]	< .001
Sleep efficiency (between)²	−0.03 [−0.04, −0.01]	< .001	−0.02 [−0.03, −0.01]	< .001
Sleep efficiency (within, prior) × Monday	0.01 [−0.01, 0.02]	.484	−0.01 [−0.03, 0.01]	.239
Sleep efficiency (within, prior) × Saturday	−0.01 [−0.03, 0.01]	.353	−0.01 [−0.03, 0.00]	.124
Sleep efficiency (within, prior) × Sunday	0.01 [−0.01, 0.03]	.285	0.00 [−0.02, 0.02]	.869
Sleep efficiency (within, prior) × Thursday	0.00 [−0.02, 0.02]	.918	−0.01 [−0.03, 0.01]	.243
Sleep efficiency (within, prior) × Tuesday	0.00 [−0.02, 0.02]	.937	−0.01 [−0.03, 0.01]	.471
Sleep efficiency (within, prior) × Wednesday	0.00 [−0.02, 0.02]	.789	−0.01 [−0.03, 0.01]	.395
Monday × Sleep efficiency (within, prior)²	0.00 [−0.02, 0.01]	.606	−0.01 [−0.02, 0.00]	.118
Saturday × Sleep efficiency (within, prior)²	−0.01 [−0.02, 0.01]	.267	−0.02 [−0.03, 0.00]	.018

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sunday × Sleep efficiency (within, prior)²	−0.01 [−0.03, 0.00]	.037	−0.01 [−0.03, 0.00]	.026
Thursday × Sleep efficiency (within, prior)²	−0.01 [−0.02, 0.01]	.322	0.00 [−0.01, 0.01]	.895
Tuesday × Sleep efficiency (within, prior)²	−0.02 [−0.03, 0.00]	.028	−0.01 [−0.03, 0.00]	.091
Wednesday × Sleep efficiency (within, prior)²	−0.01 [−0.02, 0.00]	.197	−0.01 [−0.03, 0.00]	.123
Monday × Sleep efficiency (between)	0.03 [0.01, 0.04]	< .001	0.04 [0.02, 0.05]	< .001
Saturday × Sleep efficiency (between)	0.04 [0.02, 0.05]	< .001	0.10 [0.09, 0.12]	< .001
Sunday × Sleep efficiency (between)	0.01 [−0.01, 0.02]	.309	0.11 [0.09, 0.12]	< .001
Thursday × Sleep efficiency (between)	0.01 [−0.01, 0.02]	.509	0.00 [−0.02, 0.02]	.885
Tuesday × Sleep efficiency (between)	0.01 [−0.01, 0.02]	.304	0.03 [0.01, 0.04]	< .001
Wednesday × Sleep efficiency (between)	0.01 [0.00, 0.03]	.143	0.02 [0.00, 0.03]	.064
Monday × Sleep efficiency (between)²	0.02 [0.00, 0.03]	.032	0.00 [−0.01, 0.02]	.795
Saturday × Sleep efficiency (between)²	0.01 [0.00, 0.02]	.139	0.02 [0.01, 0.03]	.004

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sunday × Sleep efficiency (between)²	0.00 [−0.01, 0.02]	.597	0.02 [0.01, 0.03]	.004
Thursday × Sleep efficiency (between)²	0.00 [−0.02, 0.01]	.553	−0.01 [−0.02, 0.01]	.233
Tuesday × Sleep efficiency (between)²	−0.01 [−0.02, 0.01]	.252	0.00 [−0.01, 0.02]	.605
Wednesday × Sleep efficiency (between)²	0.01 [0.00, 0.02]	.172	0.01 [0.00, 0.02]	.179
Sleep onset (z)				
(Intercept)	0.20 [−0.12, 0.52]	.223	−0.05 [−0.18, 0.08]	.467
Sleep onset (within, prior)	0.00 [−0.02, 0.02]	.908	0.02 [0.00, 0.04]	.039
Monday	−0.06 [−0.08, −0.05]	< .001	−0.01 [−0.03, 0.00]	.040
Saturday	−0.11 [−0.12, −0.09]	< .001	−0.10 [−0.11, −0.08]	< .001
Sunday	−0.17 [−0.18, −0.16]	< .001	−0.10 [−0.11, −0.08]	< .001
Thursday	0.00 [−0.01, 0.01]	.966	0.01 [0.00, 0.03]	.096
Tuesday	−0.01 [−0.03, 0.00]	.050	0.02 [0.00, 0.03]	.009
Wednesday	−0.04 [−0.05, −0.02]	< .001	0.01 [−0.01, 0.02]	.219
Sleep onset (within, prior)²	−0.01 [−0.02, 0.00]	.005	−0.01 [−0.02, 0.00]	.019
Sleep onset (between)	−0.15 [−0.17, −0.13]	< .001	−0.14 [−0.15, −0.12]	< .001
Sleep onset (between)²	0.00 [−0.01, 0.01]	.475	0.02 [0.01, 0.03]	< .001
Sleep onset (within, prior) × Monday	0.04 [0.02, 0.07]	< .001	0.00 [−0.02, 0.03]	.786

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep onset (within, prior) × Saturday	0.01 [−0.01, 0.03]	.465	0.02 [0.00, 0.04]	.105
Sleep onset (within, prior) × Sunday	0.00 [−0.03, 0.02]	.781	0.00 [−0.02, 0.02]	.938
Sleep onset (within, prior) × Thursday	0.01 [−0.01, 0.04]	.402	−0.01 [−0.04, 0.01]	.273
Sleep onset (within, prior) × Tuesday	0.01 [−0.02, 0.03]	.616	−0.01 [−0.04, 0.02]	.461
Sleep onset (within, prior) × Wednesday	0.02 [−0.01, 0.04]	.206	−0.01 [−0.04, 0.01]	.364
Monday × Sleep onset (within, prior)²	0.01 [0.00, 0.02]	.124	0.01 [0.00, 0.02]	.085
Saturday × Sleep onset (within, prior)²	−0.01 [−0.03, 0.00]	.078	−0.01 [−0.03, 0.01]	.179
Sunday × Sleep onset (within, prior)²	−0.01 [−0.03, 0.00]	.107	0.01 [−0.01, 0.02]	.281
Thursday × Sleep onset (within, prior)²	0.00 [−0.02, 0.02]	.756	0.00 [−0.01, 0.02]	.701
Tuesday × Sleep onset (within, prior)²	−0.01 [−0.02, 0.01]	.424	0.00 [−0.01, 0.02]	.569
Wednesday × Sleep onset (within, prior)²	−0.01 [−0.03, 0.01]	.291	0.00 [−0.02, 0.02]	.940
Monday × Sleep onset (between)	0.03 [0.02, 0.04]	< .001	0.05 [0.03, 0.06]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Saturday × Sleep onset (between)	0.09 [0.08, 0.10]	< .001	0.14 [0.13, 0.15]	< .001
Sunday × Sleep onset (between)	0.07 [0.06, 0.08]	< .001	0.16 [0.15, 0.17]	< .001
Thursday × Sleep onset (between)	−0.01 [−0.02, 0.01]	.241	−0.01 [−0.02, 0.01]	.332
Tuesday × Sleep onset (between)	0.01 [0.00, 0.02]	.179	0.03 [0.02, 0.04]	< .001
Wednesday × Sleep onset (between)	0.01 [−0.01, 0.02]	.232	0.01 [−0.01, 0.02]	.252
Monday × Sleep onset (between)²	0.01 [−0.01, 0.02]	.361	0.00 [−0.01, 0.01]	.724
Saturday × Sleep onset (between)²	−0.01 [−0.02, 0.00]	.052	−0.05 [−0.06, −0.04]	< .001
Sunday × Sleep onset (between)²	0.00 [−0.01, 0.01]	.807	−0.05 [−0.06, −0.04]	< .001
Thursday × Sleep onset (between)²	0.00 [−0.01, 0.01]	.985	0.00 [−0.01, 0.01]	.763
Tuesday × Sleep onset (between)²	−0.01 [−0.02, 0.00]	.207	−0.01 [−0.02, 0.00]	.038
Wednesday × Sleep onset (between)²	0.01 [−0.01, 0.02]	.362	0.01 [0.00, 0.02]	.093
Sleep regularity (z)				
(Intercept)	0.09 [−0.21, 0.39]	.565	−0.12 [−0.26, 0.01]	.061

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep regularity (within, prior)	−0.01 [−0.02, 0.01]	.497	0.02 [0.00, 0.04]	.053
Monday	−0.07 [−0.09, −0.05]	< .001	−0.01 [−0.03, 0.01]	.254
Saturday	−0.12 [−0.14, −0.10]	< .001	−0.15 [−0.17, −0.13]	< .001
Sunday	−0.16 [−0.18, −0.14]	< .001	−0.17 [−0.18, −0.15]	< .001
Thursday	0.00 [−0.02, 0.02]	.766	0.02 [0.00, 0.04]	.063
Tuesday	−0.03 [−0.05, −0.01]	.010	0.02 [0.00, 0.04]	.129
Wednesday	−0.02 [−0.04, 0.00]	.076	0.04 [0.01, 0.06]	.002
Sleep regularity (within, prior)²	0.02 [0.00, 0.03]	.010	0.00 [−0.02, 0.01]	.482
Sleep regularity (between)	0.19 [0.17, 0.21]	< .001	0.05 [0.04, 0.07]	< .001
Sleep regularity (between)²	0.00 [−0.02, 0.01]	.670	−0.03 [−0.05, −0.01]	< .001
Sleep regularity (within, prior) × Monday	−0.07 [−0.09, −0.04]	< .001	−0.06 [−0.08, −0.03]	< .001
Sleep regularity (within, prior) × Saturday	−0.02 [−0.05, 0.00]	.048	−0.02 [−0.05, 0.00]	.082
Sleep regularity (within, prior) × Sunday	0.00 [−0.03, 0.02]	.806	0.03 [0.01, 0.06]	.011
Sleep regularity (within, prior) × Thursday	0.01 [−0.02, 0.03]	.688	−0.01 [−0.03, 0.02]	.679
Sleep regularity (within, prior) × Tuesday	0.01 [−0.02, 0.03]	.625	−0.04 [−0.07, −0.01]	.005

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep regularity (within, prior) × Wednesday	−0.02 [−0.04, 0.01]	.232	0.00 [−0.03, 0.02]	.723
Monday × Sleep regularity (within, prior)²	−0.05 [−0.07, −0.03]	< .001	−0.01 [−0.03, 0.01]	.283
Saturday × Sleep regularity (within, prior)²	−0.03 [−0.05, −0.01]	.005	0.01 [−0.01, 0.02]	.598
Sunday × Sleep regularity (within, prior)²	−0.04 [−0.06, −0.02]	< .001	0.02 [0.00, 0.04]	.121
Thursday × Sleep regularity (within, prior)²	−0.01 [−0.03, 0.01]	.286	0.01 [−0.01, 0.04]	.223
Tuesday × Sleep regularity (within, prior)²	−0.02 [−0.04, 0.00]	.104	−0.01 [−0.03, 0.01]	.287
Wednesday × Sleep regularity (within, prior)²	−0.01 [−0.03, 0.01]	.491	0.00 [−0.02, 0.03]	.637
Monday × Sleep regularity (between)	−0.03 [−0.05, −0.01]	.002	0.00 [−0.02, 0.02]	.969
Saturday × Sleep regularity (between)	−0.03 [−0.05, −0.01]	.002	0.02 [0.00, 0.04]	.040
Sunday × Sleep regularity (between)	−0.04 [−0.06, −0.02]	< .001	0.01 [−0.01, 0.03]	.242
Thursday × Sleep regularity (between)	−0.02 [−0.04, 0.00]	.126	0.00 [−0.02, 0.02]	.916
Tuesday × Sleep regularity (between)	0.01 [−0.01, 0.03]	.351	0.03 [0.01, 0.05]	.014

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Wednesday × Sleep regularity (between)	−0.04 [−0.06, −0.02]	< .001	−0.01 [−0.03, 0.02]	.567
Monday × Sleep regularity (between)²	0.02 [0.00, 0.03]	.093	0.02 [0.00, 0.04]	.077
Saturday × Sleep regularity (between)²	−0.01 [−0.03, 0.01]	.192	0.01 [−0.01, 0.03]	.360
Sunday × Sleep regularity (between)²	−0.01 [−0.03, 0.01]	.320	0.02 [0.00, 0.04]	.068
Thursday × Sleep regularity (between)²	0.00 [−0.02, 0.02]	.986	0.01 [−0.01, 0.03]	.446
Tuesday × Sleep regularity (between)²	0.02 [0.01, 0.04]	.013	0.03 [0.01, 0.05]	.011
Wednesday × Sleep regularity (between)²	−0.01 [−0.03, 0.02]	.640	0.01 [−0.01, 0.03]	.493

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

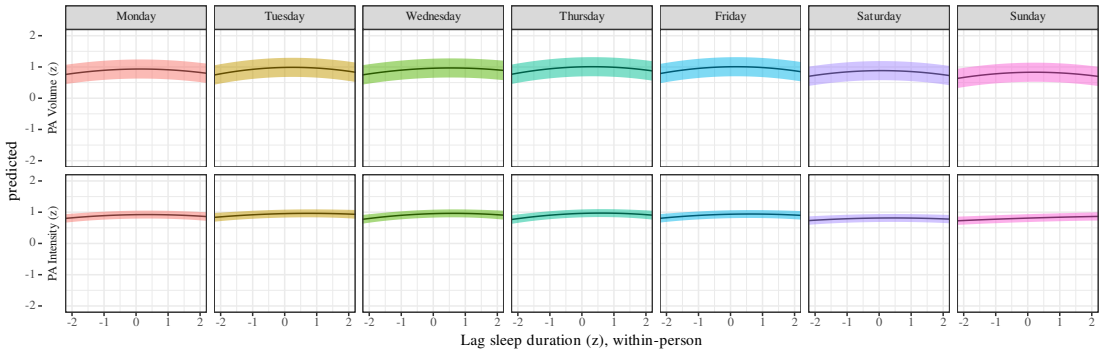


Figure 49: Physical activity by sleep duration moderated by weekday

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 50.

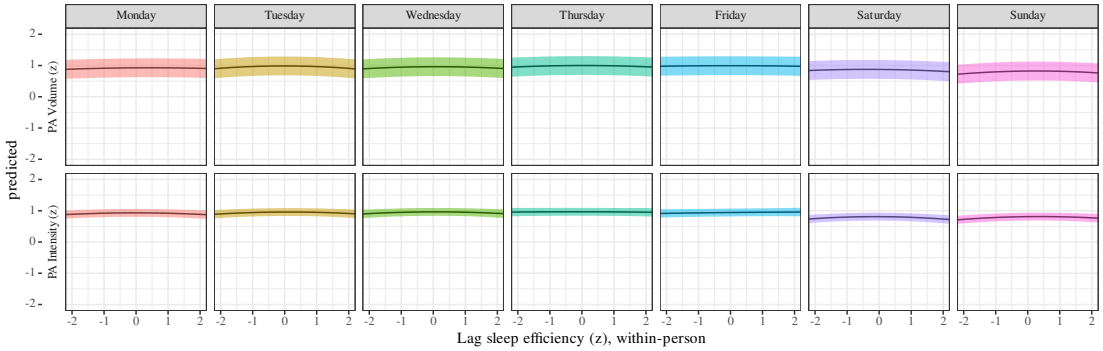


Figure 50: Physical activity by sleep efficiency moderated by weekday

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by weekday. Results, controlling for SES, age, sex and BMI z-score are presented in Table 27 and Figure 51.

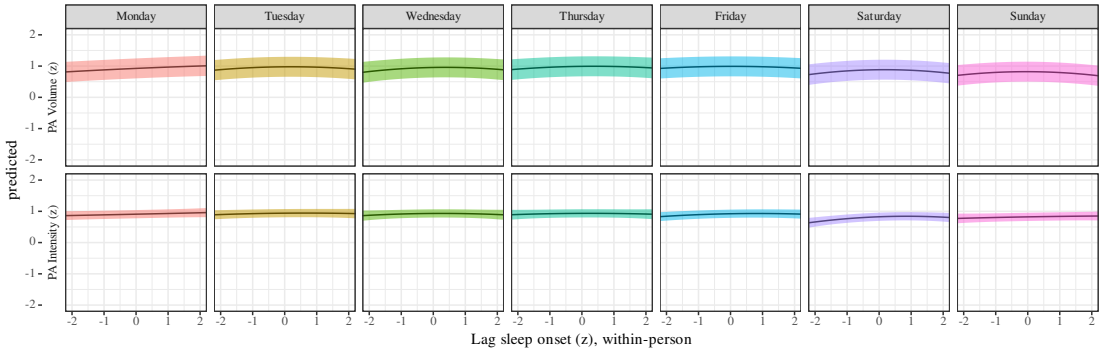


Figure 51: Physical activity by sleep onset moderated by weekday

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by weekday. Results are presented in Table 27 and Figure 52.

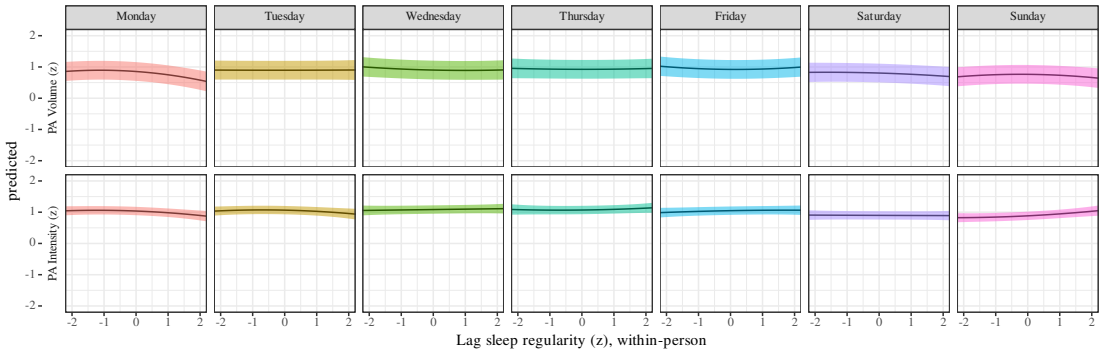


Figure 52: Physical activity by sleep regularity moderated by weekday

season

The association between physical activity volume and sleep by season

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by season, and the results are presented in Table 28 and Figure 53

Table 28: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.01 [−0.14, 0.17]	.855	0.01 [−0.14, 0.15]	.910
Physical activity (within)	0.14 [0.12, 0.15]	< .001	0.07 [0.05, 0.08]	< .001
Spring	−0.03 [−0.05, 0.00]	.039	−0.01 [−0.04, 0.01]	.330
Summer	−0.10 [−0.13, −0.08]	< .001	−0.08 [−0.10, −0.05]	< .001
Winter	0.08 [0.05, 0.10]	< .001	0.07 [0.04, 0.09]	< .001
Physical activity (within)²	0.00 [−0.01, 0.02]	.868	0.00 [−0.02, 0.02]	.940
Physical activity (between)	0.08 [0.05, 0.10]	< .001	0.09 [0.07, 0.11]	< .001
Physical activity (between)²	0.01 [0.00, 0.02]	.089	0.03 [0.02, 0.05]	< .001
Physical activity (within) × Spring	−0.04 [−0.06, −0.02]	< .001	−0.02 [−0.04, 0.00]	.097
Physical activity (within) × Summer	−0.02 [−0.04, 0.01]	.135	−0.02 [−0.04, 0.00]	.114
Physical activity (within) × Winter	−0.06 [−0.08, −0.04]	< .001	0.00 [−0.02, 0.02]	.934
Spring × Physical activity (within)²	−0.01 [−0.03, 0.01]	.338	−0.01 [−0.03, 0.02]	.633
Summer × Physical activity (within)²	−0.01 [−0.04, 0.01]	.311	−0.03 [−0.06, 0.00]	.035
Winter × Physical activity (within)²	−0.01 [−0.03, 0.01]	.318	0.00 [−0.02, 0.02]	.977

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Spring × Physical activity (between)	−0.02 [−0.05, 0.02]	.341	0.01 [−0.01, 0.03]	.380
Summer × Physical activity (between)	−0.02 [−0.05, 0.01]	.201	0.02 [−0.01, 0.04]	.154
Winter × Physical activity (between)	0.03 [0.00, 0.05]	.090	0.03 [0.01, 0.05]	.014
Spring × Physical activity (between)²	0.00 [−0.02, 0.02]	.885	−0.01 [−0.03, 0.01]	.208
Summer × Physical activity (between)²	0.01 [−0.01, 0.03]	.233	−0.01 [−0.03, 0.01]	.288
Winter × Physical activity (between)²	−0.01 [−0.02, 0.01]	.277	0.00 [−0.02, 0.02]	.691
Sleep efficiency (z)				
(Intercept)	0.20 [0.05, 0.35]	.010	0.19 [0.02, 0.35]	.025
Physical activity (within)	0.03 [0.02, 0.05]	< .001	0.01 [0.00, 0.02]	.131
Spring	0.00 [−0.03, 0.02]	.898	0.00 [−0.02, 0.03]	.848
Summer	−0.02 [−0.05, 0.01]	.127	−0.01 [−0.04, 0.02]	.370
Winter	0.03 [0.01, 0.05]	.012	0.04 [0.01, 0.06]	.005
Physical activity (within)²	−0.02 [−0.04, −0.01]	< .001	−0.04 [−0.05, −0.02]	< .001
Physical activity (between)	0.10 [0.08, 0.13]	< .001	0.04 [0.02, 0.07]	< .001
Physical activity (between)²	−0.04 [−0.05, −0.03]	< .001	−0.01 [−0.03, 0.00]	.109
Physical activity (within) × Spring	−0.02 [−0.04, 0.00]	.062	−0.01 [−0.03, 0.01]	.190

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (within) × Summer	−0.02 [−0.05, 0.00]	.043	−0.02 [−0.04, 0.00]	.062
Physical activity (within) × Winter	−0.02 [−0.04, 0.00]	.039	−0.01 [−0.03, 0.01]	.399
Spring × Physical activity (within)²	0.01 [−0.01, 0.03]	.415	0.01 [−0.01, 0.04]	.264
Summer × Physical activity (within)²	0.01 [−0.01, 0.04]	.296	0.02 [−0.01, 0.05]	.175
Winter × Physical activity (within)²	0.01 [0.00, 0.03]	.127	0.00 [−0.02, 0.03]	.859
Spring × Physical activity (between)	−0.03 [−0.06, 0.00]	.081	−0.04 [−0.06, −0.01]	.004
Summer × Physical activity (between)	0.00 [−0.03, 0.04]	.796	0.00 [−0.02, 0.02]	.999
Winter × Physical activity (between)	0.01 [−0.02, 0.04]	.561	0.01 [−0.02, 0.03]	.650
Spring × Physical activity (between)²	0.01 [−0.01, 0.03]	.279	0.00 [−0.02, 0.02]	.802
Summer × Physical activity (between)²	0.00 [−0.02, 0.02]	.844	0.00 [−0.03, 0.02]	.792
Winter × Physical activity (between)²	0.01 [−0.01, 0.02]	.529	0.00 [−0.02, 0.02]	.708
Sleep onset (z)				
(Intercept)	0.11 [−0.16, 0.38]	.436	0.13 [−0.13, 0.38]	.335

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (within)	−0.13 [−0.14, −0.12]	< .001	−0.08 [−0.09, −0.07]	< .001
Spring	0.02 [0.00, 0.04]	.058	0.02 [−0.01, 0.04]	.167
Summer	0.05 [0.03, 0.07]	< .001	0.02 [−0.01, 0.04]	.154
Winter	−0.01 [−0.03, 0.01]	.478	−0.01 [−0.03, 0.01]	.466
Physical activity (within)²	0.01 [0.00, 0.03]	.014	0.00 [−0.01, 0.01]	.817
Physical activity (between)	−0.11 [−0.13, −0.09]	< .001	−0.09 [−0.11, −0.08]	< .001
Physical activity (between)²	−0.02 [−0.03, −0.01]	< .001	−0.05 [−0.06, −0.04]	< .001
Physical activity (within) × Spring	0.04 [0.03, 0.06]	< .001	0.01 [−0.01, 0.02]	.440
Physical activity (within) × Summer	0.01 [−0.01, 0.03]	.313	−0.01 [−0.02, 0.01]	.368
Physical activity (within) × Winter	0.03 [0.02, 0.05]	< .001	0.00 [−0.02, 0.01]	.594
Spring × Physical activity (within)²	0.00 [−0.02, 0.01]	.697	0.00 [−0.02, 0.02]	.848
Summer × Physical activity (within)²	0.02 [0.00, 0.03]	.095	0.03 [0.01, 0.05]	.002
Winter × Physical activity (within)²	−0.01 [−0.02, 0.01]	.323	0.00 [−0.02, 0.02]	.793
Spring × Physical activity (between)	0.00 [−0.03, 0.03]	.977	−0.01 [−0.03, 0.01]	.376
Summer × Physical activity (between)	0.06 [0.03, 0.08]	< .001	0.06 [0.04, 0.08]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Winter × Physical activity (between)	−0.01 [−0.03, 0.02]	.546	−0.02 [−0.03, 0.00]	.088
Spring × Physical activity (between)²	0.02 [0.00, 0.03]	.025	0.01 [0.00, 0.03]	.086
Summer × Physical activity (between)²	0.02 [0.01, 0.04]	< .001	0.07 [0.05, 0.09]	< .001
Winter × Physical activity (between)²	0.01 [0.00, 0.02]	.064	0.02 [0.00, 0.03]	.042
Sleep regularity (z)				
(Intercept)	0.22 [0.08, 0.36]	.002	0.23 [0.06, 0.41]	.008
Physical activity (within)	0.19 [0.17, 0.20]	< .001	0.11 [0.09, 0.13]	< .001
Spring	−0.07 [−0.10, −0.04]	< .001	−0.13 [−0.16, −0.09]	< .001
Summer	0.04 [0.01, 0.08]	.004	−0.01 [−0.05, 0.02]	.544
Winter	−0.01 [−0.04, 0.01]	.314	−0.04 [−0.07, −0.01]	.017
Physical activity (within)²	−0.13 [−0.15, −0.11]	< .001	−0.14 [−0.16, −0.12]	< .001
Physical activity (between)	0.43 [0.40, 0.46]	< .001	0.15 [0.13, 0.18]	< .001
Physical activity (between)²	−0.07 [−0.09, −0.06]	< .001	−0.04 [−0.06, −0.02]	< .001
Physical activity (within) × Spring	0.02 [−0.01, 0.04]	.199	−0.02 [−0.05, 0.00]	.038
Physical activity (within) × Summer	−0.02 [−0.04, 0.01]	.208	−0.02 [−0.05, 0.00]	.111
Physical activity (within) × Winter	−0.01 [−0.03, 0.01]	.358	0.01 [−0.01, 0.03]	.283

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Spring × Physical activity (within)²	−0.02 [−0.05, 0.00]	.056	0.01 [−0.02, 0.04]	.470
Summer × Physical activity (within)²	0.02 [−0.01, 0.05]	.116	0.04 [0.00, 0.07]	.023
Winter × Physical activity (within)²	0.04 [0.01, 0.06]	< .001	−0.03 [−0.06, 0.00]	.028
Spring × Physical activity (between)	0.07 [0.03, 0.11]	< .001	−0.03 [−0.05, 0.00]	.059
Summer × Physical activity (between)	0.13 [0.09, 0.17]	< .001	0.04 [0.01, 0.07]	.006
Winter × Physical activity (between)	0.04 [0.01, 0.07]	.021	−0.01 [−0.03, 0.02]	.558
Spring × Physical activity (between)²	−0.03 [−0.05, −0.01]	.002	0.04 [0.01, 0.06]	.002
Summer × Physical activity (between)²	−0.05 [−0.07, −0.03]	< .001	0.05 [0.02, 0.07]	< .001
Winter × Physical activity (between)²	−0.03 [−0.05, −0.01]	< .001	0.00 [−0.02, 0.02]	.991

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

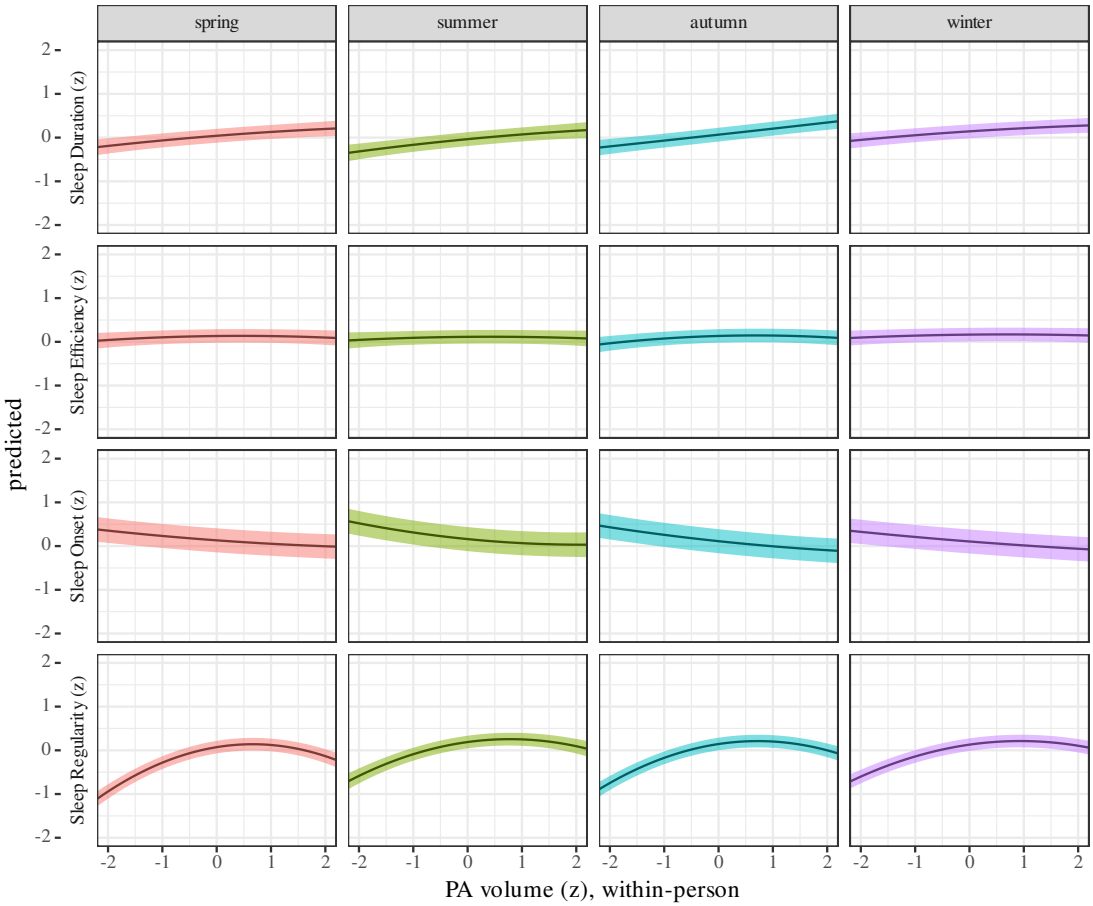


Figure 53: Sleep metrics on Physical activity volume by season

The association between physical activity intensity and sleep by season

We estimated how physical activity intensity is associated with sleep across season. We present the results in Table 28 and Figure 54.

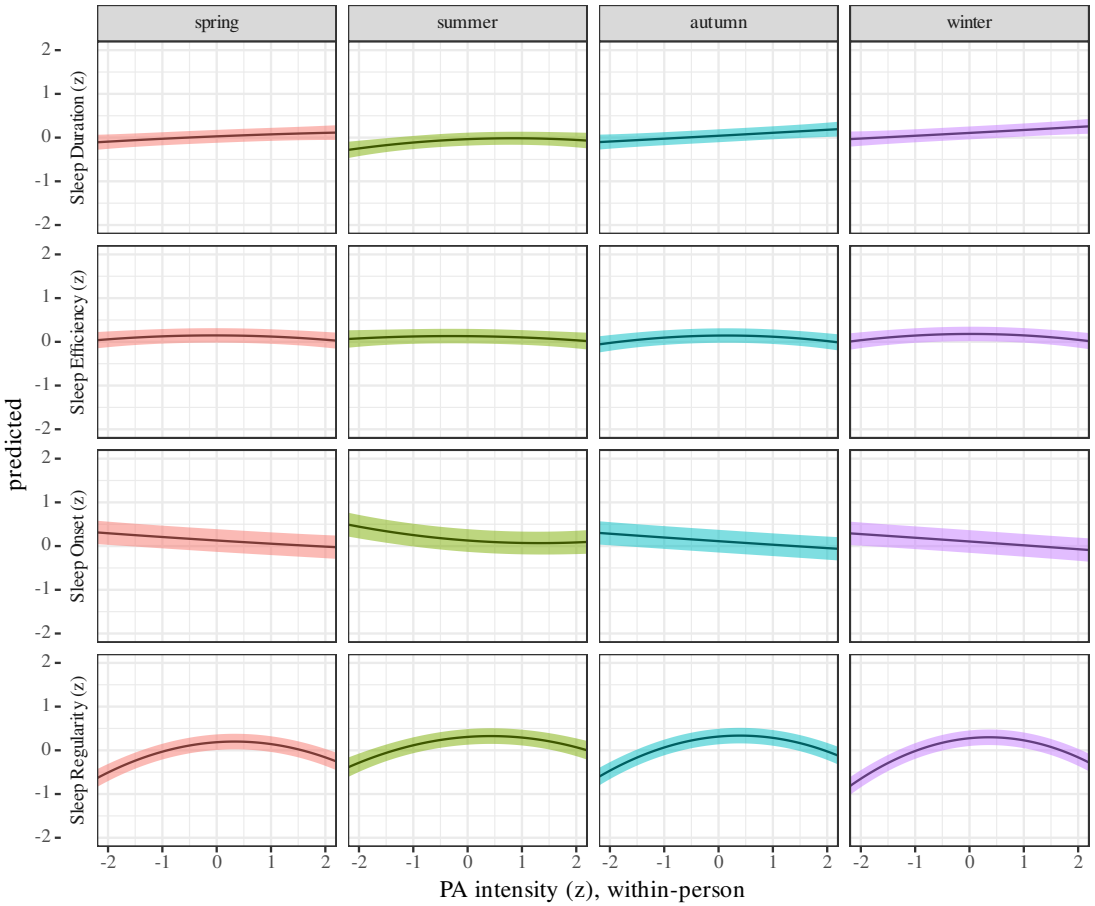


Figure 54: Sleep metrics on Physical activity intensity moderated by season

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 55.

Table 29: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.15 [−0.16, 0.46]	.336	−0.06 [−0.18, 0.07]	.371
Sleep duration (within, prior)	0.03 [0.03, 0.04]	< .001	0.03 [0.02, 0.04]	< .001
Spring	0.00 [−0.02, 0.03]	.786	−0.01 [−0.03, 0.01]	.412
Summer	0.01 [−0.02, 0.03]	.535	−0.03 [−0.05, −0.01]	.005
Winter	−0.04 [−0.06, −0.02]	< .001	−0.02 [−0.04, 0.00]	.033
Sleep duration (within, prior)²	−0.04 [−0.05, −0.03]	< .001	−0.03 [−0.03, −0.02]	< .001
Sleep duration (between)	0.09 [0.08, 0.11]	< .001	0.08 [0.07, 0.10]	< .001
Sleep duration (between)²	−0.03 [−0.05, −0.01]	< .001	0.01 [−0.01, 0.02]	.407
Sleep duration (within, prior) × Spring	−0.06 [−0.07, −0.04]	< .001	−0.03 [−0.04, −0.02]	< .001
Sleep duration (within, prior) × Summer	−0.03 [−0.04, −0.02]	< .001	−0.01 [−0.03, 0.00]	.037
Sleep duration (within, prior) × Winter	−0.01 [−0.02, 0.00]	.178	0.00 [−0.01, 0.01]	.640
Spring × Sleep duration (within, prior)²	0.00 [−0.01, 0.01]	.461	0.00 [−0.01, 0.01]	.490
Summer × Sleep duration (within, prior)²	0.00 [−0.01, 0.02]	.558	0.00 [−0.01, 0.01]	.849
Winter × Sleep duration (within, prior)²	−0.01 [−0.02, 0.00]	.200	0.01 [0.00, 0.02]	.027

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Spring × Sleep duration (between)	−0.04 [−0.07, −0.02]	< .001	−0.02 [−0.04, 0.00]	.049
Summer × Sleep duration (between)	−0.08 [−0.11, −0.06]	< .001	−0.06 [−0.09, −0.04]	< .001
Winter × Sleep duration (between)	−0.01 [−0.03, 0.01]	.407	−0.01 [−0.03, 0.01]	.521
Spring × Sleep duration (between)²	0.01 [−0.02, 0.03]	.479	−0.01 [−0.04, 0.01]	.281
Summer × Sleep duration (between)²	0.02 [0.00, 0.05]	.099	−0.03 [−0.05, 0.00]	.037
Winter × Sleep duration (between)²	0.00 [−0.02, 0.02]	.925	−0.03 [−0.05, −0.01]	.007
Sleep efficiency (z)				
(Intercept)	0.14 [−0.16, 0.44]	.375	−0.04 [−0.17, 0.08]	.485
Sleep efficiency (within, prior)	0.01 [0.01, 0.02]	.002	0.02 [0.01, 0.03]	< .001
Spring	0.01 [−0.01, 0.03]	.463	−0.02 [−0.04, 0.00]	.024
Summer	0.03 [0.01, 0.05]	.014	−0.05 [−0.08, −0.03]	< .001
Winter	−0.04 [−0.06, −0.02]	< .001	−0.04 [−0.05, −0.02]	< .001
Sleep efficiency (within, prior)²	−0.02 [−0.03, −0.01]	< .001	−0.03 [−0.03, −0.02]	< .001
Sleep efficiency (between)	0.01 [−0.01, 0.03]	.193	0.01 [−0.01, 0.03]	.158
Sleep efficiency (between)²	−0.01 [−0.03, 0.00]	.152	−0.01 [−0.03, 0.00]	.081

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency (within, prior) × Spring	0.00 [−0.02, 0.01]	.611	0.00 [−0.02, 0.01]	.727
Sleep efficiency (within, prior) × Summer	−0.01 [−0.02, 0.00]	.150	−0.01 [−0.03, 0.00]	.120
Sleep efficiency (within, prior) × Winter	−0.01 [−0.03, 0.00]	.017	−0.02 [−0.03, 0.00]	.006
Spring × Sleep efficiency (within, prior)²	0.01 [0.00, 0.02]	.091	0.02 [0.01, 0.03]	< .001
Summer × Sleep efficiency (within, prior)²	0.01 [0.00, 0.02]	.232	0.02 [0.01, 0.03]	< .001
Winter × Sleep efficiency (within, prior)²	0.01 [0.00, 0.01]	.296	0.01 [0.00, 0.02]	.003
Spring × Sleep efficiency (between)	−0.04 [−0.06, −0.01]	.006	−0.03 [−0.06, −0.01]	.009
Summer × Sleep efficiency (between)	−0.02 [−0.05, 0.01]	.160	0.01 [−0.01, 0.04]	.275
Winter × Sleep efficiency (between)	0.01 [−0.02, 0.03]	.598	0.02 [−0.01, 0.04]	.167
Spring × Sleep efficiency (between)²	−0.01 [−0.03, 0.01]	.309	0.00 [−0.02, 0.02]	.823
Summer × Sleep efficiency (between)²	−0.02 [−0.04, 0.00]	.100	0.00 [−0.02, 0.02]	.886
Winter × Sleep efficiency (between)²	−0.01 [−0.03, 0.01]	.200	0.00 [−0.02, 0.02]	.921

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep onset (z)				
(Intercept)	0.15 [−0.17, 0.47]	.354	−0.06 [−0.19, 0.07]	.389
Sleep onset (within, prior)	0.02 [0.01, 0.03]	.002	0.01 [0.00, 0.02]	.122
Spring	0.00 [−0.02, 0.02]	.995	−0.01 [−0.03, 0.01]	.518
Summer	0.00 [−0.02, 0.02]	.929	−0.04 [−0.06, −0.02]	< .001
Winter	−0.05 [−0.07, −0.02]	< .001	−0.03 [−0.05, −0.01]	.012
Sleep onset (within, prior)²	−0.03 [−0.04, −0.02]	< .001	−0.02 [−0.03, −0.01]	.001
Sleep onset (between)	−0.13 [−0.15, −0.11]	< .001	−0.09 [−0.10, −0.07]	< .001
Sleep onset (between)²	−0.01 [−0.02, 0.01]	.279	0.02 [0.00, 0.03]	.012
Sleep onset (within, prior) × Spring	−0.04 [−0.05, −0.02]	< .001	0.00 [−0.01, 0.02]	.638
Sleep onset (within, prior) × Summer	−0.01 [−0.03, 0.01]	.337	0.01 [−0.01, 0.02]	.585
Sleep onset (within, prior) × Winter	−0.04 [−0.06, −0.03]	< .001	−0.02 [−0.04, −0.01]	.006
Spring × Sleep onset (within, prior)²	0.01 [−0.01, 0.02]	.360	0.01 [−0.01, 0.02]	.235
Summer × Sleep onset (within, prior)²	0.02 [0.00, 0.03]	.012	0.02 [0.00, 0.03]	.028
Winter × Sleep onset (within, prior)²	0.00 [−0.02, 0.01]	.894	0.01 [−0.01, 0.02]	.281
Spring × Sleep onset (between)	−0.01 [−0.03, 0.01]	.324	−0.01 [−0.03, 0.01]	.346

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Summer × Sleep onset (between)	0.08 [0.06, 0.10]	< .001	0.04 [0.02, 0.06]	< .001
Winter × Sleep onset (between)	−0.02 [−0.04, 0.00]	.070	0.00 [−0.01, 0.02]	.765
Spring × Sleep onset (between)²	0.01 [−0.01, 0.03]	.166	−0.01 [−0.03, 0.00]	.164
Summer × Sleep onset (between)²	0.02 [0.00, 0.04]	.059	−0.01 [−0.03, 0.01]	.198
Winter × Sleep onset (between)²	0.01 [−0.01, 0.03]	.344	0.00 [−0.02, 0.01]	.635
Sleep regularity (z)				
(Intercept)	0.01 [−0.29, 0.31]	.936	−0.17 [−0.30, −0.04]	.011
Sleep regularity (within, prior)	0.01 [0.00, 0.02]	.042	0.03 [0.02, 0.04]	< .001
Spring	0.01 [−0.01, 0.04]	.313	0.00 [−0.02, 0.03]	.866
Summer	0.06 [0.04, 0.09]	< .001	−0.02 [−0.04, 0.01]	.221
Winter	−0.05 [−0.08, −0.03]	< .001	−0.02 [−0.04, 0.00]	.100
Sleep regularity (within, prior)²	0.01 [0.00, 0.03]	.011	0.01 [0.00, 0.02]	.042
Sleep regularity (between)	0.16 [0.14, 0.18]	< .001	0.05 [0.04, 0.07]	< .001
Sleep regularity (between)²	−0.01 [−0.03, 0.01]	.380	−0.01 [−0.03, 0.01]	.181
Sleep regularity (within, prior) × Spring	−0.02 [−0.03, 0.00]	.028	−0.03 [−0.05, −0.02]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep regularity (within, prior) × Summer	−0.04 [−0.06, −0.02]	< .001	−0.03 [−0.05, −0.01]	.002
Sleep regularity (within, prior) × Winter	−0.02 [−0.04, −0.01]	.004	−0.03 [−0.05, −0.02]	< .001
Spring × Sleep regularity (within, prior)²	−0.02 [−0.04, −0.01]	.004	−0.01 [−0.03, 0.00]	.074
Summer × Sleep regularity (within, prior)²	−0.04 [−0.05, −0.02]	< .001	−0.02 [−0.04, 0.00]	.029
Winter × Sleep regularity (within, prior)²	−0.01 [−0.02, 0.01]	.372	−0.01 [−0.02, 0.01]	.257
Spring × Sleep regularity (between)	0.02 [−0.01, 0.04]	.207	0.02 [−0.01, 0.04]	.143
Summer × Sleep regularity (between)	−0.04 [−0.07, −0.01]	.010	0.01 [−0.02, 0.04]	.530
Winter × Sleep regularity (between)	0.03 [0.00, 0.06]	.035	0.02 [0.00, 0.05]	.058
Spring × Sleep regularity (between)²	0.01 [−0.01, 0.04]	.360	−0.01 [−0.03, 0.01]	.355
Summer × Sleep regularity (between)²	−0.01 [−0.04, 0.01]	.356	−0.03 [−0.05, 0.00]	.044
Winter × Sleep regularity (between)²	0.02 [0.00, 0.05]	.072	0.01 [−0.01, 0.04]	.257

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

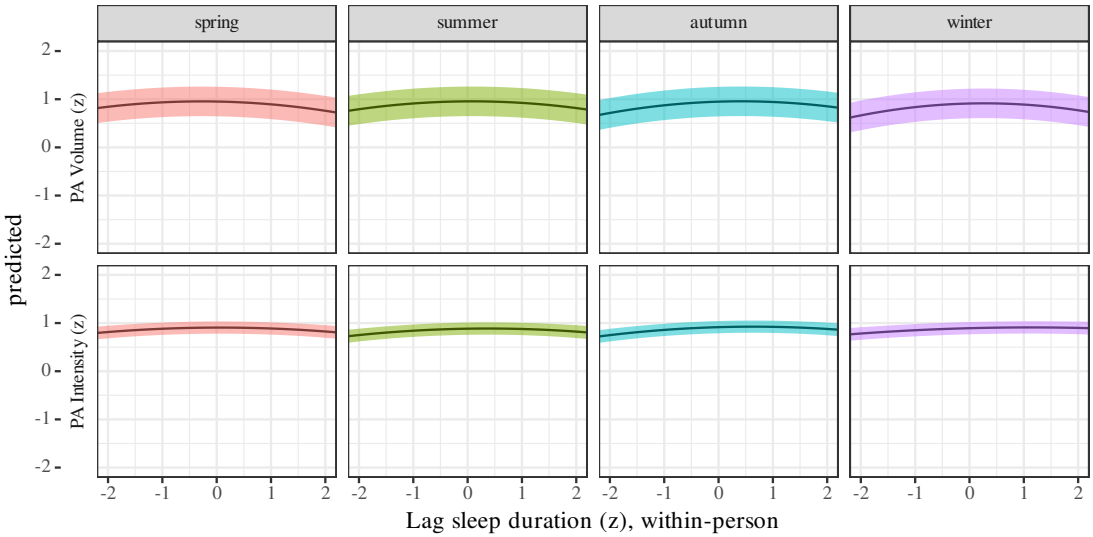


Figure 55: Physical activity by sleep duration moderated by season

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 56.

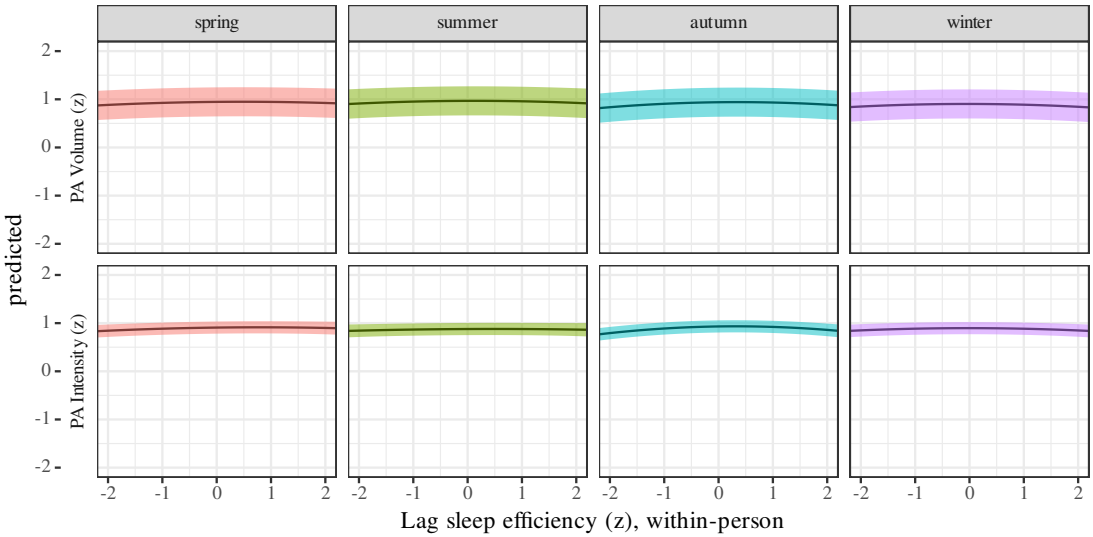


Figure 56: Physical activity by sleep efficiency moderated by season

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by season. Results, controlling for SES, age, sex and BMI z-score are presented in Table 29 and Figure 57.

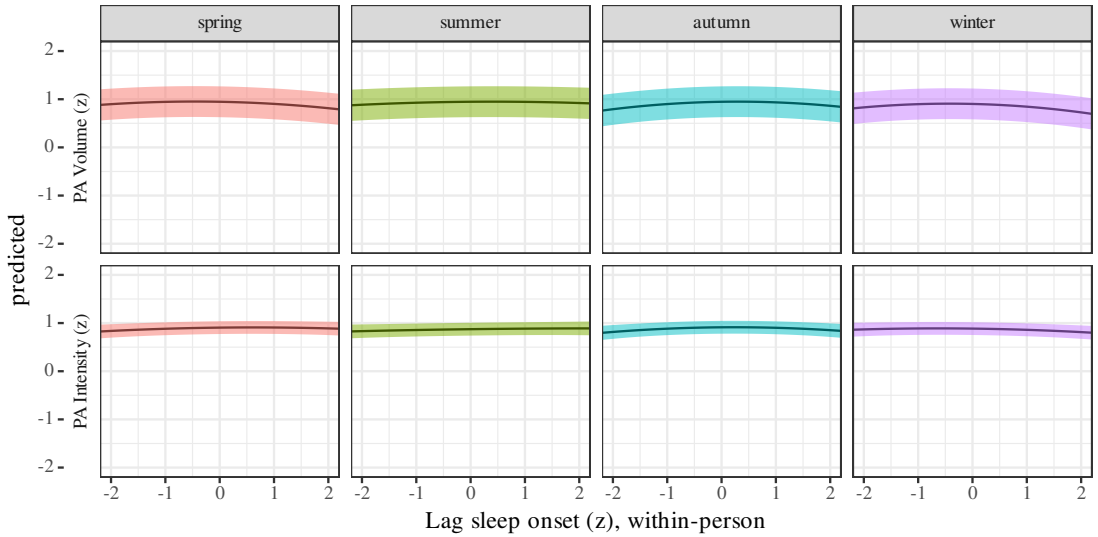


Figure 57: Physical activity by sleep onset moderated by season

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by season. Results are presented in Table 29 and Figure 58.

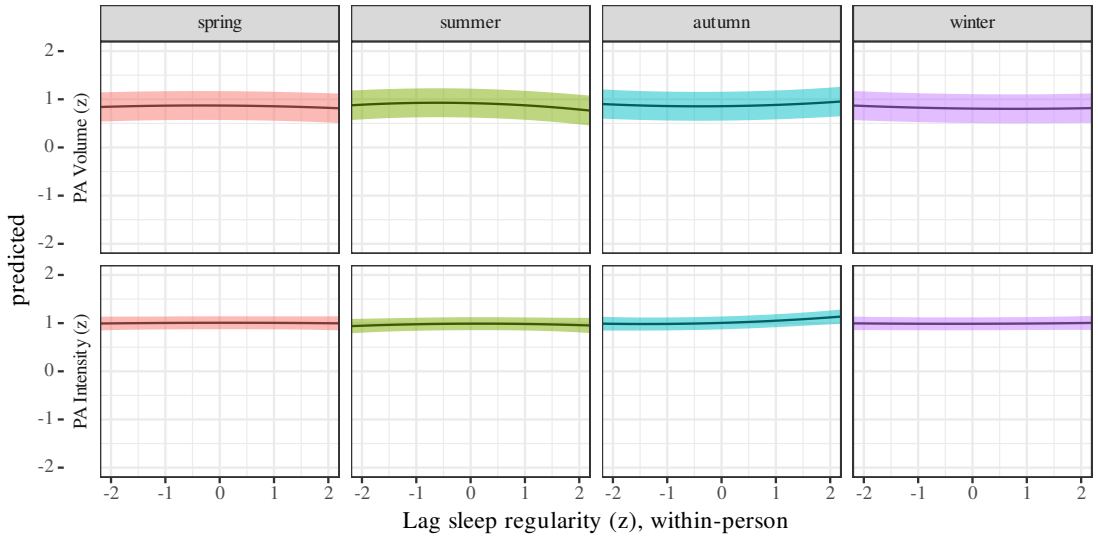


Figure 58: Physical activity by sleep regularity moderated by season

region

The association between physical activity volume and sleep by region

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by region, and the results are presented in Table 30 and Figure 59

Table 30: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.21 [0.04, 0.38]	.017	0.19 [0.03, 0.35]	.023
Physical activity (within)	0.03 [0.02, 0.05]	< .001	0.03 [0.01, 0.05]	.012
Europe	−0.24 [−0.31, −0.18]	< .001	−0.17 [−0.24, −0.10]	< .001
Africa	−0.23 [−0.31, −0.15]	< .001	−0.14 [−0.23, −0.06]	.001
Asia	−0.36 [−0.48, −0.24]	< .001	−0.47 [−0.55, −0.39]	< .001
North America	−0.04 [−0.13, 0.05]	.343	−0.06 [−0.14, 0.01]	.098
South America	−0.40 [−0.47, −0.34]	< .001	−0.40 [−0.47, −0.33]	< .001
Physical activity (within)²	0.00 [−0.01, 0.01]	.751	0.02 [−0.01, 0.05]	.170
Physical activity (between)	0.08 [0.02, 0.13]	.007	0.07 [0.01, 0.13]	.014
Physical activity (between)²	0.00 [−0.01, 0.02]	.824	0.07 [0.03, 0.11]	< .001
Physical activity (within) × Europe	0.07 [0.05, 0.09]	< .001	0.02 [0.00, 0.05]	.083
Physical activity (within) × Africa	0.08 [0.02, 0.13]	.008	0.03 [−0.01, 0.08]	.188
Physical activity (within) × Asia	0.09 [0.01, 0.17]	.029	0.02 [−0.03, 0.08]	.429
Physical activity (within) × North America	0.06 [0.01, 0.11]	.031	0.01 [−0.03, 0.05]	.609
Physical activity (within) × South America	0.15 [0.13, 0.17]	< .001	0.08 [0.05, 0.11]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Europe × Physical activity (within)²	0.00 [−0.02, 0.02]	.735	−0.04 [−0.07, −0.01]	.022
Africa × Physical activity (within)²	0.01 [−0.08, 0.11]	.763	0.01 [−0.05, 0.06]	.782
Asia × Physical activity (within)²	−0.11 [−0.25, 0.04]	.155	−0.03 [−0.10, 0.04]	.404
North America × Physical activity (within)²	−0.01 [−0.08, 0.06]	.865	0.00 [−0.05, 0.06]	.885
South America × Physical activity (within)²	−0.04 [−0.06, −0.02]	< .001	−0.01 [−0.05, 0.02]	.469
Europe × Physical activity (between)	−0.06 [−0.12, 0.00]	.037	0.02 [−0.04, 0.08]	.587
Africa × Physical activity (between)	−0.03 [−0.18, 0.12]	.714	−0.07 [−0.26, 0.11]	.449
Asia × Physical activity (between)	0.33 [0.02, 0.64]	.035	0.12 [−0.01, 0.26]	.078
North America × Physical activity (between)	0.07 [−0.04, 0.18]	.205	0.02 [−0.12, 0.15]	.792
South America × Physical activity (between)	0.06 [−0.01, 0.13]	.077	0.02 [−0.07, 0.10]	.713
Europe × Physical activity (between)²	0.03 [0.01, 0.05]	.007	−0.05 [−0.09, −0.01]	.006
Africa × Physical activity (between)²	−0.01 [−0.19, 0.17]	.892	−0.07 [−0.23, 0.09]	.369

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Asia × Physical activity (between)²	0.16 [−0.11, 0.42]	.251	−0.10 [−0.30, 0.10]	.320
North America × Physical activity (between)²	−0.02 [−0.06, 0.03]	.489	−0.06 [−0.18, 0.05]	.270
South America × Physical activity (between)²	0.00 [−0.02, 0.02]	.746	−0.07 [−0.12, −0.01]	.019
Sleep efficiency (z)				
(Intercept)	0.26 [0.09, 0.43]	.003	0.37 [0.20, 0.54]	< .001
Physical activity (within)	0.00 [−0.02, 0.01]	.898	0.00 [−0.02, 0.02]	.980
Europe	−0.09 [−0.16, −0.02]	.008	−0.20 [−0.27, −0.13]	< .001
Africa	0.06 [−0.03, 0.14]	.177	−0.06 [−0.15, 0.03]	.206
Asia	−0.06 [−0.19, 0.06]	.328	−0.40 [−0.49, −0.32]	< .001
North America	0.13 [0.03, 0.23]	.008	−0.17 [−0.25, −0.09]	< .001
South America	−0.14 [−0.21, −0.07]	< .001	−0.34 [−0.42, −0.27]	< .001
Physical activity (within)²	0.01 [0.00, 0.02]	.076	−0.01 [−0.03, 0.02]	.697
Physical activity (between)	−0.08 [−0.14, −0.02]	.006	−0.04 [−0.10, 0.02]	.225
Physical activity (between)²	0.01 [−0.01, 0.02]	.368	−0.03 [−0.07, 0.00]	.073
Physical activity (within) × Europe	0.00 [−0.02, 0.02]	.891	−0.01 [−0.03, 0.02]	.663
Physical activity (within) × Africa	0.03 [−0.02, 0.09]	.223	−0.01 [−0.05, 0.04]	.785
Physical activity (within) × Asia	0.15 [0.07, 0.23]	< .001	−0.01 [−0.06, 0.05]	.798

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (within) × North America	0.10 [0.05, 0.15]	< .001	0.02 [−0.03, 0.06]	.473
Physical activity (within) × South America	0.06 [0.04, 0.08]	< .001	0.02 [−0.01, 0.05]	.138
Europe × Physical activity (within)²	−0.04 [−0.06, −0.02]	< .001	−0.02 [−0.05, 0.01]	.247
Africa × Physical activity (within)²	0.02 [−0.07, 0.11]	.681	0.00 [−0.05, 0.06]	.882
Asia × Physical activity (within)²	−0.30 [−0.45, −0.16]	< .001	−0.10 [−0.17, −0.04]	.003
North America × Physical activity (within)²	−0.05 [−0.11, 0.02]	.199	−0.04 [−0.09, 0.01]	.154
South America × Physical activity (within)²	−0.05 [−0.06, −0.03]	< .001	−0.06 [−0.10, −0.02]	.002
Europe × Physical activity (between)	0.15 [0.09, 0.21]	< .001	0.06 [−0.01, 0.12]	.086
Africa × Physical activity (between)	0.13 [−0.04, 0.29]	.126	0.06 [−0.14, 0.25]	.569
Asia × Physical activity (between)	0.42 [0.09, 0.75]	.012	0.39 [0.25, 0.54]	< .001
North America × Physical activity (between)	0.34 [0.23, 0.46]	< .001	0.20 [0.06, 0.35]	.006
South America × Physical activity (between)	0.27 [0.20, 0.34]	< .001	0.23 [0.15, 0.32]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Europe × Physical activity (between)²	−0.04 [−0.06, −0.02]	< .001	0.02 [−0.02, 0.06]	.232
Africa × Physical activity (between)²	−0.14 [−0.34, 0.05]	.142	−0.01 [−0.18, 0.16]	.877
Asia × Physical activity (between)²	0.00 [−0.29, 0.28]	.975	−0.05 [−0.26, 0.16]	.627
North America × Physical activity (between)²	−0.06 [−0.11, −0.01]	.023	0.04 [−0.09, 0.16]	.571
South America × Physical activity (between)²	−0.07 [−0.09, −0.05]	< .001	−0.11 [−0.17, −0.05]	< .001
Sleep onset (z)				
(Intercept)	0.01 [−0.26, 0.28]	.948	0.14 [−0.12, 0.40]	.294
Physical activity (within)	−0.07 [−0.08, −0.05]	< .001	−0.09 [−0.10, −0.07]	< .001
Europe	0.12 [0.07, 0.17]	< .001	−0.01 [−0.06, 0.05]	.817
Africa	0.14 [0.07, 0.20]	< .001	−0.09 [−0.16, −0.01]	.021
Asia	0.26 [0.16, 0.36]	< .001	0.10 [0.03, 0.17]	.006
North America	0.13 [0.05, 0.21]	.001	−0.13 [−0.19, −0.06]	< .001
South America	0.12 [0.07, 0.18]	< .001	−0.03 [−0.09, 0.03]	.315
Physical activity (within)²	0.00 [−0.01, 0.01]	.888	−0.01 [−0.03, 0.01]	.224
Physical activity (between)	−0.22 [−0.27, −0.18]	< .001	−0.16 [−0.20, −0.11]	< .001
Physical activity (between)²	0.02 [0.00, 0.03]	.010	−0.09 [−0.12, −0.06]	< .001
Physical activity (within) × Europe	−0.03 [−0.04, −0.02]	< .001	0.01 [−0.01, 0.03]	.366

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (within) × Africa	0.03 [−0.01, 0.07]	.110	0.03 [0.00, 0.06]	.095
Physical activity (within) × Asia	0.04 [−0.02, 0.10]	.184	0.02 [−0.02, 0.06]	.393
Physical activity (within) × North America	0.06 [0.02, 0.10]	.003	0.05 [0.02, 0.09]	< .001
Physical activity (within) × South America	−0.12 [−0.14, −0.10]	< .001	−0.06 [−0.08, −0.04]	< .001
Europe × Physical activity (within)²	0.00 [−0.01, 0.02]	.790	0.03 [0.01, 0.05]	.007
Africa × Physical activity (within)²	−0.01 [−0.08, 0.06]	.692	0.02 [−0.02, 0.06]	.421
Asia × Physical activity (within)²	−0.08 [−0.18, 0.03]	.168	−0.03 [−0.09, 0.02]	.192
North America × Physical activity (within)²	−0.02 [−0.07, 0.03]	.423	−0.02 [−0.06, 0.02]	.279
South America × Physical activity (within)²	0.05 [0.03, 0.06]	< .001	−0.01 [−0.04, 0.02]	.485
Europe × Physical activity (between)	0.12 [0.07, 0.17]	< .001	0.07 [0.02, 0.12]	.007
Africa × Physical activity (between)	0.25 [0.12, 0.38]	< .001	0.16 [0.00, 0.32]	.054
Asia × Physical activity (between)	0.06 [−0.20, 0.33]	.634	0.16 [0.04, 0.28]	.008

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
North America × Physical activity (between)	0.32 [0.22, 0.41]	< .001	0.19 [0.08, 0.31]	< .001
South America × Physical activity (between)	0.16 [0.10, 0.21]	< .001	0.16 [0.09, 0.23]	< .001
Europe × Physical activity (between)²	−0.02 [−0.04, −0.01]	.004	0.07 [0.04, 0.10]	< .001
Africa × Physical activity (between)²	−0.04 [−0.20, 0.12]	.617	0.06 [−0.07, 0.20]	.371
Asia × Physical activity (between)²	−0.16 [−0.39, 0.07]	.181	−0.02 [−0.19, 0.14]	.782
North America × Physical activity (between)²	0.01 [−0.03, 0.05]	.626	0.06 [−0.04, 0.15]	.268
South America × Physical activity (between)²	−0.03 [−0.05, −0.01]	.002	0.01 [−0.04, 0.06]	.733
Sleep regularity (z)				
(Intercept)	−0.01 [−0.16, 0.14]	.919	0.09 [−0.09, 0.26]	.329
Physical activity (within)	0.14 [0.13, 0.16]	< .001	0.02 [0.00, 0.05]	.067
Europe	0.34 [0.27, 0.40]	< .001	0.18 [0.11, 0.26]	< .001
Africa	0.57 [0.49, 0.66]	< .001	0.35 [0.25, 0.45]	< .001
Asia	0.35 [0.22, 0.48]	< .001	0.46 [0.36, 0.55]	< .001
North America	0.28 [0.18, 0.38]	< .001	0.01 [−0.08, 0.10]	.853
South America	0.11 [0.03, 0.18]	.004	−0.03 [−0.11, 0.05]	.410
Physical activity (within)²	−0.09 [−0.10, −0.07]	< .001	−0.23 [−0.26, −0.20]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (between)	0.46 [0.39, 0.52]	< .001	0.26 [0.19, 0.33]	< .001
Physical activity (between)²	−0.10 [−0.11, −0.08]	< .001	−0.12 [−0.16, −0.07]	< .001
Physical activity (within) × Europe	0.08 [0.05, 0.10]	< .001	0.06 [0.04, 0.09]	< .001
Physical activity (within) × Africa	−0.07 [−0.13, −0.01]	.016	0.09 [0.04, 0.14]	< .001
Physical activity (within) × Asia	0.14 [0.06, 0.22]	< .001	0.18 [0.13, 0.24]	< .001
Physical activity (within) × North America	0.16 [0.11, 0.22]	< .001	0.08 [0.03, 0.12]	< .001
Physical activity (within) × South America	0.04 [0.02, 0.06]	< .001	0.13 [0.10, 0.17]	< .001
Europe × Physical activity (within)²	−0.06 [−0.08, −0.04]	< .001	0.14 [0.10, 0.17]	< .001
Africa × Physical activity (within)²	−0.46 [−0.56, −0.36]	< .001	−0.04 [−0.10, 0.02]	.236
Asia × Physical activity (within)²	−0.35 [−0.50, −0.21]	< .001	0.11 [0.03, 0.18]	.004
North America × Physical activity (within)²	−0.16 [−0.24, −0.09]	< .001	0.04 [−0.01, 0.10]	.133
South America × Physical activity (within)²	−0.03 [−0.05, −0.01]	.002	0.03 [−0.01, 0.07]	.149
Europe × Physical activity (between)	0.06 [−0.01, 0.13]	.084	−0.15 [−0.22, −0.07]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Africa × Physical activity (between)	−0.19 [−0.36, −0.02]	.032	0.31 [0.09, 0.53]	.006
Asia × Physical activity (between)	−0.59 [−0.93, −0.25]	< .001	0.02 [−0.14, 0.18]	.813
North America × Physical activity (between)	0.13 [0.00, 0.25]	.041	0.20 [0.04, 0.36]	.012
South America × Physical activity (between)	0.03 [−0.04, 0.11]	.396	0.04 [−0.06, 0.14]	.468
Europe × Physical activity (between)²	−0.03 [−0.05, 0.00]	.017	0.09 [0.05, 0.14]	< .001
Africa × Physical activity (between)²	−0.35 [−0.55, −0.15]	< .001	−0.25 [−0.44, −0.07]	.008
Asia × Physical activity (between)²	−0.19 [−0.49, 0.11]	.208	−0.19 [−0.42, 0.04]	.103
North America × Physical activity (between)²	−0.05 [−0.11, 0.00]	.049	−0.19 [−0.33, −0.04]	.010
South America × Physical activity (between)²	0.00 [−0.02, 0.03]	.873	0.01 [−0.06, 0.07]	.866

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

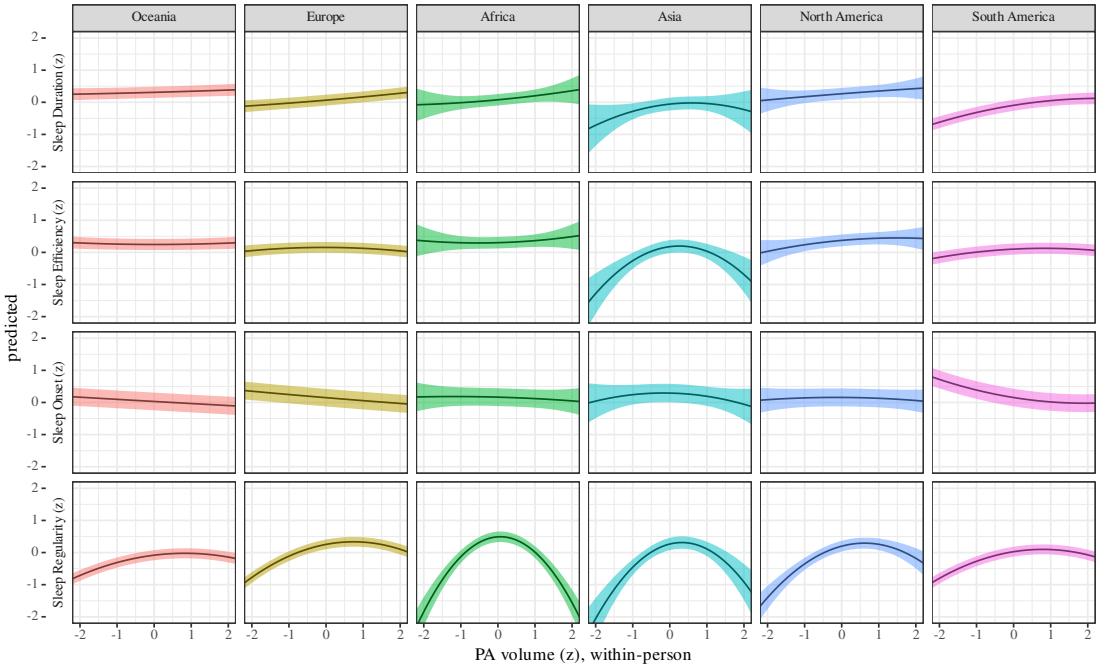


Figure 59: Sleep metrics on Physical activity volume by region

The association between physical activity intensity and sleep by region

We estimated how physical activity intensity is associated with sleep across region. We present the results in Table 30 and Figure 60.

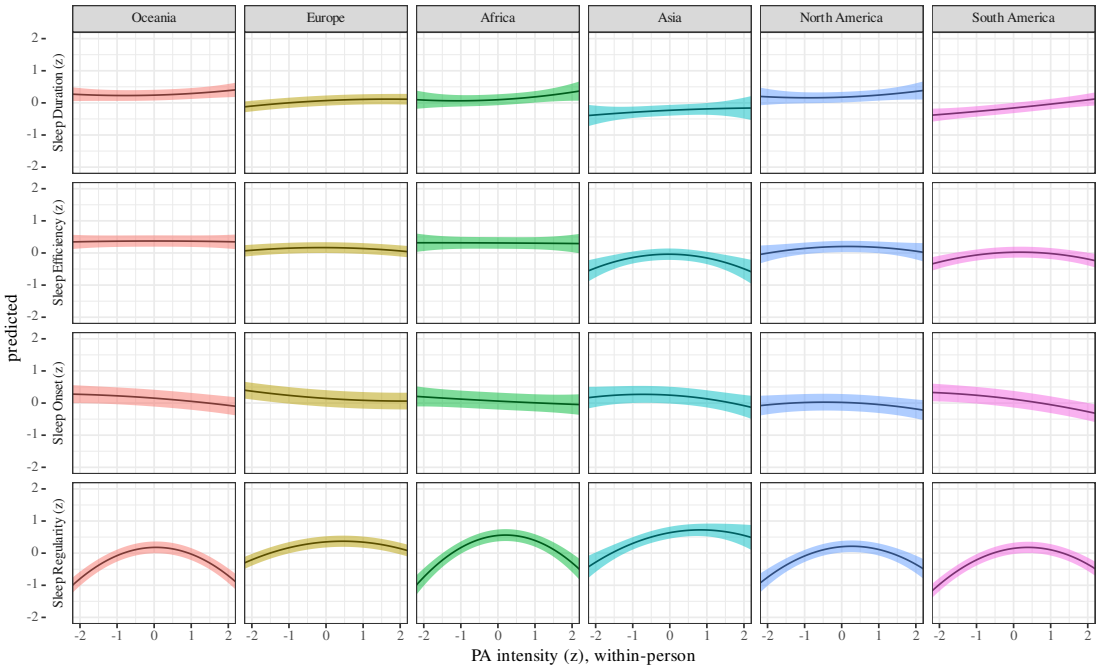


Figure 60: Sleep metrics on Physical activity intensity moderated by region

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 61.

Table 31: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.26 [−0.05, 0.56]	.098	0.04 [−0.09, 0.16]	.590
Sleep duration (within, prior)	0.08 [0.07, 0.10]	< .001	0.05 [0.03, 0.06]	< .001
Europe	−0.08 [−0.16, −0.01]	.021	−0.07 [−0.14, −0.01]	.028
Africa	−0.08 [−0.17, 0.01]	.086	0.06 [−0.02, 0.14]	.166
Asia	−0.35 [−0.44, −0.27]	< .001	−0.40 [−0.48, −0.32]	< .001
North America	−0.28 [−0.37, −0.19]	< .001	−0.21 [−0.29, −0.12]	< .001
South America	−0.17 [−0.24, −0.10]	< .001	−0.22 [−0.28, −0.15]	< .001
Sleep duration (within, prior)²	−0.07 [−0.08, −0.06]	< .001	−0.03 [−0.04, −0.02]	< .001
Sleep duration (between)	0.13 [0.09, 0.17]	< .001	0.13 [0.10, 0.17]	< .001
Sleep duration (between)²	0.00 [−0.04, 0.04]	.896	−0.01 [−0.04, 0.03]	.657
Sleep duration (within, prior) × Europe	−0.08 [−0.10, −0.07]	< .001	−0.02 [−0.03, 0.00]	.030
Sleep duration (within, prior) × Africa	−0.07 [−0.11, −0.04]	< .001	−0.10 [−0.14, −0.07]	< .001
Sleep duration (within, prior) × Asia	−0.09 [−0.13, −0.05]	< .001	−0.03 [−0.07, 0.01]	.177
Sleep duration (within, prior) × North America	−0.05 [−0.08, −0.02]	.001	−0.04 [−0.07, −0.01]	.016
Sleep duration (within, prior) × South America	−0.07 [−0.08, −0.05]	< .001	−0.06 [−0.08, −0.04]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Europe × Sleep duration (within, prior)²	0.03 [0.02, 0.04]	< .001	0.01 [0.00, 0.02]	.154
Africa × Sleep duration (within, prior)²	0.04 [0.01, 0.07]	.013	−0.05 [−0.08, −0.02]	.002
Asia × Sleep duration (within, prior)²	0.08 [0.03, 0.12]	< .001	−0.04 [−0.09, 0.01]	.088
North America × Sleep duration (within, prior)²	0.04 [0.02, 0.07]	.002	−0.02 [−0.04, 0.01]	.273
South America × Sleep duration (within, prior)²	0.02 [0.01, 0.04]	< .001	0.00 [−0.01, 0.02]	.685
Europe × Sleep duration (between)	−0.12 [−0.17, −0.07]	< .001	−0.09 [−0.14, −0.05]	< .001
Africa × Sleep duration (between)	−0.05 [−0.19, 0.09]	.496	−0.17 [−0.30, −0.04]	.009
Asia × Sleep duration (between)	−0.01 [−0.15, 0.13]	.897	0.03 [−0.10, 0.15]	.644
North America × Sleep duration (between)	0.05 [−0.06, 0.16]	.371	−0.10 [−0.20, 0.00]	.058
South America × Sleep duration (between)	−0.02 [−0.07, 0.03]	.402	−0.10 [−0.15, −0.06]	< .001
Europe × Sleep duration (between)²	−0.03 [−0.07, 0.01]	.172	−0.02 [−0.06, 0.01]	.214
Africa × Sleep duration (between)²	−0.03 [−0.13, 0.07]	.511	0.00 [−0.09, 0.09]	.981

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Asia × Sleep duration (between)²	0.00 [−0.15, 0.14]	.947	−0.04 [−0.17, 0.09]	.542
North America × Sleep duration (between)²	−0.07 [−0.15, 0.02]	.123	0.05 [−0.03, 0.12]	.234
South America × Sleep duration (between)²	−0.07 [−0.11, −0.02]	.004	0.01 [−0.03, 0.05]	.760
Sleep efficiency (z)				
(Intercept)	0.31 [0.02, 0.61]	.038	0.11 [−0.01, 0.24]	.076
Sleep efficiency (within, prior)	0.02 [0.00, 0.04]	.039	0.03 [0.01, 0.05]	.001
Europe	−0.18 [−0.24, −0.12]	< .001	−0.18 [−0.23, −0.12]	< .001
Africa	−0.13 [−0.20, −0.06]	< .001	−0.13 [−0.19, −0.07]	< .001
Asia	−0.38 [−0.46, −0.29]	< .001	−0.42 [−0.49, −0.34]	< .001
North America	−0.25 [−0.31, −0.18]	< .001	−0.24 [−0.30, −0.18]	< .001
South America	−0.25 [−0.31, −0.19]	< .001	−0.33 [−0.38, −0.27]	< .001
Sleep efficiency (within, prior)²	−0.02 [−0.03, 0.00]	.024	−0.01 [−0.02, 0.01]	.431
Sleep efficiency (between)	−0.16 [−0.20, −0.12]	< .001	−0.04 [−0.08, −0.01]	.019
Sleep efficiency (between)²	−0.13 [−0.17, −0.09]	< .001	−0.09 [−0.12, −0.05]	< .001
Sleep efficiency (within, prior) × Europe	0.00 [−0.02, 0.02]	.888	−0.02 [−0.04, 0.00]	.086
Sleep efficiency (within, prior) × Africa	−0.02 [−0.05, 0.01]	.156	−0.04 [−0.07, −0.02]	.001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency (within, prior) × Asia	−0.02 [−0.05, 0.01]	.216	−0.02 [−0.04, 0.01]	.266
Sleep efficiency (within, prior) × North America	−0.02 [−0.05, 0.00]	.070	−0.04 [−0.07, −0.01]	.004
Sleep efficiency (within, prior) × South America	−0.04 [−0.06, −0.02]	< .001	−0.03 [−0.05, −0.01]	.004
Europe × Sleep efficiency (within, prior)²	0.01 [−0.01, 0.02]	.454	−0.01 [−0.03, 0.01]	.272
Africa × Sleep efficiency (within, prior)²	0.01 [−0.01, 0.03]	.357	−0.01 [−0.03, 0.01]	.302
Asia × Sleep efficiency (within, prior)²	0.03 [0.00, 0.05]	.018	−0.01 [−0.04, 0.01]	.263
North America × Sleep efficiency (within, prior)²	0.00 [−0.02, 0.02]	.976	−0.01 [−0.03, 0.01]	.221
South America × Sleep efficiency (within, prior)²	−0.01 [−0.02, 0.01]	.538	0.00 [−0.02, 0.01]	.667
Europe × Sleep efficiency (between)	0.18 [0.13, 0.22]	< .001	0.06 [0.02, 0.09]	.005
Africa × Sleep efficiency (between)	0.21 [0.11, 0.30]	< .001	0.02 [−0.07, 0.11]	.648
Asia × Sleep efficiency (between)	0.18 [0.05, 0.32]	.008	0.13 [0.00, 0.25]	.042
North America × Sleep efficiency (between)	0.26 [0.16, 0.36]	< .001	0.14 [0.05, 0.23]	.003

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
South America × Sleep efficiency (between)	0.12 [0.07, 0.17]	< .001	0.06 [0.01, 0.10]	.018
Europe × Sleep efficiency (between)²	0.12 [0.08, 0.16]	< .001	0.07 [0.04, 0.11]	< .001
Africa × Sleep efficiency (between)²	0.13 [0.08, 0.19]	< .001	0.09 [0.04, 0.14]	< .001
Asia × Sleep efficiency (between)²	0.11 [0.03, 0.19]	.005	0.08 [0.01, 0.15]	.021
North America × Sleep efficiency (between)²	0.14 [0.08, 0.21]	< .001	0.10 [0.04, 0.16]	< .001
South America × Sleep efficiency (between)²	0.11 [0.07, 0.15]	< .001	0.09 [0.05, 0.13]	< .001
Sleep onset (z)				
(Intercept)	0.09 [−0.22, 0.40]	.589	−0.03 [−0.16, 0.10]	.645
Sleep onset (within, prior)	−0.14 [−0.16, −0.13]	< .001	−0.08 [−0.10, −0.06]	< .001
Europe	0.07 [−0.01, 0.16]	.072	−0.01 [−0.08, 0.06]	.799
Africa	0.31 [−0.16, 0.77]	.198	0.20 [−0.22, 0.63]	.351
Asia	−0.02 [−0.25, 0.22]	.881	−0.26 [−0.47, −0.04]	.018
North America	0.29 [0.13, 0.44]	< .001	0.01 [−0.13, 0.15]	.886
South America	0.05 [−0.04, 0.14]	.235	−0.13 [−0.21, −0.05]	.001
Sleep onset (within, prior)²	−0.02 [−0.03, −0.01]	< .001	−0.01 [−0.02, 0.00]	.002
Sleep onset (between)	−0.27 [−0.31, −0.24]	< .001	−0.17 [−0.20, −0.13]	< .001
Sleep onset (between)²	0.03 [0.02, 0.04]	< .001	0.02 [0.01, 0.03]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep onset (within, prior) × Europe	0.13 [0.11, 0.15]	< .001	0.09 [0.07, 0.11]	< .001
Sleep onset (within, prior) × Africa	0.15 [0.10, 0.21]	< .001	0.07 [0.01, 0.13]	.014
Sleep onset (within, prior) × Asia	0.17 [0.12, 0.22]	< .001	0.11 [0.06, 0.17]	< .001
Sleep onset (within, prior) × North America	0.17 [0.12, 0.22]	< .001	0.06 [0.01, 0.11]	.013
Sleep onset (within, prior) × South America	0.23 [0.20, 0.25]	< .001	0.10 [0.08, 0.13]	< .001
Europe × Sleep onset (within, prior)²	−0.01 [−0.02, 0.00]	.031	0.00 [−0.01, 0.01]	.854
Africa × Sleep onset (within, prior)²	0.04 [−0.05, 0.14]	.383	−0.07 [−0.17, 0.03]	.184
Asia × Sleep onset (within, prior)²	−0.03 [−0.12, 0.06]	.570	−0.08 [−0.17, 0.01]	.099
North America × Sleep onset (within, prior)²	−0.09 [−0.15, −0.03]	.004	−0.05 [−0.11, 0.01]	.095
South America × Sleep onset (within, prior)²	−0.01 [−0.03, 0.00]	.072	0.00 [−0.02, 0.01]	.629
Europe × Sleep onset (between)	0.19 [0.14, 0.23]	< .001	0.11 [0.07, 0.15]	< .001
Africa × Sleep onset (between)	0.38 [−0.56, 1.33]	.425	0.38 [−0.48, 1.24]	.383

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Asia × Sleep onset (between)	0.43 [−0.12, 0.98]	.128	0.14 [−0.36, 0.64]	.588
North America × Sleep onset (between)	0.52 [0.34, 0.69]	< .001	0.16 [0.01, 0.32]	.041
South America × Sleep onset (between)	0.21 [0.16, 0.26]	< .001	0.13 [0.09, 0.18]	< .001
Europe × Sleep onset (between)²	−0.05 [−0.06, −0.03]	< .001	−0.03 [−0.05, −0.01]	< .001
Africa × Sleep onset (between)²	−0.03 [−0.50, 0.45]	.910	0.10 [−0.33, 0.54]	.639
Asia × Sleep onset (between)²	0.09 [−0.24, 0.42]	.600	−0.04 [−0.34, 0.26]	.798
North America × Sleep onset (between)²	0.00 [−0.10, 0.10]	.993	−0.07 [−0.16, 0.03]	.163
South America × Sleep onset (between)²	−0.06 [−0.08, −0.04]	< .001	−0.02 [−0.04, 0.00]	.027
Sleep regularity (z)				
(Intercept)	0.22 [−0.07, 0.51]	.143	0.00 [−0.13, 0.13]	.947
Sleep regularity (within, prior)	−0.02 [−0.04, 0.00]	.026	0.00 [−0.01, 0.02]	.768
Europe	−0.26 [−0.32, −0.19]	< .001	−0.18 [−0.24, −0.12]	< .001
Africa	−0.25 [−0.32, −0.18]	< .001	−0.12 [−0.18, −0.05]	< .001
Asia	−0.48 [−0.55, −0.40]	< .001	−0.51 [−0.58, −0.44]	< .001
North America	−0.33 [−0.40, −0.26]	< .001	−0.28 [−0.34, −0.21]	< .001
South America	−0.24 [−0.30, −0.17]	< .001	−0.33 [−0.39, −0.27]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep regularity (within, prior)²	0.01 [0.00, 0.02]	.048	0.01 [0.00, 0.02]	.091
Sleep regularity (between)	0.16 [0.12, 0.19]	< .001	0.07 [0.04, 0.10]	< .001
Sleep regularity (between)²	−0.03 [−0.06, 0.00]	.053	−0.02 [−0.04, 0.01]	.203
Sleep regularity (within, prior) × Europe	0.03 [0.01, 0.05]	.003	0.01 [−0.01, 0.03]	.318
Sleep regularity (within, prior) × Africa	0.05 [0.02, 0.08]	< .001	0.04 [0.00, 0.07]	.025
Sleep regularity (within, prior) × Asia	0.03 [−0.01, 0.06]	.139	0.01 [−0.03, 0.05]	.576
Sleep regularity (within, prior) × North America	0.01 [−0.02, 0.04]	.343	0.02 [−0.01, 0.05]	.152
Sleep regularity (within, prior) × South America	−0.01 [−0.03, 0.01]	.215	0.00 [−0.02, 0.02]	.769
Europe × Sleep regularity (within, prior)²	−0.01 [−0.02, 0.01]	.409	−0.01 [−0.03, 0.00]	.145
Africa × Sleep regularity (within, prior)²	0.01 [−0.02, 0.04]	.530	0.00 [−0.03, 0.03]	.992
Asia × Sleep regularity (within, prior)²	−0.02 [−0.06, 0.01]	.203	−0.02 [−0.05, 0.02]	.341
North America × Sleep regularity (within, prior)²	−0.02 [−0.05, 0.00]	.088	0.00 [−0.03, 0.02]	.815
South America × Sleep regularity (within, prior)²	−0.03 [−0.04, −0.01]	< .001	−0.01 [−0.03, 0.00]	.098

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Europe × Sleep regularity (between)	0.02 [−0.01, 0.06]	.242	0.00 [−0.03, 0.04]	.962
Africa × Sleep regularity (between)	−0.01 [−0.08, 0.05]	.756	0.00 [−0.06, 0.06]	.955
Asia × Sleep regularity (between)	−0.17 [−0.24, −0.09]	< .001	−0.03 [−0.10, 0.04]	.384
North America × Sleep regularity (between)	−0.04 [−0.11, 0.04]	.311	−0.03 [−0.10, 0.04]	.352
South America × Sleep regularity (between)	0.05 [0.01, 0.09]	.020	0.00 [−0.04, 0.04]	.815
Europe × Sleep regularity (between)²	0.03 [0.00, 0.07]	.034	−0.01 [−0.04, 0.02]	.650
Africa × Sleep regularity (between)²	0.01 [−0.04, 0.07]	.628	0.00 [−0.05, 0.06]	.876
Asia × Sleep regularity (between)²	0.03 [−0.03, 0.09]	.387	0.02 [−0.04, 0.08]	.548
North America × Sleep regularity (between)²	0.03 [−0.03, 0.09]	.312	−0.01 [−0.07, 0.04]	.629
South America × Sleep regularity (between)²	0.02 [−0.01, 0.06]	.188	0.02 [−0.02, 0.05]	.291

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

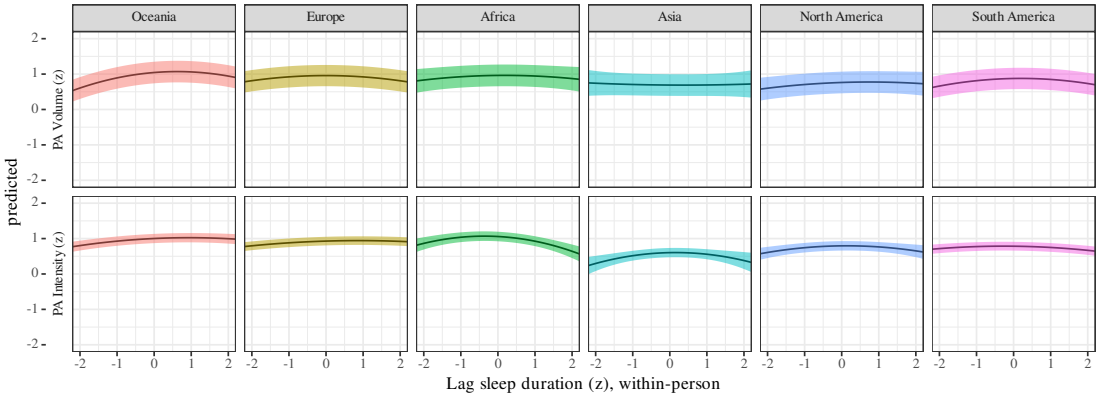


Figure 61: Physical activity by sleep duration moderated by region

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 62.

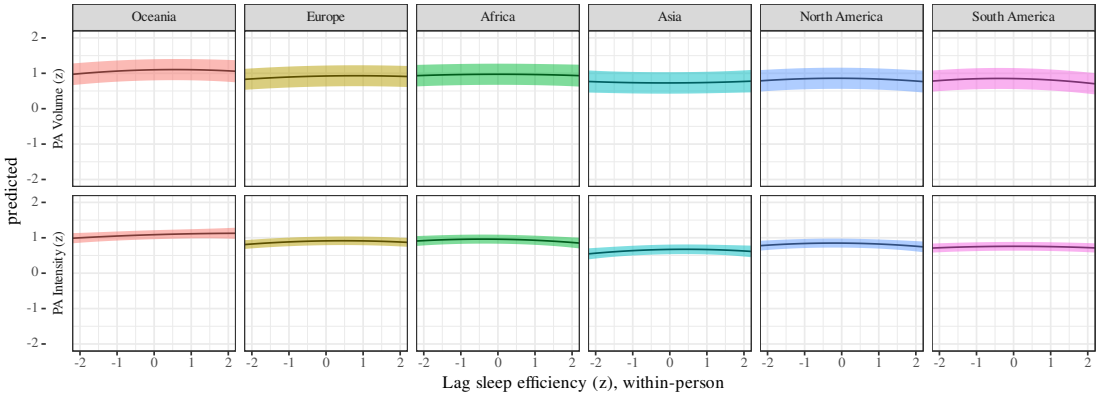


Figure 62: Physical activity by sleep efficiency moderated by region

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by region. Results, controlling for SES, age, sex and BMI z-score are presented in Table 31 and Figure 63.

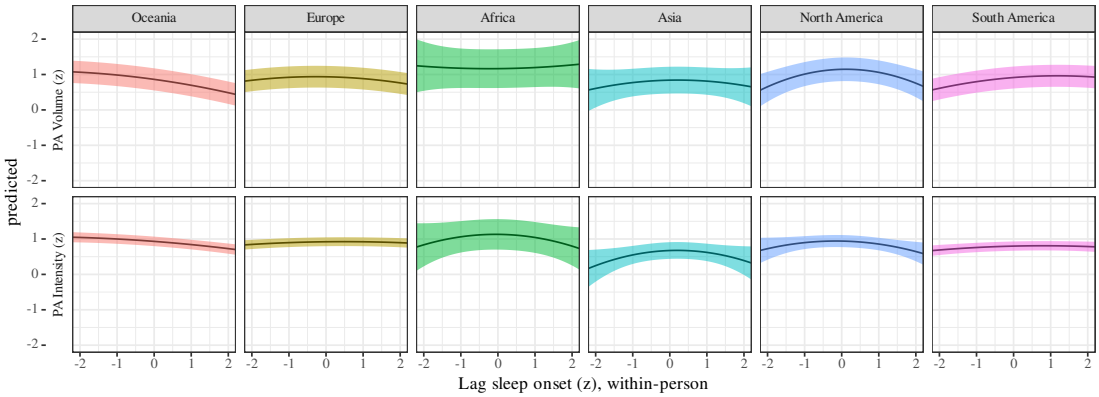


Figure 63: Physical activity by sleep onset moderated by region

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by region. Results are presented in Table 31 and Figure 64.

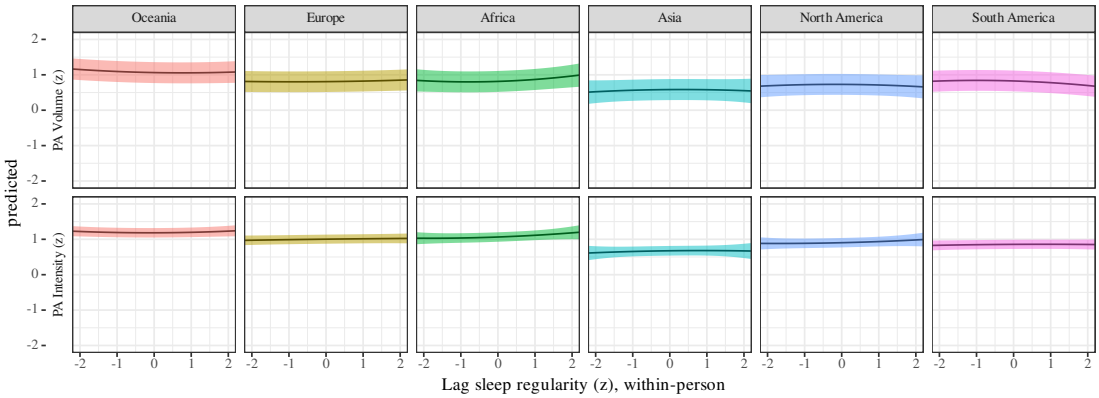


Figure 64: Physical activity by sleep regularity moderated by region

daylight hours

The association between physical activity volume and sleep by daylight hours

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by daylight hours, and the results are presented in Table 32 and Figure 65

Table 32: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.38 [0.22, 0.54]	< .001	0.32 [0.17, 0.47]	< .001
Physical activity (within)	0.09 [0.04, 0.13]	< .001	0.11 [0.07, 0.15]	< .001
Daylight hours	−0.03 [−0.03, −0.03]	< .001	−0.03 [−0.03, −0.02]	< .001
Physical activity (within)²	0.01 [−0.04, 0.06]	.779	0.01 [−0.03, 0.06]	.565
Physical activity (between)	0.26 [0.19, 0.33]	< .001	0.17 [0.13, 0.22]	< .001
Physical activity (between)²	−0.03 [−0.07, 0.01]	.124	0.04 [0.00, 0.08]	.067
Physical activity (within) × Daylight hours	0.00 [0.00, 0.01]	.415	0.00 [−0.01, 0.00]	.007
Daylight hours × Physical activity (within)²	0.00 [−0.01, 0.00]	.575	0.00 [−0.01, 0.00]	.389
Daylight hours × Physical activity (between)	−0.01 [−0.02, −0.01]	< .001	−0.01 [−0.01, 0.00]	.005
Daylight hours × Physical activity (between)²	0.00 [0.00, 0.01]	.050	0.00 [0.00, 0.00]	.471
Sleep efficiency (z)				
(Intercept)	0.28 [0.12, 0.44]	< .001	0.28 [0.11, 0.45]	.001
Physical activity (within)	0.06 [0.02, 0.11]	.005	0.04 [0.00, 0.08]	.038
Daylight hours	−0.01 [−0.01, 0.00]	.002	−0.01 [−0.01, 0.00]	.002
Physical activity (within)²	−0.03 [−0.07, 0.02]	.291	−0.07 [−0.11, −0.03]	.001
Physical activity (between)	0.17 [0.10, 0.25]	< .001	0.08 [0.03, 0.13]	.003

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (between)²	−0.04 [−0.08, 0.00]	.052	−0.03 [−0.08, 0.01]	.145
Physical activity (within) × Daylight hours	0.00 [−0.01, 0.00]	.043	0.00 [−0.01, 0.00]	.039
Daylight hours × Physical activity (within)²	0.00 [0.00, 0.00]	.661	0.00 [0.00, 0.01]	.051
Daylight hours × Physical activity (between)	−0.01 [−0.01, 0.00]	.045	0.00 [−0.01, 0.00]	.090
Daylight hours × Physical activity (between)²	0.00 [0.00, 0.00]	.905	0.00 [0.00, 0.01]	.405
Sleep onset (z)				
(Intercept)	−0.04 [−0.32, 0.23]	.762	0.02 [−0.24, 0.28]	.858
Physical activity (within)	−0.13 [−0.16, −0.09]	< .001	−0.09 [−0.12, −0.06]	< .001
Daylight hours	0.01 [0.01, 0.02]	< .001	0.01 [0.01, 0.01]	< .001
Physical activity (within)²	−0.04 [−0.07, 0.00]	.042	−0.04 [−0.07, −0.01]	.015
Physical activity (between)	−0.29 [−0.35, −0.23]	< .001	−0.24 [−0.28, −0.20]	< .001
Physical activity (between)²	−0.05 [−0.08, −0.02]	.002	−0.11 [−0.15, −0.07]	< .001
Physical activity (within) × Daylight hours	0.00 [0.00, 0.00]	.304	0.00 [0.00, 0.00]	.770
Daylight hours × Physical activity (within)²	0.00 [0.00, 0.01]	.005	0.00 [0.00, 0.01]	.004
Daylight hours × Physical activity (between)	0.02 [0.01, 0.02]	< .001	0.01 [0.01, 0.02]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Daylight hours × Physical activity (between)²	0.00 [0.00, 0.01]	.004	0.01 [0.00, 0.01]	< .001
Sleep regularity (z)				
(Intercept)	0.10 [−0.06, 0.25]	.223	0.10 [−0.09, 0.28]	.307
Physical activity (within)	0.27 [0.21, 0.32]	< .001	0.23 [0.18, 0.27]	< .001
Daylight hours	0.01 [0.00, 0.01]	.004	0.01 [0.00, 0.01]	.038
Physical activity (within)²	−0.01 [−0.06, 0.04]	.702	−0.26 [−0.30, −0.21]	< .001
Physical activity (between)	0.35 [0.26, 0.43]	< .001	0.14 [0.08, 0.20]	< .001
Physical activity (between)²	−0.09 [−0.14, −0.05]	< .001	−0.09 [−0.14, −0.04]	< .001
Physical activity (within) × Daylight hours	−0.01 [−0.01, 0.00]	.002	−0.01 [−0.01, −0.01]	< .001
Daylight hours × Physical activity (within)²	−0.01 [−0.01, 0.00]	< .001	0.01 [0.01, 0.01]	< .001
Daylight hours × Physical activity (between)	0.01 [0.00, 0.02]	.002	0.00 [0.00, 0.01]	.751
Daylight hours × Physical activity (between)²	0.00 [0.00, 0.00]	.890	0.01 [0.00, 0.01]	.012

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

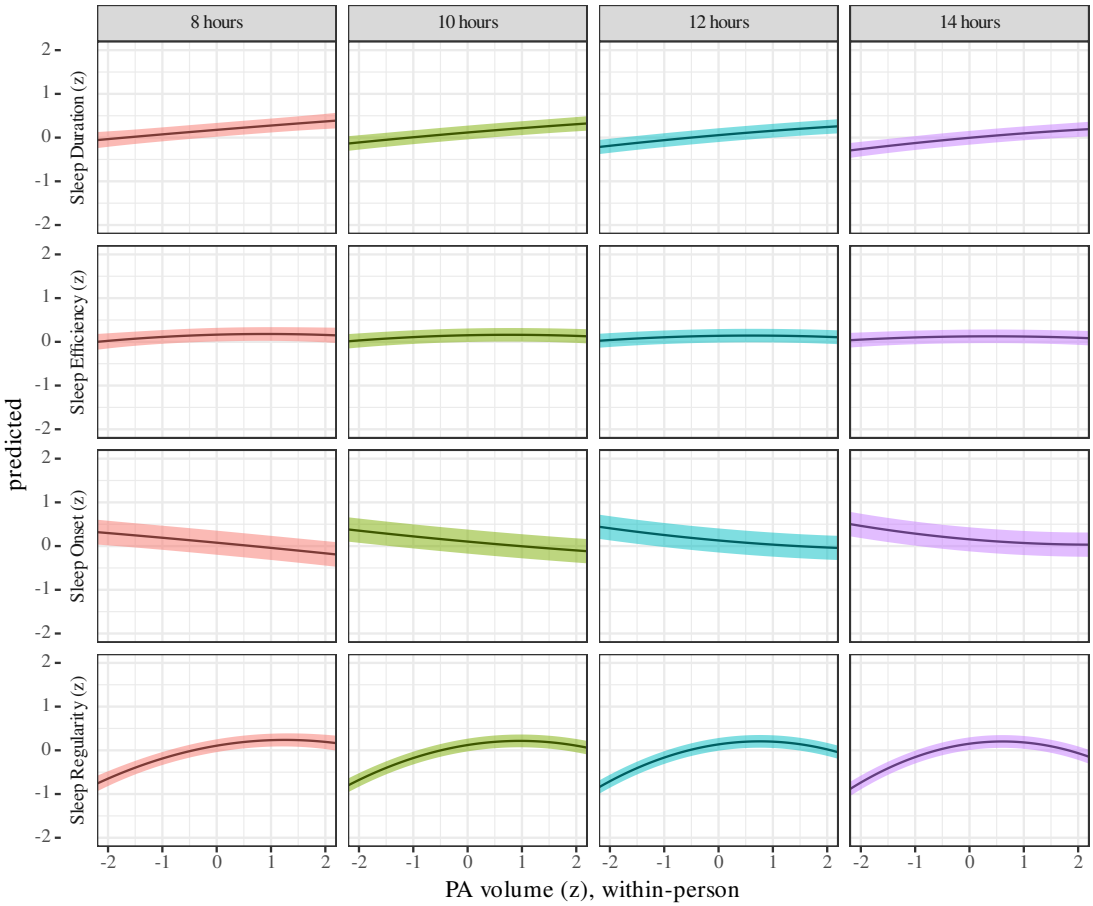


Figure 65: Sleep metrics on Physical activity volume by daylight hours

The association between physical activity intensity and sleep by daylight hours

We estimated how physical activity intensity is associated with sleep across daylight hours. We present the results in Table 32 and Figure 66.

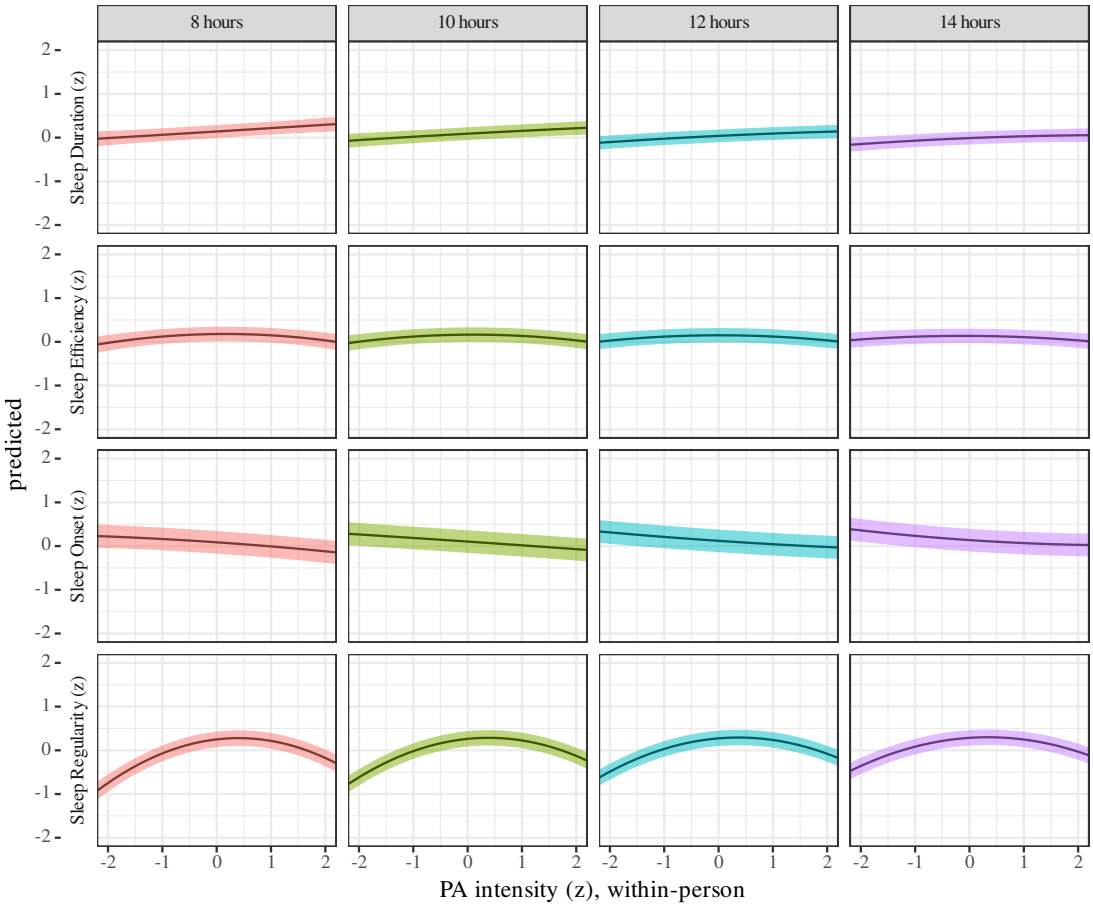


Figure 66: Sleep metrics on Physical activity intensity moderated by daylight hours

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 33 and Figure 67.

Table 33: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−0.04 [−0.35, 0.26]	.778	−0.09 [−0.22, 0.04]	.154
Sleep duration (within, prior)	0.10 [0.08, 0.12]	< .001	0.08 [0.06, 0.10]	< .001
Daylight hours	0.02 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.242
Sleep duration (within, prior)²	−0.06 [−0.08, −0.04]	< .001	−0.01 [−0.03, 0.02]	.569
Sleep duration (between)	0.11 [0.06, 0.16]	< .001	0.09 [0.05, 0.14]	< .001
Sleep duration (between)²	−0.06 [−0.11, −0.01]	.013	−0.01 [−0.06, 0.03]	.545
Sleep duration (within, prior) × Daylight hours	−0.01 [−0.01, −0.01]	< .001	0.00 [−0.01, 0.00]	< .001
Daylight hours × Sleep duration (within, prior)²	0.00 [0.00, 0.00]	.097	0.00 [0.00, 0.00]	.184
Daylight hours × Sleep duration (between)	0.00 [−0.01, 0.00]	.090	0.00 [−0.01, 0.00]	.222
Daylight hours × Sleep duration (between)²	0.00 [0.00, 0.01]	.092	0.00 [0.00, 0.00]	.798
Sleep efficiency (z)				
(Intercept)	−0.10 [−0.41, 0.20]	.506	−0.07 [−0.20, 0.06]	.260
Sleep efficiency (within, prior)	0.02 [−0.01, 0.04]	.195	0.02 [−0.01, 0.04]	.233
Daylight hours	0.02 [0.02, 0.02]	< .001	0.00 [0.00, 0.00]	.891

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency (within, prior)²	−0.02 [−0.04, 0.00]	.050	−0.04 [−0.06, −0.02]	< .001
Sleep efficiency (between)	0.16 [0.11, 0.22]	< .001	0.07 [0.02, 0.12]	.005
Sleep efficiency (between)²	−0.01 [−0.06, 0.04]	.805	−0.02 [−0.07, 0.03]	.418
Sleep efficiency (within, prior) × Daylight hours	0.00 [0.00, 0.00]	.417	0.00 [0.00, 0.00]	.609
Daylight hours × Sleep efficiency (within, prior)²	0.00 [0.00, 0.00]	.481	0.00 [0.00, 0.00]	.014
Daylight hours × Sleep efficiency (between)	−0.01 [−0.02, −0.01]	< .001	0.00 [−0.01, 0.00]	.015
Daylight hours × Sleep efficiency (between)²	0.00 [−0.01, 0.00]	.588	0.00 [0.00, 0.00]	.748
Sleep onset (z)				
(Intercept)	−0.05 [−0.37, 0.27]	.772	−0.10 [−0.23, 0.04]	.166
Sleep onset (within, prior)	−0.02 [−0.05, 0.01]	.245	−0.04 [−0.08, −0.01]	.009
Daylight hours	0.01 [0.01, 0.02]	< .001	0.00 [0.00, 0.01]	.282
Sleep onset (within, prior)²	−0.06 [−0.09, −0.03]	< .001	−0.03 [−0.06, 0.00]	.024
Sleep onset (between)	−0.14 [−0.19, −0.09]	< .001	−0.06 [−0.10, −0.01]	.008
Sleep onset (between)²	−0.04 [−0.08, 0.00]	.068	0.02 [−0.02, 0.06]	.329
Sleep onset (within, prior) × Daylight hours	0.00 [0.00, 0.00]	.427	0.00 [0.00, 0.01]	.003
Daylight hours × Sleep onset (within, prior)²	0.00 [0.00, 0.01]	.005	0.00 [0.00, 0.00]	.106

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Daylight hours × Sleep onset (between)	0.00 [0.00, 0.01]	.460	0.00 [−0.01, 0.00]	.226
Daylight hours × Sleep onset (between)²	0.00 [0.00, 0.01]	.035	0.00 [0.00, 0.00]	.696
Sleep regularity (z)				
(Intercept)	−0.23 [−0.53, 0.08]	.141	−0.21 [−0.35, −0.07]	.003
Sleep regularity (within, prior)	0.02 [−0.02, 0.05]	.341	0.05 [0.01, 0.08]	.006
Daylight hours	0.02 [0.02, 0.02]	< .001	0.00 [0.00, 0.01]	.162
Sleep regularity (within, prior)²	0.05 [0.02, 0.09]	.002	0.05 [0.01, 0.08]	.006
Sleep regularity (between)	0.24 [0.19, 0.29]	< .001	0.10 [0.05, 0.15]	< .001
Sleep regularity (between)²	0.00 [−0.05, 0.05]	.970	0.02 [−0.03, 0.07]	.469
Sleep regularity (within, prior) × Daylight hours	0.00 [0.00, 0.00]	.189	0.00 [−0.01, 0.00]	.032
Daylight hours × Sleep regularity (within, prior)²	0.00 [−0.01, 0.00]	.002	0.00 [−0.01, 0.00]	.008
Daylight hours × Sleep regularity (between)	−0.01 [−0.01, 0.00]	.007	0.00 [−0.01, 0.00]	.132
Daylight hours × Sleep regularity (between)²	0.00 [0.00, 0.00]	.907	0.00 [−0.01, 0.00]	.151

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

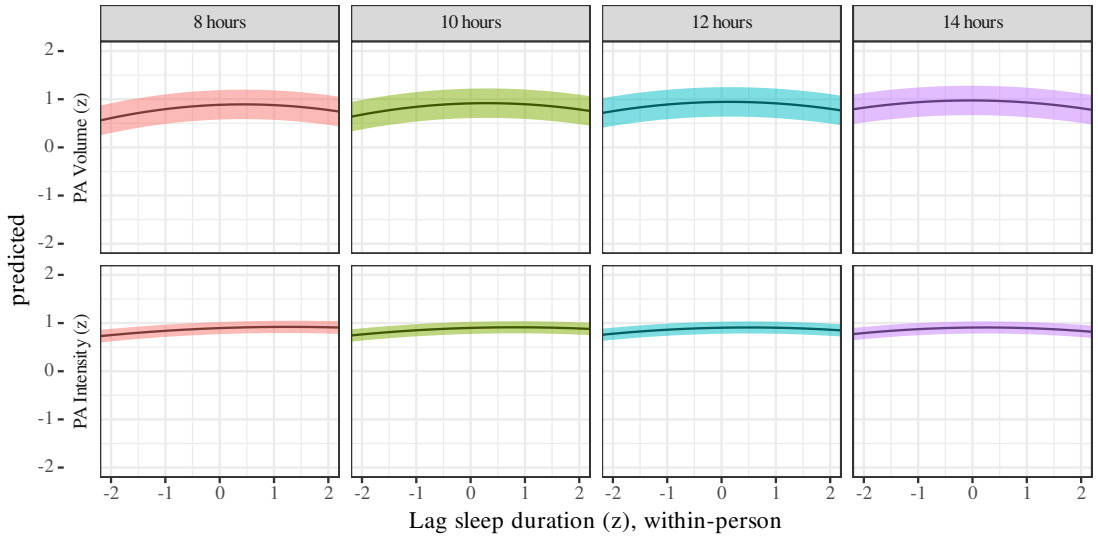


Figure 67: Physical activity by sleep duration moderated by daylight hours

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 33 and Figure 68.

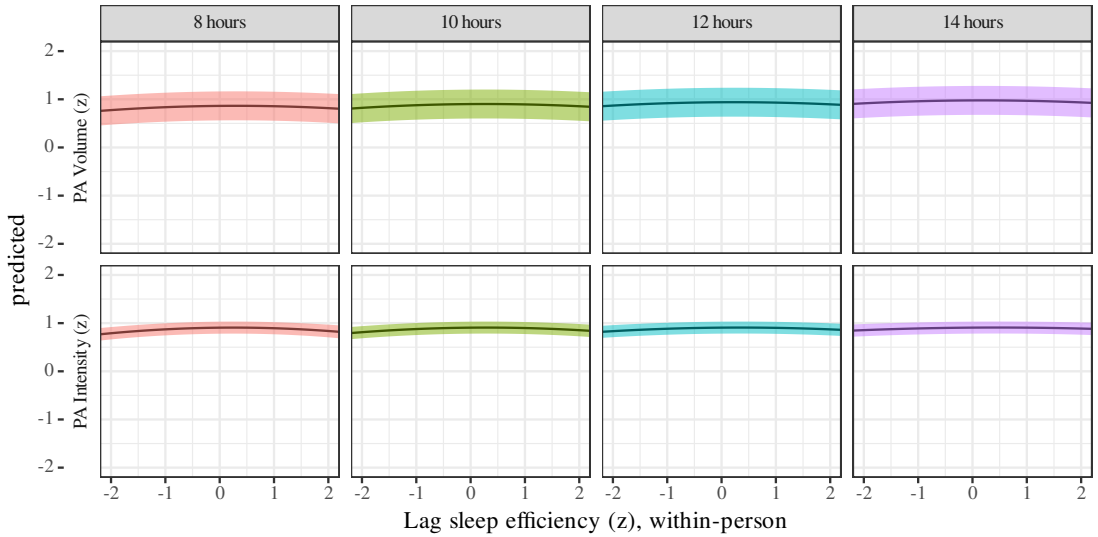


Figure 68: Physical activity by sleep efficiency moderated by daylight hours

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by daylight hours. Results, controlling for SES, age, sex and BMI z-score are presented in Table 33 and Figure 69.

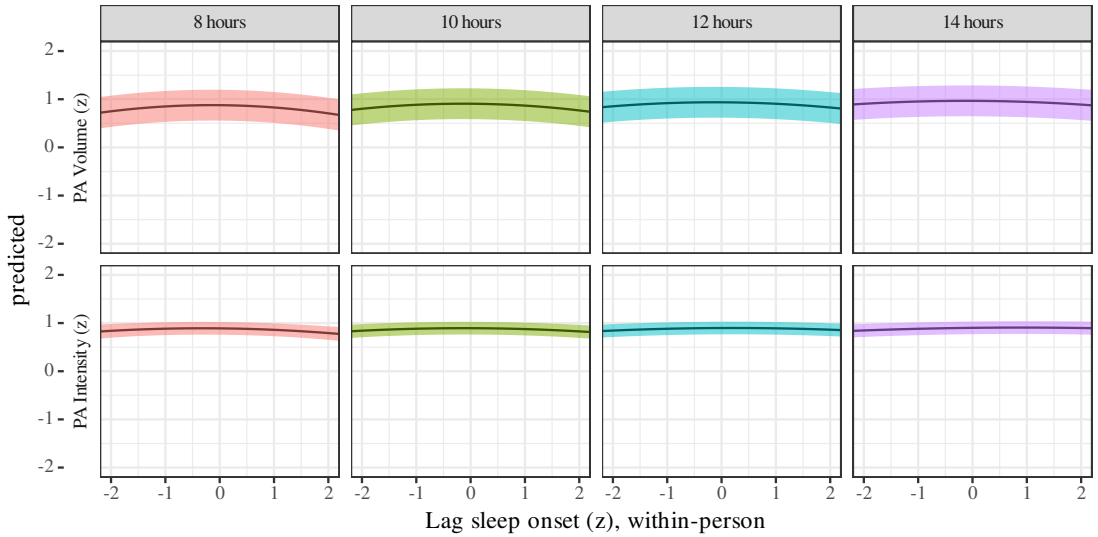


Figure 69: Physical activity by sleep onset moderated by daylight hours

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by daylight hours. Results are presented in Table 33 and Figure 70.

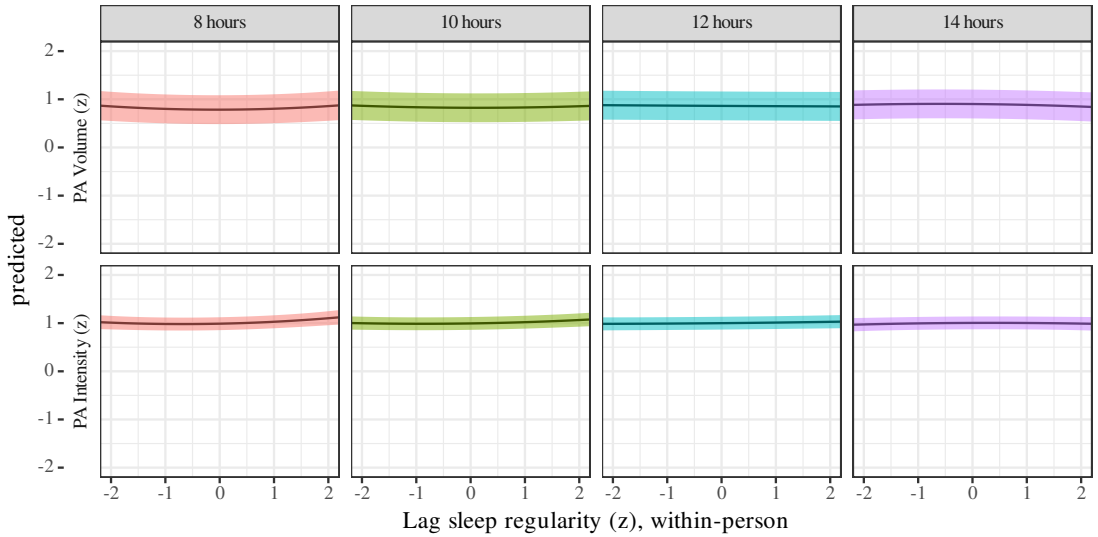


Figure 70: Physical activity by sleep regularity moderated by daylight hours

wear location

The association between physical activity volume and sleep by wear location

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by wear location, and the results are presented in Table 34 and Figure 71

Table 34: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.55 [0.31, 0.78]	< .001	0.45 [0.22, 0.68]	< .001
Physical activity (within)	0.10 [0.08, 0.12]	< .001	0.07 [0.06, 0.08]	< .001
Wrist	−0.67 [−0.93, −0.41]	< .001	−0.56 [−0.81, −0.31]	< .001
Physical activity (within)²	−0.01 [−0.04, 0.02]	.641	0.01 [−0.01, 0.03]	.201
Physical activity (between)	0.12 [0.07, 0.17]	< .001	0.19 [0.14, 0.24]	< .001
Physical activity (between)²	−0.13 [−0.19, −0.07]	< .001	−0.05 [−0.09, 0.00]	.035
Physical activity (within) × Wrist	0.01 [−0.02, 0.03]	.620	−0.01 [−0.03, 0.00]	.113
Wrist × Physical activity (within)²	0.00 [−0.03, 0.03]	.956	−0.02 [−0.04, 0.00]	.019
Wrist × Physical activity (between)	−0.11 [−0.17, −0.05]	< .001	−0.09 [−0.14, −0.04]	< .001
Wrist × Physical activity (between)²	0.16 [0.10, 0.22]	< .001	0.08 [0.03, 0.12]	.001
Sleep efficiency (z)				
(Intercept)	−0.28 [−0.48, −0.08]	.006	−0.48 [−0.68, −0.29]	< .001
Physical activity (within)	0.16 [0.14, 0.17]	< .001	0.04 [0.02, 0.05]	< .001
Wrist	0.62 [0.40, 0.84]	< .001	0.84 [0.62, 1.05]	< .001
Physical activity (within)²	−0.14 [−0.16, −0.11]	< .001	−0.06 [−0.08, −0.04]	< .001
Physical activity (between)	0.15 [0.10, 0.21]	< .001	0.28 [0.23, 0.33]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (between)²	−0.27 [−0.33, −0.21]	< .001	−0.16 [−0.21, −0.12]	< .001
Physical activity (within) × Wrist	−0.17 [−0.19, −0.15]	< .001	−0.06 [−0.07, −0.04]	< .001
Wrist × Physical activity (within)²	0.13 [0.10, 0.16]	< .001	0.05 [0.04, 0.07]	< .001
Wrist × Physical activity (between)	−0.14 [−0.20, −0.08]	< .001	−0.29 [−0.34, −0.24]	< .001
Wrist × Physical activity (between)²	0.26 [0.19, 0.32]	< .001	0.14 [0.09, 0.19]	< .001
Sleep onset (z)				
(Intercept)	−0.83 [−1.19, −0.46]	< .001	−0.81 [−1.17, −0.45]	< .001
Physical activity (within)	0.00 [−0.02, 0.01]	.497	−0.05 [−0.06, −0.04]	< .001
Wrist	1.20 [0.79, 1.60]	< .001	1.18 [0.77, 1.58]	< .001
Physical activity (within)²	−0.05 [−0.07, −0.03]	< .001	−0.03 [−0.04, −0.02]	< .001
Physical activity (between)	−0.02 [−0.06, 0.02]	.353	0.00 [−0.04, 0.04]	.888
Physical activity (between)²	−0.05 [−0.10, 0.00]	.065	−0.05 [−0.09, −0.01]	.007
Physical activity (within) × Wrist	−0.13 [−0.14, −0.11]	< .001	−0.06 [−0.07, −0.05]	< .001
Wrist × Physical activity (within)²	0.07 [0.04, 0.09]	< .001	0.06 [0.05, 0.08]	< .001
Wrist × Physical activity (between)	−0.12 [−0.17, −0.08]	< .001	−0.12 [−0.17, −0.08]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Wrist × Physical activity (between)²	0.05 [0.00, 0.10]	.039	0.01 [−0.03, 0.05]	.530
Sleep regularity (z)				
(Intercept)	0.16 [−0.14, 0.46]	.293	−0.31 [−0.62, 0.00]	.050
Physical activity (within)	0.32 [0.30, 0.34]	< .001	0.17 [0.16, 0.18]	< .001
Wrist	0.04 [−0.29, 0.38]	.808	0.58 [0.24, 0.93]	< .001
Physical activity (within)²	−0.40 [−0.43, −0.37]	< .001	−0.21 [−0.23, −0.19]	< .001
Physical activity (between)	0.34 [0.28, 0.40]	< .001	0.41 [0.36, 0.47]	< .001
Physical activity (between)²	−0.23 [−0.30, −0.16]	< .001	−0.23 [−0.29, −0.18]	< .001
Physical activity (within) × Wrist	−0.17 [−0.19, −0.15]	< .001	−0.13 [−0.14, −0.11]	< .001
Wrist × Physical activity (within)²	0.30 [0.27, 0.33]	< .001	0.13 [0.11, 0.15]	< .001
Wrist × Physical activity (between)	0.18 [0.11, 0.24]	< .001	−0.31 [−0.37, −0.25]	< .001
Wrist × Physical activity (between)²	0.12 [0.05, 0.19]	< .001	0.21 [0.15, 0.26]	< .001

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

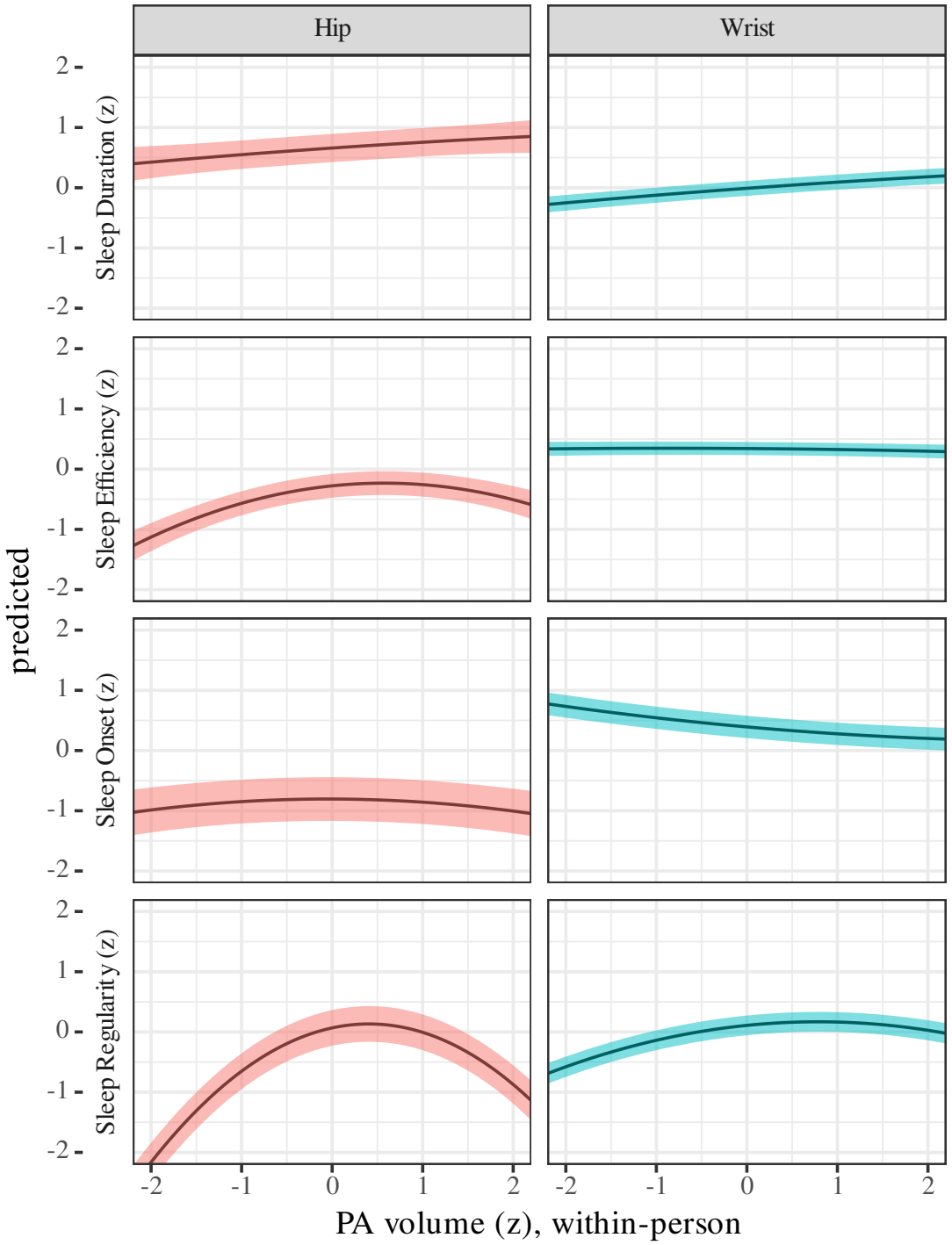


Figure 71: Sleep metrics on Physical activity volume by wear location

The association between physical activity intensity and sleep by wear location

We estimated how physical activity intensity is associated with sleep across wear location. We present the results in Table 34 and Figure 72.

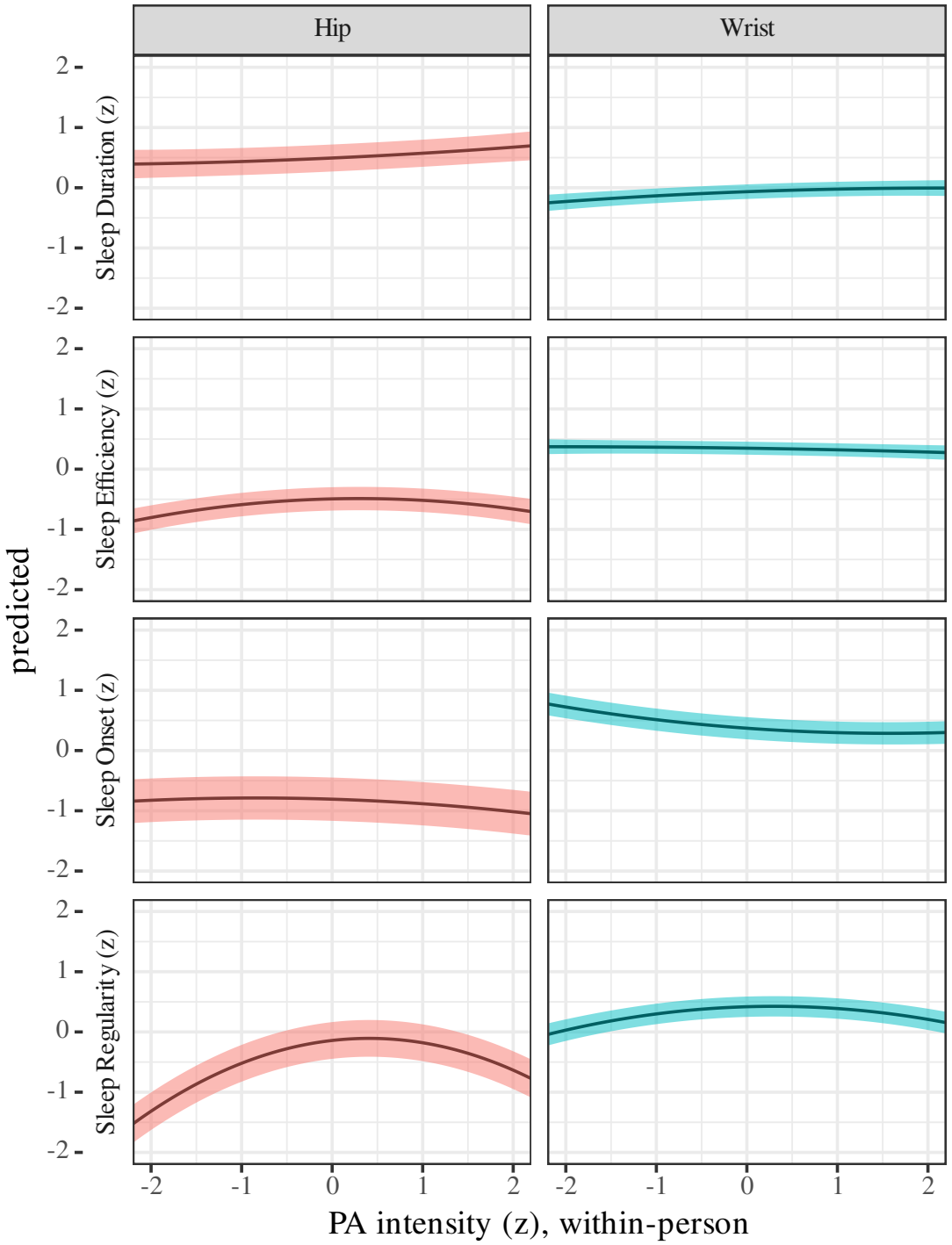


Figure 72: Sleep metrics on Physical activity intensity moderated by wear location

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 35 and Figure 73.

Table 35: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	−1.00 [−1.39, −0.61]	< .001	−0.42 [−0.64, −0.20]	< .001
Sleep duration (within, prior)	0.03 [0.02, 0.04]	< .001	0.02 [0.01, 0.03]	< .001
Wrist	1.43 [1.00, 1.86]	< .001	0.44 [0.19, 0.69]	< .001
Sleep duration (within, prior)²	−0.05 [−0.06, −0.03]	< .001	−0.07 [−0.08, −0.05]	< .001
Sleep duration (between)	0.12 [0.08, 0.15]	< .001	0.08 [0.04, 0.11]	< .001
Sleep duration (between)²	−0.03 [−0.06, 0.00]	.074	0.02 [−0.01, 0.05]	.129
Sleep duration (within, prior) × Wrist	−0.02 [−0.03, −0.01]	< .001	0.00 [−0.01, 0.01]	.761
Wrist × Sleep duration (within, prior)²	0.00 [−0.01, 0.02]	.470	0.05 [0.04, 0.06]	< .001
Wrist × Sleep duration (between)	−0.07 [−0.11, −0.03]	< .001	−0.03 [−0.07, 0.01]	.108
Wrist × Sleep duration (between)²	−0.01 [−0.04, 0.03]	.737	−0.05 [−0.08, −0.02]	.002
Sleep efficiency (z)				
(Intercept)	−0.93 [−1.33, −0.53]	< .001	−0.36 [−0.59, −0.12]	.003
Sleep efficiency (within, prior)	0.02 [0.01, 0.03]	< .001	0.02 [0.02, 0.03]	< .001
Wrist	1.34 [0.89, 1.78]	< .001	0.36 [0.10, 0.62]	.007

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency (within, prior)²	−0.01 [−0.02, −0.01]	< .001	−0.02 [−0.02, −0.01]	< .001
Sleep efficiency (between)	0.06 [0.02, 0.09]	< .001	0.07 [0.04, 0.10]	< .001
Sleep efficiency (between)²	−0.01 [−0.03, 0.01]	.184	0.01 [−0.01, 0.02]	.218
Sleep efficiency (within, prior) × Wrist	−0.03 [−0.04, −0.02]	< .001	−0.02 [−0.03, −0.01]	< .001
Wrist × Sleep efficiency (within, prior)²	−0.01 [−0.02, 0.00]	.041	0.01 [0.00, 0.02]	.003
Wrist × Sleep efficiency (between)	−0.07 [−0.10, −0.04]	< .001	−0.06 [−0.09, −0.03]	< .001
Wrist × Sleep efficiency (between)²	0.00 [−0.02, 0.02]	.748	−0.02 [−0.04, 0.00]	.026
Sleep onset (z)				
(Intercept)	−0.99 [−1.38, −0.60]	< .001	−0.41 [−0.64, −0.19]	< .001
Sleep onset (within, prior)	0.01 [−0.01, 0.03]	.254	0.01 [−0.01, 0.02]	.394
Wrist	1.43 [0.99, 1.86]	< .001	0.43 [0.19, 0.68]	< .001
Sleep onset (within, prior)²	−0.05 [−0.08, −0.03]	< .001	−0.10 [−0.13, −0.07]	< .001
Sleep onset (between)	−0.13 [−0.24, −0.01]	.031	−0.05 [−0.15, 0.05]	.346
Sleep onset (between)²	−0.07 [−0.13, 0.00]	.040	0.01 [−0.05, 0.07]	.778
Sleep onset (within, prior) × Wrist	−0.02 [−0.03, 0.00]	.040	0.00 [−0.02, 0.02]	.899
Wrist × Sleep onset (within, prior)²	0.03 [0.00, 0.06]	.033	0.09 [0.07, 0.12]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Wrist × Sleep onset (between)	−0.01 [−0.13, 0.10]	.804	−0.03 [−0.14, 0.07]	.521
Wrist × Sleep onset (between)²	0.08 [0.02, 0.15]	.012	0.00 [−0.06, 0.06]	.906
Sleep regularity (z)				
(Intercept)	−1.01 [−1.38, −0.64]	< .001	−0.54 [−0.75, −0.32]	< .001
Sleep regularity (within, prior)	0.02 [0.01, 0.03]	< .001	0.03 [0.02, 0.04]	< .001
Wrist	1.30 [0.88, 1.72]	< .001	0.46 [0.22, 0.70]	< .001
Sleep regularity (within, prior)²	0.00 [−0.01, 0.01]	.468	0.01 [0.00, 0.02]	.014
Sleep regularity (between)	0.09 [0.07, 0.11]	< .001	0.06 [0.04, 0.08]	< .001
Sleep regularity (between)²	−0.02 [−0.04, 0.00]	.061	−0.02 [−0.04, 0.00]	.015
Sleep regularity (within, prior) × Wrist	−0.05 [−0.06, −0.03]	< .001	−0.04 [−0.05, −0.02]	< .001
Wrist × Sleep regularity (within, prior)²	0.00 [−0.02, 0.01]	.414	−0.02 [−0.03, −0.01]	.005
Wrist × Sleep regularity (between)	0.12 [0.09, 0.14]	< .001	0.00 [−0.02, 0.03]	.713
Wrist × Sleep regularity (between)²	0.00 [−0.02, 0.03]	.713	0.00 [−0.02, 0.02]	.662

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

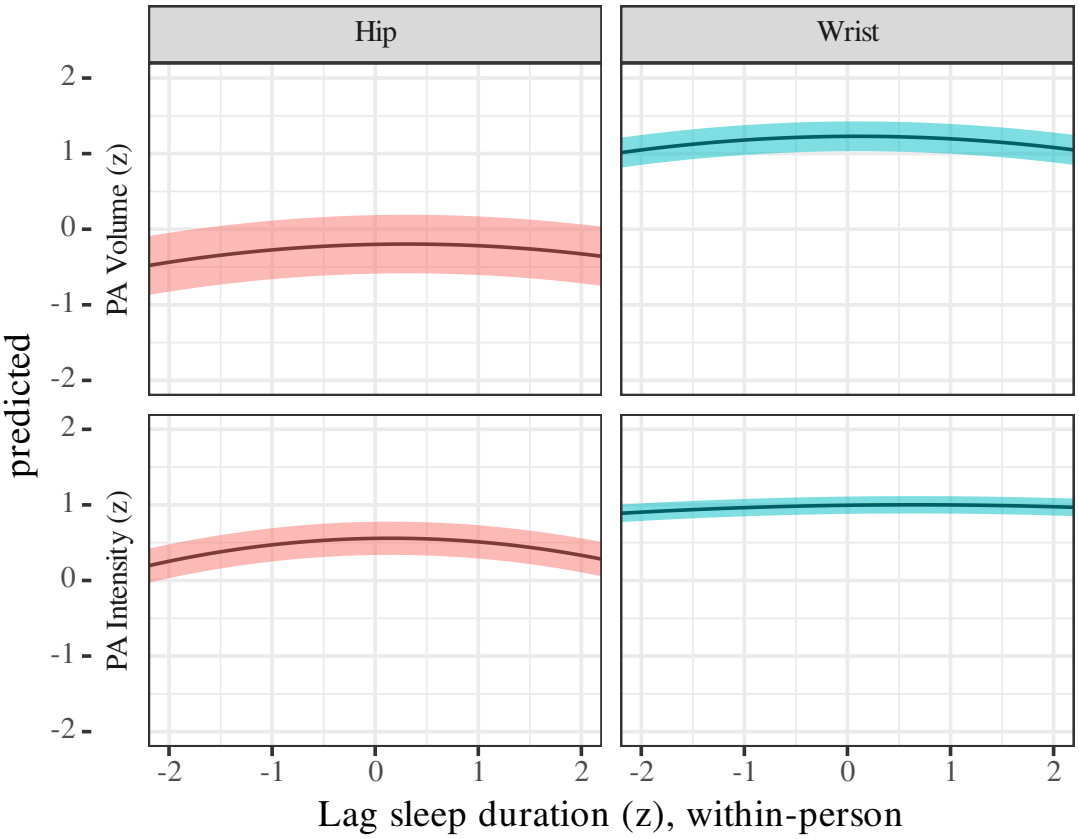


Figure 73: Physical activity by sleep duration moderated by wear location

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 35 and Figure 74.

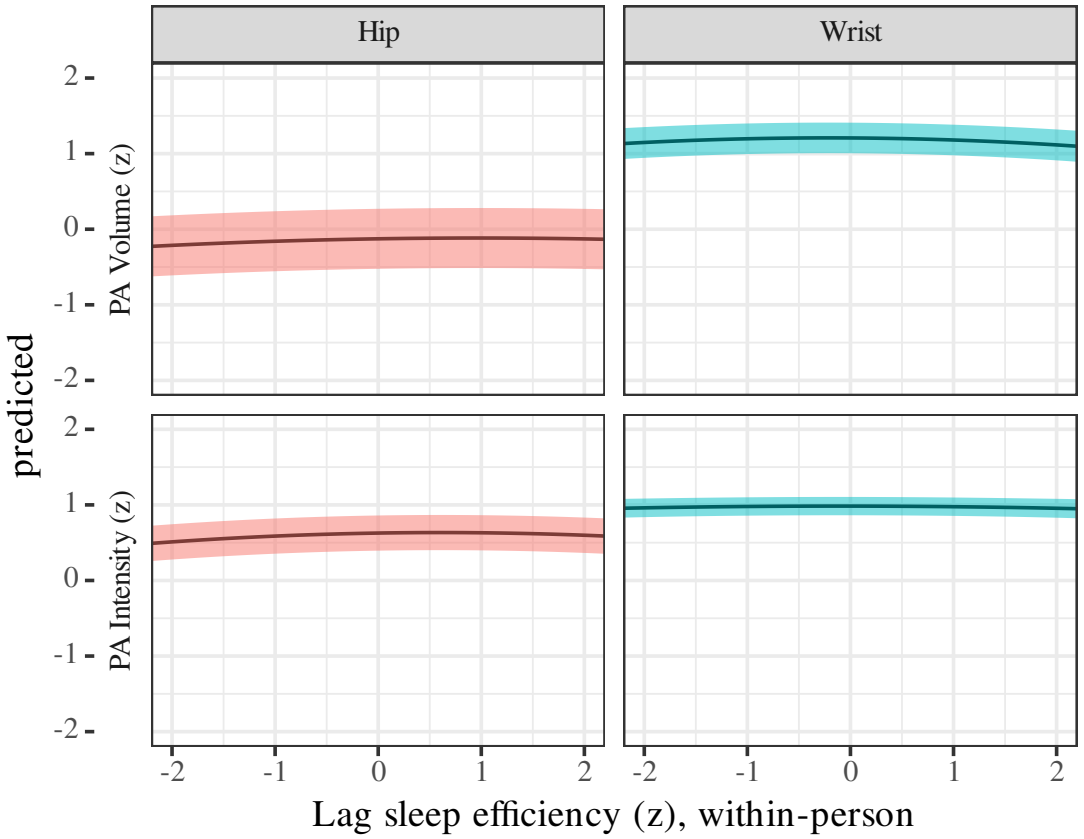


Figure 74: Physical activity by sleep efficiency moderated by wear location

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by wear location. Results, controlling for SES, age, sex and BMI z-score are presented in Table 35 and Figure 75.

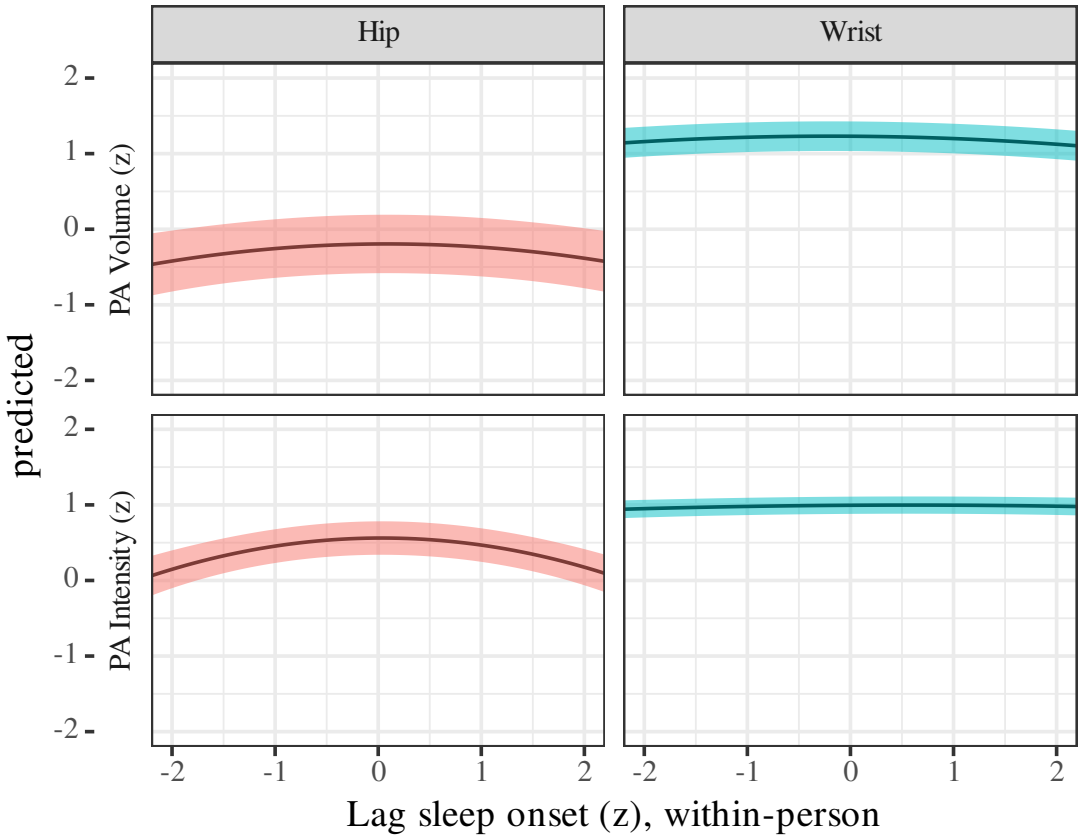


Figure 75: Physical activity by sleep onset moderated by wear location

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by wear location. Results are presented in Table 35 and Figure 76.

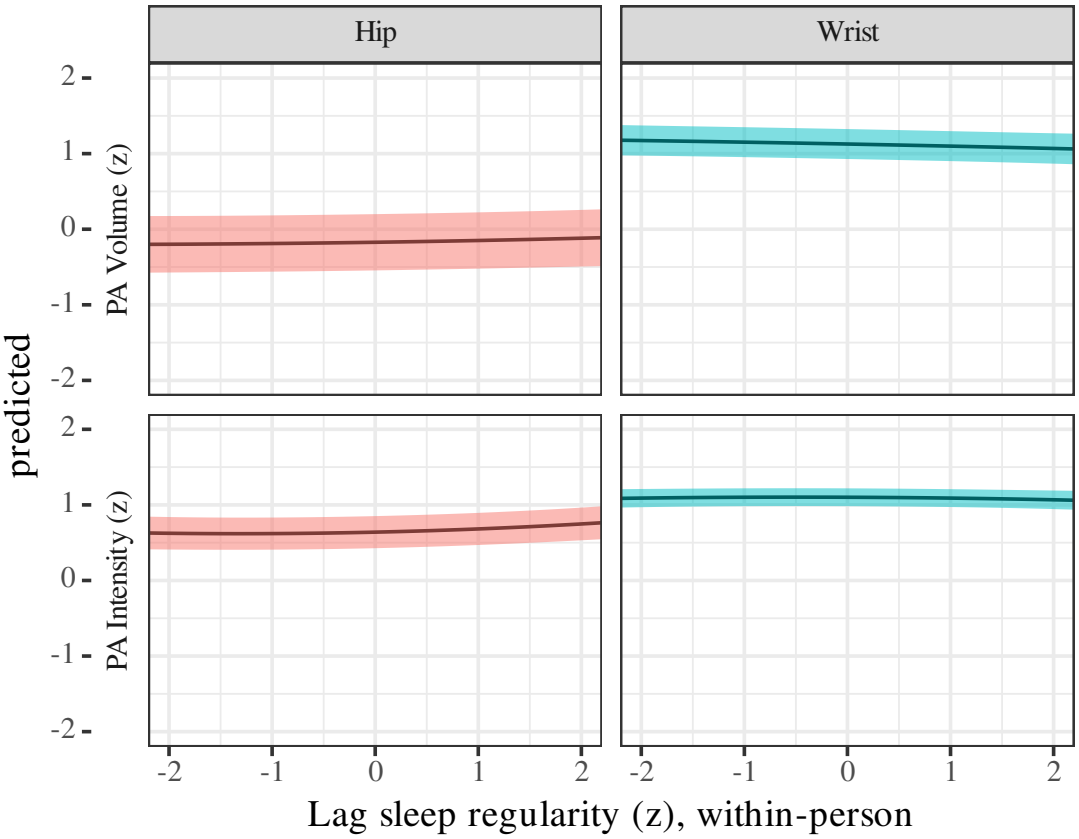


Figure 76: Physical activity by sleep regularity moderated by wear location

most active hour

The association between physical activity volume and sleep by most active hour

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by most active hour, and the results are presented in Table 36 and Figure 77

Table 36: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.07 [−0.09, 0.22]	.404	0.08 [−0.06, 0.23]	.265
Physical activity (within)	0.06 [0.03, 0.09]	< .001	0.03 [0.00, 0.06]	.052
Most active hour	0.00 [−0.01, 0.00]	< .001	−0.01 [−0.01, 0.00]	< .001
Physical activity (within)²	0.03 [0.00, 0.06]	.027	0.03 [0.00, 0.06]	.041
Physical activity (between)	0.02 [−0.01, 0.05]	.242	0.12 [0.09, 0.14]	< .001
Physical activity (between)²	0.03 [0.01, 0.04]	< .001	0.01 [0.00, 0.03]	.154
Physical activity (within) × Most active hour	0.00 [0.00, 0.01]	.002	0.00 [0.00, 0.00]	.024
Most active hour × Physical activity (within)²	0.00 [0.00, 0.00]	.006	0.00 [0.00, 0.00]	.012
Most active hour × Physical activity (between)	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.205
Most active hour × Physical activity (between)²	0.00 [0.00, 0.00]	.012	0.00 [0.00, 0.00]	.071
Sleep efficiency (z)				
(Intercept)	0.21 [0.05, 0.36]	.008	0.22 [0.05, 0.38]	.009
Physical activity (within)	−0.08 [−0.11, −0.05]	< .001	−0.07 [−0.10, −0.04]	< .001
Most active hour	0.00 [0.00, 0.00]	.308	0.00 [0.00, 0.00]	.009
Physical activity (within)²	−0.02 [−0.04, 0.01]	.273	0.01 [−0.02, 0.04]	.368
Physical activity (between)	0.03 [0.00, 0.06]	.024	0.02 [0.00, 0.04]	.084

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Physical activity (between)²	−0.02 [−0.04, −0.01]	.003	−0.04 [−0.06, −0.02]	< .001
Physical activity (within) × Most active hour	0.01 [0.00, 0.01]	< .001	0.01 [0.00, 0.01]	< .001
Most active hour × Physical activity (within)²	0.00 [0.00, 0.00]	.926	0.00 [−0.01, 0.00]	.002
Most active hour × Physical activity (between)	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.181
Most active hour × Physical activity (between)²	0.00 [0.00, 0.00]	.036	0.00 [0.00, 0.00]	.001
Sleep onset (z)				
(Intercept)	−0.04 [−0.31, 0.23]	.761	−0.02 [−0.28, 0.23]	.853
Physical activity (within)	−0.17 [−0.19, −0.14]	< .001	−0.09 [−0.12, −0.07]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.01 [0.01, 0.01]	< .001
Physical activity (within)²	−0.02 [−0.04, 0.00]	.088	0.05 [0.03, 0.07]	< .001
Physical activity (between)	−0.17 [−0.19, −0.14]	< .001	−0.14 [−0.16, −0.12]	< .001
Physical activity (between)²	−0.01 [−0.02, 0.00]	.129	−0.05 [−0.07, −0.04]	< .001
Physical activity (within) × Most active hour	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.266
Most active hour × Physical activity (within)²	0.00 [0.00, 0.00]	.008	0.00 [0.00, 0.00]	< .001
Most active hour × Physical activity (between)	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	< .001

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Most active hour × Physical activity (between)²	0.00 [0.00, 0.00]	.744	0.00 [0.00, 0.00]	.003
Sleep regularity (z)				
(Intercept)	0.28 [0.14, 0.42]	< .001	0.26 [0.09, 0.43]	.003
Physical activity (within)	0.15 [0.12, 0.19]	< .001	0.03 [0.00, 0.06]	.053
Most active hour	−0.01 [−0.01, −0.01]	< .001	−0.01 [−0.01, 0.00]	< .001
Physical activity (within)²	−0.18 [−0.21, −0.15]	< .001	−0.15 [−0.18, −0.11]	< .001
Physical activity (between)	0.46 [0.43, 0.50]	< .001	0.12 [0.09, 0.15]	< .001
Physical activity (between)²	−0.10 [−0.12, −0.08]	< .001	−0.05 [−0.07, −0.03]	< .001
Physical activity (within) × Most active hour	0.00 [0.00, 0.00]	.086	0.01 [0.00, 0.01]	< .001
Most active hour × Physical activity (within)²	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.583
Most active hour × Physical activity (between)	0.00 [0.00, 0.00]	.121	0.00 [0.00, 0.00]	.044
Most active hour × Physical activity (between)²	0.00 [0.00, 0.00]	.923	0.00 [0.00, 0.00]	.006

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

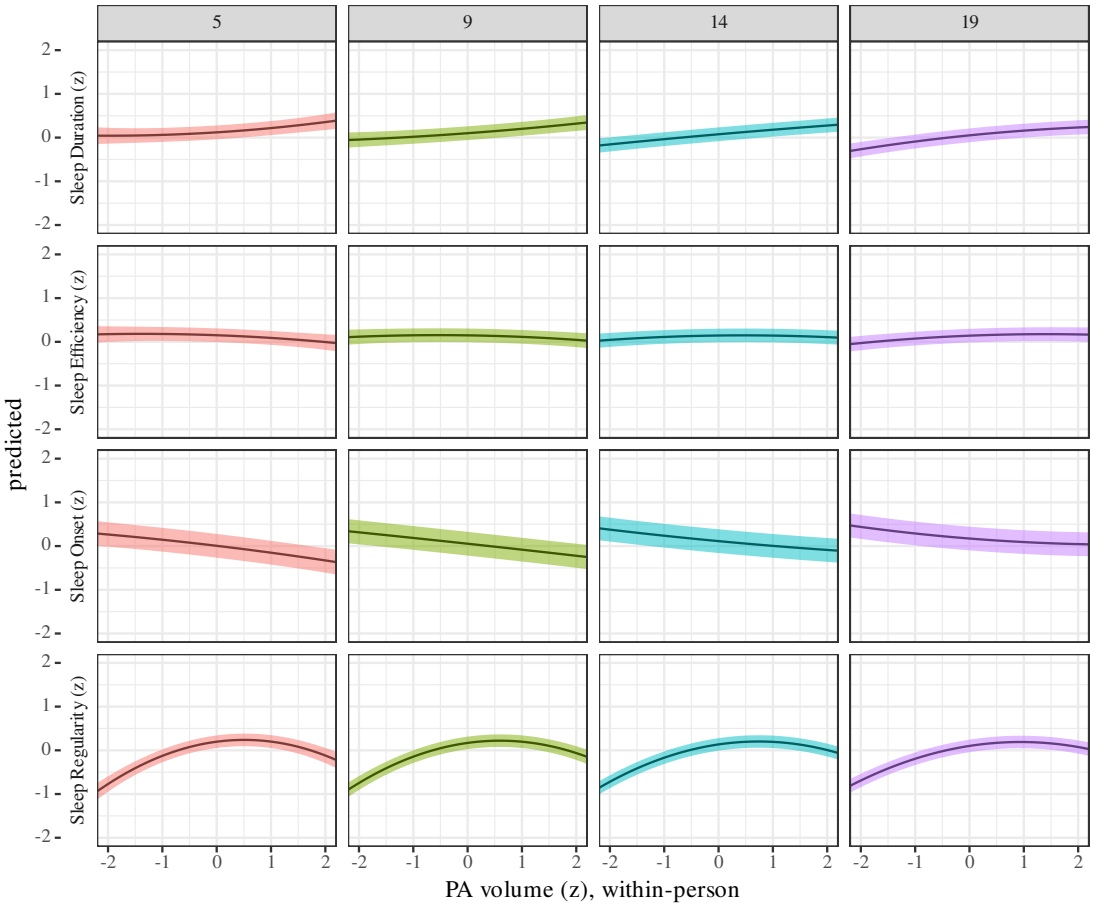


Figure 77: Sleep metrics on Physical activity volume by most active hour

The association between physical activity intensity and sleep by most active hour

We estimated how physical activity intensity is associated with sleep across most active hour. We present the results in Table 36 and Figure 78.

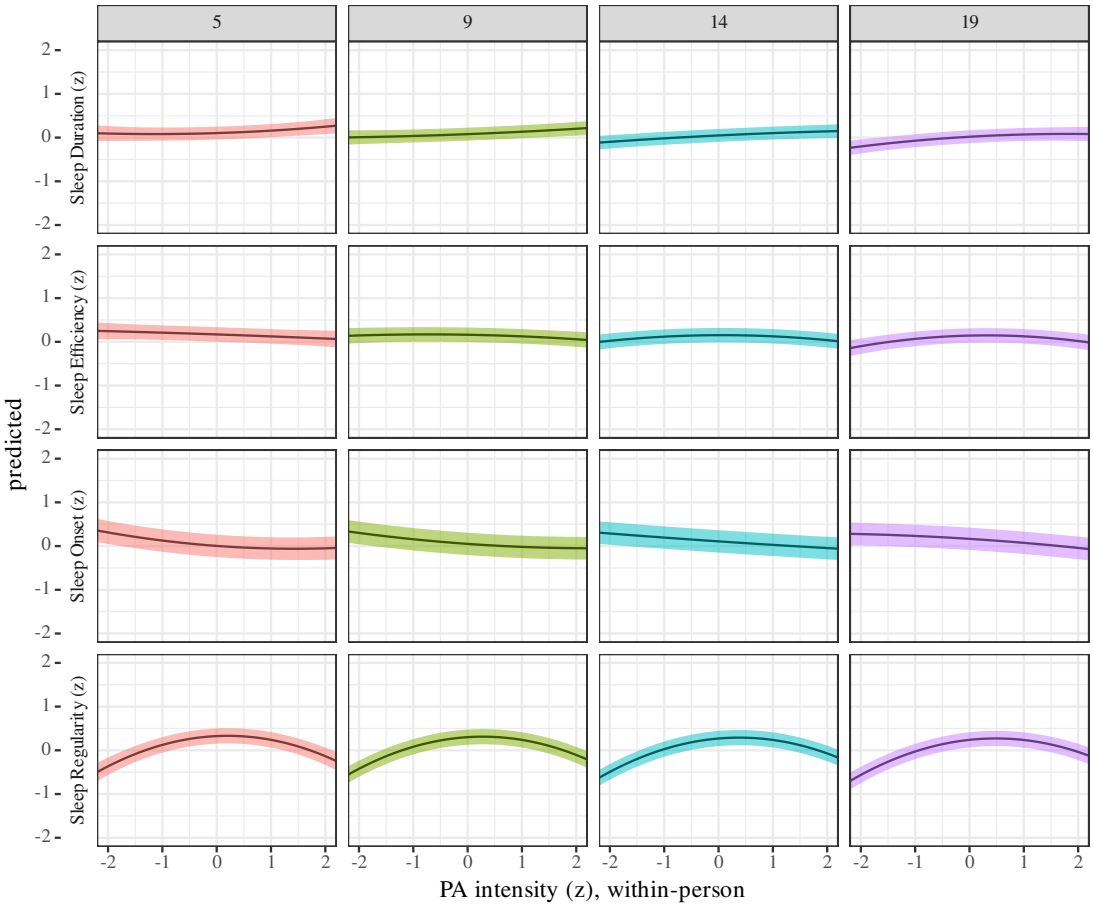


Figure 78: Sleep metrics on Physical activity intensity moderated by most active hour

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 37 and Figure 79.

Table 37: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.03 [−0.27, 0.33]	.846	−0.09 [−0.21, 0.03]	.155
Sleep duration (within, prior)	−0.02 [−0.03, 0.00]	.060	0.03 [0.02, 0.05]	< .001
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	.005
Sleep duration (within, prior)²	−0.01 [−0.02, 0.00]	.049	−0.02 [−0.03, −0.01]	.005
Sleep duration (between)	0.05 [0.03, 0.07]	< .001	0.12 [0.10, 0.14]	< .001
Sleep duration (between)²	0.00 [−0.02, 0.02]	.812	0.00 [−0.02, 0.02]	.993
Sleep duration (within, prior) × Most active hour	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.064
Most active hour × Sleep duration (within, prior)²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	.438
Most active hour × Sleep duration (between)	0.00 [0.00, 0.00]	.147	0.00 [−0.01, 0.00]	< .001
Most active hour × Sleep duration (between)²	0.00 [0.00, 0.00]	.033	0.00 [0.00, 0.00]	.460
Sleep efficiency (z)				
(Intercept)	0.02 [−0.27, 0.32]	.878	−0.10 [−0.22, 0.03]	.126
Sleep efficiency (within, prior)	0.01 [−0.01, 0.03]	.219	0.00 [−0.02, 0.02]	.964
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	.002

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep efficiency (within, prior)²	0.01 [−0.01, 0.02]	.429	0.00 [−0.02, 0.01]	.594
Sleep efficiency (between)	−0.03 [−0.05, −0.01]	.003	−0.07 [−0.09, −0.05]	< .001
Sleep efficiency (between)²	0.00 [−0.01, 0.02]	.632	−0.01 [−0.03, 0.01]	.338
Sleep efficiency (within, prior) × Most active hour	0.00 [0.00, 0.00]	.666	0.00 [0.00, 0.00]	.255
Most active hour × Sleep efficiency (within, prior)²	0.00 [0.00, 0.00]	.003	0.00 [0.00, 0.00]	.110
Most active hour × Sleep efficiency (between)	0.00 [0.00, 0.00]	< .001	0.01 [0.00, 0.01]	< .001
Most active hour × Sleep efficiency (between)²	0.00 [0.00, 0.00]	.001	0.00 [0.00, 0.00]	.847
Sleep onset (z)				
(Intercept)	0.02 [−0.30, 0.34]	.908	−0.11 [−0.24, 0.01]	.080
Sleep onset (within, prior)	−0.03 [−0.05, −0.01]	.002	0.01 [−0.01, 0.03]	.275
Most active hour	0.01 [0.01, 0.01]	< .001	0.00 [0.00, 0.00]	< .001
Sleep onset (within, prior)²	−0.02 [−0.04, −0.01]	< .001	0.00 [−0.01, 0.01]	.859
Sleep onset (between)	−0.14 [−0.16, −0.12]	< .001	−0.15 [−0.17, −0.14]	< .001
Sleep onset (between)²	0.03 [0.02, 0.05]	< .001	0.03 [0.02, 0.04]	< .001
Sleep onset (within, prior) × Most active hour	0.00 [0.00, 0.00]	.016	0.00 [0.00, 0.00]	.441
Most active hour × Sleep onset (within, prior)²	0.00 [0.00, 0.00]	.840	0.00 [0.00, 0.00]	.068

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Most active hour × Sleep onset (between)	0.00 [0.00, 0.00]	.012	0.01 [0.00, 0.01]	< .001
Most active hour × Sleep onset (between)²	0.00 [0.00, 0.00]	< .001	0.00 [0.00, 0.00]	< .001
Sleep regularity (z)				
(Intercept)	−0.07 [−0.36, 0.23]	.664	−0.17 [−0.30, −0.04]	.010
Sleep regularity (within, prior)	0.02 [0.00, 0.04]	.092	0.02 [0.00, 0.04]	.080
Most active hour	0.01 [0.00, 0.01]	< .001	0.00 [0.00, 0.00]	.767
Sleep regularity (within, prior)²	0.01 [−0.01, 0.03]	.393	0.00 [−0.02, 0.02]	.852
Sleep regularity (between)	0.11 [0.09, 0.14]	< .001	0.02 [−0.01, 0.04]	.143
Sleep regularity (between)²	0.01 [−0.02, 0.03]	.611	−0.02 [−0.04, 0.00]	.042
Sleep regularity (within, prior) × Most active hour	0.00 [0.00, 0.00]	.023	0.00 [0.00, 0.00]	.380
Most active hour × Sleep regularity (within, prior)²	0.00 [0.00, 0.00]	.431	0.00 [0.00, 0.00]	.690
Most active hour × Sleep regularity (between)	0.00 [0.00, 0.01]	< .001	0.00 [0.00, 0.01]	< .001
Most active hour × Sleep regularity (between)²	0.00 [0.00, 0.00]	.561	0.00 [0.00, 0.00]	.533

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

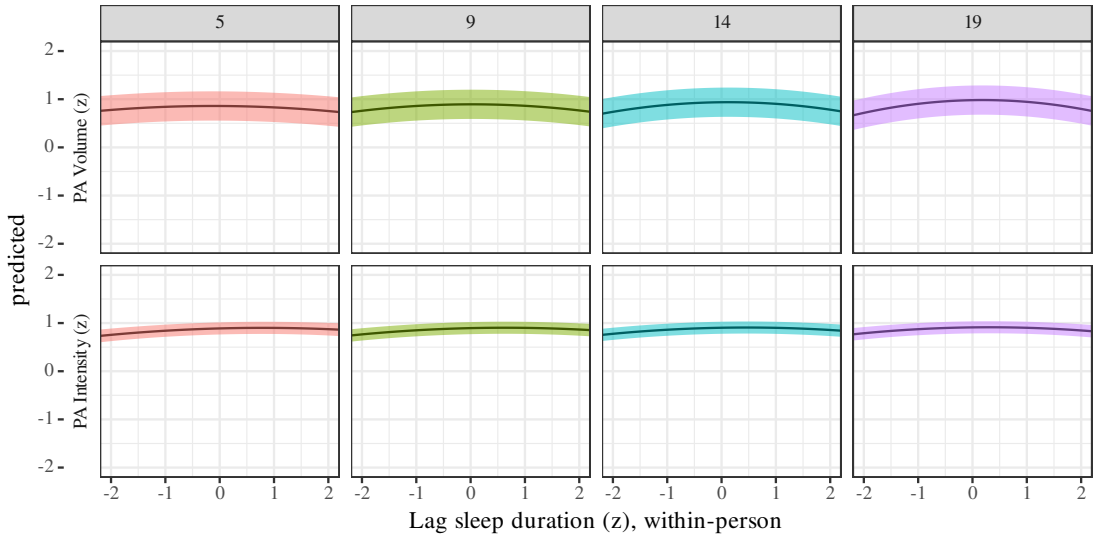


Figure 79: Physical activity by sleep duration moderated by most active hour

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 37 and Figure 80.

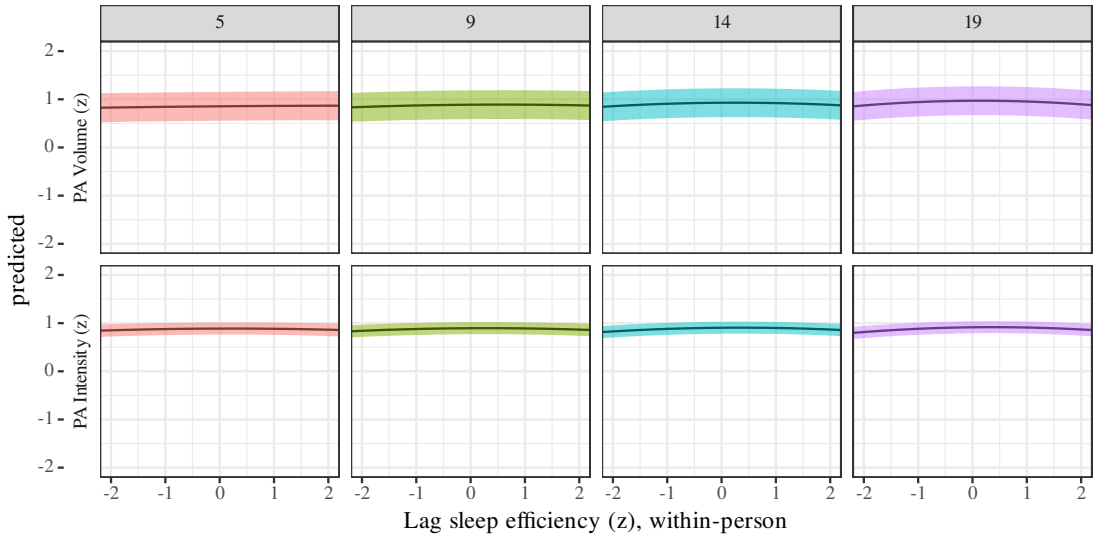


Figure 80: Physical activity by sleep efficiency moderated by most active hour

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by most active hour. Results, controlling for SES, age, sex and BMI z-score are presented in Table 37 and Figure 81.

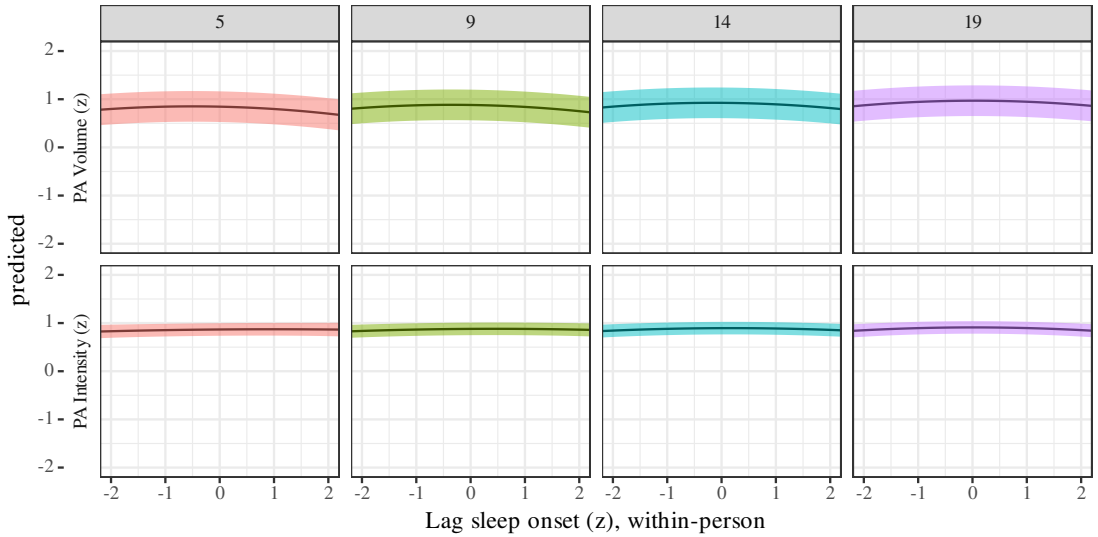


Figure 81: Physical activity by sleep onset moderated by most active hour

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by most active hour. Results are presented in Table 37 and Figure 82.

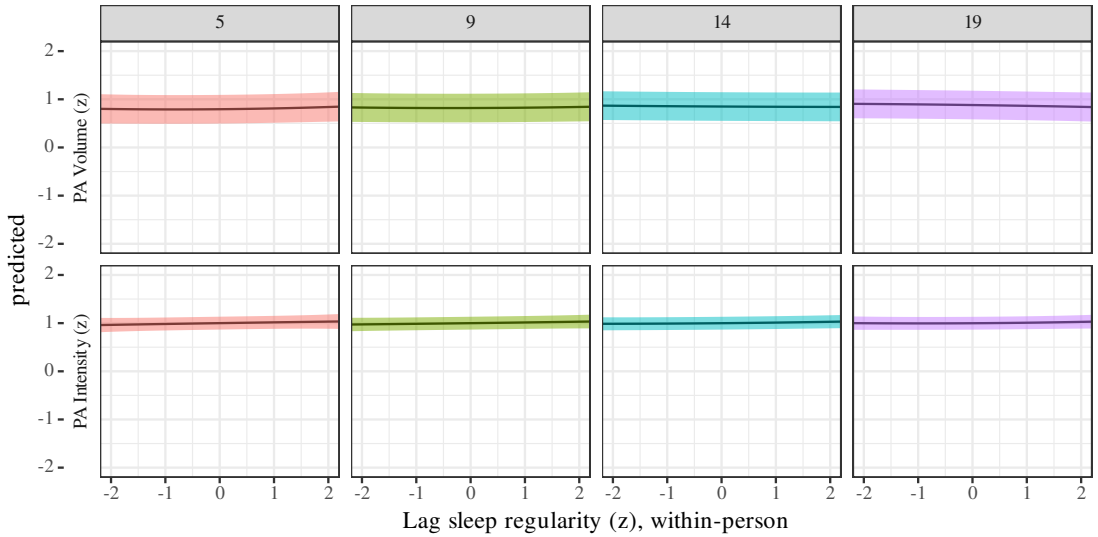


Figure 82: Physical activity by sleep regularity moderated by most active hour

ethnicity

The association between physical activity volume and sleep by ethnicity

We estimated the associations between physical activity and sleep (RQ1) using mixed-effects models. We estimated the association between physical activity volume and sleep by ethnicity, and the results are presented in Table 38 and Figure 83

Table 38: Physical activity predicting sleep controlling for SES, age, sex, and BMI z-score.

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.06 [−0.09, 0.22]	.416	0.07 [−0.08, 0.22]	.338
Physical activity (within)	0.12 [0.11, 0.13]	< .001	0.07 [0.06, 0.08]	< .001
Non-white	−0.13 [−0.16, −0.10]	< .001	−0.18 [−0.22, −0.15]	< .001
Unclear	−0.11 [−0.14, −0.08]	< .001	−0.11 [−0.14, −0.08]	< .001
Physical activity (within)²	−0.02 [−0.03, −0.01]	< .001	−0.01 [−0.02, 0.00]	.043
Physical activity (between)	0.05 [0.03, 0.07]	< .001	0.09 [0.08, 0.11]	< .001
Physical activity (between)²	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.03]	.033
Physical activity (within) × Non-white	0.03 [0.00, 0.06]	.039	0.01 [−0.02, 0.04]	.429
Physical activity (within) × Unclear	−0.05 [−0.06, −0.03]	< .001	−0.03 [−0.05, −0.02]	< .001
Non-white × Physical activity (within)²	0.02 [−0.01, 0.05]	.118	0.01 [−0.03, 0.04]	.705
Unclear × Physical activity (within)²	0.02 [0.00, 0.04]	.011	0.01 [−0.01, 0.03]	.197
Non-white × Physical activity (between)	0.09 [0.06, 0.13]	< .001	0.07 [0.04, 0.10]	< .001
Unclear × Physical activity (between)	0.01 [−0.02, 0.04]	.629	0.00 [−0.02, 0.03]	.772
Non-white × Physical activity (between)²	−0.02 [−0.04, 0.00]	.038	0.02 [−0.01, 0.05]	.221

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Unclear × Physical activity (between)²	−0.01 [−0.02, 0.00]	.212	0.02 [0.00, 0.04]	.019
Sleep efficiency (z)				
(Intercept)	0.21 [0.06, 0.36]	.007	0.20 [0.04, 0.36]	.013
Physical activity (within)	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.01]	.301
Non-white	−0.06 [−0.09, −0.02]	.002	−0.13 [−0.16, −0.09]	< .001
Unclear	−0.02 [−0.05, 0.02]	.309	0.01 [−0.03, 0.04]	.764
Physical activity (within)²	−0.03 [−0.04, −0.02]	< .001	−0.03 [−0.05, −0.02]	< .001
Physical activity (between)	0.09 [0.06, 0.11]	< .001	0.04 [0.02, 0.06]	< .001
Physical activity (between)²	−0.03 [−0.04, −0.02]	< .001	−0.02 [−0.03, −0.01]	.003
Physical activity (within) × Non-white	0.04 [0.01, 0.07]	.004	0.02 [−0.01, 0.04]	.156
Physical activity (within) × Unclear	−0.01 [−0.03, 0.00]	.081	−0.02 [−0.03, 0.00]	.028
Non-white × Physical activity (within)²	−0.01 [−0.04, 0.01]	.322	−0.01 [−0.04, 0.02]	.421
Unclear × Physical activity (within)²	0.03 [0.01, 0.04]	< .001	0.02 [0.00, 0.04]	.083
Non-white × Physical activity (between)	0.08 [0.04, 0.12]	< .001	0.12 [0.08, 0.15]	< .001
Unclear × Physical activity (between)	−0.01 [−0.04, 0.02]	.533	−0.04 [−0.06, −0.01]	.016

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Non-white × Physical activity (between)²	−0.03 [−0.05, 0.00]	.021	−0.01 [−0.04, 0.02]	.560
Unclear × Physical activity (between)²	0.00 [−0.01, 0.02]	.631	0.01 [−0.01, 0.02]	.441
Sleep onset (z)				
(Intercept)	0.09 [−0.18, 0.36]	.505	0.10 [−0.16, 0.35]	.454
Physical activity (within)	−0.13 [−0.14, −0.13]	< .001	−0.09 [−0.10, −0.08]	< .001
Non-white	0.07 [0.04, 0.10]	< .001	0.08 [0.05, 0.11]	< .001
Unclear	0.05 [0.02, 0.08]	< .001	0.07 [0.05, 0.10]	< .001
Physical activity (within)²	0.02 [0.01, 0.03]	< .001	0.01 [0.00, 0.02]	.009
Physical activity (between)	−0.10 [−0.12, −0.08]	< .001	−0.07 [−0.08, −0.05]	< .001
Physical activity (between)²	0.00 [−0.01, 0.01]	.717	−0.02 [−0.03, −0.01]	< .001
Physical activity (within) × Non-white	0.00 [−0.02, 0.02]	.864	0.01 [−0.01, 0.03]	.238
Physical activity (within) × Unclear	0.07 [0.05, 0.08]	< .001	0.01 [0.00, 0.02]	.131
Non-white × Physical activity (within)²	−0.01 [−0.03, 0.01]	.375	−0.02 [−0.05, 0.00]	.039
Unclear × Physical activity (within)²	−0.01 [−0.02, 0.00]	.048	−0.01 [−0.02, 0.01]	.421
Non-white × Physical activity (between)	0.01 [−0.02, 0.04]	.395	−0.03 [−0.06, −0.01]	.018

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Unclear × Physical activity (between)	0.00 [−0.03, 0.02]	.788	−0.07 [−0.09, −0.05]	< .001
Non-white × Physical activity (between)²	−0.02 [−0.03, 0.00]	.069	−0.02 [−0.04, 0.01]	.208
Unclear × Physical activity (between)²	−0.01 [−0.02, 0.00]	.227	−0.02 [−0.04, −0.01]	.001
Sleep regularity (z)				
(Intercept)	0.25 [0.10, 0.40]	< .001	0.22 [0.04, 0.40]	.014
Physical activity (within)	0.21 [0.20, 0.23]	< .001	0.13 [0.11, 0.14]	< .001
Non-white	0.03 [−0.01, 0.06]	.179	0.06 [0.02, 0.10]	.005
Unclear	−0.11 [−0.15, −0.08]	< .001	−0.09 [−0.13, −0.05]	< .001
Physical activity (within)²	−0.12 [−0.13, −0.11]	< .001	−0.13 [−0.14, −0.12]	< .001
Physical activity (between)	0.49 [0.46, 0.52]	< .001	0.13 [0.10, 0.15]	< .001
Physical activity (between)²	−0.10 [−0.11, −0.09]	< .001	−0.01 [−0.03, 0.00]	.163
Physical activity (within) × Non-white	−0.05 [−0.08, −0.03]	< .001	−0.01 [−0.03, 0.02]	.690
Physical activity (within) × Unclear	−0.06 [−0.08, −0.05]	< .001	−0.07 [−0.08, −0.05]	< .001
Non-white × Physical activity (within)²	−0.01 [−0.04, 0.02]	.598	−0.07 [−0.10, −0.03]	< .001
Unclear × Physical activity (within)²	0.01 [−0.01, 0.02]	.462	−0.01 [−0.03, 0.01]	.282

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Non-white × Physical activity (between)	−0.07 [−0.11, −0.03]	< .001	0.17 [0.13, 0.20]	< .001
Unclear × Physical activity (between)	0.02 [−0.02, 0.05]	.398	0.01 [−0.02, 0.05]	.528
Non-white × Physical activity (between)²	0.01 [−0.01, 0.04]	.260	−0.13 [−0.17, −0.09]	< .001
Unclear × Physical activity (between)²	−0.01 [−0.02, 0.01]	.282	−0.01 [−0.03, 0.01]	.218

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by sleep outcome; columns give the physical activity predictor. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

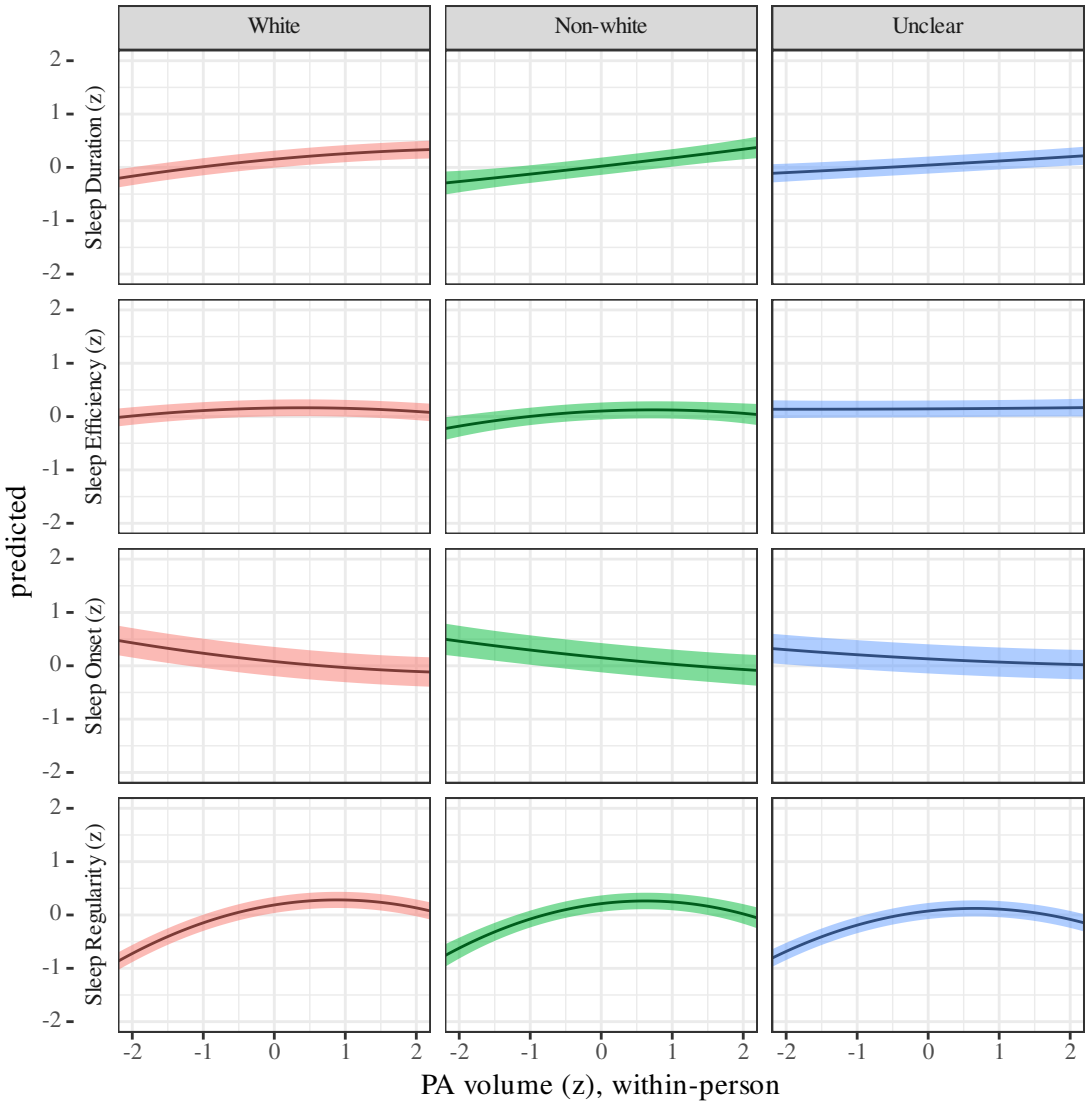


Figure 83: Sleep metrics on Physical activity volume by ethnicity

The association between physical activity intensity and sleep by ethnicity

We estimated how physical activity intensity is associated with sleep across ethnicity. We present the results in Table 38 and Figure 84.

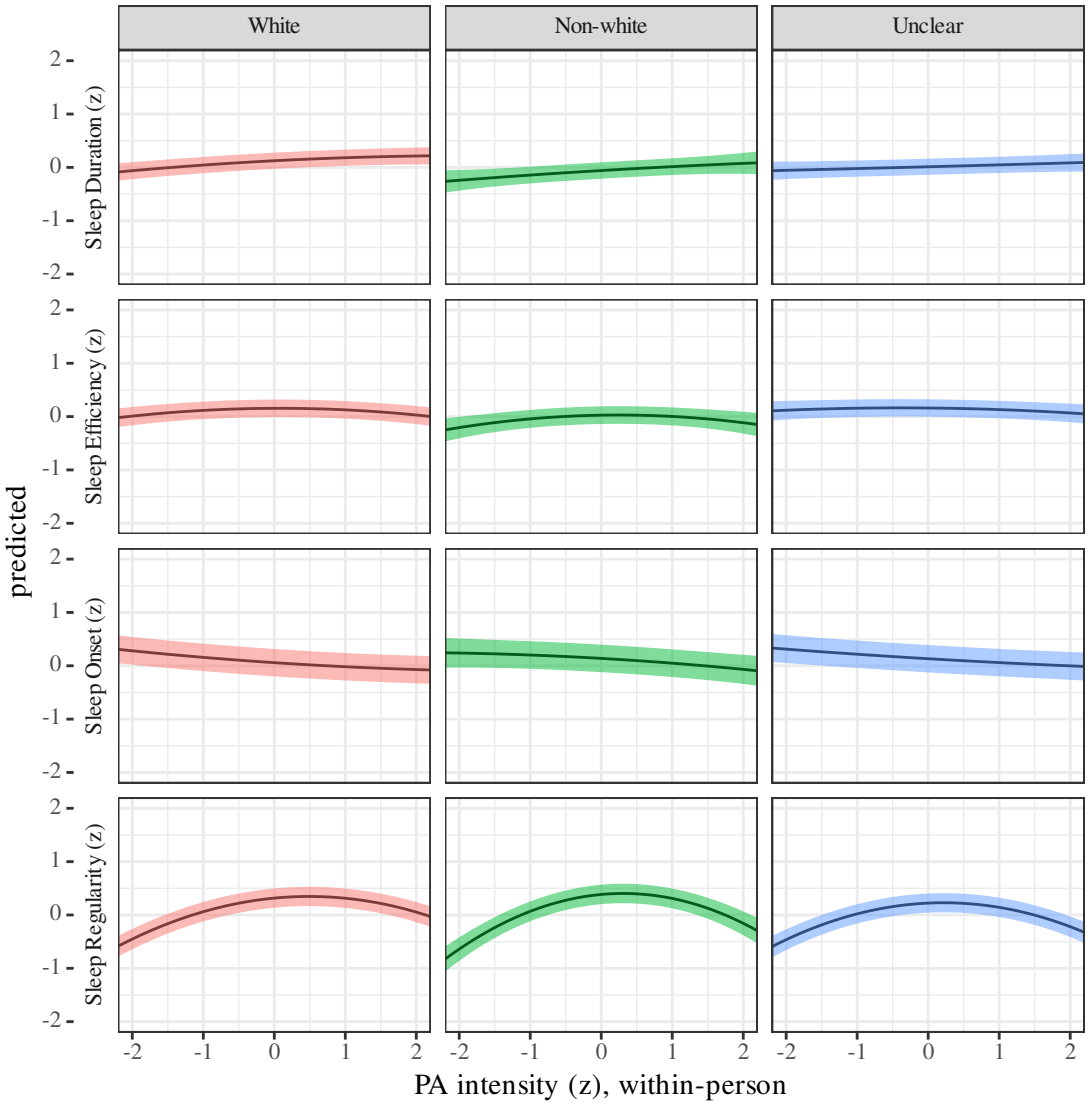


Figure 84: Sleep metrics on Physical activity intensity moderated by ethnicity

The association between sleep duration and next-day physical activity

We estimated the association between sleep duration and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 39 and Figure 85.

Table 39: Sleep predicting physical activity controlling for SES, age, sex, and BMI z-score

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep duration (z)				
(Intercept)	0.14 [−0.17, 0.44]	.376	−0.05 [−0.17, 0.07]	.428
Sleep duration (within, prior)	0.02 [0.01, 0.02]	< .001	0.03 [0.03, 0.04]	< .001
Non-white	−0.04 [−0.07, −0.01]	.020	−0.05 [−0.08, −0.02]	< .001
Unclear	0.03 [0.00, 0.06]	.067	−0.03 [−0.06, 0.00]	.031
Sleep duration (within, prior)²	−0.04 [−0.04, −0.03]	< .001	−0.02 [−0.02, −0.01]	< .001
Sleep duration (between)	0.06 [0.05, 0.08]	< .001	0.07 [0.06, 0.09]	< .001
Sleep duration (between)²	−0.02 [−0.04, −0.01]	< .001	−0.01 [−0.02, 0.00]	.159
Sleep duration (within, prior) × Non-white	−0.01 [−0.03, 0.00]	.068	−0.04 [−0.06, −0.03]	< .001
Sleep duration (within, prior) × Unclear	−0.02 [−0.03, −0.01]	< .001	−0.02 [−0.03, −0.01]	< .001
Non-white × Sleep duration (within, prior)²	0.00 [−0.02, 0.01]	.722	−0.02 [−0.03, 0.00]	.008
Unclear × Sleep duration (within, prior)²	−0.01 [−0.02, 0.00]	.008	0.00 [−0.01, 0.00]	.415
Non-white × Sleep duration (between)	−0.03 [−0.06, 0.00]	.056	−0.03 [−0.06, 0.00]	.035
Unclear × Sleep duration (between)	−0.01 [−0.03, 0.01]	.387	−0.03 [−0.05, −0.01]	.005

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Non-white × Sleep duration (between)²	0.02 [−0.01, 0.05]	.191	0.04 [0.01, 0.07]	.008
Unclear × Sleep duration (between)²	0.01 [−0.01, 0.02]	.612	−0.01 [−0.03, 0.01]	.392
Sleep efficiency (z)				
(Intercept)	0.13 [−0.17, 0.43]	.392	−0.03 [−0.15, 0.09]	.582
Sleep efficiency (within, prior)	0.01 [0.00, 0.02]	.005	0.02 [0.01, 0.02]	< .001
Non-white	−0.03 [−0.06, 0.00]	.029	−0.04 [−0.06, −0.01]	.003
Unclear	0.01 [−0.01, 0.04]	.312	−0.07 [−0.09, −0.04]	< .001
Sleep efficiency (within, prior)²	−0.02 [−0.02, −0.01]	< .001	−0.02 [−0.02, −0.01]	< .001
Sleep efficiency (between)	0.01 [−0.01, 0.02]	.291	0.02 [0.01, 0.03]	.005
Sleep efficiency (between)²	−0.01 [−0.02, 0.00]	.053	−0.01 [−0.02, 0.00]	.031
Sleep efficiency (within, prior) × Non-white	−0.02 [−0.03, 0.00]	.010	−0.02 [−0.04, −0.01]	< .001
Sleep efficiency (within, prior) × Unclear	0.00 [−0.01, 0.01]	.937	−0.01 [−0.02, 0.00]	.095
Non-white × Sleep efficiency (within, prior)²	0.01 [0.00, 0.02]	.151	0.00 [−0.01, 0.01]	.678
Unclear × Sleep efficiency (within, prior)²	0.00 [−0.01, 0.01]	.441	0.01 [0.00, 0.01]	.170

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Non-white × Sleep efficiency (between)	−0.01 [−0.05, 0.03]	.739	0.04 [0.00, 0.08]	.036
Unclear × Sleep efficiency (between)	−0.03 [−0.05, 0.00]	.033	−0.04 [−0.06, −0.02]	< .001
Non-white × Sleep efficiency (between)²	−0.02 [−0.04, 0.01]	.120	0.01 [−0.01, 0.03]	.501
Unclear × Sleep efficiency (between)²	−0.02 [−0.03, 0.00]	.102	0.00 [−0.01, 0.02]	.828
Sleep onset (z)				
(Intercept)	0.15 [−0.17, 0.46]	.366	−0.05 [−0.18, 0.08]	.454
Sleep onset (within, prior)	0.00 [−0.01, 0.01]	.570	0.01 [0.00, 0.02]	.003
Non-white	−0.01 [−0.05, 0.03]	.569	−0.03 [−0.06, 0.01]	.136
Unclear	−0.01 [−0.04, 0.02]	.468	−0.05 [−0.08, −0.02]	.002
Sleep onset (within, prior)²	−0.03 [−0.04, −0.02]	< .001	−0.01 [−0.02, 0.00]	.002
Sleep onset (between)	−0.12 [−0.14, −0.10]	< .001	−0.09 [−0.10, −0.07]	< .001
Sleep onset (between)²	−0.01 [−0.02, 0.01]	.346	0.02 [0.01, 0.03]	.004
Sleep onset (within, prior) × Non-white	0.06 [0.03, 0.08]	< .001	0.01 [−0.01, 0.03]	.258
Sleep onset (within, prior) × Unclear	−0.03 [−0.04, −0.02]	< .001	−0.03 [−0.04, −0.01]	< .001
Non-white × Sleep onset (within, prior)²	0.00 [−0.02, 0.03]	.801	−0.01 [−0.04, 0.01]	.258

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Unclear × Sleep onset (within, prior)²	0.01 [0.00, 0.02]	.025	0.00 [−0.01, 0.01]	.530
Non-white × Sleep onset (between)	0.07 [0.05, 0.10]	< .001	0.05 [0.02, 0.07]	< .001
Unclear × Sleep onset (between)	−0.01 [−0.03, 0.01]	.343	0.00 [−0.02, 0.02]	.804
Non-white × Sleep onset (between)²	0.00 [−0.02, 0.03]	.743	−0.01 [−0.03, 0.02]	.641
Unclear × Sleep onset (between)²	0.02 [0.00, 0.03]	.038	−0.01 [−0.02, 0.00]	.174
Sleep regularity (z)				
(Intercept)	−0.01 [−0.30, 0.29]	.962	−0.15 [−0.28, −0.02]	.022
Sleep regularity (within, prior)	−0.01 [−0.02, 0.00]	.126	0.02 [0.01, 0.02]	< .001
Non-white	−0.02 [−0.05, 0.01]	.205	−0.04 [−0.07, −0.01]	.003
Unclear	0.05 [0.02, 0.08]	.003	−0.04 [−0.07, −0.01]	.008
Sleep regularity (within, prior)²	−0.01 [−0.02, 0.00]	.186	0.01 [0.00, 0.02]	.193
Sleep regularity (between)	0.17 [0.16, 0.19]	< .001	0.06 [0.05, 0.08]	< .001
Sleep regularity (between)²	0.01 [−0.01, 0.02]	.468	−0.02 [−0.03, −0.01]	.006
Sleep regularity (within, prior) × Non-white	0.01 [−0.01, 0.03]	.323	0.00 [−0.02, 0.02]	.894

Term	PA volume (z) β [95% CI]	PA volume (z) p	PA intensity (z) β [95% CI]	PA intensity (z) p
Sleep regularity (within, prior) × Unclear	0.00 [−0.01, 0.01]	.936	−0.01 [−0.03, 0.00]	.052
Non-white × Sleep regularity (within, prior)²	0.00 [−0.02, 0.02]	.978	−0.01 [−0.03, 0.01]	.176
Unclear × Sleep regularity (within, prior)²	0.01 [0.00, 0.03]	.023	−0.01 [−0.02, 0.01]	.365
Non-white × Sleep regularity (between)	−0.05 [−0.09, −0.02]	.003	0.01 [−0.02, 0.04]	.426
Unclear × Sleep regularity (between)	0.01 [−0.01, 0.04]	.305	0.01 [−0.01, 0.03]	.417
Non-white × Sleep regularity (between)²	−0.03 [−0.06, 0.00]	.071	0.00 [−0.03, 0.03]	.870
Unclear × Sleep regularity (between)²	−0.01 [−0.03, 0.01]	.395	0.00 [−0.02, 0.02]	.747

Adjusted for SES, age, sex, and BMI z-score. Rows are grouped by the previous night's sleep predictor; columns give the physical activity outcome. Variables marked (z) are standardised; ² denotes a quadratic term and × an interaction. (within) is the day's deviation from the participant's mean, (between) is that mean, and (within, prior) is the previous day's deviation (for sleep regularity, the index for the two preceding days).

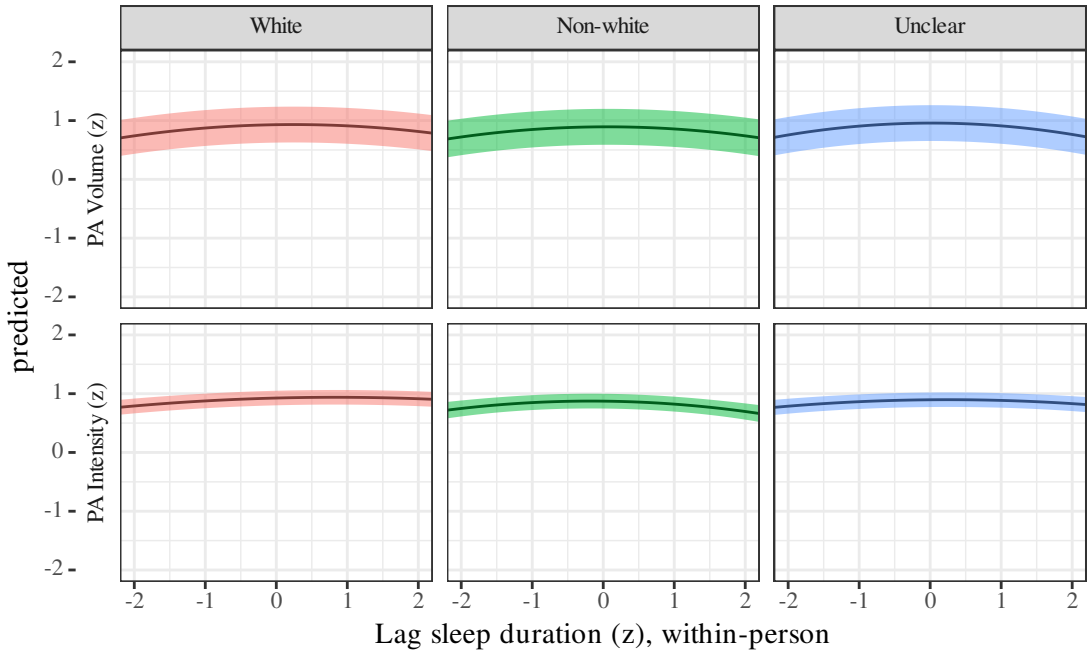


Figure 85: Physical activity by sleep duration moderated by ethnicity

The association between sleep efficiency and next-day physical activity

We estimated the association between sleep efficiency and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 39 and Figure 86.

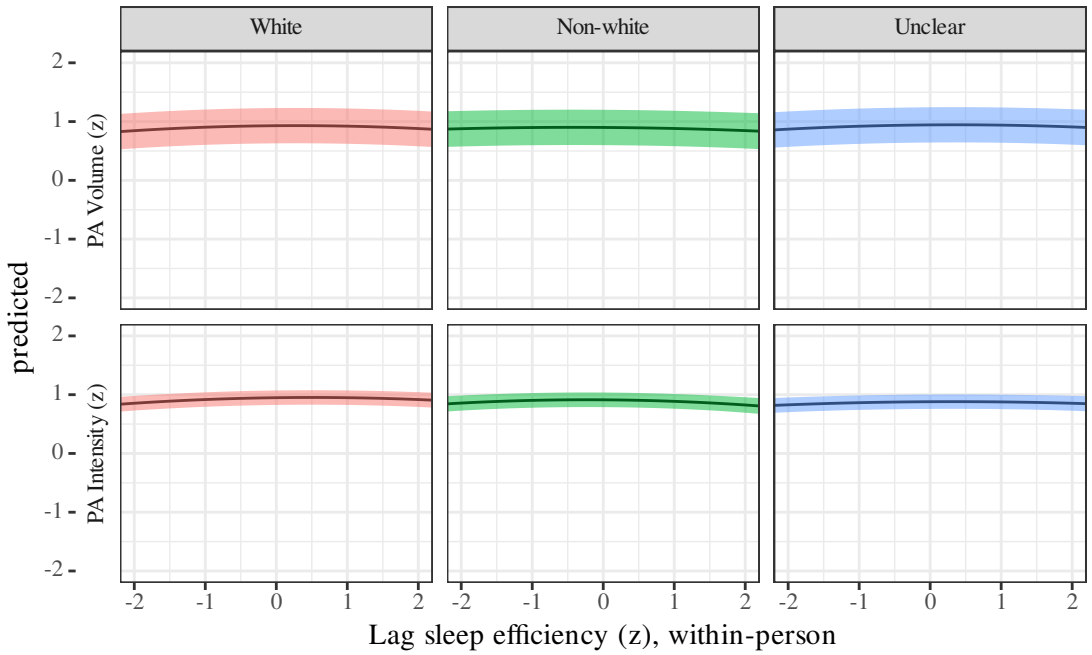


Figure 86: Physical activity by sleep efficiency moderated by ethnicity

The association between sleep onset and next-day physical activity

We estimated the association between sleep onset and next-day physical activity by ethnicity. Results, controlling for SES, age, sex and BMI z-score are presented in Table 39 and Figure 87.

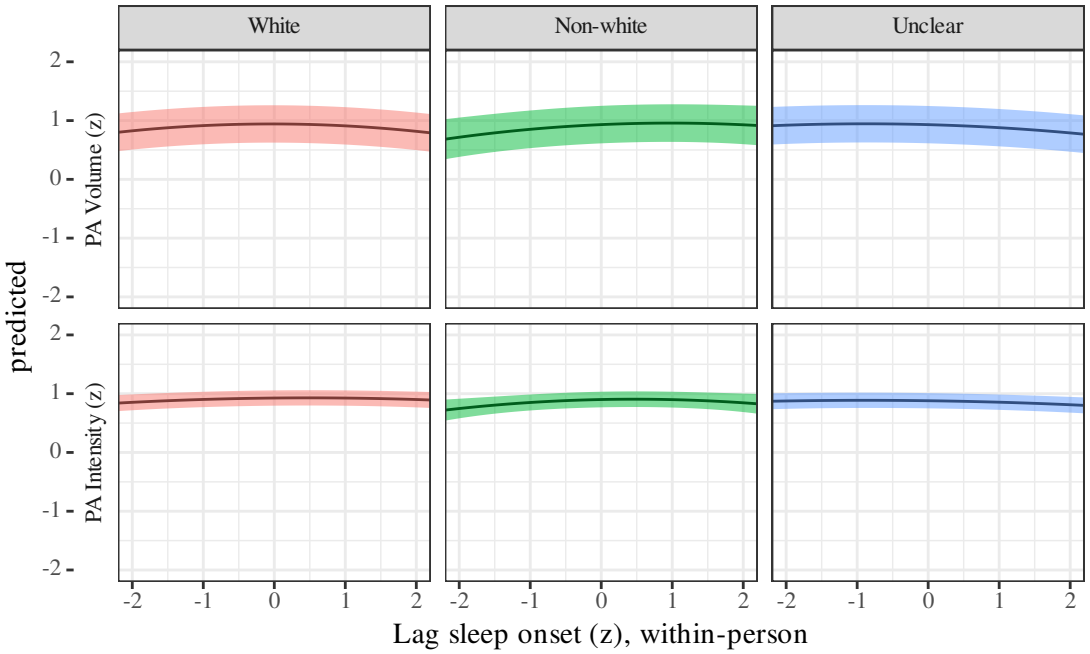


Figure 87: Physical activity by sleep onset moderated by ethnicity

The association between sleep regularity and next-day physical activity

We estimated the association between sleep regularity and next-day physical activity by ethnicity. Results are presented in Table 39 and Figure 88.

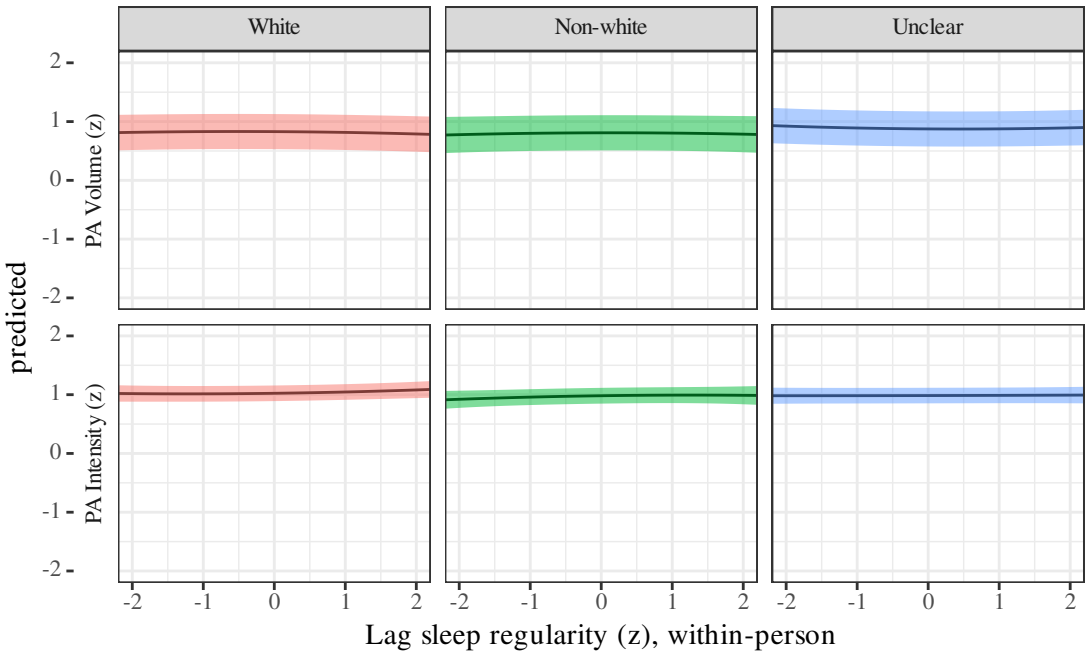


Figure 88: Physical activity by sleep regularity moderated by ethnicity